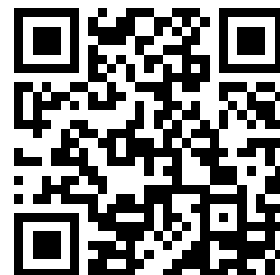

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Guidelines

for

Blood-Material

Interactions

Report of the
National Heart, Lung,
and Blood Institute
Working Group

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES
Public Health Service
National Institutes of Health

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Chapter 1

INTRODUCTION

GENERAL OVERVIEW AND BACKGROUND

It is estimated that 1.8 million cardiovascular procedures are performed annually in the United States. About 500,000 are cardiac catheterizations, 220,000 involve peripheral vascular procedures, and 241,000 involve open heart surgery (1). Virtually every one of these procedures requires short-term blood contact with several different biomaterials, and a large percentage of these procedures involve implantation of a permanent prosthetic device. As a result of the increasing importance of biomaterials in clinical medicine, it would be useful to provide guidelines for both the physicochemical characterization and biological evaluation of materials and devices.

This report is a second edition and consolidation of the Guidelines for Physicochemical Characterization of Biomaterials (NIH Publication No. 80-2186) and the Guidelines for Blood-Material Interactions (NIH Publication No. 80-2185), published in 1980 by the Devices and Technology Branch of the National Heart, Lung, and Blood Institute. The purpose of this publication is to provide the biomaterials community with a compendium of appropriate test methods and related information for characterizing and understanding materials properties and associated blood interactions. Valuable references in the first edition may not have been included here.

The tests outlined in the following paragraphs are not intended to be comprehensive nor should they be considered a dictate from NHLBI. Rather, they should be used as a guide for planning biomaterial research projects and improving the overall quality and consistency of data generated by biomaterials and clinical scientists. Since individuals using this report will come from widely different fields, it may not always be apparent to the user what information is lacking in the test description. For this reason, the reader should always consult the referenced texts and journal articles before attempting to use these test procedures. In many cases, it would not be practical for the user to perform the tests in-house, but investigators should be aware that private-service laboratories are available nationwide. Frequently, they can be found locally in the yellow pages of the telephone book under "laboratories--testing."

Materials characterization can range from simple to complex, correlating generally with increases in cost, sophistication of equipment, and in the expertise necessary for appropriate analysis and interpretation of results. It is clear that not all laboratories are equipped to perform each of these tests and that each test is not necessary for all applications. Therefore, the physicochemical test methods have been divided into categories referred to as level I and level II. Level I tests are generally not difficult to perform, require readily available equipment, and are considered necessary for minimal characterization of

materials and their interaction with blood. These distinguish broad categories of behavior, with low resolution. It is recommended that level I tests be done during the early stages of materials development and continued as a quality assurance measure on new lots of materials.

The remaining test methods that are considered to be well-defined and meaningful are classified as level II tests. We envision that the use of these tests in materials development will be determined by an iterative process. That is, more sensitive and sophisticated characterization procedures would be employed when less sensitive ones fail to discriminate among materials that respond differently to the blood-material interaction tests or fail in distinguishable ways.

Level III tests have been described in some of the chapters of these guidelines. These are tests that show potential for biomaterials applications and probably will be important in the near future. However, at this time, either some controversy exists concerning the methodology or data interpretation, or the test is so complicated and/or expensive that only specialists can perform it.

From a functional point of view, the materials properties to be measured include those related to the bulk chemical and physical characteristics as well as the surface properties of materials. Blood-material interactions tests are recommended for in vitro and in vivo (animal and human) testing. The controversy over appropriate blood tests for materials evaluation will not be solved by the recommendations. However, it is necessary for the biomaterials community to standardize blood testing protocols and to provide appropriate statistical verification for claims of improved "blood compatibility."

The term "blood compatibility" should be used only in a carefully defined context. There are at least two separate coagulation mechanisms in the blood, one that dominates in arterial (high wall shear rate) flow and the other that is important in venous or low wall shear rate flows. Few materials have consistently shown good "blood compatibility" in both blood flow environments. In addition, it is not yet clear if results obtained with one species apply to other species or if results obtained with one test system apply to those obtained in a different test system. Therefore, any statement about blood compatibility should include a description of the conditions under which that statement is known to be valid. For example, to state that "polyethylene is blood compatible" would support the work of some investigators and strongly contradict work by other groups. A more meaningful statement might be "by the criteria of the baboon arteriovenous shunt system, polyethylene (NHLBI reference tubing) is found to be blood compatible," or "by the criteria of the Lee-White test using fresh human blood, polyethylene (NHLBI reference films) is not found to be blood compatible." The word "blood compatible" by itself lacks precision and induces misunderstandings in those who are not familiar with the recent developments in this field. Thus, a useful general guideline would be to avoid using the term "blood compatibility" without clearly placing it in the proper context.

This consolidation edition of the "guidelines" has attempted to provide information on current test methods with added emphasis on the need for a quantitative data base, including statistical analysis. An overview of blood coagulation and thrombosis has been added along with information on the fluid dynamics of blood as it relates to materials interactions. A chapter on reference materials and fabrication methods has also been added.

Other new information includes chapters on fabrics in medicine, pathological evaluation of implants, and calcification. More detailed information is included on the preparation, care, and handling of blood and materials to be exposed to blood. Also, a chapter on pharmacologically active surfaces has been added.

A list of level I and level II tests for physicochemical characterization is provided in table 1.1. Accuracy, reproducibility, and interpretability are the key elements in the value of any measurement. These depend on the method employed, the equipment used, and the investigator's skill and experience. The source of error and the degree of inherent variability in the various physical and biological measurements should be identified if possible. Attempts to correlate the physical data with the biological data should include all appropriate statistical information and should identify assumptions of the statistical treatments employed.

Three interacting factors control normal hemostasis, on the one hand, and a tendency towards excessive thrombosis, on the other. These are 1) the thromboresistance of blood contacting surfaces, 2) the coagulability of the blood, and 3) the presence of smooth, nonturbulent flow. Deviations from the normal physiologic limits in one or more of these three related qualities (Virchow's triad of stasis, endothelial injury, and hypercoagulability) may lead to thrombosis. Clearly, prosthetic medical devices often violate these principles: surfaces are generally nonendothelialized and thrombogenic, flow through and around devices is often marked by turbulence and focal eddies, and a combination of these factors may cause local hypercoagulability in the form of activation of platelets or coagulation proteins. Since biomaterials themselves are not thromboresistant, prevention of thrombosis in functional devices has generally been accomplished by a judicious combination of 1) use of materials with superior thromboresistance, 2) generation of flow regimens which enhance washing of component surfaces, minimize points of stasis, and eliminate buildup of macroscopic thrombus, and 3) alterations of blood chemistry by antithrombotic agents (e.g., coumadin and heparin) to minimize activation of the coagulation cascade and by antiplatelet agents (e.g., aspirin, dipyridamole, and sulfinpyrazone) to minimize platelet participation. These concepts emphasize the interaction of biomaterials properties, device design, and proper patient management.

NEED FOR STATISTICAL EVALUATION

Perhaps the most neglected, misused, and abused area in biomaterials research is statistical treatment of accumulated data or lack of sta-

Table 1.1--Tests Recommended for Physicochemical Characterization
of Materials

	Bulk Properties	Surface Properties
LEVEL I	Polymer Identification Infrared Spectrum Hardness Porosity Density Mechanical Tests Thermal Analysis Molecular Weight	Light Microscopy Infrared Spectrum X-ray Photoelectron Spectroscopy Contact Angles
LEVEL II	Solvent Responses Elemental Composition and Extractables Water Diffusivity and Sorption Molecular Weight Dis- tribution Extractability/Perme- ability Wide Angle X-ray Spectrum Optical Birefringence NMR Spectrum Morphology (SEM) Electrical Conductivity Creep Stress Relaxation Oscillatory Response Fatigue Testing	Morphology (High resolu- tion SEM) Contact Angles (Zisman Plot) Charge (Zeta Potential) ESCA Protein Adsorption

tistical treatment. Modern statistical approaches applied during the early stages of experimental design can significantly increase the amount of useful information and at the same time reduce the number of experiments and ultimately the costs of a project. It is strongly recommended that a biostatistician be consulted on numerous occasions from the planning through the analysis stages of the project.

The scientific approach to solving problems requires a statement of the problem, formulation of hypotheses which account for the problem, an experimental design to test the hypotheses, interpretation of the experimental data, and conclusions concerning the hypotheses tested. This process is frequently an interactive one which requires statistics not only to interpret the data, but also to help formulate and design the experiments.

Data analysis frequently uses descriptive statistics to define cumulative observations, such as the mean or median and the standard deviation, of small samples taken from a large population. Inferential statistics, such as correlation coefficients, t-tests, and analysis of variance, are utilized to provide the experimenter with probabilities that differences in the small samples are due to unique population differences.

Most of the probability tables used for inferential statistics are based on comparisons planned before the results are obtained, i.e., a priori comparison (2). It is acceptable to make unplanned (a posteriori) comparisons after collecting the data, but the requirements for significance are more stringent (3).

Probabilities generated in hypotheses testing are always in reference to the null hypothesis which assumes there is no real difference between the sample mean and true population mean, or it assumes that there are no real differences among several sample means. The use of probability does provide a degree of comfort that the conclusions drawn are correct, but it does not provide absolute certitude (4). Two types of errors can occur in hypothesis testing. A type I error assumes that a real difference exists when, in fact, there is no difference among the sample groups. The assumption that there is not a difference in treatment groups when a real difference does exist is a type II error. Restated in terms of the null hypothesis, a type I error rejects a true null hypothesis, and a type II error accepts a false null hypothesis (5). The relative importance of type I versus type II errors will vary from study to study. Therefore the investigator has to make a decision on protection against erroneously declaring a difference or lose the power of the significance test. Generally, the power of an experiment increases with the sample size so some compensation for loss of power can be made by increasing the sample size.

Sample Size

It is difficult to provide an exact number for the sample size required to meet the objectives of a study. However, it is possible to approximate the required number of replications needed for a test if one can determine the following information: 1) the population standard

deviation σ (estimated from a preliminary study), 2) the smallest true difference to be detected δ , and 3) the desired probability that a significant difference will be found.

An example of this process is as follows (6): You plan to analyze platelet retention of four different polymers (treatment groups) using analysis of variance (ANOVA). The coefficient of variation (CV) is known from preliminary studies to be about 20 percent. How many measurements (n) are required in each treatment group to be 80 percent certain of detecting a 15 percent difference between the means of two of the four groups at a 5-percent level of significance? Since the number of groups (a) equals four, the error mean square (MS) will have $v = 4(n-1)$ degrees of freedom. The formula for calculating n is

$$n \geq 2 (\sigma/\delta)^2 (t_{\alpha} [v] + t_{2(1-P)} [v])^2$$

where n = number of replications

σ = true standard deviation

δ = the smallest true difference to be detected. (Only the ratio of σ to δ is required, not their actual values.)

v = degrees of freedom of the sample standard deviation (MS within) with " a " groups and " n " replications per group.

α = significance level

P = desired probability that a difference will be found to be significant

$t_{\alpha} [v]$ and $t_{2(1-P)} [v]$ = values from a two tailed t -table with v degrees of freedom corresponding to probabilities of α and $2(1-P)$, respectively.

The value for " n " is determined by an iterative process by guessing a value for " n ." In this example a guess of $n = 20$ is reasonable. Then $v = 4(20-1) = 76$. Since $CV = 20$ percent, $s = 20 \times \bar{x}/100$. The δ is to be 15 percent of the mean, therefore, $\delta = 15 \bar{x}/100$. Using s (sample standard deviation) as an estimate of σ the ratio is $\sigma/\delta = (0.20 \bar{x})/(0.15 \bar{x}) = 0.20/0.15$:

$$n \geq 2 (0.20/0.15)^2 (t_{0.05}[76] + t_{2(1-0.80)}[76])^2$$

$$n = 2 (0.20/0.15)^2 [1.991 + 0.847]^2 = 2(1.78) 8.05 = 28.7$$

next try $n = 29$, making $v = 4(29-1) = 112$

$$n \geq 2(1.78) [1.982 + 0.846]^2 = 3.56(8.00) = 28.5$$

Since our first solution to the problem was 28.7 and the second solution was 28.5, it appears that 29 samples should be run in each polymer group.

Experience with this equation will demonstrate that changes in σ and P are not very sensitive in determining changes in " n ," but the

ratio of σ/δ can drastically affect the number of required replicates. Therefore, improvements in experimental technique that reduce σ will result in smaller sample sizes (6).

Multiparameter Testing

Frequently, biomaterials studies require that a multitude of parameters be considered at one time. This necessitates the use of two-way or three-way analysis of variance or analysis of covariance, or the use of factorial experiments (5,7,8). Target factor analysis and pattern recognition are also useful techniques for solving multidimensional problems (9,10).

Table 1.2 illustrates a typical data layout for a model I two-way ANOVA with replicates. Test tubes coated with four radial triblock copolymers each with a different styrene concentration ($b = 4$) were evaluated using the whole blood clotting time recorded in minutes. Each polymer was evaluated in triplicate ($n = 3$) using rabbit and human blood ($a = 2$). After making the appropriate calculations, an ANOVA table can be generated (table 1.3) which provides information about the source of variation in the data (i.e., species, styrene concentration, interaction between A and B, and the error within subgroups), the degrees of freedom (v), the mean square (MS), and the F statistic to be compared with table values for $F_p[v_1, v_2]$.

It can be concluded from the information contained in table 1.3 that the clotting times are not significantly different between the two species but are affected by the polymer composition. Clotting times are prolonged at 20 percent and 40 percent styrene. Polymer composition appears to affect the blood from rabbits and humans equally since the interaction "species X concentration" is not significant.

Other examples of factorial statistics can be found in the work of Brash and Whicker (11), Mason and Sinoda (12), and Fischer et al. (13).

Nonparametric Tests

Classical methods use statistics to estimate certain population parameters. Comparative tests assume the sample data fit a distribution, i.e., normal, Poisson, etc. Frequently these assumptions are clearly untrue; therefore, to circumvent this problem, data are transformed (log, reciprocal, etc.) to make the assumptions appear more reasonable. However, statistical methods are available to avoid distributional assumptions and are referred to as distribution-free methods or nonparametric methods.

Nonparametric methods are justified if: 1) there is obvious non-normality, 2) the sample sizes are too small to clearly establish a normal distribution, 3) a rapid experimental approach is required ("quick and dirty" methods), and 4) ordered or semiquantitative scoring is used, e.g., 0, +1, +2, . . . , +5 (7). Such tests include the Mann-Whitney U-test, Wilcoxon two-sample test, Kendall's S-test, Friedman's method for randomized blocks (in lieu of two-way ANOVA), and the Kruskal-Wallis test (in lieu of a one-way ANOVA) (7).

Table 1.2--Data Matrix for a Two-Way ANOVA With Replicates*

Factor B: Styrene Concentration (b = 4)	Factor A: Species (a = 2)		Σ
	Rabbit	Human	
20%	29.5	30.5	191.5
	32.5	33.0	
	40.5	25.5	
	$\Sigma=102.5$	$\Sigma=89.0$	
30%	24.5	29.5	147.0
	22.5	26.5	
	20.0	23.5	
	$\Sigma=67.5$	$\Sigma=79.5$	
40%	32.0	28.5	198.5
	28.5	33.0	
	42.0	34.5	
	$\Sigma=102.5$	$\Sigma=96.0$	
75%	29.5	29.5	179.5
	25.0	33.5	
	28.0	34.0	
	$\Sigma=82.5$	$\Sigma=97.0$	
	355.0	361.5	716.5

Table 1.3--Two-Way ANOVA Table

Source of Variation	v	MS	Fs
A (Columns; species)	1	1.7596	0.117 ns
B (Rows; styrene conc.)	3	86.7049	5.782**
A X B (Interaction)	3	31.5662	2.105 ns
Within subgroups (Error)	16	14.9948	
Total	23		
$F_{0.05}[1,16]=4.49$ $F_{0.05}[3,16]=3.24$ $**F_{0.01}[3,16]=5.29$			

* Unpublished data, D.L. Coleman, D.E. Gregonis, and J.D. Andrade (1979).

To illustrate nonparametric methods, consider the following situation. You have synthesized a new polymer coating that you believe should have superior blood compatibility properties. However, you only have enough to coat three platelet retention columns, and you want to compare your new polymer coating with the untreated substrate, pre-albuminated beads, and a commercially available polymer coating. The number of data points is too small to justify using the F-distribution required for the normal one-way ANOVA approach. Therefore, the Kruskal-Wallis method (7) is used to test for differences among the groups based on ranking the data.

X = percent platelets retained by the column

a = 4 treatment groups

N_i = number of data points in groups

1. Pool all observations into one group, and rank them from smallest to largest. In the case of ties, calculate the average rank.

<u>Rank</u>	<u>X</u>	<u>Rank</u>	<u>X</u>	<u>Rank</u>	<u>X</u>
1	-6.5	9	12.9	17	45.4
2	-2.9	10	14.8	18	56.1
3	3.6	11	16.7	19	63.6
4	8.7	12	18.4	20	67.3
5	9.0	13	27.3	21	75.5
6	9.7	14	28.9	22	78.5
7	10.7	15	29.1	23	84.2
8	12.2	16	31.0		

2. Replace each data point in the original table of observations by its rank or average rank.

<u>Untreated</u>		<u>Albuminated</u>		<u>Commercial Polymer</u>		<u>New Polymer</u>	
<u>X</u>	<u>Rank</u>	<u>X</u>	<u>Rank</u>	<u>X</u>	<u>Rank</u>	<u>X</u>	<u>Rank</u>
75.5	21	-2.9	2	27.3	13	9.7	6
45.4	17	-6.5	1	18.4	12	9.0	5
56.1	18	10.7	7	14.8	10	12.2	8
63.6	19	28.9	14	12.9	9		
84.2	23	8.7	4	16.7	11		
67.3	20	3.6	3	31.0	16		
78.5	22	29.1	15				
N_i							
$(\sum R)_i$	140		46		71		19
n	7		7		6		3

3. Sum the ranks for each group, $(\sum R)_i$

4. Compute the following H value, where 12 and 3 are constants and

$\sum N_i$ = sum of the sample sizes for all groups = 7+7+6+3=23.

$$\begin{aligned}
 H &= \left(\frac{12}{(\sum N_i)(\sum N_i+1)} \sum_{i=1}^a \frac{(\sum R_i)^2}{N_i} \right) - 3(\sum N_i+1) \\
 &= (12/(23)(23+1)) ((140^2/7)+(46^2/7)+(71^2/6)+(19^2/3)) - 3(23+1) \\
 &= (0.217)(4062.8) - 72 = 16.16
 \end{aligned}$$

If the null hypothesis is true, i.e., there is no difference in location of the four groups, then H approximates a chi square distribution ($\chi^2_{[a-1]}$). Since H in this example is greater than $\chi^2_{0.01[3]} = 11.34$, the null hypothesis is rejected, and we conclude that treating the beads does reduce the percent platelets retained by the column.

Is there a possibility that the new polymer coating reduces platelet retention more than the commercial polymer? Since we wish to compare just two treatment groups, the Wilcoxon test can be used (5).

<u>Commercial Polymer</u>		<u>New Polymer</u>	
<u>X</u>	<u>Rank</u>	<u>X</u>	<u>Rank</u>
27.3	8	9.7	2
18.4	7	9.0	1
14.8	5	12.2	3
12.9	4		
16.7	6		
31.0	9		
$n_2=6$		$n_1=3$	$\sum R=6$

Compute the Wilcoxon statistic (5):

$$T_1 = \sum_{i=1}^{n_1} R_i = 6$$

$$T_2 = n_1(n_1 + n_2 + 1) - T_1 = 24$$

The smaller of T_1 and T_2 should be used for comparison with table values of Wilcoxon's two-sample rank test. Values of T below the table value would indicate rejection of the null hypothesis. In this example, 6 would be used to compare to the table value of $T_{0.05[3,6]} = 7$ which indicates that the new polymer does have a lower platelet retention than the commercial polymer. Therefore, we can conclude that it is

worth the investigator's time and money to synthesize more polymer and continue with an expanded blood evaluation program.

Other statistical tools are available to investigators, and with the ready accessibility of computers, statistics packages (14,15) are not difficult to find. However, it is important to review the assumptions made by the specific test being used. The numbers generated by the computer are the result of force fitting the data to these assumptions. If the assumptions do not apply to your data, you will be misled.

Investigators should be familiar with the statistical concepts listed in table 1.4. A review of these concepts can be found in references 2,4,5,8, and 16.

RATIONALE FOR MULTIPLICITY OF TESTS AND TEST SPECIES

Numerous hypotheses have been proposed to explain the apparent success or failure of materials contacting blood. Surface wettability (17), surface free energy (18), critical surface tension (19), interfacial free energy (20), charge (21), hydrophilic-hydrophobic ratio (22), surface mobility (23), and surface microdomains (24) have all been proposed to account for blood compatibility. This list is by no means exhaustive, but does illustrate the fact that the blood-material interaction (BMI) field is still controversial and in many respects immature. State-of-the-art BMI tests are fairly primitive compared with most areas of science (25). Ideal BMI assays should be accurate, significant, specific, and economical, but no one test will meet all of these criteria.

Although the hypotheses listed above have been and continue to be useful to stimulate discussion and to focus scientific thinking, they can also become a dangerous trap if we become too attached to them.

There is one great difficulty with a good hypothesis. When it is completed and rounded, the corners smooth and the content cohesive and coherent, it is likely to become a thing in itself, a work of art. It is then like a finished sonnet or a painting completed. One hates to disturb it. Even if subsequent information should put a hole in it, one hates to tear it down because it once was beautiful and whole.

--John Steinbeck
Log from the Sea of Cortez

It is highly unlikely that a single parameter hypothesis will be able to account for general blood-material interactions (26). Material properties act in concert to direct the eventual blood response. However, it is difficult to change any one material parameter without also changing other properties of the material (26). Therefore, it is necessary to monitor as many materials properties as possible. Certainly, the relevant level I physicochemical tests listed in table 1.1 should be evaluated.

Table 1.4 Basic Statistical Concepts

Sampling Problems

Populations and Frequency Distributions

Mean and Standard Deviation (Population vs. Sample)

Normal Distribution

Hypotheses Testing

Comparison of Two Samples

Binomial Distribution

Regression Analysis

Correlation Analysis

Analysis of Variance

One-way Classifications

Two-way Classifications

Factorial Design

Analysis of Covariance

Nonparametric Methods

The reaction of blood to a material can range from minimal protein adsorption to activation of coagulation proteins, complement activation, platelets and/or white blood cell activation, and destruction of red blood cells. That is to say, several complicated mechanisms exist in blood to interfere with, or alter, prostheses or devices required to interface with the cardiovascular system. Therefore, it is important to emphasize that blood compatibility should not be defined by a single blood test.

The multidisciplinary nature of scientists studying BMI's simply reflects the importance of a multiparameter approach to BMI testing. The common concern of clinicians, materials scientists, design engineers, and regulatory scientists is medical acceptability expressed as a benefit-to-risk ratio. The benefit-to-risk ratio is the summation of all prior in vitro, ex vivo, and in vivo animal testing, as well as the clinical circumstances that may suggest the use of a particular device and alternative risks associated with the disease process it is intended to treat.

Existing and proposed medical devices vary enormously in type, function, and duration of blood-materials contact. It is unlikely that any one test system or species could realistically accommodate all applications. As a result of such multifaceted concerns, the information in this volume is organized to define and interpret the results of useful in vitro, ex vivo, and in vivo tests of materials to be used in acute, chronic, or permanent situations.

It should be emphasized that correlations of materials used in a test configuration versus a device configuration have not been clearly established. Since fabrication procedures may alter surface chemistry or morphology, careful characterization is necessary before such correlations can be made.

In Vitro Tests

These include static tests (e.g., to assess protein adsorption or activation of the intrinsic clotting system), centrifugal assays (assess platelet adhesion), and various dynamic procedures to assess thrombogenesis and effects on circulating components. Advantages of such tests are that they may reveal the primary events in the BMI processes, are quantifiable, can be performed with human blood, and are sufficiently fast and inexpensive to use for checking on the many material variables, including fabrication effects, that might be encountered during a development program. Disadvantages are that they miss most manifestations of chronic incompatibility or compatibility (e.g., passivation) and most "downstream" effects, such as cell damage or embolism. In addition, their relevance to in vivo events cannot be taken for granted, but must be proved in each case. Studies to establish a firm relationship between in vitro, ex vivo, and in vivo tests require verification from many laboratories before the scientific community will accept any protocol to evaluate BMI's. It is hoped that this document will stimulate efforts to correlate physicochemical data with BMI data, both in vitro and in vivo.

Acute Extracorporeal Tests

In these tests, native or anticoagulated blood is passed from an animal to a special test chamber, generally designed for controlling the flow regimen. Such tests offer the best available way to assess the role of hemodynamic facts in the initial events of cellular adhesion and activation, but suffer the disadvantage that they miss most chronic and "downstream" effects, since the blood used is normally not returned to the animal.

Tests of Tubular Devices in Animals

For these tests, the material to be evaluated is fabricated into a ring, catheter, cannula, or vascular shunt, which is exposed to flowing blood either extracorporeally or intracorporeally for days or weeks. Such tests offer the easiest route assessing chronic thrombotic effects such as progressive occlusion, passivation, or embolism. Disadvantages are that the initial and subsequent events in the BMI process can no longer be measured directly, the blood flow is usually in a laminar regimen that minimizes material-cell interactions, most "downstream" effects are undetectable because of the small area of the test surface, and particularly in the case of the ring tests, the reproducibility has been poor.

Tests of Membrane Devices in Animals

If the test material can be fabricated into a sheet-like form, an extracorporeal test using a membrane oxygenator or hemodialyzer offers the easiest way to bring a large area of test surface into contact with circulating blood, and hence to assess "downstream" effects such as protein alteration, hemolysis, and other forms of cell damage. Disadvantages of such tests are that they cannot be used to pick up either the initial events or the long-term outcomes of the blood-material interactions, and they are restricted to materials that can be fabricated into membranes. Also, such tests would usually be performed with anticoagulation of the host during the test period. Therefore, the results would not be indicative of how such a material would perform in other configurations and with anticoagulation.

Tests of Cardiovascular Devices in Animals

The ultimate test of any proposed material is to fabricate it into a prototype device, implant it into an animal, and observe the results over a period of weeks to months. The great advantage of such tests is in offering the maximum possible (though not complete) assurance that no deleterious effects will result if the material-device system is used in humans. The disadvantage is that the high costs of such testing limits the number and duration of such tests which can be performed, thereby limiting the extent of scientific data which can be obtained. While animals can be highly instrumented and certain indirect tests can be performed, the BMI must be deduced at the end of the study from careful examination of the device and autopsy material. Device testing is usually used primarily for certifying acceptability of device performance. In addition, there are some real problems in determining which animal

systems, if any, can serve as a valid model for predicting long-term effects in the human.

Clinical Tests

A properly designed clinical trial is obviously the ultimate test of a proposed clinical device. Such tests are primarily of value in establishing medical acceptability. Scientific information on the details of BMI is usually difficult to obtain from clinical tests, particularly those involving chronic implants, because of the inaccessibility of the device after implantation, possible confusion with effects introduced by that patient's disease, and the concern of the physician with patient care and ethical considerations, which limit the opportunity to accumulate invasive data.

The proposed guidelines for blood-material interactions are intended to provide academic, regulatory, and industrial scientists with a common set of useful test methods, by no means comprehensive, in the hope that various laboratories will be more consistent for a given material. A general approach to device development utilizing in vitro, ex vivo, and in vivo testing is outlined in figure 1.1.

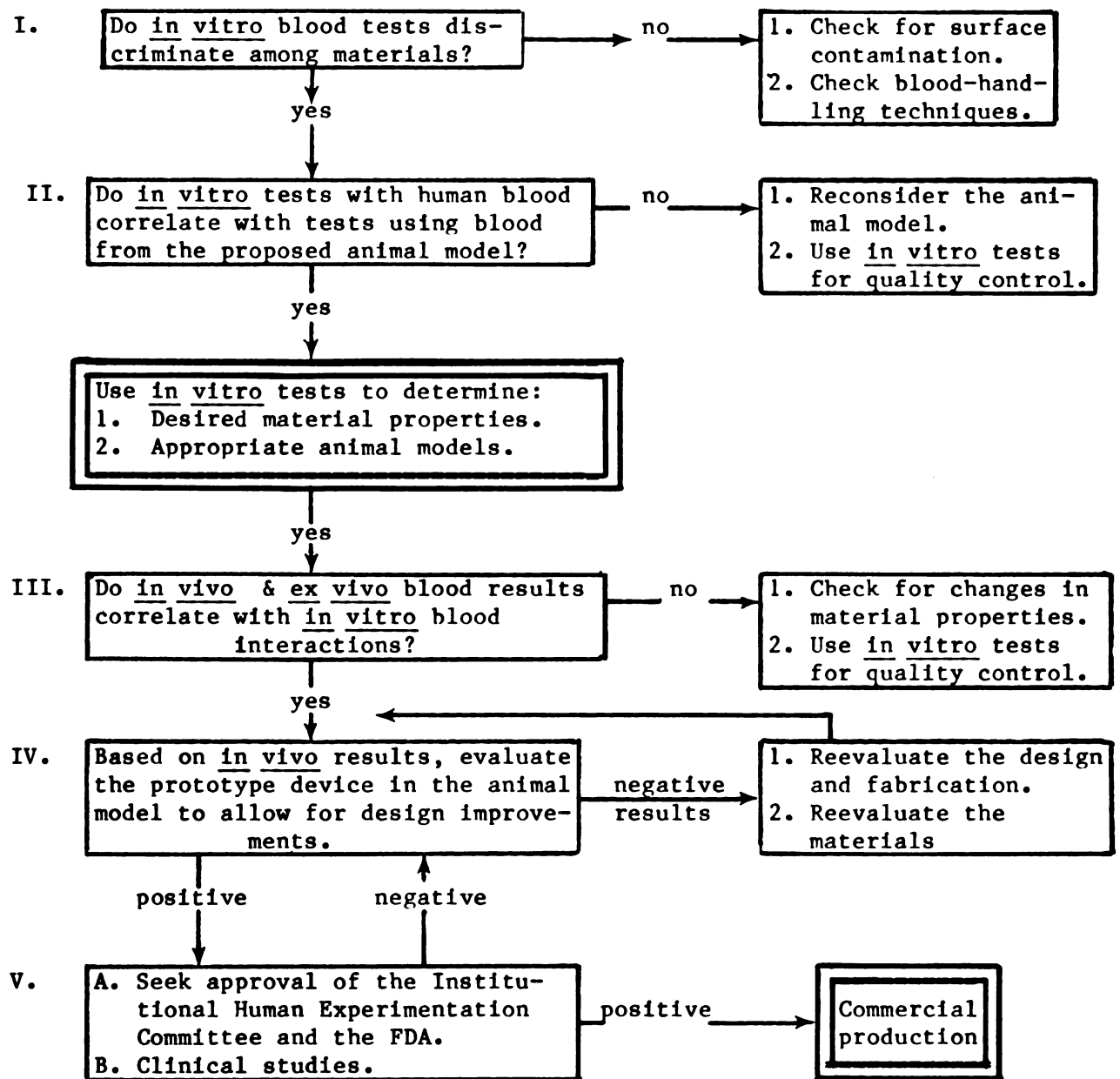


Figure 1.1 A generalized flow diagram showing the potential use of in vitro, ex vivo, and in vivo blood evaluation in the development of a cardiovascular device.

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Chapter 2

CONCEPTS OF THROMBUS FORMATION, DISSOLUTION, AND ANTITHROMBOTIC THERAPY

Thrombus, a blood-derived mass, forms on altered endovascular surfaces or cardiovascular devices anywhere in the circulation as a pathological consequence of activating hemostatic mechanisms under variable flow conditions. Local thrombus may interrupt blood flow to tissues, causing ischemia or infarction, or a thrombus may embolize downstream to occlude blood vessels distally.

The thrombotic process is generally rapid and localized without compromising fluidity of the blood elsewhere in the circulation. It involves a complex integrated interaction among 1) abnormal surface, 2) platelets, and 3) coagulation factors to form a localized stable mechanical mass that subsequently undergoes slow dissolution by 4) fibrinolysis. Rapid, localized thrombus formation within a fluid medium is achieved by complicated systems of activation and inhibition whereby generalized thrombosis is avoided. In this discussion, the process of thrombosis will be considered with regard to the four components, their normal structure, function interrelationships, activation, and inhibition mechanisms.

THROMBOGENESIS

Thrombogenesis in association with prosthetic cardiovascular devices includes the following sequence of events:

1. Exposure of the thrombogenic surface to circulating blood.
2. Platelet adherence, aggregation, and release of platelet components.
3. Thrombin generation and fibrin formation.
4. Thrombus dissolution requires plasmin generation and fibrinolysis.

These processes will be discussed individually and then will be integrated with their control systems in an overall scheme of thrombosis.

Surface Exposure to Blood

When blood contacts an artificial surface, the hemostatic system may respond in diverse ways depending upon the nature and duration of the stimulus.

Protein Adsorption

It is widely believed that artificial surfaces placed in contact with blood will rapidly acquire a layer of adsorbed plasma proteins which will mediate subsequent thrombotic events. This hypothesis has prompted many studies of protein adsorption to surfaces, and several reviews are available (1-5).

In vitro studies of protein adsorption from buffered solutions have generally been limited to the major plasma proteins albumin, fibrinogen, and various globulin fractions. The composition of the adsorbed protein layer has been shown to depend on solution pH, temperature, exposure time, the types of proteins studied, and their concentrations (1,2,6-18). Adsorption is strongly influenced by the nature of the substrate surface (6-11,16-18) and may be quite selective (10,11,18). Fluid motion may affect the pattern of protein deposition which may be nonuniform on a macroscopic scale (19). Albumin may desorb completely from silica (20), but adsorption to polyethylene can be irreversible (15). The presence of red blood cells in the suspending medium has been shown to influence protein adsorption through an undefined mechanism (16,21). Conformational changes may occur upon protein adsorption (13). Such changes can cause an increase or loss of enzymatic activity (13) or the exposure of immunogenic sites (22).

There are relatively few studies of protein adsorption from more complex media such as blood or plasma although this is the focus of much recent interest and work. Using ellipsometry and antisera staining techniques, Vroman (23-25) observed that blood or plasma will deposit a film of fibrinogen about 50 Å thick within 3 seconds. The plasma then "converts, masks, or replaces" the film within about 20 seconds. If the plasma contained intact factor XII, it then removes part of the film (26). Vroman was also able to detect gamma globulin, but albumin was never detected.

Lyman et al. (27,28) studied adsorption onto arteriovenous shunts in dogs. After exposure times up to 45 minutes, adsorbed proteins were eluted, concentrated, and examined electrophoretically. Albumin, globulins, and possibly fibrinogen were identified; relative concentrations varied considerably with exposure time and depended on shunt material used.

Limber et al. (29,30) exposed particles of glass and Cuprophane to plasma and analyzed eluates by electrophoresis and immunodiffusion. Albumin, factor XII, glycoproteins, and various globulins were recovered, but fibrinogen could not be detected even when it was known to be present.

Weathersby et al. (8) exposed films of various materials to plasma in vitro. Adsorbed proteins were radioiodinated in situ, eluted, and separated electrophoretically. Fibrinogen and albumin were always found. As many as six other iodinated species were also observed but could not be readily identified. Adsorbed protein patterns varied with plasma exposure time and type of material studied. One of the proteins was subsequently shown to be hemoglobin, which appears to have very high surface activity (31-34). In a much more extensive study with the in situ iodination technique, the distribution of iodine into proteins adsorbed to a series of hydrophobic-hydrophilic model surface was shown to differ markedly (35). The large differences in the composition of the adsorbed film's protein layer on these surfaces were confirmed with the use of ¹²⁵I prelabeled proteins added to plasma (36). Horbett and Hoffman (6) had previously observed that fibrinogen adsorption to silicone

rubber from plasma was much reduced compared to that adsorbed from simple protein solution.

Brash and coworkers have also used the prelabeled protein technique to demonstrate major differences in the composition of the protein layer formed on various polymers exposed to plasma (37). Maxima in the amount of fibrinogen and other proteins adsorbed from plasma or serum to various surfaces have also been reported by several groups (33,38,39). This phenomenon is apparently closely related to the "conversion" reaction observed by Vroman. Vroman's group has recently shown this reaction seems to be related to displacement by HMWK since it is reduced in plasma deficient in this factor (40).

The preadsorption of proteins to surfaces prior to blood exposure may strongly affect subsequent platelet adhesion, release, and aggregation (9-11,28,41-45). In general, these studies have shown that albuminated surfaces inhibit platelet adhesion, whereas surfaces coated with fibrinogen or gamma globulin tend to promote platelet adhesion, and in some cases, release and aggregation. Air denaturation of adsorbed proteins will also promote platelet adhesion (10,46). Platelets do not adhere to glass when the platelet-rich plasma is obtained from subjects with afibrinogenemia (45) or when washed platelets are suspended in serum (42). These studies suggest that fibrinogen may strongly mediate platelet-surface interactions.

The effects of protein adsorptive phenomena in determining thrombotic events subsequent to initial platelet adhesion have not been extensively studied. Evans and Mustard (41) measured thrombus formation on a polyethylene bifurcation inserted into an arteriovenous shunt in the rabbit. The weight of thrombus formed on albumin-coated bifurcations was less than that formed with gamma globulin-coated bifurcations. Albuminated surfaces have also been reported to perform well as vena cava rings in the dog (46,47). The effect of protein preadsorption on the rate of thromboembolization in baboon blood measured with a laser light scattering device was reported to range from passivation (hemoglobin, immunoglobulin G) to activation (fibrinogen, plasma) (48).

It has become apparent that the earliest events following blood contact with artificial surfaces may involve complex and time-dependent interactions among a number of plasma proteins. More recent investigations have included other blood proteins such as the contact coagulation factors (26,49) (coagulation factor XII, high molecular weight kininogen, prekallekrein, and factor XI) and proteins thought to be involved in cell adhesion processes (see p. 25, Activation of Coagulation Factors) including fibrinogen, von Willebrand's factor (50), fibronectin (51,52), and thrombospondin (53-55). When preadsorbed to polymeric surfaces, some of these proteins enhance acute thrombus formation in the dog shunt model while others reduce it (56). However, since the adsorption of such trace proteins from whole blood is much lower due to competition from the major proteins, it is not clear what role the adsorption of these trace proteins actually plays in thrombosis on polymers. Thus while von Willebrand's factor is taken up from blood by vascular grafts (57), the amount of its adsorption to smooth surfaces is very small (58).

Platelet Adherence, Aggregation, and Release

Exposure of nonendothelialized surfaces to circulating blood induces platelets to adhere. Following vascular injury, the process of platelet adhesion involves interaction of the platelet membrane glycoprotein (GPI_b), subendothelial collagen, and plasma von Willebrand factor (vWF) (59-61). vWF is synthesized by endothelial cells and megakaryocytes, is present in plasma, on platelets, in platelet alpha granules, and is adsorbed to exposed receptors for vWF. Hereditary absence of platelet GPI_b (Bernard-Soulier syndrome) or plasma vWF (von Willebrand's disease) produces defective platelet adhesion (59-61). The process of platelet adhesion to artificial surfaces has also not been adequately established to clarify relevant differences from those events occurring between platelets and vascular subendothelium. Similarly, the roles of other attachment proteins with respect to artificial surfaces such as fibronectin and thrombospondin have not been defined.

Platelet adhesion initiates a series of complex, interactive platelet reactions including 1) release of dense granule adenosine diphosphate (ADP) from adherent platelets, 2) release of alpha granule constituents, including fibrinogen, vWF, factor V, platelet factor 4 (PF4), beta thromboglobulin (BTG), platelet-derived growth factor (PDGF), and 3) activation of platelet membrane phospholipase complex leading to the generation of thromboxane A₂ (TxA₂) (see figure 2.1 and general references).

Released ADP and TxA₂ act synergistically to recruit circulating platelets, causing them to change shape and attach both to each other and to already adherent platelets in the process of enlarging the accumulating thrombotic mass. Platelet aggregation requires the rapid mobilization of a platelet-membrane fibrinogen receptor complex (GPIIb, GPIIIa) and calcium-dependent interplatelet bridging by fibrinogen (64-66). This bridging reaction may also involve other platelet alpha granule proteins, vWF, fibronectin and thrombospondin (55,67,68).

Thrombus growth and stabilization depend in large part on the formation of thrombin (figure 2.2). Thrombin is generated by means of the intrinsic coagulation pathway through contact factor activation, by subendothelial collagen (69) or when tissue thromboplastin from disrupted endothelial cells activates the extrinsic coagulation cascade (70).

Platelets also contribute significantly to thrombin generation by providing membrane phospholipid and receptors for the molecular assembly of prothrombinase complex. Platelets may promote the early stages of intrinsic coagulation by a process that involves a factor XI receptor and high molecular weight kininogen (71). Thrombin modified or activated factors VIII and V also bind specifically to platelet membrane receptors to facilitate conversion of factor X to factor X_a and the final conversion of prothrombin to thrombin, respectively (72). Concentrations of thrombin that are sufficient to stimulate platelet alpha granule release are formed within seconds after blood is exposed to nonendothelial surfaces and long before fibrin is formed (73-75). Thrombin-modified factor V serves as a high-affinity platelet receptor of factor X_a (72, 76). Thrombin generation is markedly amplified in the presence of

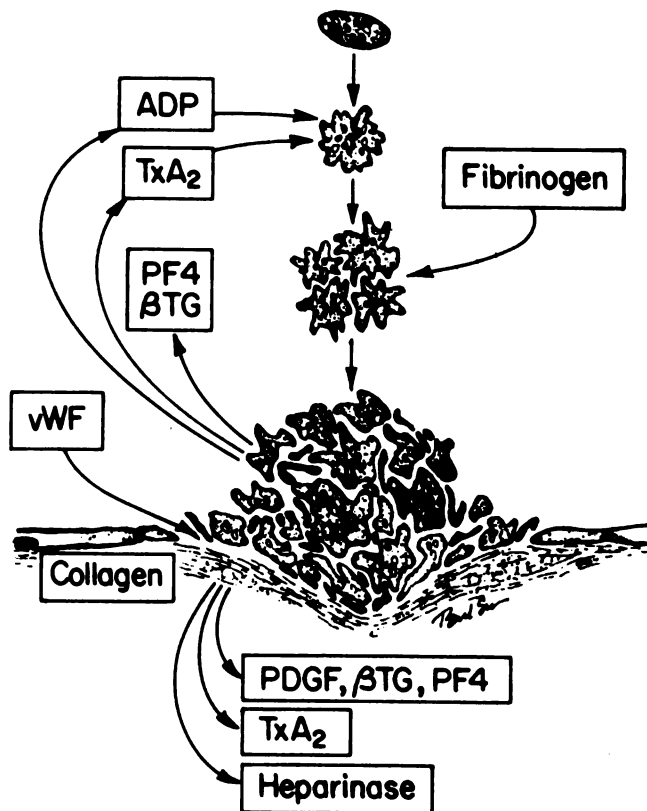


Figure 2.1--Platelet adhesion and aggregation.

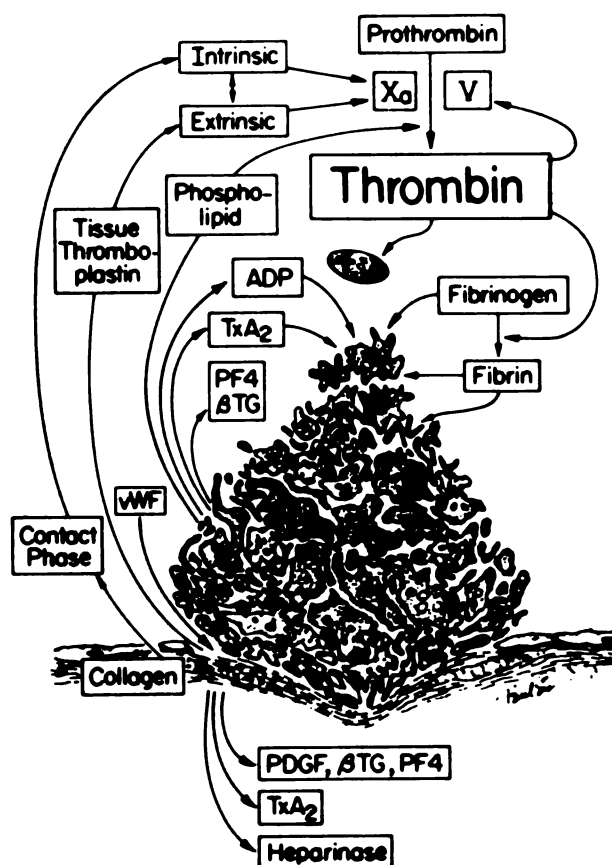


Figure 2.2--Thrombotic platelet functions. Platelets are normally non-reactive to intact vascular endothelium. Vessel injury initiates platelet adherence with von Willebrand factor (vWF) as an important plasma cofactor. Adherent platelets release dense granule contents, including ADP and α -granule constituents, including platelet factor 4 (PF₄), β -thromboglobulin (β TG), and platelet-derived growth factor (PDGF). Thrombin is generated locally through tissue factor, factor XII_a and platelet procoagulant activity. Thromboxane A₂ (TxA₂) is synthesized from arachidonic acid derived from membrane phospholipases. Released ADP, TxA₂, and thrombin recruit additional circulating platelets to the enlarging platelet mass. Thrombin-generated fibrin stabilizes the platelet mass. PGI₂ released by the vessel wall in response to thrombin limits thrombus formation by inhibiting further platelet deposition.

modified factor V bound to the platelet surface (76). Once formed, thrombin stimulates the synthesis of TxA_2 and the release of platelet ADP, thus promoting platelet aggregation. Subsequently, thrombin-generated fibrin stabilizes the growing platelet mass; without fibrin stabilization, platelet thrombi readily disaggregate (77).

Blood platelets have a primary role in the thrombogenic response to artificial surfaces as shown by: 1) direct visualization of platelets on prosthetic surfaces following exposure to circulating blood (78-83); 2) renal microinfarction produced by platelet microemboli following surgical placement of prosthetic grafts proximal to renal arteries (84-86); 3) selective platelet consumption regularly observed in patients and animals with cardiovascular devices (86-91); and 4) interruption of platelet consumption and thromboembolism by antiplatelet drugs (89-95).

Activation of Coagulation Factors

Contact activation in the intrinsic system (table 2.1) refers to reactions following adsorption of contact factors to a highly negatively charged surface (96,97). Although the reactions are well understood in vitro, the physiologic or pathologic significance remains to be confirmed. Involved are factors XII, XI, prekallikrein and high-molecular-weight kininogen. All of the contact activation reactions take place in the absence of calcium. Kallikrein, it should be noted, is also involved in the fibrinolytic system, inflammation, and has additional physiologic roles (98).

Prekallikrein and high-molecular-weight kininogen are intimately involved in the activation of surface-bound factor XII (figure 2.3). Activation is localized at the surface and requires adsorption of not only factor XII or Hageman factor (HF) but also factor XI and prekallikrein in close proximity, to induce intrinsic coagulation (100).

Factor XII undergoes a conformational change when activated (99). This change in conformation exposes hydrophobic active sites previously unavailable to the external environment. Surface-bound factor XII or Hageman factor is cleaved to produce $\alpha\text{-HF}_a$, which remains surface bound and can activate prekallikrein and factor XI only if they are also surface bound. The $\beta\text{-HF}_a$ apparently leaves the surface and, as a soluble molecule, will activate prekallikrein, whether bound or in solution, but cannot activate factor XI either in solution or bound to a surface. Once factor XIIa ($\alpha\text{-HF}_a$) has been formed, it acts as a serine protease (101) to activate factor XI, and the coagulation cascade begins. The fibrinolytic system is also activated by either surface-bound $\alpha\text{-HF}_a$ or in solution by $\beta\text{-HF}_a$ (99). In addition, factor XII activation can also promote coagulation of the extrinsic pathway by direct action of factor VII (102), and can activate the fibrinolytic pathway via plasminogen proactivator (103). Other intrinsic factors that may be adsorbed by an artificial surface include factors XI, V, and vWF (104).

A middle phase of intrinsic clotting begins with the first calcium-dependent step, activation of factor IX by factor XIa. Factor IX is

Table 2.1--Properties of Human Clotting Factors

<u>Clotting Factor (Synonym)</u>	<u>Molecular Weight (Number of Chains)</u>	<u>Normal Plasma Concentration ($\mu\text{g/ml}$)</u>	<u>T_{1/2} Elimination (hours)</u>	<u>Active Fo</u>
<u>INTRINSIC SYSTEM</u>				
Factor XII (Hageman Factor)	80,000 (1)	29	60	Serine protease
Prekallikrein (Fletcher Factor)	80,000 (1)	50	?	Serine protease
High-Molecular-Weight Kininogen (Fitzgerald Factor)	120,000 (1)	70	?	Cofactor
Factor XI (Plasma Thromboplastin Antecedent)	160,000 (2, dimer)	4	65	Serine protease
Factor IX (Christmas Factor)	57,000 (1)	4	20	Cofactor
von Willebrand Factor (vWF)	1-2,000,000* (Series of G-10 subunits)	7	10	
Factor VIII:C (Antihemophilic Factor)	200,000-350	0.1	10	Cofactor
<u>EXTRINSIC SYSTEM</u>				
Factor VII (Proconvertin)	55,000 (1)	1	5	Serine protease
Tissue Factor (Tissue Thromboplastin)	45,000 (1)	1	5	Serine protease
<u>COMMON PATHWAY</u>				
Factor X (Stuart-Prower Factor)	59,000 (2)	5	65	Serine protease
Factor V (Proaccelerin)	330,000 (1)	5-12	25	Cofactor
Prothrombin (Factor II)	70,000 (1)	100	100	Serine protease
Fibrinogen (Factor I)	340,000 (6: A α ₂ , B β ₂ , γ ₂)	2,500 (250 mg/dl)	120	Clot structure
Factor XIII (Fibrin Stabilizing Factor)	300,000 (4: a ₂ , b ₂)	10	150	Transglutaminase

*Subunit molecular weight of factor VIII/vWF is around 200,000 with a series of multimers found in circulation.

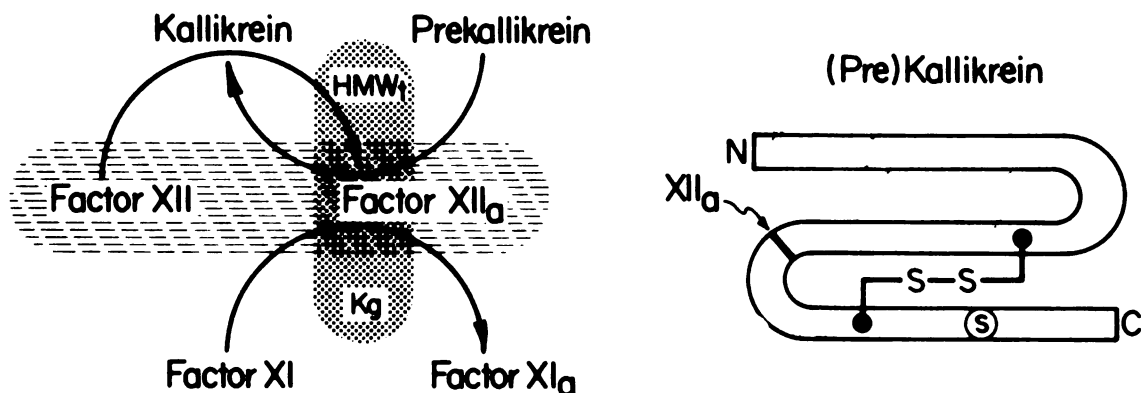


Figure 2.3--Contact activation. Left: The initial event in vitro is the adsorption of factor XII to a negatively charged surface (hatched, horizontal ovoid) where it undergoes a conformational change to expose its active site. Factor XII converts prekallikrein to kallikrein. Additional factor XII_a and kallikrein are then generated by reciprocal activation. Factor XII_a also activates factor XI. Both prekallikrein and factor XI bind to a cofactor, high-molecular-weight kininogen (HMWt Kg; dotted, vertical ovoid), which serves to anchor them to the charged surface. Right: Prekallikrein, as a typical contact activation factor, is activated by cleavage of an Arg-Ile bond by factor XII_a to form an active serine protease of two chains with the amino-terminal (N) larger fragment being disulfide bonded (S-S) to the carboxy-terminal (C) smaller fragment. The latter contains the active center Ser (S). The structures of other contact factors (not shown) include factor XI, which is highly homologous to prekallikrein but circulates as disulfide bonded dimer of twice the molecular weight. Factor XII is of similar size, but the active form can also degrade itself to a lower molecular-weight species that no longer binds to the negatively charged surface. High-molecular-weight kininogen has a different structure, including a nine amino acid, vasoactive peptide, bradykinin, which can be released after specific cleavages at each end by kallikrein.

specifically cleaved by factor XIa at two sites to produce factor IXa. Factor IX, with factors X, VII, and prothrombin, are the vitamin K-dependent clotting factors which have strikingly homologous sequences of amino acid residues (105). These four, after polypeptide chain synthesis but prior to secretion from the hepatocyte, are carboxylated at 10-12 specific glutamyl residues. The residues occur near the aminoterminal portion of the molecules to form gamma-carboxy-glutamyl side chains, abbreviated as Gla. Due to the tertiary conformation, pairs of gamma carboxyl groups form a charge density that tightly chelates calcium ions (106).

Factor IXa, together with calcium ions, phospholipid, and the protein cofactor, factor VIII:C, converts factor X to its activated form, factor Xa (figure 2.4). Factor VIII:C is an essential cofactor in intrinsic factor X activation (107-109). It circulates as a complex with vWF, a series of multimers of at least several disulfide-bonded subunit chains which are of the same basic size (110). Interestingly enough, factor VIII:C (as with factor V) first requires modification (m) by a serine protease, such as thrombin, to exert its reactivity independently (111). These two cofactors bind independently to platelets, without ionic bridges, to facilitate the reaction between enzyme and substrate (factor IXa and factor X; or factor Xa and prothrombin).

The extrinsic system is initiated by activation of factor VII, another vitamin K-dependent protein, which primarily differs from the others by its extremely low plasma concentration (112). When factor VII interacts with tissue factor, an intracellular (hence extrinsic to plasma) microsomal lipoprotein, factor VIIa becomes a serine protease which is the extrinsic factor X activator (113). Tissue factor is present in large amounts in the brain, lung, and placenta. It is also secreted by stimulated leukocytes (114), and has been found in the intima of large blood vessels. Lipid comprises one-third of the molecular weight of this factor and is essential for activity, providing the surface for calcium-vitamin K-dependent factor binding.

The common pathway begins as factor X and is activated by either factor VIIa-tissue factor or factors IXa-VIII_m. Factor X is also vitamin K-dependent, but circulates as a disulfide-bonded two-chained zymogen (105). Activation by either pathway generates factor Xa.

After formation of factor Xa, the next step involves factor V, a cofactor, which (like factor VIII) has its activity manifest after modification by a proteolytic enzyme such as thrombin (115,116). Factor V is a large protein but its lability in the human system has made its isolation and subsequent characterization difficult. It markedly facilitates the activation of prothrombin, the final vitamin K-dependent factor in this cascade (figure 2.5).

The higher plasma concentration of prothrombin as well as a biologic amplification of the clotting system allow a few molecules of activated initiator or intermediate factors to generate a large burst of thrombin activity. Thrombin, in addition to its ability to modify factors V and VIII, is able to induce platelet aggregation. Its best characterized

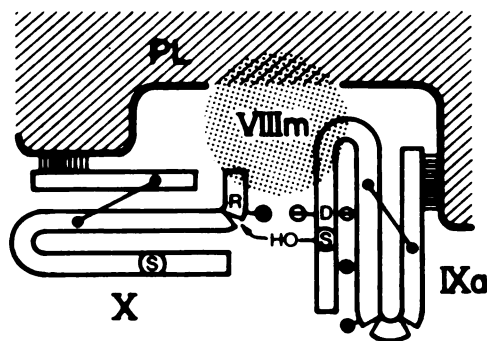


Figure 2.4--Intrinsic factor X activation. Factors IX_a and X are bound to the phospholipid surface of the platelet membrane (PL; cross-hatched area) by calcium bridges through their γ -carboxyglutamic acid residues (Gla). Factor VIII, having been modified by thrombin (VIII_m; stippled circle), binds nonionically to the phospholipid surface and serves as the cofactor in the activation by an as-yet undefined mechanism. The steps of catalysis established for tryptic-like cleavage involves binding of substrate (factor X) with an Arg (R) group ionically attaching to an Asp (D) residue in the bottom of the binding pocket near the active center Ser (S) of the enzyme (factor IX_a). The active center Ser forms an unstable acyl intermediate which is hydrolyzed, cleaving the Arg-Ile peptide bond to generate factor X_a. In the conversion of prothrombin to thrombin (not shown), factor X_a is the enzyme and factor V_m the modified cofactor.

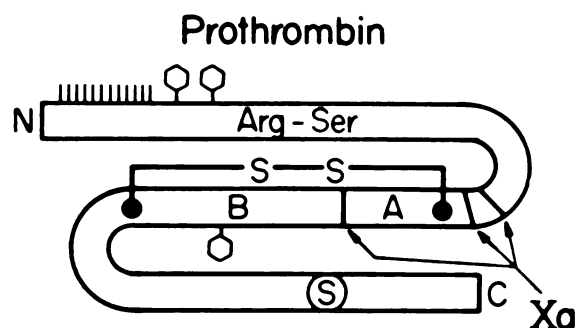


Figure 2.5--Structure of prothrombin. Prothrombin is the largest of the vitamin K-dependent proteins and is depicted from its amino terminus (N) to its carboxy terminus (C). The amino-terminal region contains 10 Gla (vertical bars). In its conversion to thrombin, factor X_a cleaves two bonds (Arg-Thr and Arg-Ile) to form disulfide bonded A and B chains, respectively, that represent the carboxy-terminal half of the precursor. In the human enzyme, a second Arg-Thr cleavage near the amino-terminal portion of the A chain occurs which does not alter its activity. Once it is generated, thrombin is no longer covalently linked to the Gla region so that it enters the soluble phase to interact with fibrinogen. The amino-terminal "pro-piece" (fragment 1 • 2) remains bound to phospholipid, where thrombin cleaves an Arg-Ser bond in this half of the molecule, generating two separate fragments; a physiologic role for these peptides remains to be identified. The hexagonal structures represent Asn-linked carbohydrate chains.

interactions, however, are with two substrates, fibrinogen and factor XIII (117).

Fibrinogen is a large, six-chained molecule (118). It is relatively cryoinsoluble (like vWF). Thrombin only removes about 3 percent of its molecular mass by releasing first the two A and then the two B peptides. The fibrin monomers so formed are able to polymerize nonenzymatically (figure 2.6), leaving the fluid phase to become a gel. Denaturing agents such as urea, however, can still make the clot soluble at this point. An insoluble fibrin polymer is formed by interaction of the fibrin polymer with factor XIIIa.

Factor XIII is either trapped in the plasma within the clot or, alternatively, provided by platelets, and it is activated by thrombin cleavage of a singular bond near the aminoterminal portion of its heavier (α) chain. This converts it to an active transamidase. In the presence of calcium, factor XIIIa catalyzes peptide-like bond formation between epsilon-amino groups of lysine from one fibrin monomer and the gamma-amide groups of glutamine from adjacent molecules, splitting out NH_3 (119). These residues are located at the C-terminal portions of the α^3 and beta chains of fibrinogen.

Regulation of Coagulation Mechanism

There are protective mechanisms to avoid massive thrombus formation once coagulation is initiated. Although fibrinogen, prothrombin and factors V and VIII are largely consumed or destroyed when blood or plasma is clotted in a test tube, cleavage of other intermediate enzymes is somewhat blocked after 5-20 percent activation has occurred. At least three types of mechanisms which contribute to and control the rates of coagulation can be considered.

First, higher blood flow reduces the chance of localized concentration of precursors and removes activated materials. This is demonstrated by the absence of fibrin thrombi in damaged, but patent, arteries and arterioles as opposed to their presence in the venous system. Coupled with blood flow is the rapid disappearance of activated or inhibited-activated clotting factors during passage through the liver. Here these altered proteins are presumably taken up by mononuclear phagocytic cells or hepatocytes, perhaps while adsorbed to surfaces or vesicles as "particle-like" materials, or to inhibitors.

Second, during the process of activation, proteolytic enzymes are generated which not only activate clotting factors, but also degrade cofactors. Plasmin, for example, degrades fibrinogen and fibrin monomers, and can rapidly inactivate cofactors V and VIII by relatively specific cleavages. Of special interest is the recently discovered "protein C," a vitamin K-dependent precursor of a serine protease (120), which is another substrate for thrombin activation in the presence of an endothelial cofactor, thrombomodulin (121). The active form, not a procoagulant, rapidly destroys factor V and probably factor VIII:C. Further physiologic studies are necessary to determine the relative importance of these and other neutral proteases (such as from neutrophils) in limiting coagulation.

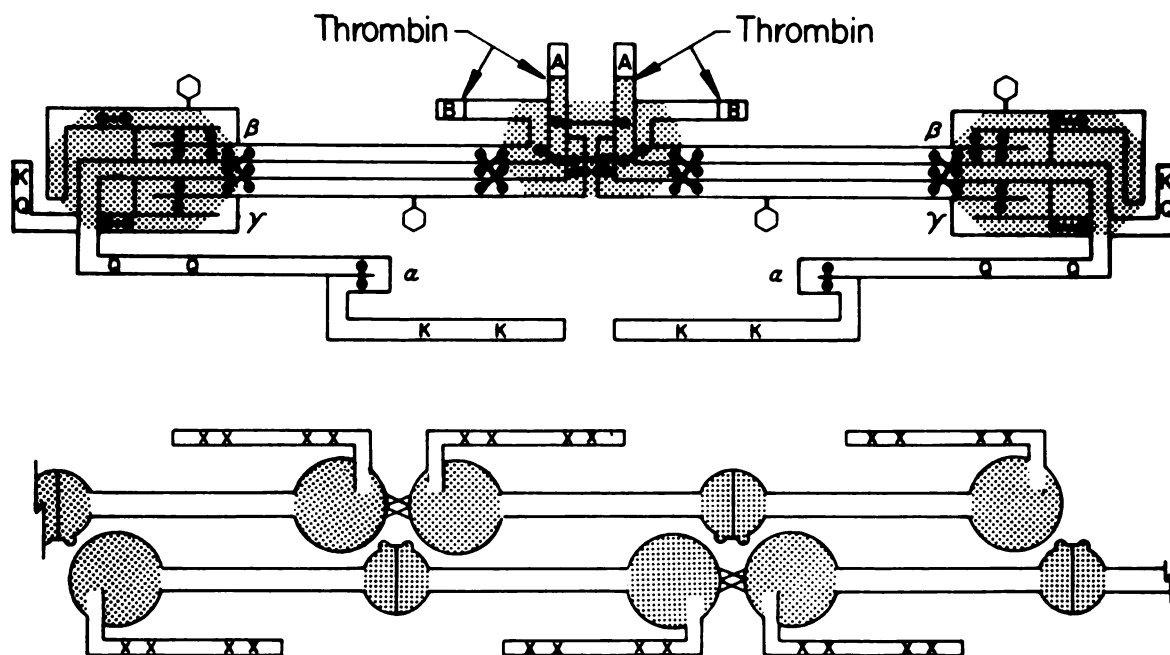


Figure 2.6--Fibrin(ogen) structure and polymerization. Upper: Depicts the primary structure of fibrinogen arranged to correspond to its image on electron microscopy, which includes a central and two terminal domains (stippled) with rod-like connections. This 340,000 mol wt glycoprotein is a paired structure of three highly homologous chains, A α , B β , and γ , respectively. Complete amino acid sequence has been solved and arrangement of the disulfide bonds (lines connecting dots) proposed. The central or "E" region contains the amino-terminal portions of each set of three chains, including three disulfide bridges between the sets. The A and B fibrinopeptide portions (16 and 14 amino acids, respectively) are also found protruding from this region. Each set of three chains then forms a rope-like rod, lashed at each end by a disulfide ring. The carboxy-terminal domains ("D" regions) are formed by the β and γ chains; the carboxy-terminal ends of the longer α chains extend out from these regions. Cross-linking sites include donor Gln (Q) and acceptor Lys (K) residues. Thrombin cleaves the A (and then the B) peptides, as indicated by arrows. Lower: Newly formed amino-terminal "knobs" of the α chains interact with and orient two D regions of different monomers to form alternate rows of a fibrin strand. Factor XIII_a covalently cross-links two γ -chain ends from adjacent D regions by reciprocal transamidation of Gln and Lys residues. Subsequent interactions and cross-linkings (x) occur between the extended ends of the α -chains to other fibrin strands to form a three-dimensional fibrin mesh.

The third category of regulation involves naturally occurring circulating protease inhibitors. These plasma proteins have been compared to families of naturally occurring tissue and plant protease inhibitors. There are five inhibitors that may inactivate the proteases involved in hemostasis (table 2.2).

Alpha-2-macroglobulin, which contains two sets of disulfide-bonded dimers, competes with macromolecular substrates and envelops a broad spectrum of serine, thiol, carboxy or metallo-proteases in a "trap" that is rapidly cleared from plasma (122). Alpha-1-antitrypsin is best known for its deficiency state in patients which leads to lung or liver diseases; its major action appears to be the inhibition of neutral proteases from inflammatory reactions including neutrophil elastase or tissue enzymes. Because of its broad specificity, it is often referred to as the alpha-1-proteinase inhibitor. Like the next three inhibitors, it inhibits serine proteases by forming 1:1 stoichiometric complexes that inactivate the inhibitor upon association; affinity of alpha-1-antitrypsin for active coagulation factors is relatively weak. Antithrombin III shares sequence homology with alpha-1-antitrypsin and, by itself, antithrombin can progressively inactivate all serine proteases of hemostasis by forming high-molecular-weight, stable complexes (123). Heparin and contact with endothelium markedly accelerate this reaction accounting for an immediate, potent, anticoagulant effect. Congenital deficiency of antithrombin III predisposes individuals to venous thromboembolism. C1 esterase inhibitor is primarily known since congenital deficiency leads to hereditary angioedema. This protease inhibitor of the complement system is a potent inhibitor of plasma kallikrein, particularly when it is not bound to high-molecular-weight kininogen (122). Although the recently described alpha-2-antiplasmin circulates in the lowest concentration of these inhibitors, its inactivation of plasmin is the most avid and immediate inhibition known (124). In purified systems, alpha-2-antiplasmin will inactivate most of the contact proteases when they are no longer surface bound.

In comparing the effect of these different inhibitors upon thrombin activity, for example, relative kinetics of inhibition show that antithrombin III is more potent than alpha-1-antitrypsin or alpha-2-macroglobulin when small synthetic substrates are used for the assay. In comparing the effect on thrombin with fibrinogen as a substrate, alpha-2-macroglobulin becomes more important (122).

THROMBUS DISSOLUTION

Plasmin Generation and Fibrinolysis

The fibrinolytic system removes unwanted fibrin deposits to reestablish flow in vessels occluded by thrombus and to facilitate the healing process following inflammation and injury. It is a multicomponent enzymic system composed of circulating zymogen (plasminogen), activators, cofactors and inhibitors (125). Although a variety of proteases are associated with fibrinolytic reactions, the best characterized is plasmin (figure 2.7). This enzyme circulates as the proenzyme plasminogen

Table 2.2--Plasma Protease Inhibitors

<u>Protein</u>	<u>Molecular Weight (Number of Chains)</u>	<u>Plasma Concentration (mg/dl)</u>	<u>Mechanism of Inhibition</u>
α_2 -macroglobulin	725,000 (4)	210*	Nonactive site binding of virtually any proteolytic enzyme
α_1 -antitrypsin	54,000 (1)	200*	
antithrombin III	62,000 (1)	29	More specific active site, stoichiometric complex with serine protease(s)
C ₁ -esterase inhibitor	105,000 (1)	18	
α_2 -antiplasmin	63,000 (1)	6	

*Because of its size, relative micromolar concentration of α_2 -macroglobulin is 1/12 that of α_1 -antitrypsin.

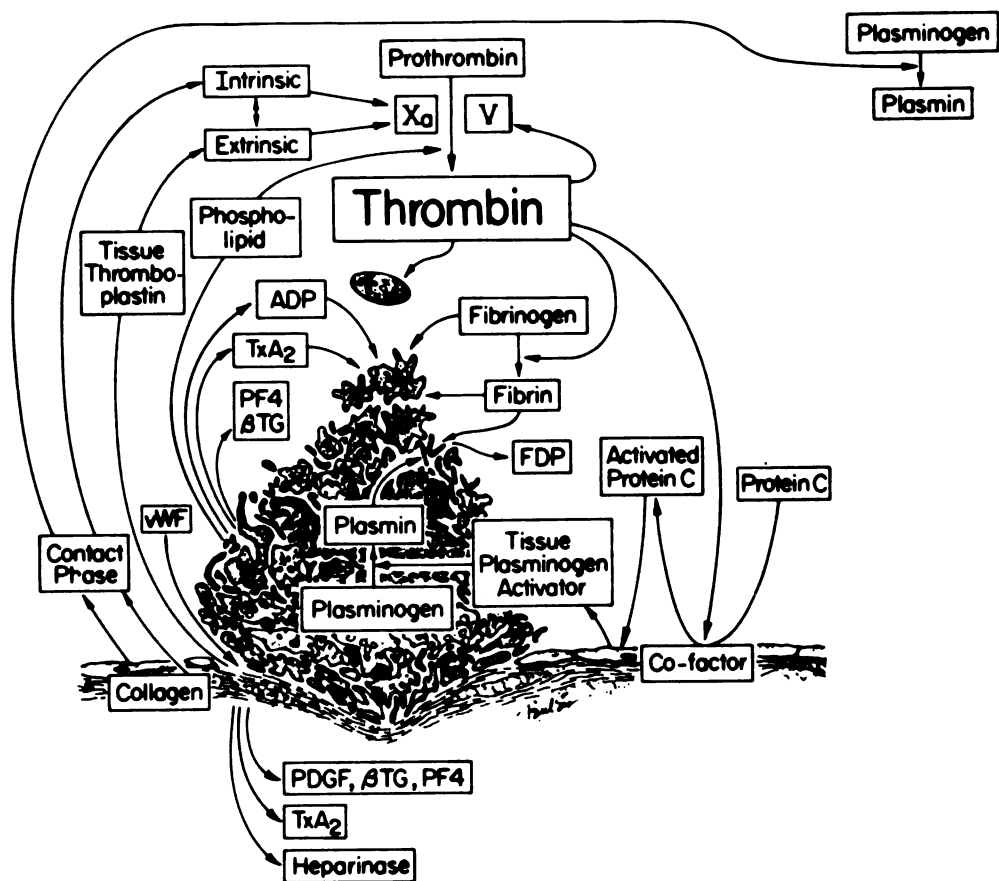


Figure 2.7--Fibrinolysis proceeds primarily within thrombus.

and is then converted to an active serine protease by limited proteolysis. The fibrinolytic system interacts with the coagulation system at the level of contact activation. After modification, plasminogen adheres to a fibrin clot, being incorporated within the fibrin mesh during polymerization. Plasminogen activation may occur by three different pathways: an intrinsic pathway in which all components involved are present in precursor form in the blood, an extrinsic pathway in which the activator originates from tissues or from the vessel wall and is released into the blood by certain stimuli or trauma, and an exogenous pathway in which the activating substances streptokinase or urokinase may be infused for therapeutic purposes.

Plasminogen is a single chain plasma glycoprotein which is similar in size to the contact activation factors. Its molecular weight is 90,000 (125). It is produced by the liver and has a biological half-life of about 2 days. Its plasma level (120 $\mu\text{g/ml}$ or comparable to that of prothrombin) increases with chronic inflammation and falls with either consumption or decreased liver production, paralleling changes in fibrinogen concentration.

Intrinsic activation of plasminogen may occur by one or more pathways involving factor XII, prekallikrein, high-molecular-weight kininogen, and a distinct plasma protein (proactivator). The plasma proactivator is converted to an active form by kallikrein, derived from contact activation. An endothelial cell protease can generate kallikrein through factor XII activation (126). An intrinsic mechanism is presumably responsible for the circulating degradation products and fibrin monomers found during disseminated intravascular coagulation.

Fibrinolysis in vivo proceeds predominantly within the thrombus (figure 2.7). During fibrin formation, plasminogen binds to fibrin and is thereby incorporated into the consolidating thrombus (127). Tissue plasminogen activator, which is derived from intact adjacent endothelium, also is specifically bound to, and activated by, fibrin (128-130). Moreover, activated protein C, which is generated by the action of thrombin and an endothelial cofactor (thrombomodulin) on protein C, markedly increases the endothelial release of tissue plasminogen activator (131). Direct activation of fibrin-bound plasminogen by fibrin-bound tissue plasminogen activator leads to the local release of plasmin (132). In this manner, plasmin generation proceeds within the thrombus where it is protected from inactivation by plasma alpha-2-antiplasmin.

Within the thrombus, plasmin digests fibrin to produce progressively smaller fibrin degradation products (FDP) and, ultimately, thrombus dissolution. Circulating plasminogen may also be converted to plasmin by components of the intrinsic coagulation cascade (133). However, plasmin generated in this manner is rapidly inactivated by plasma protease inhibitors, especially alpha-2-antiplasmin, and probably plays little role in fibrinolysis in vivo.

Once plasmin is formed, it becomes an active serine protease with relatively broader specificity than thrombin. It hydrolyzes not only Arg-X but Lys-X bonds and can dissolve insoluble fibrin polymers (134). Cleavages of fibrin monomers occur in restricted regions and produce

first the larger X and Y fragments and then smaller D and E fragments. Soluble X and Y fragments and intermediate complexes produce anticoagulant effects reflected in screening tests such as the thrombin and bleeding times. Once the lower molecular weight fragments are formed, they are rapidly cleared from plasma. In vivo, it appears that a fibrin mesh can be solubilized with incomplete digestion, since cleavage of only some of the chains within a given strand is needed, producing a variety of cross-linked complexes (134). On electron microscopy, images of X, Y and D fragments, DD dimers and DED complexes have been visualized (135).

Factors V and VIII:C also serve as substrates for plasmin degradation in the circulation. Indeed, in secondary fibrinolysis associated with severe disseminated intravascular coagulation, circulating levels of these factors may become sufficiently low to compromise hemostasis. In such situations, the fibrinogen levels are also severely depressed due to simultaneous consumption and degradation.

REGULATORY REACTIONS AMONG HEMOSTATIC COMPONENTS

Endothelial cells, platelets, and the coagulation and fibrinolytic systems interrelate in a number of ways to promote localized thrombus formation while preventing generalized thrombosis (figure 2.8).

The luminal surface of normal endothelium is an inert barrier preventing constituents of the blood from interacting with subendothelial structures and forming thrombus on the vessel wall (136). Moreover, intact endothelium actively resists thrombus formation locally whenever activated species of the platelet or coagulation systems circulate.

For example, platelet aggregate formation is prevented by the endothelium despite the presence of ADP, epinephrine, or thrombin, through the inactivation of ADP (137), active clearance of vasoactive amines (138), facilitated complexing of thrombin with antithrombin III (139), and the explosive thrombin-mediated synthesis and release of inhibitory PGI₂ (prostacyclin) (140). These mechanisms markedly decrease the possibility of thrombus forming at distal sites and limit the local extension of thrombus.

The fact that platelets do not react with normal vascular endothelium has been ascribed in part to the extracellular matrix (including membrane proteoglycans) of the endothelium (136). In addition to this passive protection, the following are mechanisms by which endothelial cells may actively reduce platelet reactivity and limit thrombus formation: a) thrombin-mediated synthesis and release of antiaggregatory PGI₂ (prostacyclin) (140); b) metabolism of proaggregatory ADP released by platelets to inactive adenosine monophosphate (AMP) or inhibitory adenosine by a membrane-associated adenosine diphosphatase (ADPase)(137); c) uptake and degradation of proaggregatory vasoactive amines (138).

Nonendothelialized surfaces initiate thrombus formation. This process initially involves the attachment of platelets to the surface through the interaction of a platelet receptor (GPI_b), subendothelial

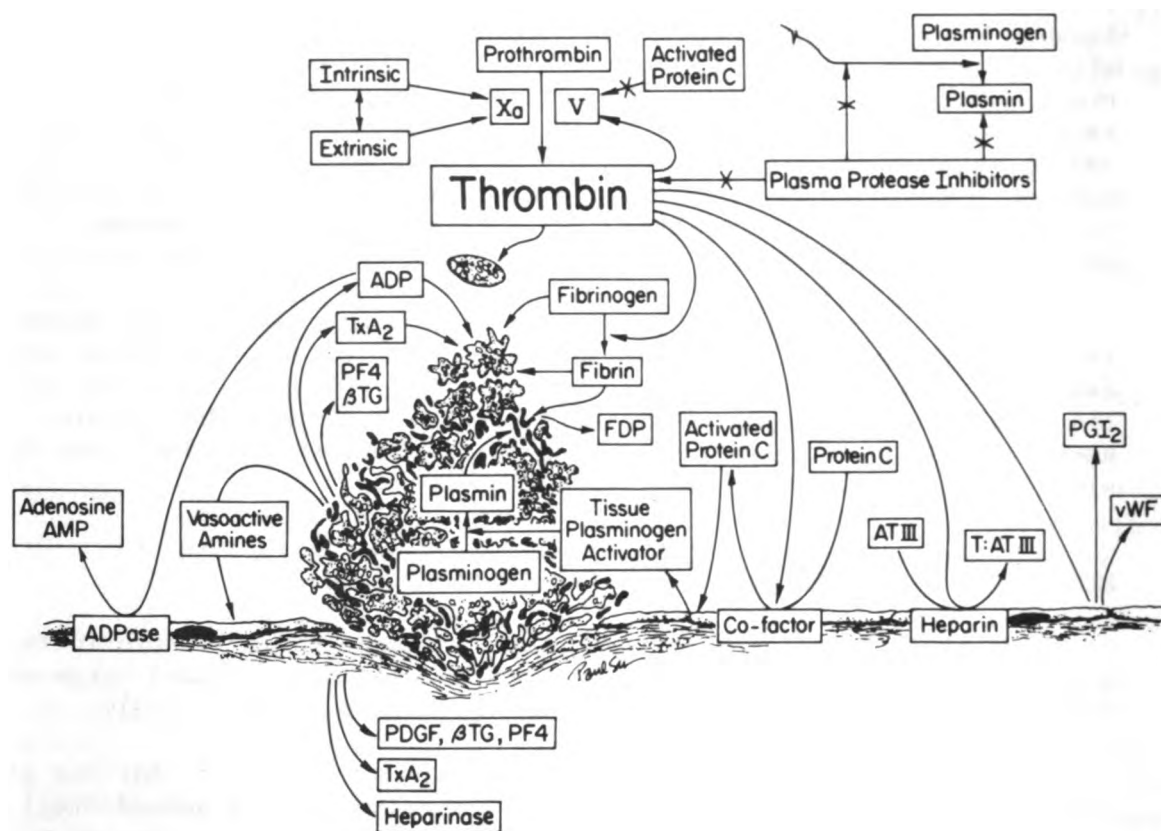


Figure 2.8--Interactions among thrombotic components and regulators.

collagen and a plasma cofactor, vWF (107-110). Fibronectin and thrombospondin derived from plasma, platelets, and endothelium may also play roles in this process (50-55). An unstable platelet mass forms through several interactive but independent recruitment mechanisms including the generation of thromboxane A_2 , release of dense granule ADP, and thrombin-mediated platelet-membrane fibrinogen receptor complex (GPII_b and III_a) and calcium-dependent interplatelet bridging by fibrinogen (64-68). This binding reaction may also involve the alpha-granule proteins thrombospondin and fibronectin (67,68). Regulation of fibrinogen receptors appears to be important in modulating platelet function. The appearance of platelet-specific alpha-granule proteins in plasma and urine serves as an indicator of platelet activation in vivo (141). The deposition of platelet factor 4 in the subendothelium may play a role in neutrophil and macrophage chemotaxis (142), and platelet-derived growth factor appears to be important in the intimal migration and proliferation of smooth muscle cells (143).

Thrombin plays a pivotal role in the formation of thrombus. Thrombin is generated locally on the platelet surface through both extrinsic and intrinsic pathways (105,113). All injured cells release tissue thromboplastin to activate extrinsic coagulation. Hageman factor is activated both by disrupted endothelial cells (126), and by subendothelial connective tissue (96). Platelets promote subsequent activation of the early stages of intrinsic coagulation (127) by a process that may involve a factor XI receptor and high-molecular-weight kininogen. Factors V and VIII also specifically bind to platelets; they act with membrane phospholipid to facilitate the activation of factor X to Xa and in the final conversion of prothrombin to thrombin (72,76). Low concentrations of thrombin are formed before fibrin is formed. These concentrations are sufficient to stimulate platelet release of alpha-granules at levels below which fibrinogen is cleaved (73-75).

The potent effects of thrombin are actively limited to the site of vascular injury by three important protective mechanisms. The first involves direct inactivation of circulating thrombin by the plasma protease inhibitors. The second reaction, preventing the escape of thrombin into the circulation, involves the enhancement by the endothelial surface of thrombin-antithrombin III complex formation (139). Third, thrombin released into the circulation binds to thrombomodulin, a receptor on the luminal surface of the endothelial cells. The resultant complex activates protein C (121) which destroys factor V and VIII. Thus thrombin initiates negative feedback mechanisms controlling its own generation.

Fibrinolysis is localized within the thrombus. Plasminogen binds to fibrin in the consolidating thrombus and tissue plasminogen activator, derived from the endothelium adjacent to the forming thrombus, is specifically coupled to fibrin. Direct activation of associated plasminogen proceeds with the local release of plasmin within the thrombus, where it escapes inactivation by circulating alpha-2-anti-plasmin. Moreover thrombin-dependent activation of protein C markedly increases the release of tissue plasminogen activator (131).

Within the coagulation system there are several lines of evidence which indicate the coagulation in the body does not proceed entirely as clotting does in the test tube, despite the clinical utility of in vitro factor tests. The initiation reactions are obvious cases in point; in addition, however, the modification of cofactors and the mechanisms limiting complete consumption of clotting factors also need to be considered.

Under certain conditions, factors XIIa and IXa can generate factor VIIa activity and factor VIIa itself may influence intrinsic clotting by activating factor IX (113). Although such interactions can be demonstrated in highly purified, in vitro systems, their relevance to complex conditions remains to be assessed. They do suggest that the simple, intrinsic-extrinsic system model of clotting is an oversimplification of a much more complex physiologic process.

It is not clear how thrombin can modify the cofactors factor V and factor VIII:C to their active forms. These reactions are difficult to study in the test tube because of susceptibility of these molecules to inactivation by further proteolytic cleavage by various plasma and cellular proteases.

Surface-mediated reactions also deserve consideration in the binding of active enzymes (beta-HFa). For example, once factor XII is cleaved to a low-molecular-weight enzyme, it is released from the portion of the molecule that binds to the negatively charged surfaces. Although it retains activity in other systems, it is no longer able to activate factor XI. Through its activation of prekallikrein and kallikrein's reciprocal activation of factor XII leading to a final, soluble (nonsurface-bound) factor XIIa, there are additional pathways involved. The first relates to the effect of kallikrein on kininogens to release kinins which mediate inflammatory responses. In this regard, the C1-esterase inhibitor functions to inhibit kallikrein thereby linking clotting to the complement system. One of the smaller factor XII fragments itself may enhance vascular permeability as well (97). Other established pathways involve the formation of a plasminogen activator from a proactivator; again this activation is kallikrein-dependent. The significance of this plasma reaction (or its relationship to tissue-derived activator substances) to in vivo clot dissolution is unclear, yet it may well explain the generation of plasmin which leads to the soluble fibrin degradation products seen in disseminated intravascular coagulation.

Interactions between inflammation and hemostasis are established (144). Activated mononuclear leukocytes release tissue factor and thus activate extrinsic coagulation (114), generate thrombin locally by secreting a direct prothrombin activator (145) and produce elastase-mediated modification of coagulation factors and plasminogen-independent fibrinolysis. Thromboxane A₂ and PGI₂ produced by activated leukocytes and other products of activated neutrophils affect platelet-endothelial interactions as well as vascular tone. The production of acetyl glyceryl ether phosphorylcholine (AGEPC) by leukocytes activates platelets (146). The mononuclear phagocytic system also functions in pathologic processes as an avid scavenger of alpha-2-macroglobulin-active

protease complexes (122), whereas granulocyte peroxides can inactivate alpha-1-antitrypsin (147). Conversely, the hemostatic mechanism affects inflammation. Granulocyte-endothelial adherence (148) and chemotaxis of leukocytes is induced by platelet released products, fibrinopeptide B (114) and platelet factor 4 (142). Platelet factor 4 also stimulates the release of elastase from leukocytes (149). Permeability is increased by platelet cationic protein (150). Platelets also possess Fc receptors and initiate complement complex formation, a process that may enhance platelet coagulant activity (151). Complement components also interact with vitamin K-dependent protein C (152) and other coagulation factors (144).

TURNOVER OF A THROMBUS

Studies of thrombus formation have shown that when a platelet-rich thrombus forms, it may undergo continuous dissolution and reformation; the constituents comprising permanent thrombi turn over. For example, as fibrin and platelets are lost from experimental pulmonary emboli through fibrinolysis, new fibrin and platelets are laid down through activation on the surface of the embolus. Also, fibrin continues to be laid down in coronary artery thrombi in man for several hours after initiation of the thrombus. The most rapid removal of such an embolus can be achieved with a combination of heparin and activated fibrinolysis (153,154).

Important distinctions between arterial and venous thrombogenesis are illustrated by kinetic studies using labeled platelets and fibrinogen. In patients with ongoing arterial thromboembolism, the principal role of platelets in the thrombotic process is evidenced as isolated platelet consumption, i.e., shortened platelet survival and increased platelet consumption. Circulating fibrinogen is not detectably consumed in this setting, presumably because procoagulant material is swept away from the thrombogenic focus by the rapid arterial flow before coagulation factors become fully activated. On the other hand, ongoing venous thrombosis is manifest kinetically as consumption of both platelets and fibrinogen at equivalent rates, since it reflects extension of thrombus under static or low shear flow conditions (155). Thus, the composition of the thrombus has important therapeutic implications, i.e., fibrin formation in venous thrombosis is most effectively inhibited by anticoagulants such as heparin or warfarin, whereas some arterial thrombi may be prevented by agents that modify platelet behavior.

The fate of a thrombus also depends on its nature and site. In an arterial thrombus, platelet deposits undergo changes of cell disruption and lysis such that the platelet mass becomes changed morphologically to a "fibrin" thrombus within 12-24 hours. The thrombus is also invaded by polymorphonuclear leukocytes and mononuclear cells, perhaps through the chemotactic effects of fibrinopeptide B and platelet factor 4 (69,90). These cells phagocytize cellular debris. Thus after 2 to 3 days, the amount of fibrin is markedly reduced, and phagocytic cells are present in increasing numbers. By 4 to 7 days, the margins of the thrombus are covered with endothelium, and elongated smooth muscle cells may appear in the thrombus. Thereafter, the amount of fibrin is

further reduced and there is evidence of collagen and elastin formation. A week or so later, the initial platelet mass has organized into a thickening of the arterial intima, rich in collagen, smooth muscle cells, and elastic fibers, i.e., a nonspecific intimal lesion. The intimal migration and proliferation of smooth muscle cells may be mediated by the platelet-derived growth factor released from platelet alpha-granules (156).

When small vessels are injured, the platelet mass formed at the site of damage readily builds up and breaks down. The mass may be so unstable that it fragments before sufficient fibrin accumulates to stabilize it or alternatively fibrin may be digested through fibrinolytic activity leading to fragmentation of the thrombus. These processes may proceed simultaneously; the sequence in any given instance depends on whether factors leading to the growth and persistence of the thrombus overcome those causing its disruption. The microcirculation could be obstructed by thrombosis or thromboembolic processes involving at least three mechanisms: a) embolic material derived from mural thrombi in larger proximal arteries; b) intravascular stimuli that cause platelet aggregation; and c) direct injury to the endothelium in the microcirculation.

DETECTION OF ONGOING THROMBUS FORMATION

Since activation of the coagulation mechanism and the fibrinolytic pathway leads to changes in plasma proteins, numerous approaches have been developed to detect activated clotting factors and fragments of fibrinogen or fibrin in blood as an indication that thromboembolic processes are occurring (figure 2.9; 157-159). In venous thrombosis, it is possible to detect increased activity of clotting factors in plasma as well as fibrinopeptides which are cleavage products of the action of thrombin on fibrinogen (158). Degradation products of fibrinogen and fibrin formed by activation of plasminogen have been found in the plasma of patients with clinical evidence of thromboembolic processes (135).

Attempts to detect altered platelets during thromboembolism have followed three main lines of study (160). These approaches include detection of hypersensitive platelets in terms of platelet aggregation and the release reaction (160), changes in platelet survival (161), and the detection of products released from platelets such as beta-thromboglobulin and platelet factor 4 (162). The involvement of the platelet release reactions in thromboembolic processes has been shown by the increased levels of platelet factor 4 and beta-thromboglobulin in the plasma of some individuals during what appear to be thromboembolic episodes.

Decreased platelet survival has been shown to correlate with clinical manifestations of arterial disease (161). The mechanisms responsible for shortened platelet survival appear to be related to the factors involved in thrombosis and loss of the endothelium.

However, our ability to detect individuals predisposed to clinical thrombosis is limited at present. Despite the large number of blood

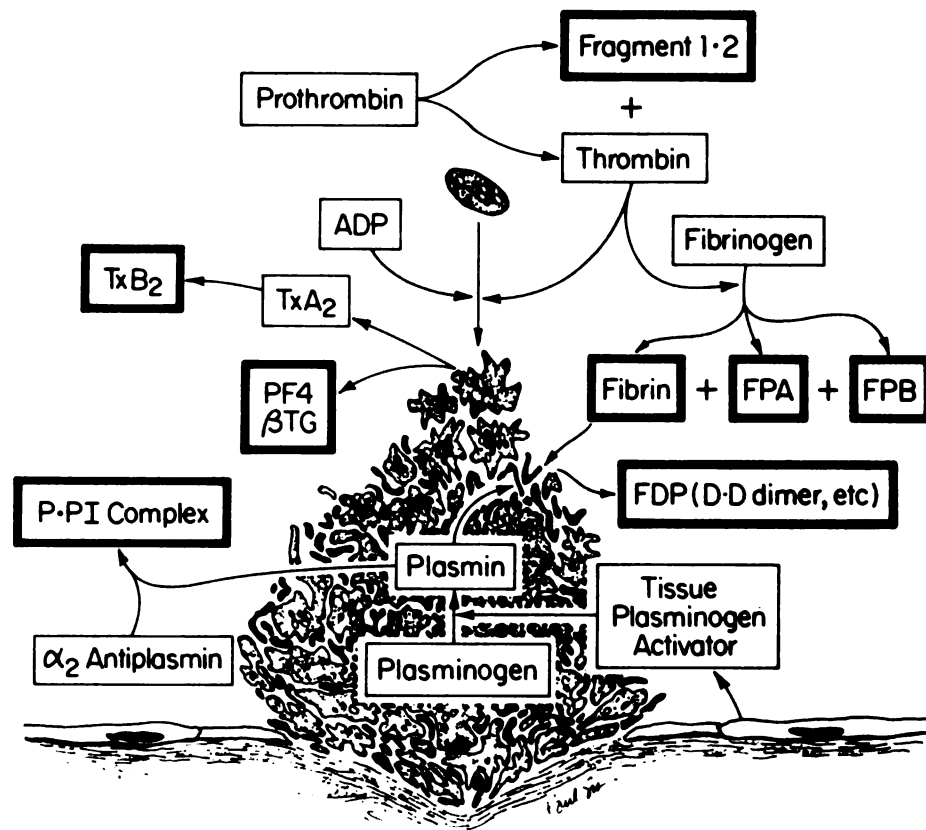


Figure 2.9--Experimental laboratory evidence of ongoing thrombotic disease. Evidence can arise from specific immunoassays of protein metabolites indicated in the heavy boxes. These include the amino-terminal half of prothrombin (fragment 1 • 2) from thrombin generation, or from the effects of thrombin to stimulate platelet release of β -thromboglobulin (β TG) or platelet factor 4 (PF4). Platelet reactivity can be further documented by measuring plasma levels of the more stable metabolite thromboxane B₂. Blood levels of fibrinopeptides A and B reflect the proteolysis of fibrinogen by thrombin, and soluble fibrin monomers can also be detected. As the result of fibrinolysis, circulating levels of plasmin- α_2 -antiplasmin complex (P-PI) as well as specific fibrin degradation products (FDP) can be shown. With the exception of the B β form of fibrinopeptide B, which is found after plasmin degradation of the amino-terminal portion of the B β -chain of fibrinogen (following thrombin cleavage of the A α chain), other tests cannot distinguish between thrombin formation in disseminated intravascular coagulation and ongoing thrombotic disorders.

tests that have been proposed, few of these are predictive or practical. Moreover, those tests that may be useful for identifying risk of venous thrombosis have little or no predictive value in individual patients with arterial thrombosis and vice versa.

Certain clinical and pathophysiological states have been associated with an increased frequency of arterial or venous thrombosis. Unfortunately, the relative importance of these various risk factors and their interactions remains unclear.

MECHANISM OF ANTITHROMBOTIC DRUGS

There are three classes of antithrombotic agents, each directed at different aspects of the thrombotic process (table 2.3). Moreover, a specific agent may affect only one of several activation pathways for a given component. This section deals with the mechanism of action for each class of agents used. Since thrombolytic therapy potentially represents the means for removing established thrombus, and therefore, is at least theoretically, the initial therapeutic consideration, it is discussed first, despite the present uncertainty regarding many of the proposed indications and risks. Anticoagulant therapy is much more clearly defined with respect to its indications and efficacy, especially in venous disease. Currently available drugs that affect platelet function have few well established clinical indications at present.

Fibrinolytic Therapy

Theoretically, thrombi and thromboemboli can be effectively removed by thrombolytic agents, and two fibrinolytic drugs, streptokinase and urokinase, have been used clinically. Urokinase is a trypsin-like enzyme found in trace amounts in human urine. Of experimental interest is the tissue-culture-derived plasminogen activator preparation that differs from urokinase in its ability to localize in established thrombus by binding to fibrin; it is currently under study in animals. Tissue activator(s) and urokinase directly activate plasminogen to plasmin by limited proteolysis. Streptokinase is a nonenzymatic protein excreted by group C beta-hemolytic streptococci. In contrast to tissue activator and urokinase, streptokinase acts by forming a complex with plasminogen, allowing its active site to become exposed, and generating plasmin from other plasminogen molecules. In contrast, urokinase and tissue activator(s) directly activate plasminogen to plasmin by limited proteolysis.

Both urokinase and streptokinase are currently approved by the Federal Drug Administration for use in massive pulmonary emboli (163) and streptokinase is also approved for use in extensive deep venous thrombosis, AV-cannula occlusion, and intracoronary administration. There is no demonstrated difference in efficacy between streptokinase and urokinase except in patients with high antistreptococcal-antibody titers in whom only urokinase is effective. Both drugs are administered by an initial loading dose followed by continuous intravenous infusion, required because of the rapid clearance from the circulation. The

Table 2.3--Antithrombotic Agents

Agent	Action	Administered Route	Major Indications
<u>Fibrinolytic</u>			
Streptokinase Urokinase (Tissue Activator)	Plasminogen activators	Intravenous, continuous infusion for 24-72 hours followed by heparin	*Acute, massive PE ?DVT (may increase risk of postphlebotic symptoms)
		Intracoronary artery infusion at low doses	Acute myocardial infarction (under study)
<u>Anticoagulant</u>			
Heparin	Markedly enhances inhibition of proteases by thrombin III	Intravenous, continuous infusion	*Deep venous thrombosis (DVT) *Pulmonary embolism (PE) (7-12 days)
		Subcutaneous Full doses	*DVT during pregnancy or for recurrence when in control on oral agents
		Low doses	Preventive use in high-risk patients during surgery or bed rest
Warfarin	Vitamin K antagonist	Oral, daily	*Longer term prevention of recurrence (2-3 months with uncomplicated, distal DVT: 3-6 months with PE or proximal DVT)
<u>Antiplatelet</u>			
Aspirin	Decreases platelet aggregation and release	Oral	*Transient ischemic attacks
Sulfipyrazone	Not established	Oral	?Prevention of recurrent myocardial infarction
Dipyridamole	Decreases platelet adhesion	Oral	*Artificial heart valves (plus warfarin) to prevent systemic and cerebral emboli *Decreases (with aspirin) occlusion of saphenous vein coronary artery bypass
Ticlopidine	Blocks ADP induced platelet interaction with fibrinogen and vWF	Oral	?Under investigation
PGI ₂	Activates platelet adenylyl cyclase	Local intravenous	Under study in hemodialysis, cardiopulmonary bypass, peripheral vascular disease

*Well established indication.

bacteria-derived product is about one-tenth as expensive for a comparable dose but patients may become transiently refractory to subsequent administration.

The amount of plasmin generated by streptokinase or urokinase must first exceed the capacity of plasma alpha-2-antiplasmin to inactivate the enzyme. Free plasmin not only digests fibrin but also destroys circulating fibrinogen, factor V, and factor VIII:C, producing a potentially serious hemostatic defect. Although several coagulation screening tests are altered by the proteolytic effects of plasmin, it is sufficient to follow therapy with the thrombin time. This test becomes prolonged by the inhibitory effects of higher molecular-weight fibrinogen degradative products and hypofibrinogenemia.

Patients receiving fibrinolytic therapy are at significant risk of bleeding if they are subjected to any traumatic procedures. Perhaps of greater importance is the risk of bleeding from sites of previous vascular injury since fibrin-stabilized platelet thrombi may be readily lysed. The latter consideration applies to postoperative wounds less than 2 weeks old.

Anticoagulants

Heparin

This sulfonated mucopolysaccharide has repeating disaccharide units which are primarily iduronic acid and N-acetylglucosamine. Two to three sulfate groups are attached per disaccharide unit, and at the reducing end of the sugar, the molecules are bound in vivo to several seryl residues of specific binding proteins. It should be noted that the molecules are "polydisperse" with a molecular-weight range between 6,000 to 14,000. A specific tetrasaccharide sequence with two nonsulfonated uronic acid residues is required for the anticoagulant activity, which is mediated through its binding to antithrombin III. The heparin-antithrombin complex is an immediate, potent inhibitor of activated coagulation proteases (123).

Anticoagulant-active fractions of heparin represent a variable, relatively small portion of commercial preparation. Organ or species of origin or the preparative process itself may produce heparins of different specific activity, although there is no clear therapeutic benefit of any of the preparations over the commonly available type from porcine intestine. Consequently, doses are standardized for anticoagulant effect, i.e., units as opposed to weight, by clotting screening tests.

The presence of heparin is manifest by the immediate inhibition of screening clotting tests. The thrombin time is the most sensitive, as it is markedly prolonged at even 0.01 units/ml of heparin. The next most sensitive screens are tests of intrinsic clotting such as the partial thromboplastin time (PTT) or recalcification time. These are usually doubled by about 0.2 units/ml of heparin; therapeutic levels are 0.2 to 0.5 units/ml of whole blood. The extrinsic screening test, the prothrombin time, is only slightly prolonged by even 0.5 unit/ml. Thus, the prothrombin time can be used to assess the status of oral anticoagulation without interrupting anticoagulation

with continuous heparin infusion. In general, the more thrombin added or generated in these tests the less sensitive that test will be to inhibition. Another important factor in screening test sensitivity to heparin may relate to the specific rates of inhibition of the various steps in each pathway; the amounts of factors Xa or IXa generated are much smaller than thrombin and the earlier enzymes may be more reactive to heparin inhibition, especially in intrinsic coagulation.

Currently, heparin is indicated acutely (and in some instances, chronically) in patients with venous thromboembolism to prevent recurrence or extension while endogenous fibrinolysis occurs. It is also used during open heart surgery or renal dialysis. Low doses have been administered in attempts to prevent venous thromboembolism.

Warfarin

The coumarin nucleus is similar chemically to the naphthoquinone derivative, vitamin K, and indirect evidence strongly suggests that they interact at a common receptor site. The four vitamin K-dependent clotting factors, II (prothrombin), VII, IX and X have specific glutamyl residues near their amino-terminal ends that are carboxylated in a post-protein synthesis reaction, forming tight calcium binding sites (164). In the absence of vitamin K or in the presence of its antagonists such as coumarin derivatives, this final carboxylation is incomplete, and abnormal proteins then circulate with markedly impaired activity.

The pharmacologic response depends upon the net effect of factor levels achieved. Since the four factors have different half-lives, depletion of factor VII, which has the shortest half-life, can prolong the prothrombin time within 24 hours following large doses of warfarin prior to a significant effect upon the other factors. It is therefore common practice to begin warfarin therapy with two daily doses of 10 mg each and reevaluate the dose based upon the results of the daily prothrombin time until it equilibrates in the therapeutic range. Prothrombin times vary considerably in their dose-response curves, however, such that they need to be carefully standardized. Approximately 1 to 2 weeks of coumarin administration is required for all four factors to reflect completely a given dosage. The range of response among different individuals is particularly wide; from 2 to 18 mg/day may be required to maintain a patient in therapeutic range once equilibration has been achieved. When monitoring, the prothrombin time reflects the dose given 36 to 48 hours previously. To prevent inordinant risks of bleeding, a partial hemostatic defect is produced, i.e., approximately 10 percent of the normal levels of factors II, VII, IX, and X, which is clinically similar to mild hemophilia.

Drug interactions play an important role in coumarin therapy. Since warfarin contains an asymmetric carbon atom, there are two optically active isomers. In the presence of some drugs the metabolic effects are different for the two isomers. Thus, the potentiating effect of phenylbutazone, for example, is, in part, due to the increase of enzymes that metabolize the less active form in preference to the more active form. This leads to no net change of plasma warfarin yet a potentiation of the anticoagulant effect. Other drugs

can compete with the hydroxylase that initiates breakdown of the active anticoagulant. Barbiturates induce microsomal enzymes in the hepatocytes which increase the metabolism of both forms of warfarin and thus antagonize the anticoagulant effect of the drug. Aspirin has a variable effect on synthesis, and occasionally some individuals on large daily doses of aspirin alone demonstrate vitamin K antagonism. Nearly any strongly acidic drug can displace albumin-bound warfarin in vitro, but other types of interactions may predominate.

Agents That Modify Platelet Behavior

Antiplatelet agents may interfere with thrombus formation at several steps. They may prevent platelet adherence to exposed subendothelium, inhibit primary platelet aggregation or secondary platelet release, and perhaps disaggregate platelets already present in a thrombus. Presently, the three agents available for clinical use are aspirin, dipyridamole, or sulfinpyrazone. All of these agents are available for oral use, are relatively safe, and have shown therapeutic benefits in some clinical trials and laboratory models of thrombosis. A more potent agent, ticlopidine, is currently being studied to assess its clinical utility versus complication rate. Infusions of the short-lived PGI₂ have also been performed experimentally in selected patient groups.

Aspirin irreversibly acetylates cyclooxygenase in platelets and thus blocks synthesis of thromboxane A₂. Inhibition occurs at very low doses; 80 mg in an adult will produce an 86 percent inactivation of cyclooxygenase, and the standard 325 mg tablet administered once daily produces a greater than 95 percent inactivation (165). Since platelets are not capable of synthesizing new protein, the inhibition is irreversible and lasts for the lifetime of the platelet (166). Megakaryocytes in the marrow may also be affected. Following a single dose of aspirin in a normal individual, the bleeding time may be prolonged to 8 or 10 minutes and remain detectably prolonged for 2 or more days until enough new, unaffected platelets are produced to normalize the bleeding time. Aspirin, however, will not correct shortened platelet survival times in patients with ongoing platelet consumption due to active thrombus formation (155).

Theoretically, single, small doses of aspirin could have a greater antithrombotic effect than larger, more frequent doses. At 80 to 325 mg daily, arachidonate metabolism is virtually blocked in all circulating platelets, but endothelial prostacyclin production is either less affected or recovers rapidly (166). Low-dose aspirin has been shown to be antithrombotic in patients with arteriovenous silastic cannulae (167). Thus, it has been suggested that large or frequent doses of aspirin could be less antithrombotic by completely inhibiting prostacyclin production by endothelium. However, this has never been directly demonstrated in vivo, and studies of patients taking large doses of aspirin for rheumatoid arthritis do not show any increase in thrombotic events. Similarly, the two British trials of secondary prevention of myocardial infarction gave similar results despite a threefold difference in the aspirin dose (168,169). Indeed, aspirin in larger

doses has been shown to have antithrombotic effects apart from inhibition of platelet thromboxane production in primate models of arterial thromboembolism. None of the controlled clinical studies available have yet resolved the question of high- versus low-dose aspirin.

Although aspirin completely blocks platelet thromboxane A₂ production, it does not block collagen or thrombin-dependent pathways of activation, and thrombin plays a critical role in thrombus production in vivo. Aspirin also does not affect platelet adherence to exposed subendothelium or release of ADP from adherent platelets.

Sulfinpyrazone, a uricosuric agent, was first used in the management of gout. Structurally, it is similar to anti-inflammatory agents such as phenylbutazone. Like other nonsteroidal anti-inflammatory agents, sulfinpyrazone acts as a competitive, i.e., reversible inhibitor of cyclooxygenase (170). Patients taking therapeutic doses of sulfinpyrazone do not show prolonged bleeding times, but, like dipyridamole, shortened platelet survival times are prolonged by sulfinpyrazone in patients with ongoing thrombus formation (170). Thus, sulfinpyrazone probably has, as yet to be characterized, antithrombotic mechanism unrelated to its effect on inhibiting cyclooxygenase. The usual dose of sulfinpyrazone in antithrombotic therapy is 200 mg four times daily (95). Sulfinpyrazone appears to have fewer gastrointestinal side effects than aspirin, perhaps because it is a less potent inhibitor of gastric prostaglandin synthesis.

Dipyridamole is officially classified as a coronary vasodilator. In vivo it inhibits platelet adhesion, and in high concentrations in vitro it inhibits platelet aggregation and release (170). It can also normalize shortened platelet survival times in patients with ongoing clinical arterial thromboembolism (155). Interestingly, in patients taking dipyridamole, the bleeding times are generally normal. Dipyridamole inhibits platelet phosphodiesterase, the enzyme which degrades cyclic AMP to AMP. It has been proposed that dipyridamole may act in vivo by potentiating the effect of endogenous prostacyclin, since prostacyclin elevates platelet cyclic AMP and dipyridamole prevents its breakdown (171). Dipyridamole, with warfarin, reduced the thromboembolic complications of old-style artificial heart valves (92-94). It is currently used only in combination with other antiplatelet agents. There is platelet survival evidence in patients and in animal models of thrombogenesis that dipyridamole and aspirin are synergistic. Normalization of platelet survival in patients with ongoing platelet consumption is optimally produced by the regimen aspirin (325 mg) and dipyridamole (75 mg) given together three times daily (172).

Dipyridamole appears to act in vivo by synergistically modifying several biochemical pathways, including: 1) inhibition of platelet cAMP-phosphodiesterase; 2) potentiation of adenosine inhibition of platelet function by blocking reuptake by vascular and blood cells, and subsequent degradation of adenosine; and possibly, 3) potentiation of PGI₂ antiaggregatory activity and enhancement of function by increasing platelet cAMP through both a reduction in enzymatic cAMP-degradation, and stimulation of cAMP formation via activation of adenylcyclase by adenosine and possibly PGI₂. Only the inhibition

of cAMP phosphodiesterase appears to be involved in the dipyridamole inhibition of isolated platelets in vitro, since adenosine and PGI₂ originate in vivo from tissues other than platelets, and any blood concentrations existing in vivo will disappear before platelet-rich plasma has been prepared for in vitro platelet studies.

Ticlopidine, a new, potent platelet function inhibitor, has recently been introduced for clinical trials. It prolongs the bleeding time and abolishes in vitro primary aggregation of platelets to nearly all agonists. The mechanism of action appears to relate to the blocking of fibrinogen and von Willebrand factor interactions with membrane glycoproteins (173). It is anticipated that the risk of hemorrhagic complications of this more potent agent may be increased, although relative risks and potential benefits remain to be clarified.

Prostacyclin (PGI₂), a biologic substance, has been chemically synthesized and is available for experimental studies. Its half-life in vivo is probably 30 seconds to 2 minutes. Unlike prostaglandins of the E series, it is not metabolized by the lung, but approximately 50 percent is lost during a single transit through the circulation, probably being metabolized in the liver and the vasculature. In vitro, it is nonenzymatically oxidized to 6-keto PGF₁ alpha, a stable, inactive metabolite unique to PGI₂. In vivo, numerous other metabolites are also produced. PGI₂ is produced by a number of cells in the body including endothelial cells, smooth muscle cells, fibroblasts and macrophages. It is the most potent platelet antiaggregating agent yet described in nature and is also a potent vasodilator in all vascular beds. While PGI₂ is probably not present as a circulating hormone in vivo, it is clear that infusions of PGI₂ have potent hemodynamic and antithrombotic effects.

PGI₂ may be stabilized for experimental use by maintaining the compound at high pH and low temperature before infusion. It has been studied experimentally as an antithrombotic agent in charcoal hemoperfusion, cardiopulmonary bypass, renal hemodialysis, and peripheral vascular disease. Whether PGI₂ has any real advantage over the more stable antiaggregating and vasodilating prostaglandin PGE₁ is not yet clear. PGI₂ is currently available only as an investigational drug.

Combination chemotherapy with multiple antiplatelet agents in experimental animals produces enhanced effects by interfering with platelet thrombus formation at multiple steps. However, controlled clinical studies to date in human arterial thrombotic disorders have not been performed.

It is apparent that the ideal antiplatelet agent has not yet been found. Current research is directed at drugs which: 1) produce selective inhibition of thromboxane synthetase without affecting prostacyclin synthetase; 2) affect platelet adherence as well as aggregation and release; 3) enhance vascular PGI₂ production; and 4) inhibit both thrombin and thromboxane-mediated platelet aggregation. There are few such candidates presently available for clinical trials.

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Chapter 3

FLUID DYNAMIC AND HEMORHEOLOGIC CONSIDERATIONS

Physiologists have long recognized that local blood flow conditions can greatly alter cellular responses to biochemical stimuli. For example, thrombus material formed under high-flow-rate conditions (arteries and arterioles) consists mainly of platelets and was called "white thrombus." That formed under low flow (venules and veins) includes more fibrin and red cells and was named "red thrombus." High fluid shear rates in prosthetic heart valves are known to cause both lytic and sublytic damage to erythrocytes and other blood cellular elements. In addition, access of platelets and leukocytes to biomaterial surfaces in all blood-contacting prostheses is primarily controlled by the local fluid mechanics. The purpose of this chapter is to give an introduction to commonly used terminology and quantitative concepts in the areas of fluid dynamics and mass transfer. There is also a discussion of areas where these ideas would be of considerable relevance to blood-material interactions. Blood flows through vessels and artificial organs in response to imposed pressure differences and boundary wall movements generated in the vascular system or by mechanical assist pumps. If we consider the simple case of pressure-driven steady flow of plasma or any other Newtonian fluid through a long tube of constant cross-sectional area, the fully developed velocity profile is given by the following equation:

$$V_z = 2V_0 \left[1 - \left(\frac{r}{R} \right)^2 \right] \quad (1)$$

where V_z is the axial velocity, V_0 the mass average velocity and R the vessel radius (1,2) (see figure 3.1). This form of velocity profile is found as long as the fluid flow rate is relatively low or, more precisely, the Reynolds number* is less than about 2,100. This criterion is satisfied almost everywhere in human blood vessels. Fluid motion obeying equation (1) is referred to as Poiseuille flow.

*The Reynolds number (N_{Re}) is a dimensionless group defined for tube

flow as $N_{Re} = \frac{2\rho V_0 R}{\mu}$ where ρ is the fluid density and μ the fluid

viscosity. Qualitatively, it is a measure of the importance of fluid momentum or inertial forces relative to viscous forces. At low Reynolds numbers, viscous forces dominate, and at high Reynolds numbers, fluid inertial forces are of greater importance in momentum transport. For very high Reynolds numbers, steady laminar Poiseuille flow does not exist, being replaced by very complex, locally time dependent turbulent flow.

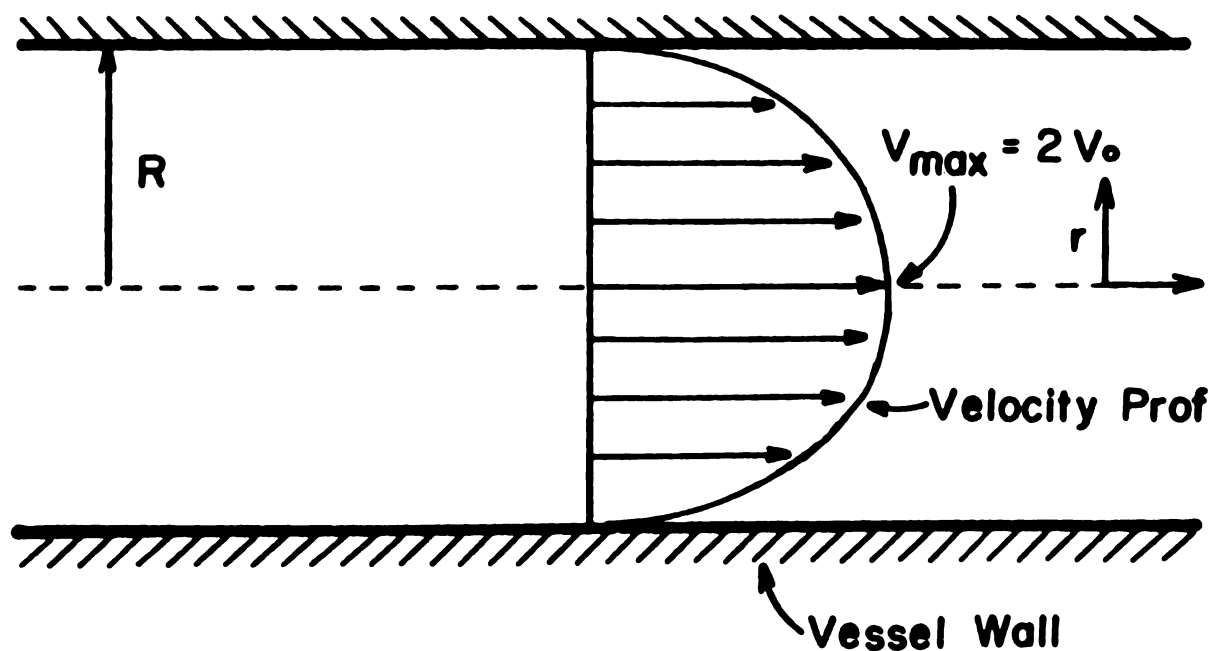


Figure 3.1--Velocity profile for steady pressure-driven flow of a Newtonian fluid in a cylindrical tube (vessel). R is the vessel radius, Z the axial (flow) direction, and the velocity profile is given by equation (1). The maximum fluid velocity is twice the mass average (V_0) and occurs along the vessel center line ($r = 0$).

A commonly used term is the shear rate ($\dot{\gamma}$). It is a measure of how rapidly adjacent fluid layers are sliding past each other (velocity gradient). For our example, the shear rate is given by (ignoring the sign)

$$\frac{dv_z}{dr} = \frac{4V_o r}{R^2} = \dot{\gamma} \quad (2)$$

and it is a linear function of radial position. The shear rate is zero at the tube centerline and maximum at the vessel wall. The wall shear rate ($\dot{\gamma}_w$) is found by setting $r = R$ in equation (2).

$$\dot{\gamma}_w = \frac{4V_o}{R} = \frac{8V_o}{D} \quad (3)$$

This is often given in terms of volumetric flow rate

$$[Q = \pi R^2 V_o = \frac{\pi D^2}{4} V_o] \text{ as}$$

$$\dot{\gamma}_w = \frac{4Q}{\pi R^3} = \frac{32Q}{\pi D^3} \quad (4)$$

$\dot{\gamma}_w$ has units of inverse time (sec^{-1}). Typical estimates for time average wall shear rates in various parts of the vascular system are shown in figure 3.2 (3,4). Actual values vary from zero in regions to stasis to 10^5 sec^{-1} in high flow arterioles and in some prosthetic heart valves (5,6).

The wall shear stress (τ_w) is a measure of the force per unit area the fluid is exerting on the vessel boundary in the flow direction and is the product of the wall shear rate and the fluid viscosity (μ).

$$\tau_w = \mu \dot{\gamma}_w = \frac{\mu 8V_o}{D} = \frac{32\mu Q}{\pi D^3} \quad (5)$$

If the units of viscosity used are poise (water has a viscosity of approximately 1 centipoise or 0.01 poise), then shear stress has units of dynes/cm². Thus a vessel containing water flowing at a wall shear rate of 10^4 sec^{-1} would experience a wall shear stress of 100 dynes/cm².

The Hagen-Poiseuille equation (1,2) relates the tube pressure drop and the average fluid velocity.

LARGE ARTERIES
 SMALL ARTERIES
 ARTERIOLES
 CAPILLARIES
 VENULES
 SMALL VEINS
 LARGE VEINS

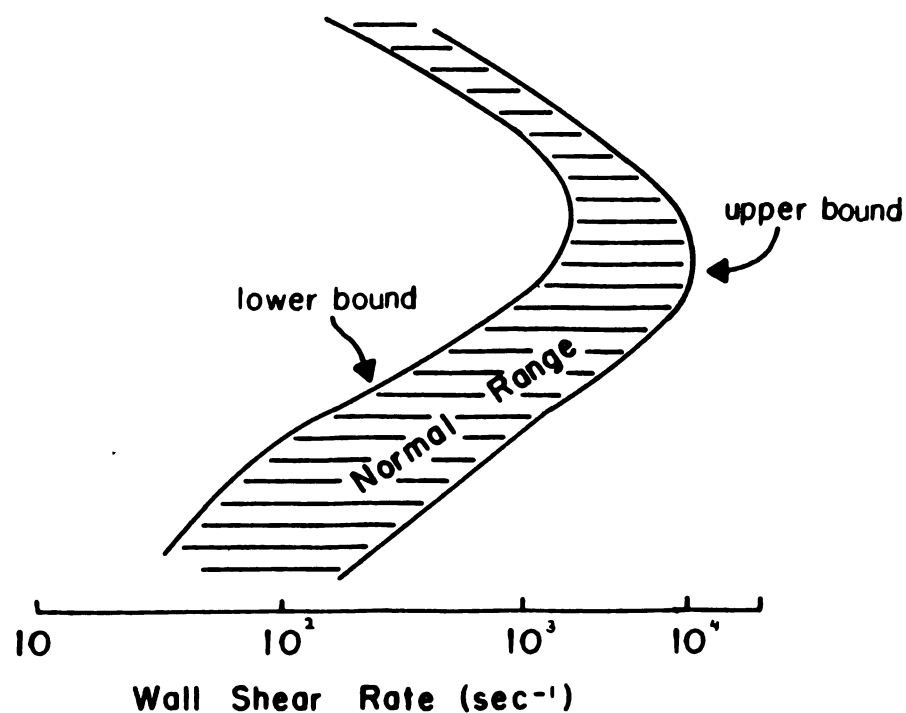


Figure 3.2--Upper and lower bounds for typical time average wall shear rates in the vascular system. Maximum values occur in the small arterioles (taken after reference (3)).

$$\Delta P/L = \frac{32V_o\mu}{D^2} = \frac{8V_o\mu}{R^2} = \frac{8Q\mu}{\pi R^4} \quad (6)$$

where μ is the fluid viscosity and $\Delta P/L$ is the local vessel pressure gradient (pressure drop per unit length). In most biological vessels, the pressure drop is pulsatile and therefore the velocities, wall shear rates, etc., are also time dependent. However, the equations given above are still a reasonable starting point if interpreted in terms of local time average values.

ARTIFICIAL ORGANS

In most blood contacting devices, the wall shear rate varies spatially across the biomaterial surface and is a function of time at a fixed spatial position. Blood that flows through a device passes through regions of different shear rates. The residence time of any cell or protein within each region of different shear rates varies, depending on its initial entry point into the device.

The local distribution of shear rates in blood flowing through a device can be calculated given the velocity field. The velocity field can be determined either by solving the conservation equations of mass and momentum, subject to proper boundary and initial conditions, or by experimental measurement. Experimental methods commonly employed include laser velocimetry, hot film anemometry and quantitative visualization (for example, tracking electrolytically generated hydrogen bubbles with cinephotography and image analysis) (7-10). Most of these methods work well in clear fluids, but have severe restrictions when applied to whole blood. This has led to most quantitative velocity and shear rate measurements in medical devices being taken with so-called blood-analog clear liquids. Hot film anemometers typically have films 1 mm long and are mounted on yet larger substrates, limiting their use to fairly large vessels or devices with fairly large clearances between biomaterial surfaces. Ultrasonic Doppler has poor spatial resolution, and thus velocity gradients (and therefore shear rates) are hard to estimate with it.

The theoretical determination of shear rates from the momentum and overall mass balance equations can also be difficult if the geometry is complex, the fluid is non-Newtonian and/or the flow is turbulent. Numerical methods and computational capabilities are improving, but the cost associated with determining three dimensional velocity field in devices with topographically complex boundaries is very substantial (11, 12). Usually a very good estimate of wall shear rates can be obtained by intelligent simplification of the physics and this may suffice for design purposes.

MASS TRANSPORT

Access of plasma proteins and blood cellular elements to biomaterial surfaces occurs by molecular diffusion or fluid convection. The diffusion process in liquids is extremely slow (molecular diffusion coefficients in the range of 10^{-5} to 10^{-8} cm^2/sec), so convection if present will dominate mass transport. This implies that small alterations in the scale and direction of fluid motion can have a very large effect on surface access of blood components. This class of mass transfer problems is called the large Peclet number limit in the engineering literature. The Peclet number ($V_0 D/D$) is a dimensionless group which gives a ratio of the importance of convection to diffusion in mass transport. Here D is the molecular diffusivity. As an example, if one considers the steady Poiseuille flow given above, the only fluid velocity component is in the axial direction. Thus mass transport is mainly by axial fluid convection and radial molecular diffusion (much slower). If one can create a radial fluid velocity component (even a very small one), radial mass transport will be greatly enhanced. Since blood components must move radially to contact the tube surface, increased radial mass transport will greatly increase blood-material interactions. This can be promoted by using curved tubes, wavy walls or permeable membranes with suction, all of which can produce secondary convective radial flows superposed on the main axial flow.

TRANSPORT AT THE BOUNDARY

Surface access can be an important or rate-controlling step in determining the rate of growth of mural thrombi, embolization, hemolysis, platelet activation, and leukocyte-surface interactions.

Material arriving at a vessel or biomaterial boundary can either react with the surface, pass through the surface (if permeable) or return to the bulk. For an impermeable reactive surface, this mass balance at the wall can be written mathematically.

$$D \frac{\partial C}{\partial r} \bigg|_{r=R} = k C_{1w}^n \quad (7)$$

The first term represents the rate of arrival of species 1 to the wall by diffusion. The right-hand side of equation (7) is the consumption of 1 by chemical reaction, where k is the reaction rate and n is the reaction order. If $k/D \gg 1$, that is, the reaction consumes material more rapidly than it arrives at the wall, we say the mass transfer to the wall is diffusion limited and the wall concentration (C_{1w}) is approximately zero. If $k/D \ll 1$, material is transported to the wall more rapidly than it can react, and we say the mass transfer is reaction limited and $\partial C / \partial r|_{r=R} \approx 0$. It becomes obvious that transport rates at the wall are profoundly affected by flow across the wall if the biomaterial is porous, since this allows convection to or through the boundary in addition to purely molecular diffusion processes. Effusion decreases transport rates to the wall and suction increases them.

EFFECTS OF FLOW UNSTEADINESS

Mass transfer effects for perfectly steady flows are generally the easiest to predict and analyze. Pulsatility in flow introduces an additional oscillating shear rate region in contact with every wall surface. In complex geometries, this may affect the details of the time average motion, in particular by creating local regions of the flow near the walls where the fluid continuously recirculates, with "little" convective mass transfer with the main flow. This is a circumstance which could possibly contribute to thrombus formation (as behind venous valves). In addition, pulsatility can complicate wall transport studies because fluid which is convected downstream at one part of the oscillation cycle may actually move upstream during another part. If back-flow is not allowed, pulsation in rigid tubes does not increase mass transfer (13,14). Calculations with compliant tubes are much more complicated, and this is still an area of active research. Qualitatively one might expect the possibility of large mass transfer effects with compliant tubes, because the wall movement in response to pulsatile pressure loading will generate a fluid radial velocity component. There is also recent evidence that pulsatile flow may substantially increase production rate of the arachidonic acid metabolite PGI_2 from attached cultured endothelial cells, when compared to steady flow or stationary tissue culture controls (15,16).

Fluid dynamic instabilities, as in Taylor vortex flow between concentric cylinders and turbulence generation, can introduce pulsatile effects even in an initially steady flow. It is worth noting that the term turbulence has special features and definitions in fluid mechanics (17). Many persons misidentify various unsteady flows as turbulent when they are not. For example, an abrupt expansion in tube diameter can often cause flow separation, with a resultant annular vortex. This separated flow can be quite complex, particularly if driven by a pulsatile pressure gradient. However, at Reynolds numbers normally found in the circulation, it is not turbulent. Karino and Goldsmith in an excellent set of papers (18,19) have mapped out the behavior of blood cells in these flows.

RHEOLOGY--MACRO VERSUS MICRO

Macrorheology is concerned with the properties of blood when flowing in conduits which have a characteristic dimension large compared to a red cell diameter. The most important physical parameter in this situation is the blood apparent viscosity. Blood is a non-Newtonian fluid, whose apparent viscosity decreases with increasing shear rate (figure 3.3). Above a shear rate of approximately 500 sec^{-1} , the apparent viscosity of blood is nearly constant, and we can safely use Newtonian fluid mechanics for analysis in large vessels. The main determinant of blood viscosity at fixed shear rate is the hematocrit. To a good approximation, viscosity increases exponentially with hematocrit (i.e., $\mu = \mu_0 e^{a \text{hct}}$, with μ_0 the plasma viscosity). Other contributing factors are various protein concentrations (particularly fibrinogen and immunoglobulins) (20-23). The Casson equation ($\eta^{1/2} \gamma^{1/2}$

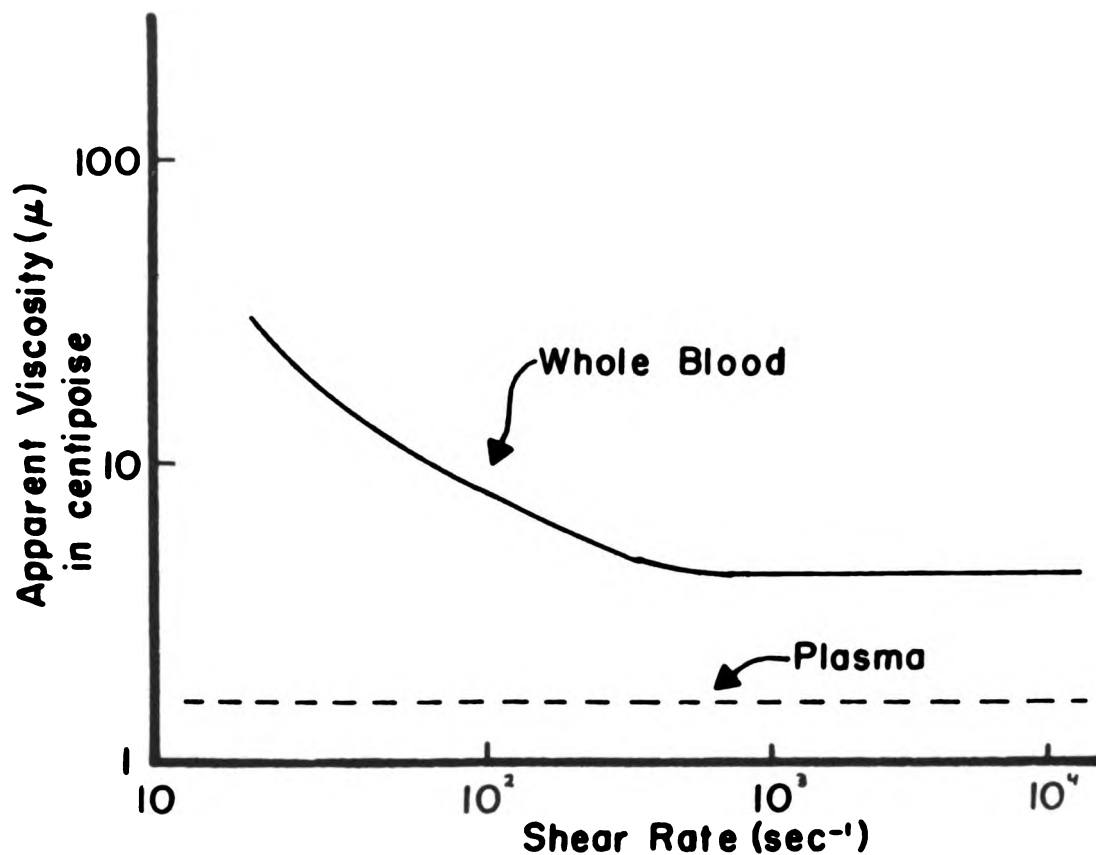


Figure 3.3--Apparent viscosity as a function of fluid shear rate for plasma and whole blood. Normal plasma is essentially Newtonian over the entire range of shear rates of physiological interest. Whole blood is shear thinning, with an apparently Newtonian region above approximately 500 sec^{-1} .

$= \tau^{1/2} + \tau_0^{1/2}$), where η is the limiting high shear rate viscosity, τ_0 is a yield stress and τ is the shear stress, gives a good fit of viscometric data over a wide range of shear rates at fixed hematocrit for both normal and pathological bloods (24). η and τ_0 are both functions of hematocrit, proteins, present and environmental conditions (temperature, pH, etc.). If present, some plasma extenders (such as some types of dextran) can also have a major effect on blood viscosity. Blood elasticity, though finite, is usually neglected. This may not always be justified in some low flow or pulsatile situations (25,26).

Microrheology is concerned with the material properties that are important in flows where wall spacing dimensions are of the order of the cells themselves. One example would be flow in the natural microcirculation. Sometimes microrheological effects can lead to alterations in macro- or continuum-scale mass transport. An example would be the greatly enhanced platelet "diffusivity" in shear flows when red cells are present. This may be due to local fluid motion generated by individual red cell rotation, which leads to greater random excursions of platelets (and thus higher diffusivity) than would be predicted from purely homogeneous fluid molecular considerations. The argument is as follows: red cells rotate in a shear flow dragging along some adjacent fluid, which leads to increased local mixing (figure 3.4). Keller has developed a model (27) which gives the effective diffusivity, D_e , as:

$$D_e = D + D_p \quad (8)$$

where D is the Brownian molecular diffusion coefficient and D_p is the rotation induced "diffusion" coefficient.

$$D_p \approx Ca^2 \dot{\gamma} \quad (9)$$

where C is a constant to be determined and a is the red cell radius. For shear rates, $\dot{\gamma}$, of the order of 500 sec^{-1} , D_p was estimated to be $10^{-5} \text{ cm}^2/\text{sec}$ in normal whole blood. Table 3.1 gives an estimate of the mass transfer augmentation to be expected for various kinds of blood components (recall equation 8) due to the presence of red cells.

Table 3.1--Augmentation-Effect of Red Cell Rotation

<u>Diffusing Species</u>	<u>D</u>	<u>Effect of Cells</u>
O_2	10^{-5}	small
Protein	$\sim 10^{-7} \text{ cm}^2/\text{sec}$	large
Platelet	$\sim 10^{-9} \text{ cm}^2/\text{sec}$	very large

The predictions are in qualitative agreement with the work of several groups on both platelet diffusion in model vessels at typical vascular shear rates (28-30) and careful measurement of effective diffusion

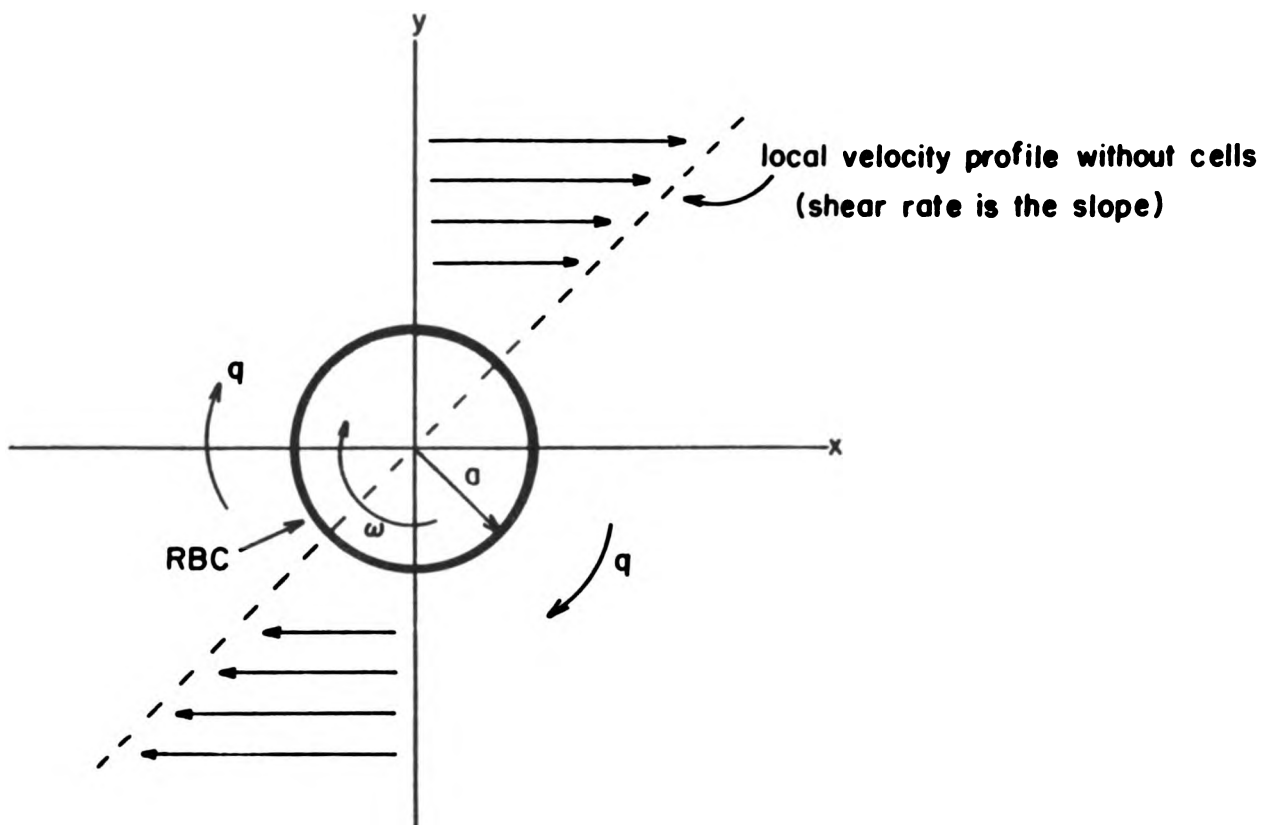


Figure 3.4--Schematic of erythrocyte rotation-induced local mixing. A red cell of radius a in a shear field will rotate with a frequency ω proportional to the macroscopic local shear rate $\dot{\gamma}$. A small amount of nearby fluid (q) will be carried along with the cell movement (due to the no-slip boundary condition at the erythrocyte surface), leading to increased mixing, and therefore an increase in effective diffusivity.

coefficients of several molecular species in Couette flow (31). It is important to note that D_p is shear-rate dependent, augmentation being important at high shear rates, but negligible at low shear rates. Baumgartner, Turitto and associates (32-34) have shown that platelet adhesion to de-endothelialized rabbit aorta from flowing whole blood is mass transfer (diffusion) limited at low wall shear rates and reaction limited at high wall shear rates, in agreement with the arguments presented above. In addition, the fluid shear stress near the wall can affect the morphology (alignment) and kinetics of growth of mural platelet aggregates forming on damaged vessel or biomaterial surfaces (35).

Confirming qualitative observations that have been made for decades, it has recently been shown (36,37) that although the red cell concentration decreases in the vicinity of the vessel wall, the platelet concentration increases in this region (figure 3.5). This has led to some reinterpretation of platelet transport in whole blood, which has not been completely resolved at this time (37).

CELL AGGREGATION

Particles traveling near each other, but at different speeds, may interact with each other. The rate of collision depends on the particle concentration, particle volume and the relative speeds. For spherical cells in a simple shearing field the collision frequency is given by (38,39)

$$\Omega_{ij} = \frac{4}{3} N_i N_j (R_i + R_j)^3 \dot{\gamma} \quad (10)$$

where Ω_{ij} is the two-body collision frequency, N_i is the number density of particle i and R_i the radius of spherical particle i . As before, $\dot{\gamma}$ is the local shear rate. If we use the subscript L to denote leukocyte and P to denote platelet and pick representative values for platelet and leukocyte number in blood (N_i) and cell volume (R_i), the most probable collision in a shear field is between a leukocyte and a platelet. In fact, the ratios of collision frequency are in the following order (39).

$$\Omega_{LP} : \Omega_{PP} : \Omega_{LL} = 10:5:1 \quad (11)$$

In whole blood shear-induced aggregation, both platelets and leukocytes appear to be important in determining the total volume of aggregates formed (40,41). To form an aggregate, the cells must not only collide, but they must stick together. Thus in aggregate formation, both the frequency of collision and the collision effectiveness are important. For given cell volume and cell numbers, the collision frequency is proportional to the local shear rate (equation 10), while the collision effectiveness will depend on the biochemical state of the cell membrane--i.e., how "sticky" or adhesive the particles are. As an example, the stirrer in a platelet aggregometer produces the high shear rates, while the ADP or other biochemical stimulus increases the effectiveness of collisions induced by the fluid motion. If $\dot{\gamma}$ goes to zero, the shear induced collision frequency drops to zero (equation 10) and aggregation,

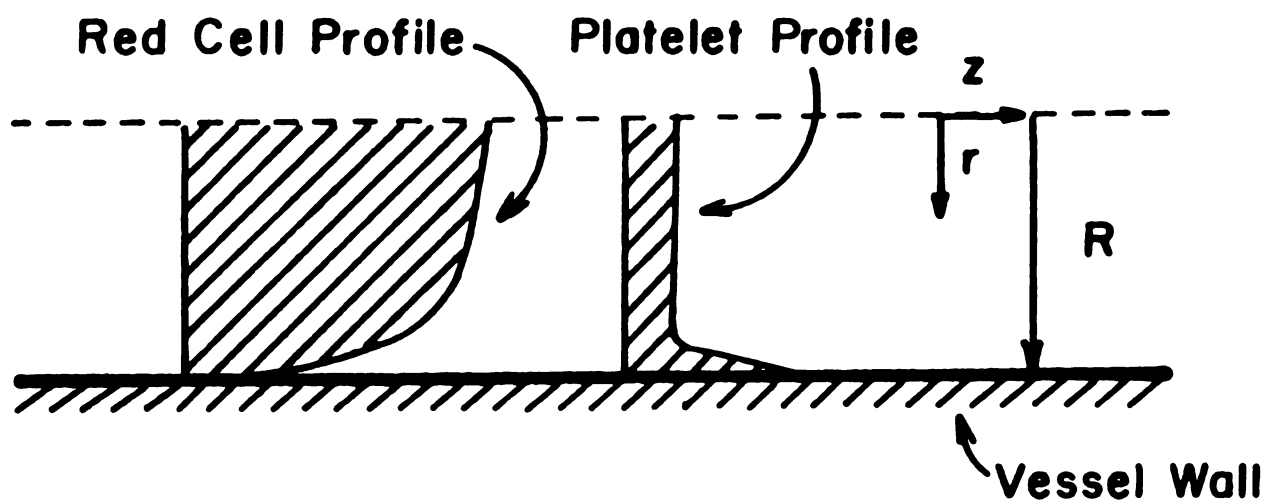


Figure 3.5--Comparison of red cell and platelet concentration profiles for whole blood flowing in a typical vessel. Near the wall the erythrocyte concentration drops dramatically, while the number of platelets per unit volume of fluid increases markedly.

even of activated platelets, can occur only by Brownian motion, an extremely slow process for particles of this size.

Disaggregating forces also increase as the shear rate increases. Above certain values of $\dot{\gamma}$ aggregates may be torn apart, or limited in maximum size attainable (40).

INDIVIDUAL CELL REACTION TO SHEAR FLOW

Cells can be distorted by immersion in a shear flow. In addition, they will rotate, which can lead to membrane stretching and compression. Erythrocyte distortion increases with the magnitude of the product of plasma viscosity and $\dot{\gamma}$ —which is the shear stress. At large values of applied fluid stress, disruption of the red cells occurs (42,43). This stress-induced lysis is also seen in platelets, PMN leukocytes and lymphocytes in plasma subjected to high shear (44,45), although the shear threshold causing damage is much lower for these cells. Lysis is caused by a combination of fluid stress application and time of exposure to these forces. Cells can tolerate very high stresses if exposure times are short enough. Platelets, PMN leukocytes and erythrocytes have quite different threshold combinations of stress-exposure time which lead to significant lysis (figure 3.6). Examples of the possible range of stress-time combinations commonly found in biomaterial applications would be artificial heart valves and hemodialysis units. The former can expose blood cells to stresses greater than 10^4 dynes/cm², but exposure times are only of the order of milliseconds. Hemodialysis units can generate stresses up to several hundred dynes/cm², with average cell transit times being several seconds. In whole blood, nucleotides (ADP) are released from erythrocytes if these cells are damaged by shear. This release may trigger subsequent platelet aggregation. Sublytic stress exposure may also lead to subtle cellular alterations and direct activation and aggregation of platelets and leukocytes (46-49).

Cells attached to a wall exposed to a shear flow are subject to forces that would carry them away if not anchored to the substrate. Hence they require adhesive forces to hold stationary. At high shear, these forces may be inadequate to retain a cell or, for that matter, a cell aggregate or aggregate fraction, leading to embolization. The forces transmitted to the endothelium by attached cells or by the flowing medium may alter cell morphology (50-52) or also cause damage to vessel linings in regions of high flow (53-56).

FAVORABLE CONDITIONS

Ideal conditions for circulation in an artificial organ are those in which cell activating or destructive forces are minimized yet blood flow is maintained. Hemodilution reduces both blood hematocrit and plasma protein levels. This effectively reduces cell collision frequency and red cell aggregation. Shear rate, optimally, should be high enough to provide disaggregating forces for weak aggregates and remove

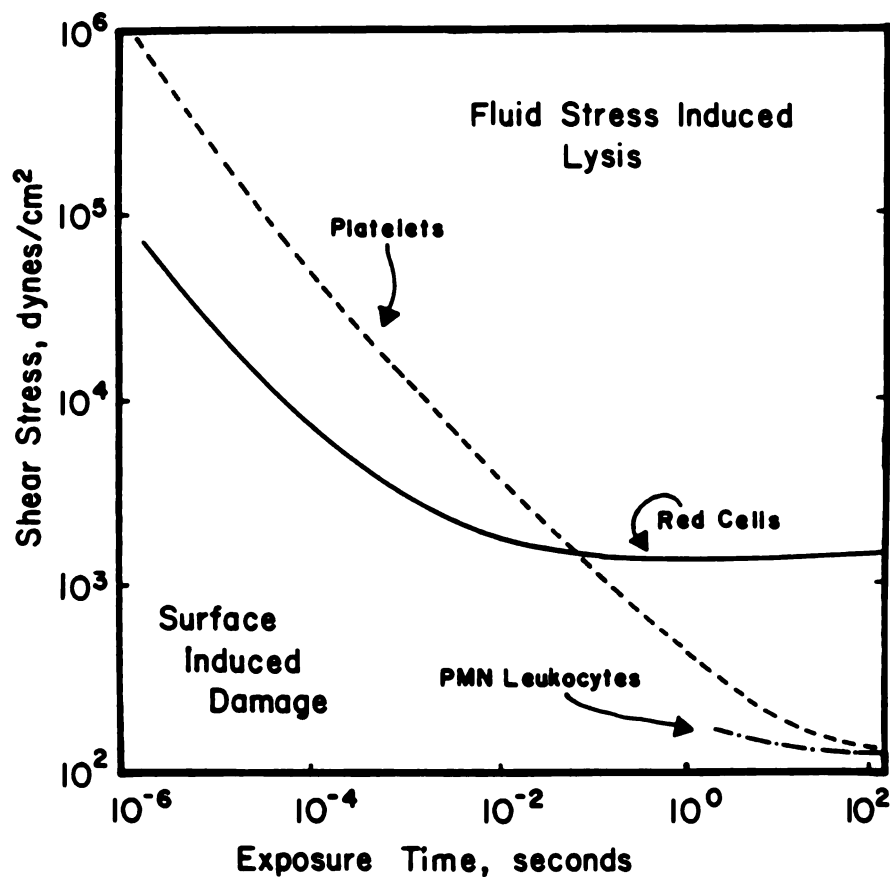


Figure 3.6--The locus in the stress exposure-time plane of shear-related damage to blood cellular elements. Regions above and to the right of each line (high stress-long exposure time) would generate significant cell lysis for the given cell type. The curve for PMN leukocytes terminates at one second exposure time because the very short time experiments have not been done as yet. Modified from reference 44.

weakly adherent cells from a wall, yet not so high as to cause appreciable hemolysis or significant sublytic damage to blood or endothelial cells.

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Chapter 4

BULK CHARACTERIZATION

INTRODUCTION

Bulk characterization of polymers is important for four reasons: It provides assurance that the material in hand is, indeed, that which the user expects it to be; it provides a basis for interpreting the results of surface analyses; it may give an indication of the effect the material is likely to have on blood; and it yields information on the basic mechanical properties which will eventually dictate the level of the structural use of the material.

As indicated in chapter 1, the test procedures to be described are divided into two levels. The level I tests constitute the absolute minimum necessary for the material user to identify the polymer being investigated and to verify that the physical and mechanical behavior of the material is in accordance with the description provided by its supplier. These tests are appropriate for all materials regardless of their final application. They are sensitive indicators of the chemistry and/or the molecular conformation and morphology, particularly as these parameters are affected by manufacturing conditions. These procedures are simple, and any laboratory involved in the development of materials or in the construction of cardiovascular devices should have access to resources for such testing.

Level II tests are more appropriate for materials development. They are employed for materials which, on the basis of preliminary toxicological and blood-material interaction testing, appear potentially useful. They provide detailed information on those elements of structure and functional behavior which are likely to relate to the performance of a material in a cardiovascular device. As such, they may not be appropriate for all applications. For example, a hardness test need not be done unless an abrasive or wear application is intended for the material. In all cases, the techniques described are capable of providing reproducible and interpretable information, although, in some cases, the sophistication of the techniques may require equipment and expertise available in only a relatively small number of laboratories.

It is advantageous to both the user and the manufacturer for the manufacturer to provide reasonable characterization information. From the manufacturer's point of view, such information serves as a quality control device; from the user's viewpoint such information serves as baseline data which may be used to "trace" the material through processing steps or subsequent aging during storage.

In this chapter, bulk characterization procedures are described with particular attention to the rationale and limitations of each. For experimental details, literature references are provided. Frequently, American Society for Testing and Materials (ASTM) standard test numbers are referenced. These tests are the results of thoroughly researched procedures which are constantly reviewed by large numbers of experts in

the field who use the tests regularly and have a wealth of experience behind them. To qualify for publication as an ASTM standard by the American Society for Testing and Materials, each written standard must receive affirmative votes after at least two levels of committee examination. The number designations, for example D-2538, serve to catalogue the tests under the main committee responsible for them as well as to locate the tests in the indexed volumes of standards. The numbers immediately following the designation indicate the year of original adoption, or in the case of revision, the year of last revision, for example D-2583-75. A number in parentheses indicates the year of last reapproval.

LEVEL I TESTS

Polymer Identification

Because of the difficulty of retrospective analysis of polymer preparations, the most direct and economical way to understand the bulk properties of a given preparation is to have information on the procedures used to prepare it. For this reason, the group that actually polymerized the material is best informed about the nature of any specimens obtained from it. Whenever possible, information should be obtained from them before any analyses are undertaken. The information should include the following, subject to purposeful omissions for the sake of keeping certain information proprietary.

1. A statement of the monomers used in the preparation, and the methods employed in monomer and polymer purification.
2. The general method of polymerization used (i.e., condensation, radical, ionic, bulk or emulsion) so that some assessment can be made by the characterization laboratory of the probable distribution of molecular weight.
3. If the synthesis involved special efforts to control the polymerization, such as the use of transfer agents, the incorporation of branching points, etc., these should be identified. Similarly, if the structure is three-dimensional, the method used to achieve cross-linking should be specified.
4. Solvents used in the purification or isolation of the polymers often are the source of residues found in the bulk material. Both the kind of solvents used and the quality should be specified, so that the evaluation laboratory can work with an understanding of the possible contaminants that may be present. Preferably, a solvent blank should be kept by the polymer preparation laboratory until level I and level II characterization tests have been completed.
5. Any additives used as antioxidants, ultraviolet or thermal stabilizers, or processing aids utilized.

Infrared Spectroscopy

An infrared spectrum of the bulk sample should be obtained (1-3) (ASTM D2238) to provide confirmation of any polymer sample data submitted with the specimens, to compare with standard catalogued polymer spectra, and to provide qualitative evidence for residual solvent, additives and plasticizers. Infrared spectrometers of relatively low resolution should be adequate for this purpose and may be expected to be available in all laboratories involved in materials development and processing. Measurements can be carried out on powder samples, in pellet matrices, on films or sheets, or in a suitable solvent for the polymer. The IR spectra so obtained will confirm the presence of expected functional groups, identify the presence of unwanted contaminants, and will serve as references for the surface characterization procedures as described in chapter 5.

Hardness

The hardness of a material reflects the compression modulus and compression yield strength and may, in some cases, be correlated with the abrasion or wear resistance, as well as the presence of plasticizers in normally rigid or glassy polymers. It is important in applications involving bearing loads where one part either slides or rolls while contacting another part of the same or dissimilar material. For softer, elastomeric-type materials, a durometer test (4) (ASTM D2240) should be used. Harder materials such as glassy polymers or reinforced plastics may be characterized with a Barcol Impressor test (5) (ASTM D2583) or a Rockwell test (6) (ASTM D785). In each of these tests, the force required to produce a given indentation in a small ($\sim 1 \text{ cm}^3$) solid sample of the material is measured using one of several commercially available devices. The thermal history and concentration of additives will affect the test results, and thus, this historical information should be provided with the test specimen to facilitate interpretation. The hardness test is an empirical test and no simple relationship exists between a hardness value of any type and any fundamental property of the material tested.

Porosity

Porosity is, on the one hand, undesirable particularly when a polymer structure is subjected to stress. In this case, voids act as stress concentrators and cause premature failures particularly in fatigue situations. On the other hand, porosity may be required as in the case of foamed polymers, of specifically tailored membranes, or of open-pore polymer implants such as polytetrafluoroethylene vascular prostheses (see chapter 6).

Two levels of porosity or void characterization are useful. The presence of voids in thick components, when they are undesirable and usually present in relatively low contents, can be verified by measuring the apparent density (7,8) or using ultrasonic detection methods. In films and sheets, optical detection is sometimes possible.

In cases where pores or voids are vital to the success of the component, the pore volume and the pore size distribution are desirable quantities to know. Mercury intrusion porosimetry (9) (ASTM D2873) and BET (Brunauer, Emmett, and Teller) methods may be used to provide quantitative results.

Density

Density measurements on solid polymers can be made quite easily with water displacement methods such as a pycnometer or with a density gradient column (7) (ASTM-D1505). They provide a simple means to compare specimens with respect to their bulk properties. Variations in crystallinity of solid plastics, void volume in cast polymers or changes in loading of filled plastics are all reflected in small changes of density. Thus this simple measurement may serve as a screening test for comparison of different specimens representative of the same general class of test materials.

Tensile Stress-Strain Properties

The minimal requirement for characterization of mechanical properties is to obtain the uniaxial tensile stress-strain curves of the materials of interest at low, medium and high rates of strain. This measurement is made on one of the standard tensile testing apparatuses, either floor or table model. The ASTM standard method for tensile testing is given in ASTM D638 (10). The specimen, in the form of a "dog bone" or other strip shape is prepared by injection molding or cutting from a plaque of material. It is usually 4-5 inches long, has a cross section sufficiently large that the extensional forces imposed can be controlled and measured, and should have a length-to-width aspect ratio of at least six to prevent grip effects. Microtensile specimens may also be used (11) (ASTM D1708). It is important not to have any notches, nicks or cracks on the test specimen. Both ends of the specimen are clamped firmly in the jaws of the tensile machine. The jaws are moved apart at the desired constant rate. Force, measured by a load cell, is automatically recorded versus elongation. This may, of course, subsequently be converted to engineering stress (force/original cross-sectional area) or true stress (force/instantaneous area) versus engineering strain, $\Delta l/l_0$, or true strain, $\ln(l/l_0)$.

Figure 4.1 shows a generalized uniaxial tensile stress-strain curve embodying the features usually seen in polymeric materials, and defining various mechanical quantities. The Young's or tensile modulus (E) (sometimes loosely called the stiffness), is the initial slope of such a plot. The reciprocal of the elastic modulus is the elastic compliance. It should be noted that the compliance thus defined applies only to the initial slope of the stress-strain curve. Only when the stress-strain curve is linear is the compliance definable at any strain other than that approaching zero. The biomaterial literature (particularly in vascular graft testing) is marred by numerous reports of compliance defined as some finite or ultimate elongation divided by load (sometimes stress). Such values taken from nonlinear load-elongation or stress-strain curves are meaningless as a means of comparing the fundamental mechanical properties of materials. A better designation for such

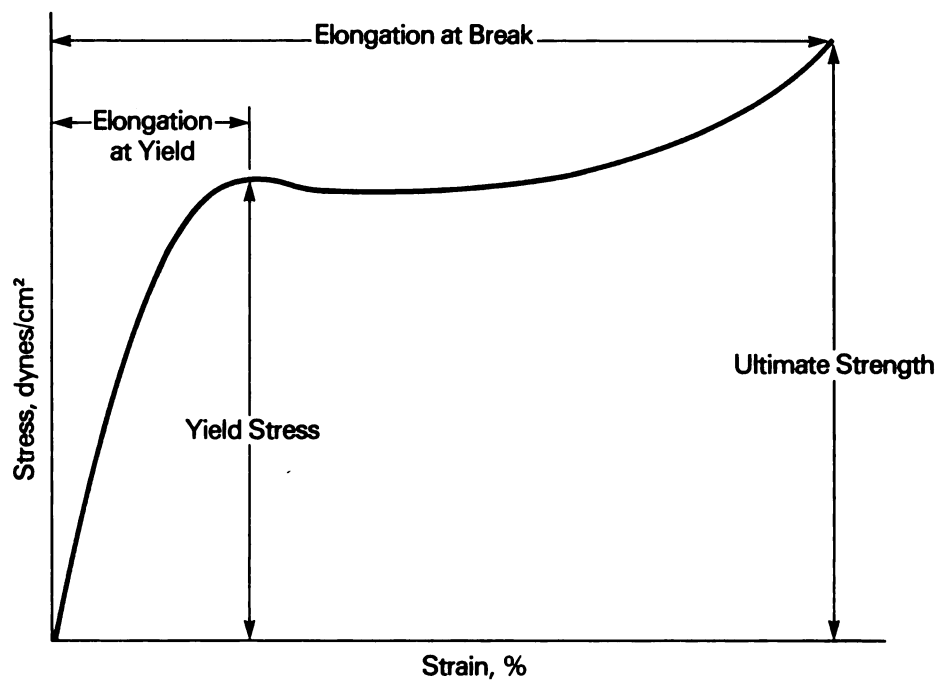


Figure 4.1--Typical stress-strain curve for plastics. From "Polymeric Materials" by Winding and Hiatt (13). Copyright 1961 by McGraw-Hill Book Company. Used with the permission of McGraw-Hill Book Company.

measurements would be "apparent compliance" or "effective compliance" at a particular designated strain. A secant modulus or secant compliance, defined as the stress divided by the strain at a designated strain is often used.

In the case of anisotropic materials such as reinforced plastics or oriented films, the modulus or compliance must be defined in the two principal directions. Simple uniaxial tensile tests on these materials in the two directions do not yield the modulus or compliance values directly because the tensile moduli in the two directions are coupled. For a more detailed discussion of characterizing the modulus and compliance of anisotropic materials, see reference 12.

The area under the stress-strain curve is a measure of the total energy input required to fracture the material, and therefore, of the toughness of the material.

Strain rates in the range 1 to 1,000 percent/min cover the usual range. Strain rates of 1 percent/min, 100 percent/min and 1,000 percent/min would be suitable as low-, medium-, and high-strain rate choices. Figure 4.2 shows typical effects of variations in strain rate (ϵ_R) and temperature (T) on the observed stress-strain behavior.

The shear modulus (G) of an isotropic material is related to the Young's modulus (E) by the relation:

$$G = \frac{E}{2(1 + \mu)}$$

where μ is Poisson's ratio:

$$\mu = \frac{1}{2} \left[1 - \frac{1}{V} \frac{\partial V}{\partial \epsilon} \right],$$

For a simple unidirectional test of a rectangular cross-sectioned, isotropic material,

$$\mu = - \frac{\epsilon_2}{\epsilon_1},$$

where ϵ_1 and ϵ_2 are the strains in the stress direction and perpendicular to it, respectively. The Poisson ratio is a measure of the dilation of the material during tensile strain. V is the test sample volume and ϵ is the strain. For incompressible materials, $\mu = 0.5$, and $G = E/3$. This is roughly true for lightly cross-linked elastomers but is, in general, not true for most plastics or for filled elastomers.

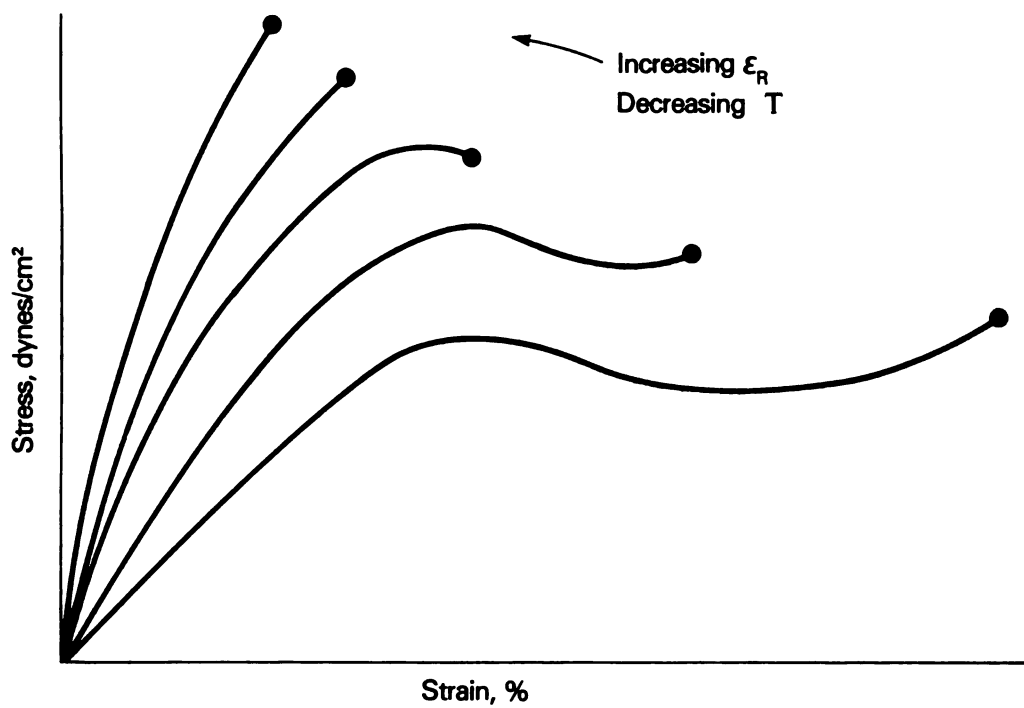


Figure 4.2--Affect of variation of strain rate ($\dot{\epsilon}_R$) or temperature (T) on stress-strain curve (14).

The minimum value of μ ordinarily observed for homogeneous isotropic materials is about 0.2, but may be smaller for certain heterogeneous materials such as sponge rubber (15). Values of μ for polymers are generally nearer to 0.5 (incompressibility) than are those for metals.

Anisotropic materials such as drawn polymer sheets or composite materials may exhibit Poisson ratio values greater than 0.5.

Thus, if Poisson's ratio for an isotropic material is known, or can confidently be taken as not significantly different from 0.5, then the shear modulus is obtainable directly from a tensile experiment. Measurement of shear modulus, however, is not recommended as an additional level 1 test since little new insight into material behavior is obtained.

Statistical analysis of the yield or ultimate strength can provide an excellent quality control, particularly on sheet or film materials, which are sensitive to flaws and other inhomogeneities arising from fabrication processes. Using a minimum of six specimens, and preferably 10 to 12, the strength data can be fit to either a normal distribution or a Weibull distribution. For a normal distribution,

$$f(\sigma_b) = \frac{1}{\sigma\sqrt{2\pi}} \exp \left[-\frac{1}{2} \left(\frac{\sigma_b - \mu}{\sigma} \right)^2 \right]$$

where σ_b is the breaking strength (or yield strength) and μ and σ^2 are the mean and variance, respectively.

For a Weibull distribution,

$$f(\sigma_b) = \alpha\beta\sigma_b^{(\beta-1)} \exp \{ -\alpha\sigma_b^\beta \}$$

where α and β are the scale parameter and shape parameter, respectively.

The standard deviation, σ , in the case of the normal distribution, and the shape parameter β , in the case of the Weibull distribution, each describe the breadth of the distribution, which is a very sensitive measure of inherent flaws in the material. In some cases, the mean strength values may be comparable, but the distributions may be entirely different. It is the distribution of strength values which will govern the material's behavior in fatigue.

It is important, when possible, to stress-strain test specimens which have been retrieved after exposure to biological environments, particularly when the samples are structural elements such as heart valve components or vascular prostheses. The modulus and the ultimate strength are very sensitive to plasticization, and chemical degradation. Any change in these properties during their performance lifetime could result in high stress concentrations, dimensional instabilities, and premature failure of the material in its device configuration.

Thermal Analysis

The response of a polymer to controlled heating processes is a rapid and valuable means of characterizing both its past history and its potential behavior, as well as its molecular configuration (chemical nature) and conformation (molecular arrangement of chains).

Differential thermal analysis (DTA) applied to polymers consists of the measurement of the change of temperature of the polymer (compared to a reference) as it is heated from a temperature characteristic of its glassy state to a point above its melting point. At a constant heating rate, the change of heat capacity resulting from some phase change in the polymer leads to a deviation in the time-temperature plot compared to a material not undergoing such a transition. Thus a change in slope is interpreted as a change in heat capacity, and this is associated with a new degree of molecular freedom in the sample.

Differential scanning calorimetry (DSC) measures the energy difference required to maintain the polymer at the same temperature as a control material, when both are heated at a uniform rate. Both DSC and DTA provide essentially the same information, i.e., the locations of discontinuities in the polymer's heat capacity versus temperature plot.

The data for DSC or DTA are presented as exotherm/endotherm plots versus temperature (figures 4.3 and 4.4). Calibration points must be marked since the detectors are subject to drift. For precise work, the heats of melting of reference compounds having melting points near the transition temperatures should be included. This calibration permits a quantitative measure of the thermal effects.

When the polymer is heated to the melting point, the quantity of heat absorbed is the sum of the energy required both to raise its temperature and to melt crystalline forms present. By measuring this energy loss for both well-annealed and quenched samples, it is possible to estimate the crystallization potential of each sample.

The DTA technique is perhaps best applied to thermally processed semicrystalline polymers to determine the crystallization history of the sample. It is also extremely useful to detect the presence of residual solvent, if the solvent has a heat of vaporization significantly greater than the heat capacity of the sample. Should residual solvent be present, the sample can be heated in a vacuum oven to remove it and then retested to ensure that the properties measured are those of the polymer itself. The DTA technique can also measure the glass transition temperature, although it is not the method of choice.

An excellent discussion of thermal analysis may be found in a book by Ke (17).

Molecular Weight Determinations

Molecular weights of two-dimensional uncross-linked polymers provide important information on the presence of oligomers (which may compromise mechanical properties), on the course of the polymerization of

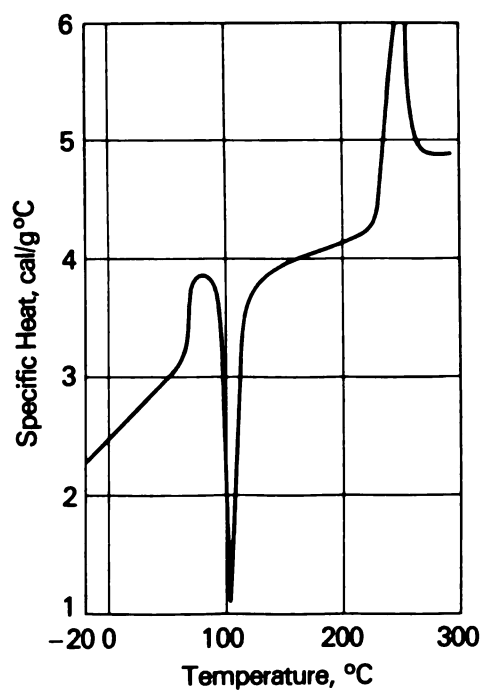


Figure 4.3--DSC analysis of amorphous poly(ethylene terephthalate).
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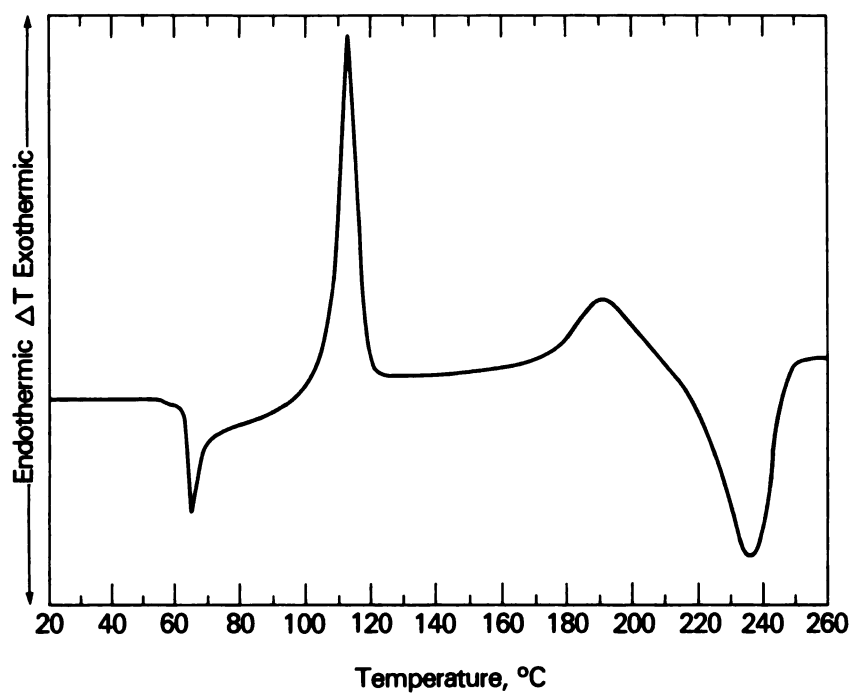


Figure 4.4--DTA analysis of amorphous poly(ethylene terephthalate).
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the specimen, and on the problems which might be encountered during processing such as thermal or ultraviolet degradation. The simplest techniques of estimating molecular weight are those based on viscosity. However, the viscosity-based value provides no information on the distribution of molecular weights, which is an important determinant of mechanical properties. Therefore, whenever possible the molecular weight should be determined by gel permeation chromatography (18) (ASTM D3593, D3586). It is necessary to calibrate the gel columns using control markers of known molecular weight. The procedure has been automated and is available through commercial laboratories, as well as in many laboratories specializing in polymer synthesis.

For monitoring the batch-to-batch variations of lots of nominally identical preparations, it is satisfactory to determine specific viscosities of dilute solutions of the polymer at five concentrations. Linear extrapolation of the normalized specific viscosity vs. concentration data to zero concentration yields the intrinsic viscosity

$$\eta_{int.} = \lim_{c \rightarrow 0} \left(\frac{\eta_{sp}}{c} \right)$$

which, in turn, can be related to the molecular weight through measurements on standard samples ($\eta_{int.} = KM^a$). These results reflect the viscosity-average molecular weight of each batch. These measurements can be made in a capillary viscometer, such as the Ubbelohde viscometer. Sample volumes of 5 to 10 ml are required. More details on the method and on viscosity nomenclature can be found in the literature (19).

For three-dimensional cross-linked polymers, the molecular weight is not an important variable. More directly applicable is a measurement of the swelling of each polymer preparation in a good solvent fluid. The latter can be selected from a knowledge of the solubility parameter space for the polymer. The swelling or the glass transition temperature of a three-dimensional polymer indicates the degree of cross-linking attained in the synthesis or subsequent processing (see 20,21).

Summary

The level I procedures will confirm the identification of the material structure and provide some indication of how the material is likely to respond chemically and mechanically. The hardness and density measurements will, in addition, provide quality-control indices. While their absolute values are not intrinsically meaningful, once an expected or control value is determined, they provide a sensitive indication of small variations in samples nominally synthesized in the same way.

It is not likely that the suggested tests will provide sufficient data to describe the material completely, since retrospective analyses of preparatory methods are always difficult and sometimes impossible.

If the information on preparatory methods is not available from the material synthesizer, the extraction and chemical analysis procedures described under level II testing would be necessary.

The stress-strain data will provide a rough estimate of the strength and flexibility of the material, and they are especially useful as a method of comparison among materials. The use of varying strain rates will provide an initial indication of the time-dependent response of the material, but will do little to elucidate creep properties and, indeed, a more complete documentation of time-dependent properties will require level II testing. A statistical analysis of the strength is a sensitive measure of flaw size distribution and hence the processing quality (level of voids, dirt, inhomogeneities) of the material.

DTA and DSC results give rapid measures of thermal characteristics such as the melting point and glass transition. The nature of these transitions, in addition to identifying characteristics unique to each polymer, can also provide information about its phase nature, thermal history, and purity.

Although gel permeation chromatography (GPC) measurements are rather time-consuming, intrinsic viscosity determinations are quick and provide good quality-control information.

LEVEL II TESTS

Solvent Responses

An understanding of the solubility of polymeric materials to be used in cardiac assist devices is required for two reasons: a) to assist in the selection of correct fabrication and cleaning techniques, and b) to provide a description of the chemical bonding potential of the polymer. The first reason is a pragmatic one, and the second may become a useful adjunct to correlate the interaction between polymer structures and blood components (proteins, lipoproteins, cell membranes) to which they may subsequently be exposed. The determination of solvents which produce maximal swelling in cross-linked polymers can provide information on the segmental length and on the chemical bonding characteristics of the polymer.

The description of polymer solubility suggested here on both applied and predictive grounds is based on the Hansen (22) modification of the Hildebrand (23) solubility parameter. The Hansen classification system uses a three-component solubility parameter to describe the chemical bonding property of any potential solvent. The three components reflect the 1) hydrogen bonding, 2) permanent dipole-permanent dipole interaction, and 3) nonpolar interaction characteristics of each solvent species. The hydrogen bonding component of the solubility parameter probably includes other "association" bonds; however, the historical use of this term favors continued usage for the purposes of this application.

The solubility parameter tabulations of Hansen can be related to the single-valued parameters of Hildebrand through the following equation:

$$\delta_T^2 = \delta_H^2 + \delta_P^2 + \delta_D^2$$

where the subscripts H, P, and D, refer to the hydrogen, polar and dispersion bonding contributions to the parameter. The solubility parameters for mixtures are additive on a volume fraction basis. Moreover, they are relatable to surface energies; Beerbower (24) has correlated the surface free energies (γ) of many organic liquids, fused salts, and liquid metals with the equation:

$$\gamma = 0.0715 V^{1/2} (\delta_D^2 + 0.632 (\delta_P^2 + \delta_H^2))$$

Polymers are not described by solubility parameter point values, but by volume elements in a three-dimensional space defined in terms of the three parameters. The boundaries of the volume element defining each polymer are determined by the location of those organic compounds which act as solvents for the polymer, i.e., polymer solvent coordinates fall within the space defined by the polymer, and nonsolvents are located outside of this volume element (see figure 4.5).

Because the differences in polarizability of many polymers are restricted, Hansen found that two-dimensional representations using only δ_H and δ_P are generally adequate. Using this two-dimensional notation, the polymer solubility is represented by the area encompassing the coordinates of its solvents. Nonsolvents have solubility coefficients whose coordinates are outside of this area. Solvent mixture coordinates prepared from two or more nonsolvents for the polymer can be calculated on the basis of the volume fraction additivity of each component's coordinates. Similarly, interaction between the polymer and any organic species can be deduced from the distance between the polymer solubility boundary and the coordinates for that compound. The application of this concept to protein interaction has not been tested, but the concept is established for predicting plasticizer effects (25), polymer compatibility (26), and the effects of solvent removal on film formation (27).

Determination of solubility parameters is an approximate function but may be useful as a diagnostic tool. Determination of polymer solubility parameters does not require special sample preparation, except to ensure that the specimen is free of extraneous solvents/nonsolvents. The specimen need only be large enough to allow rough observations of dimensional changes. The measurements consist of equilibrium determinations of solubility at room temperature, and at elevated temperature

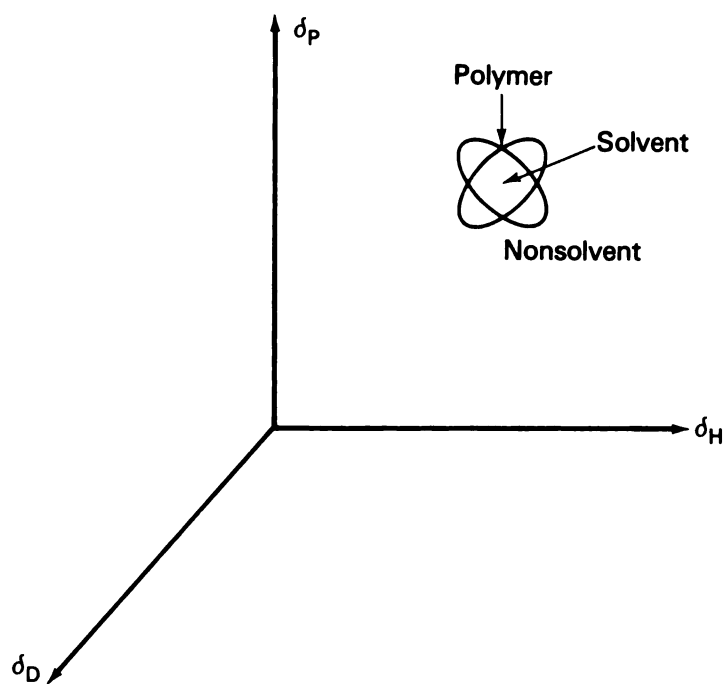


Figure 4.5--Solubility parameter space.

in the case of semicrystalline polymers. Solubility is not quantified; a determination of whether the specimen is soluble, swollen, or insoluble at 10 percent w/w is generally adequate to establish the profile. The data are reported in terms of the solubility parameters provided by Hansen (22). A sufficient number of test solvents must be used to establish the boundary of the polymer coordinates. Where necessary, solvent mixtures are used to provide additional data points. When data on new materials are being compiled, similar studies should be performed on a reference material whose characteristics have been previously described.

The data should be presented in terms of a two-dimensional plot indicating those coordinates which have been tested, and the values assigned to each solvent. For solvent mixtures, the volume fractions should be reported.

The test procedure is quite simple and can be performed in any chemical laboratory. Computations of coordinates can be carried out with pocket calculators, even if the full three-dimensional coordinates are used.

For purposes of relative comparisons and qualitative evaluations, it is possible to calculate the solubility parameter of the repeat unit in a given polymer by utilizing a group contribution approach as described by Small (28). Thus, by summing from a table the contributions of each atomic group in the repeat unit, an estimate of the polymer solubility parameter can be made.

Extractables

Solvents along with contaminants therein, reagents, and catalysts used for polymer synthesis as well as adventitious contamination from polymerization vessels, storage, or shipping containers may contribute impurities to a final product. Low-molecular-weight fractions and additives such as plasticizers, stiffeners (carbon black), antioxidants and UV stabilizers might also be present. Since such impurities may be extractable and may eventually interact with the physiological fluids, it is desirable to determine their presence in the polymers prior to use. In addition, the loss of significant quantities of such additives might lead to premature failure. This mechanical problem is quite independent of any biologic (toxic or beneficial) effect which might ensue from the leaching of the plasticizer.

Sample preparation for characterization should be based on minimal handling techniques in order to reduce contamination potential. Samples should never be touched directly, should be stored and manipulated in a clean environment, and should be subjected to a minimum number of manipulations and transfers between receipt and testing. However, for speed of analysis, a finely divided sample is generally desirable. Where simple elemental analyses are adequate, the sample should be washed in platinum vessels and the residues examined by X-ray fluorescence or some other multielement procedure to identify the presence of potentially toxic metals. Quantitation may require more sensitive procedures such

as atomic absorption. The precise methodology should be left to the discretion of a competent analyst. Many commercial and university laboratories have appropriate equipment and expertise to perform these measurements. The method of standard additions should be employed for quantitative analyses to compensate for possible matrix effects. In the decision to use a particular procedure, consideration should be given to the possibility that the contaminants are not uniformly distributed throughout the bulk of the sample. Thus, localized analytical procedures which do not incorporate a significant sample mass should be avoided.

The determination of extractable molecular components remaining in the polymerization product is not a routine procedure. The difficulty of each characterization is a function of the kind of polymer under investigation, and of the prior purification processes used by the synthesis chemist. As a general rule, it is preferable that a precise characterization of all the components of the reaction mixture be available. This narrows the analytical problem to those species known to be present, or their known reaction products. For example, in a solution polymerization process for the preparation of condensation polymers, it is preferable to have data from well-characterized solvents and reagents rather than to attempt to identify residual impurities still present in the final polymer.

In the absence of such information, one must know whether or not the polymer can be dissolved or melted, or if it is a three-dimensional structure. For polymers soluble in volatile solvents, the preferred procedure is to dissolve the sample in either pesticide-grade or spectral-grade solvents, followed by precipitation from dilute solution with a nonsolvent of similar quality. Solvent blanks of equal volume must be carried through the procedure to provide a control. Concentration of the remaining solvent/nonsolvent mixture and analysis of the residues can then be performed. If possible, gel chromatography/mass spectrometry procedures are recommended, since these lead to both qualitative and quantitative results. However, infrared spectra and NMR spectra of the residue may provide equally useful information.

For polymers insoluble in the available high-purity solvents, but with melting points significantly below their decomposition temperature, the polymer can be extruded in filament form into an excess of organic solvent selected to provide the same qualities described above. The workup can then follow as previously discussed.

For three-dimensional polymers the extraction must rely on the use of slow leaching procedures which permit adequate time for diffusion of contaminants from the bulk material. For that reason, the specimens should be finely divided. Extraction should preferably be carried out with both a highly polar hydrogen bonding solvent such as ethanol and with a less polar solvent such as methylene chloride. Both of these solvents can be obtained in high purity and can be removed from the extraction fluid by clean, dry nitrogen. As indicated above, the final identification of contaminants can be done by a combined spectral and chromatographic procedure. The appropriate method will have to be judged on the basis of the particular situation.

No simple extraction procedure would simulate the effect of extended exposure to physiological fluids. Plasma contains not only electrolytes, but also a variety of fats, surfactants, and nitrogen-based compounds. For the purposes of these experiments the extraction fluid should be chemically uncomplicated to facilitate subsequent analyses. If analysis of the extract is not possible, an alternative procedure is to characterize the residual polymer structure for changes in mechanical properties or phase relationships. The route to be taken must be left to the discretion of a competent investigator. Examples of each approach are presented here.

An aqueous alkaline emulsion of oleic acid/sodium oleate containing 150 mM NaCl provides a medium which is capable of emulsifying extractable fat-soluble components, and which is suitable for subsequent analyses. Acidification of the emulsion with dilute HCl breaks the emulsion, and leads to a two-phase system. The aqueous components will be present in the electrolyte solution and the oil-soluble components will be present in the oleic acid layer. The former can be evaporated, and the usual inorganic or organic analysis scheme conducted on the concentrated solution. If the oleic acid layer is esterified, for example with MeOH, the resultant solution can be separated by liquid or gas chromatographic techniques to facilitate identification of components extracted from the polymer.

If the suspected extractables are sensitive to the alkaline conditions required to maintain the oleic acid/sodium oleate emulsion, the sample can be extracted instead with an aqueous solution of sodium dodecyl sulfate or other equivalent surfactant. The subsequent analysis is then best carried out on the polymer itself, following extensive water extraction to remove the surfactant. Water extraction should continue until the analytical results are not affected by further water extractions. At that point surfactant removal is assumed to be complete but can be assured by appropriate analysis of the polymer for traces of surfactant. If the polymer has undergone loss of a component which significantly affects its mechanical potential, measurement of the torsional modulus glass transition temperature (T_g) (see next section) or extension to break (see above) will generally reflect such losses. If the polymer has undergone loss of an important chemical stabilizer, this should be reflected in oxidative stability tests.

Liquid Water Diffusivity and Sorption

With polymers being increasingly used as structural (load bearing) components in cardiovascular prostheses (heart valves) and blood pump bladders, water uptake during long-term *in vivo* exposure is becoming more and more important. In addition to affecting the mechanical properties of the material and lowering its glass transition temperature, transport of water through the material may jeopardize other components or cause local chemical imbalances. In other applications such as controlled drug release, the transport of water may be a governing factor in the success or failure of the material.

The easiest method of determining the water transport characteristics is to measure the amount of water absorbed as a weight percentage

of the dry sample as a function of time. For sheet geometries, the diffusivity is given by Stannett and Yasuda (29).

$$D = 0.04919 / (t/\ell^2)_{1/2}$$

where D is the diffusivity in cm^2/sec and $(t/\ell^2)_{1/2}$ is the reduced half-time, t is the time in seconds, and ℓ is the specimen thickness in cm. The reduced half-time is obtained from a plot of the percent absorbed versus t/ℓ^2 and the diffusivity calculated using the above equation. For geometries other than a sheet, different equations apply (30).

From the equilibrium water uptake, the solubility of water can be obtained and the permeability coefficient calculated from the product of the solubility and the diffusivity.

Nonpolymer Species Mobility (Permeability)

The mobility of nonpolymer species, such as oligomers or other toxic or bioactive by-products, in polymers proposed for various applications in cardiac assist devices is an important factor both in the processing of the material and in its behavior in a device. Mobility of solvents for the polymer and for other processing aids in the polymer must be great enough to permit total removal of these species when a component is solvent-cast so that the components are contaminant-free. Low mobility of solvents (for a particular contaminating species) through a polymer can lead to retention of the contaminant, and its slow release over a period of years may lead to toxic or other biologic responses quite unrelated to the basic polymer chemical properties. Thus, high mobility of both kinds of fabricating solvents is desirable.

On the other hand, high mobility of physiological fluid components in the polymers of the device in contact with blood or other fluids may lead to loss of physical properties through plasticization, induced crystallization, or alteration of phase relationships. Thus an understanding of the relative mobilities of compounds, classified according to their potential interaction with the polymer will be a necessary factor in predicting the useful life of cardiac assist devices.

No single set of test species will provide the necessary mobility data for every polymer potentially useful in cardiac assist devices. Selection of suitable test species for each polymer can be made once the two-dimensional solubility parameter coordinate system has been established for the polymer (see above). Using this coordinate system, it is possible to select species from the solvent (coordinates within the solubility area) and nonsolvent (coordinates outside the solubility area but between those of the polymer and water) groups. At least one species should be selected from each category.

Mobility data using the species representing the "solvent" category of compounds will provide data on the removal rates of residual solvents following processing. If solvent mixtures containing a nonsolvent are

used in processing, the test should be performed with the poorer solvent. If the polymer is hydrophilic, the solubility coordinates of the wet polymer must be determined.

In order to study the mobility of physiological species, the choice from the nonsolvent category should be representative of the types of solutes found in blood or plasma, i.e., lipids, steroids, amines, etc. Butyl alcohol could serve to represent the amphipathic nature of many biologic compounds. The use of butyl alcohol also offers the potential for correlation with the usual pharmacologic characterization of drug distribution in body fluids.

The techniques for measuring mobility suggested here are quite simple, and rely on the use of standard water transmission rate cups (31) in a vacuum chamber. The cups clamp film samples across a reservoir opening. When operated upright, only the vapor from the cup contacts the polymer specimen. When operated in an inverted position, the entire fluid contacts the polymer sample. The two results are generally not equivalent: the vapor transmission rate is for the polymer in equilibrium with the partial pressure of the permeant component; the liquid permeation rate is for a polymer swollen with the liquid phase. Measurements are performed by storing the cups at 37°C in a vacuum chamber operated at a pressure sufficiently low (approximately 1 mm Hg) to provide a constant pressure difference corresponding to the partial pressure of the permeant at 37°C. For single component permeants, the analysis of mass transferred as a function of exposure time is sufficient to provide a permeation rate. For binary, or multicomponent permeant mixtures, the residual composition on the source side must be analyzed as a function of time. The time span of the experiment can be adjusted by use of thin or thick polymer films and by the selection of permeants with high partial pressures at 37°C.

Even for the nonsolvent outside the solubility zone of the material, the material should be exposed to both the vapor and the liquid phases. The permeation rate of an organic solute is often quite different in the presence or absence of water even if the polymer is hydrophilic in character.

The data should be reported in terms of the permeation rate as a function of time (or mass of solute transported versus time). For nonsolvent/polymer combinations it will be possible to establish the lag time period required for the permeant to achieve a steady-state distribution in the polymer. These data will provide information on the diffusivity of the solute in the polymer (see above discussion for water diffusivity). When specially selected nonsolvents are used, some explanation for their selection should be given. For example, if polysiloxanes were to be tested for potential application as a bladder material, it would be appropriate to test the permeation rate of a fatty acid ester, such as methyl oleate, in order to find some predictive data for the permeation of apolar plasma or blood components.

Wide-Angle X-ray Analysis

The X-ray analysis of either powdered or formed components (films, sheets, etc.) yields information on the kind and extent of crystal formation present in the sample at the conditions of measurement, as well as the orientation of the crystals and/or the noncrystalline material (32). Crystallinity is important not only for its possible effect on blood-material interactions, but also because of the dominant effects it exerts on mechanical responses of materials, and on the sorption and diffusion responses of the polymer to its environment. Because crystallinity is affected by a wide variety of variables reflecting the prior history of the specimen, it can be used to follow structural changes within a polymer as it is subjected to various processing steps. Determination of orientation distribution is usually done with a pole-figure goniometer and is important in structural components when the component depends upon sufficient orientation for certain of its mechanical properties. X-ray analysis requires sophisticated equipment usually available only in specialized laboratories. The spectrum of a well-annealed sample of the material being studied should be obtained for reference purposes. Once this is available, comparisons can then be made to the analysis obtained following various processing steps, as well as after implantation or use.

Optical Birefringence

The optical birefringence of many polymer specimens is a useful indicator of orientation applied to the specimen during its fabrication steps. The birefringence is a direct measure of anisotropy of the structural properties, and may serve as a useful guide in orienting the specimen during application experiments. For example, in membranes, the birefringence can be related to the mechanical moduli in the casting and cross-casting directions. In many solid parts, birefringence can be used to detect stresses frozen into specimens during fabrication.

Equipment to measure birefringence is simple and widely available. A petrographic microscope is required, along with either a half-wave plate or a calibrated wedge. In the latter case, the wedge is calibrated in terms of the optical retardation of one plane of polarization compared to its perpendicular rays. The total retardation is a product of optical path length (i.e., specimen thickness) and birefringence. Further details may be found in reference 17.

NMR Spectroscopy

Nuclear magnetic resonance (NMR) spectra can be readily obtained on soluble, noncross-linked polymers (17,33), although the required instrumentation is relatively expensive, and it may be necessary to contract with a commercial or university laboratory to perform this analysis. The data provide information on the chemical structure of the product, and some data as to its conformation. For copolymers it provides information on possible monomer sequences in the material. Furthermore, it will provide information on the presence of gross contaminants which are not readily separated from the polymer by fractional precipitation purification methods.

Scanning Electron Microscopy (SEM)

SEM viewing of cross-sections of polymer specimens obtained by fracture of frozen pieces offers the investigator a means to establish the gross uniformity of the material and a measure of the morphological character of the specimen. Phase separations, void structures, inclusion of particulate contaminants, fatigue cracking, environmental corrosion, and distribution of fillers can be determined by this technique.

The specific technique used in the preparation of the sample for viewing must be determined for each specimen type. Some polymers may require staining in addition to the overcoating of a conductive film. The fragmentation method used can affect the morphology of the observed sample. Elastomers with a very low glass transition temperature (T_g) may undergo flow if the specimen is sliced at room temperature. For many specimens, fracture at liquid nitrogen temperature avoids most artifact formation.

Many scanning electron microscopes are equipped with energy dispersive X-ray (EDAX) spectrometers. In addition to identifying inclusions and other surface morphological features from the X-ray emission wavelengths, intensity plots may be obtained and photographs made showing precisely from where on the surface the emission is coming. The chapter on surface characterization provides more details on SEM sample preparation and the EDAX technique.

Electrical Conductivity

Electrical conductivities of polymers can and have been measured by many techniques (34) (ASTM D257). They all involve some direct measurement of the resistivity (reciprocal conductivity) of the material when placed in a series circuit. Typical values for polymers span the range 10^{-14} to $10^{-9} (\Omega\text{cm})^{-1}$, although research toward conductive polymers has now produced values of $10^3 (\Omega\text{cm})^{-1}$ and higher. The Shallcross milli-ohmmeter has been used for these measurements. The most difficult problem to overcome in making reproducible measurements, especially from laboratory to laboratory, is that of establishing good (low resistance) electrical contact between the material and the measuring circuit. ASTM D150 is the standard test for insulating properties (dielectric constant) (35).

Dynamic Mechanical Testing: General

Dynamic mechanical testing consists of two classes of tests: static transient tests such as creep and stress relaxation, and periodic or oscillatory tests. To characterize the full range of dynamic response of a candidate material with respect to frequency (or time) and temperature dependencies requires utilization of more than one type of test, since every specific mechanical testing device has a limited frequency range with respect to the range of interest. Ferry's text (15) is the best single reference for information on this type of testing, and is especially thorough in describing the interrelations among the material functions obtained by various dynamic tests done within

the linear response range of the material. The level I tests should be completed before initiating level II testing since some preliminary information is necessary. These level II tests in many cases require somewhat more specialized equipment than that required in level I.

Creep

The creep experiment consists of applying a stress (tensile or shear) to a sample within a very brief time ($t = 0$) and maintaining it constant thereafter. The time-dependent strain of the sample is measured. Typical plots of stress (σ) and strain (ϵ) are shown in figure 4.6. The property measured is the creep compliance, $D(t)$, defined by:

$$D(t) = \frac{\epsilon(t)}{\sigma_0}$$

For cross-linked materials the strain, ϵ , will reach some steady-state value (ϵ_g). For uncross-linked materials, the strain will continue to increase until failure, often in such a way that a steady-state flow develops and the strain as a function of time may be written:

$$\epsilon(t) = \sigma_0 (D_e^0 + t/\eta)$$

where η is the viscosity of the material (extensional if measured in tension) and D_e^0 is a measure of the elastic recovery of material known as the steady-state recoverable compliance. This is also the ratio of the amount of recoverable strain to applied stress determined by rapidly unloading the sample. Figure 4.7 illustrates these concepts.

Because of the simplicity of the creep experiment, a wide variety of arrangements can and have been used. The simple way of applying the force is by means of a deadweight; displacement is conveniently measured by a differential transformer, traveling microscope or cathetometer, or deflection of a light beam.

For comparatively soft materials, sample gripping may be difficult. Torsional shear creep measurement in an apparatus such as that described by Plazek (36) is recommended. For materials with tensile moduli greater than 10^7 Pascals, the tensile creep mode is most convenient, but because of the large σ_0 which must be applied, a rather massive testing instrument will be required.

In all tensile testing modes, corrections will have to be made for the changing cross-sectional area. Constant σ_0 can be maintained by reducing the applied force continuously. Temperature dependence of the creep compliance should be studied by performing the measurements in a temperature-controlled environment. Such environmental chambers are commercially available.

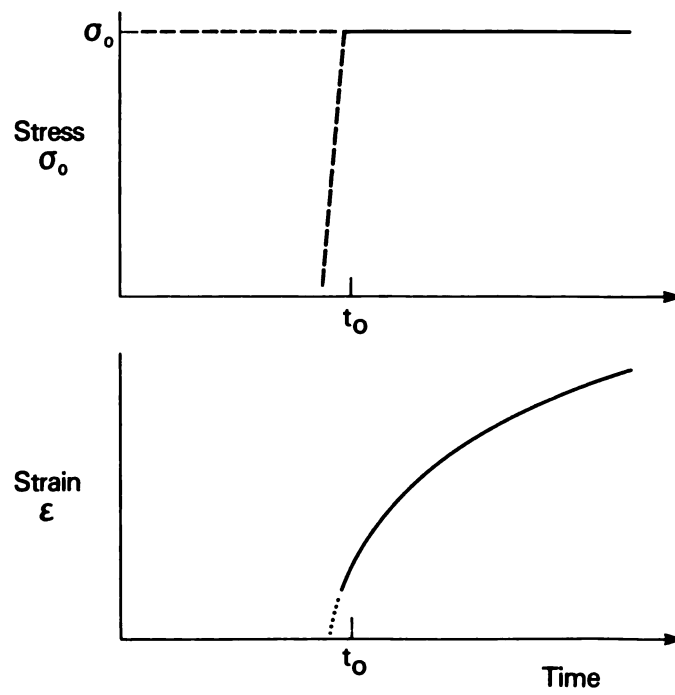


Figure 4.6--Demonstration of creep. At a constant stress (σ_0) the strain (ϵ) of a viscoelastic material will increase with time.

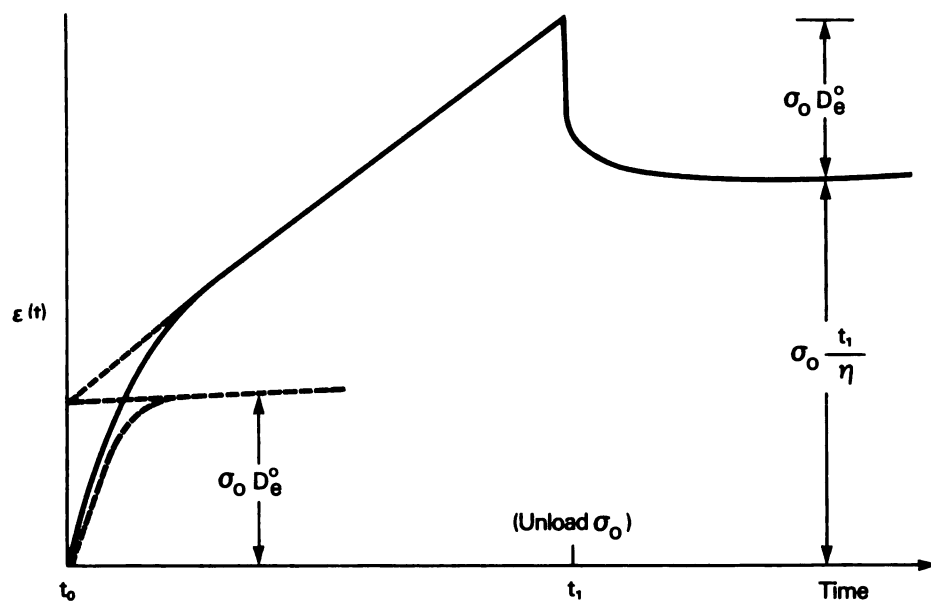


Figure 4.7--Permanent and recoverable creep.

For most compliant materials such as elastomers and the polyurethanes, it is important to condition the material prior to carrying out the creep test so that reproducible creep results can be obtained. This is done by performing a series of creep and recovery cycles with the recovery (no load) time being at least 10 times the creep time. The sample should be strained to a point below that where any irreversible (plastic flow) deformation occurs. Several cycles of creep and recovery may be needed before reproducible behavior is achieved. Annealing without load both before and after the creep-recovery conditioning also helps to prepare the sample for reproducible creep experiments. The stress and thermal conditioning eliminates effects of fabrication (frozen in stresses, etc.) and enables the material to exhibit its pristine mechanical response as it would in a performance situation.

The time-temperature superposition principle (37,38) can be applied to shift curves obtained at different temperatures to form a master curve. A vertical shift factor $\rho(T)T/\rho(T_0)T_0$ has to be applied to each individual compliance curve to account for the inherent dependence of compliance on temperature. Here, T_0 refers to the reference temperature for which the master curve will be constructed, $\rho(T)$ is the density at any temperature T and $\rho(T_0)$ is the density at T_0 .

The creep compliance curves determined at different temperatures are illustrated in figure 4.8 for the case of Avcothane in an air environment with an initial undeformed stress of 2.25 MPa. All the specimens used were preconditioned as described previously. Taking the curve at 37°C as the reference curve, all the other curves were shifted horizontally to form a master curve as shown in figure 4.9, which also includes master curves at initial stress levels of 1.67 MPa and 1.0 MPa. With the definition of engineering stress and engineering strain, the creep compliance curves of Avcothane suggest viscoelastic nonlinearity; that is, the compliance is dependent on stress level. The data have been computer-shifted, point by point, to yield the discontinuous master curves.

Stress Relaxation

The stress relaxation experiment consists of rapidly straining (tensile or shear) the sample to some constant strain ϵ_0 and subsequently monitoring stress as a function of time. Typical plots of strain and stress are shown in figure 4.10.

The material function measured is the stress relaxation modulus, $E(t)$. Experimental methods and procedures for stress relaxation have been reviewed in detail by Bergen (39). Most measurements are made in extension. There are, in general, three ways of keeping strain constant: by continuous adjustment of the force as the stress in the sample relaxes using an automatic servomechanism; by applying more than enough force with a stop to limit the deformation; and by straining a stiff spring element in series with the sample.

When stress relaxation experiments are performed on uncross-linked

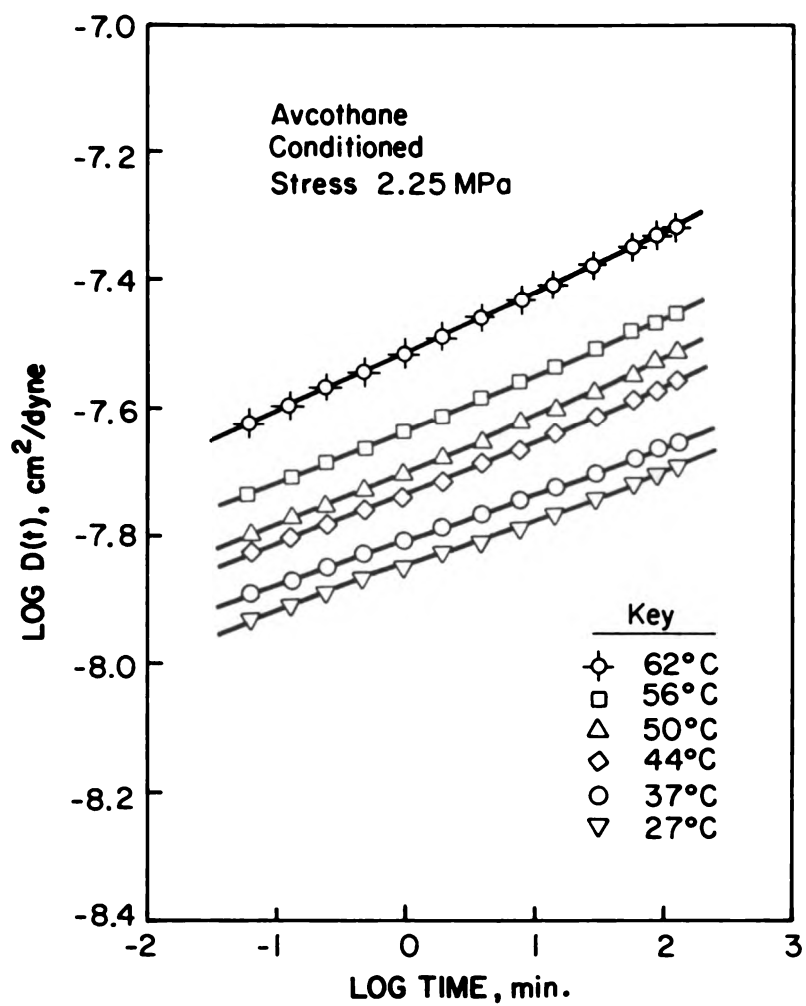


Figure 4.8--Creep compliance curves for Avcothane in air.

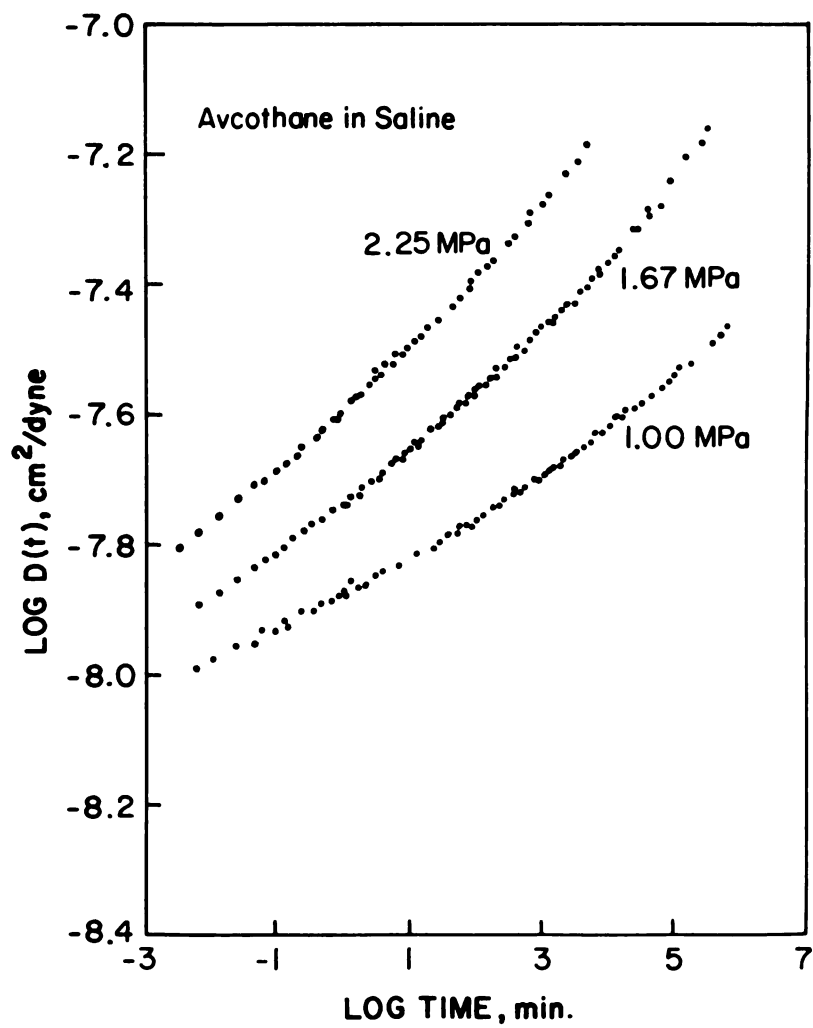


Figure 4.9--Creep compliance master curves for Avcothane in air.

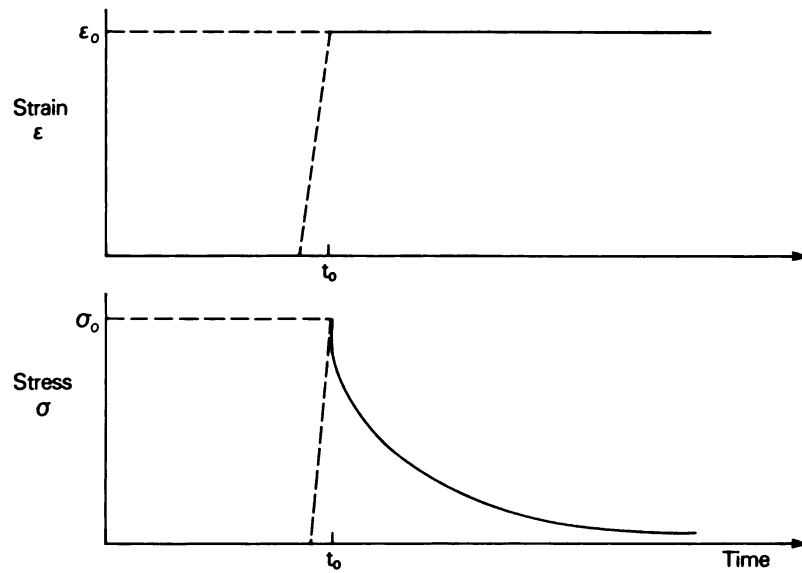


Figure 4.10--Demonstration of stress relaxation. At a constant strain (ϵ_0), the stress (σ) of a viscoelastic material will decrease with time.

fitted by a series of exponentials to give an approximate analytical expression for the time-dependent stress relaxation modulus, i.e.,

$$E(t) = \frac{\sigma(t)}{\epsilon_0} = \sum_{i=1}^N E_i e^{-t/\tau_i}$$

where the τ_i are relaxation times for the material. Usually only two or three of the τ_i can be extracted from the stress relaxation data directly with any claim to accuracy. The small forces must be measured with a sensitive detector toward the end of the experiment. Several stress measuring devices with different sensitivities should be available for use in this situation. For stiffer samples, commercial testing devices can automate the stress relaxation experiment and provide a temperature-controlled environment.

Oscillatory Testing

The protocol here is to subject a sample of the test material to an oscillatory strain and observe the oscillatory stress response. Or, one can subject the sample to a controlled oscillatory stress and observe the oscillatory strain response.

If the applied strain has the form

$$\epsilon(t) = \epsilon_0 \sin \omega t$$

it is found that for a viscoelastic material in the linear response range, the stress response may be written:

$$\sigma(t) = \sigma_0 \sin(\omega t + \delta)$$

where δ is the phase lag of the stress expressed as an angle. The stress may be rewritten, defining two material functions, as:

$$\sigma(t) = \epsilon_0 [E' (\sin \omega t) + E'' (\cos \omega t)]$$

where E' is the storage modulus and E'' is the loss modulus. The loss tangent, $\tan \delta$ is equal to E''/E' . For a linear viscoelastic material the dynamic moduli are related to the stress relaxation modulus by the Fourier transforms:

$$E'(\omega) = E_e + \omega \int_0^{\infty} [E(t) - E_e] \sin \omega t \, dt$$

$$E''(\omega) = \omega \int_0^{\infty} [E(t) - E_e] \cos \omega t \, dt$$

where E_e is the equilibrium static modulus. A similar analysis may be made for an oscillatory strain response. For uncross-linked materials which are not very viscous, E_e is zero; otherwise it is the long-time stress relaxation plateau modulus.

Oscillatory testing is most conveniently performed with one of the many commercial oscillatory testers available. Tensile oscillation, torsional pendulum and forced torsional oscillation are the most common measurement modes (the latter two giving dynamic shear moduli G' and G''). With a temperature-controlled commercial apparatus operating in the tensile mode, moduli and $\tan \delta$ are obtained directly. Disadvantages of such an apparatus are: 1) the possibility of background stress relaxation due to a small tensile strain which must be applied to prevent buckling during testing and 2) the limited number of fixed frequencies available. Advantages include the ease of operation and the low cost which makes it feasible in many laboratories.

The torsional testing mode is superior from the testing point of view. A forced torsional oscillation device with environmental control (mechanical spectrometer) is available. Such instrumentation has a continuously variable frequency range but the high cost (\$100,000) makes it less accessible to many laboratories. A freely oscillating system has a more limited frequency range, but can be constructed for less than \$15,000 (40).

Both types of dynamic mechanical testing instruments require strips of the material. If the material is very soft, serrated platens can be used to hold and subject a thin disk to torsional oscillatory strain in the mechanical spectrometer.

From this mode of testing one obtains the dynamic moduli as a function of temperature and frequency. There is a correspondence between the temperature and frequency dependence of material functions for many polymers which is expressed analytically by the Williams-Landel-Ferry (WLF) equation (19):

$$\log a_T = \frac{C_1(T-T_g)}{C_2 + (T-T_g)}$$

where a_T is a so-called WLF shift factor. This factor is the amount by which mechanical property data obtained at temperature T (such as the creep data, $D(T)$) must be shifted horizontally along the time (or frequency) axis with respect to data obtained at some reference temperature (here taken to be the glass transition temperature, T_g) in order to form a so-called "master curve." The constants C_1 and C_2 are empirically determined for every material. A construction of such a master curve is illustrated in figure 4.10 for stress relaxation. Similar master curves can be produced for any linear viscoelastic material function.

Such data on temperature dependence have been tremendously valuable in studying glass transition and other structural transitions in the

amorphous and crystalline phases of semicrystalline polymers. The frequency dependence gives important information on how the material responds to rapidly applied deformations (high frequency) relative to more slowly applied deformations.

The T_g of a homopolymer reflects the temperature at which segmental mobility of the polymer, as expressed by its relaxation time, corresponds to the time scale of the measurement. For a polymer undergoing cyclic stress at ever-increasing temperatures, there will be an observed temperature at which a discontinuity occurs in the curve relating modulus to temperature. This temperature will also be one at which a sharp increase in the viscous damping of any applied cyclic force occurs.

Since these effects reflect the action of only segmental lengths of the polymer (consisting perhaps of less than 20 repeating units) a copolymer having distinct domains may exhibit more than one transition temperature. Such measurements are, in fact, the best evidence that independently acting volume elements are present in the polymer. The effects of temperature, or contamination by components from the polymer's environment, will alter the transition temperature as well as the magnitude of the damping. Thus, the measurement of modulus as a function of temperature is an extremely sensitive descriptor of polymer morphology.

Fatigue Testing

Many applications of polymers as biomaterials require large cyclic deformation for long lifetimes. This is particularly true of applications in the cardiovascular area such as LVA pump bladders, heart valves, and vascular grafts. The actual performance condition may be quite complicated because of the final part geometry and environment. For example, cardiovascular grafts will be exposed to a biaxial stress situation in which the cycling internal pressure is superimposed on a static internal pressure, and all of this is superimposed on a static axial load; moreover, the material is subjected to different liquid environments on its inner and outer surfaces.

In order to qualify polymers for clinical usage in which they will be cyclically deformed, two major levels of fatigue testing will be required--a uniaxial screening test to rank candidate materials and provide preliminary information for design, and a performance test which mimics the actual use conditions as closely as possible. Regardless of the type of test used, the statistical spread of the cycles-to-failure data at any one stress condition and frequency must be taken into account to provide a measure of the reliability of the material.

Two major types of uniaxial tests are useful (41). Most of the polymer fatigue data in the literature have been generated in a three-point flexural mode or in a fixed-cantilever mode (42) (ASTM D671). In either of these modes the sample is bent and either the load limits or the deformation limits are controlled at constant values. Flexural fatigue tests alternately place the sample surfaces under maximum tension and then compression. Because of this, any surface flaws tend to

cause scatter in the resulting data. However, the test can usually be carried out with less expensive equipment and, if the final application is to involve flexural deformation, a flexural screening test should be used. If the material is very compliant, a flexural test (particularly with three-point loading) will be difficult because prohibitively thick specimens may be necessary. A more useful uniaxial fatigue test is the tension-tension test in which a rectangular strip is pulled from a no-load or very low load to some maximum load condition. This mode of testing is less sensitive to surface flaws, places the entire specimen in a uniform state of stress, and may be run under either load or deformation control; however, the equipment is generally more expensive than in the case of flexure. The ASTM tension-tension fatigue test for reinforced plastics may also be followed for unreinforced systems (43) (ASTM D3479). Both types of uniaxial fatigue tests (tension-tension and flexure) provide the number of cycles to failure as a function of the maximum stress level or strain level and the cyclic frequency, and both may be carried out under either load or strain control.

The various material factors that affect polymer fatigue life, such as molecular weight distribution, crystallinity, etc., are discussed in some detail in reference 44. Also discussed are such other important factors as frequency, temperature, and environment.

Cyclic biaxial testing of polymeric materials is an area in which little has been published, even though in many cases it much more closely matches performance loading conditions. A bubble-type experiment (45) has been devised to measure anisotropic stiffnesses and biaxial stress-strain curves. This type of experiment is also used in the rubber industry to characterize effects of biaxial deformation. The biaxial test has been used much more extensively in reinforced plastics research because of the necessity to evaluate coupling terms in such anisotropic materials (see above discussion of modulus and compliance measurement from stress-strain data). There are many ways to set up a biaxial test configuration. An excellent review of this type of testing is given by Bert (46). He reviews the latest configurations for static multiaxial loading using both plate and shell types of specimens. One of the simplest configurations to use is a simple tube, which allows states of pure shear, tension, or combinations thereof to be applied and analyzed in a straightforward way. A variation of the Williams spherical bubble test is an elliptical geometry which allows easy analysis of material stiffness or compliance anisotropy (45).

One of the most versatile pieces of equipment for measuring cyclic biaxial fatigue resistance, as well as for a variety of other cyclic performance tests, is the Swanson tuned fluid oscillator (47). This apparatus uses a positive displacement bellows drive to oscillate fluid between two legs connected by a high impedance shunt. This unit is capable of frequencies up to 33 Hz. Different test cell geometries can be inserted for testing heart valves and polymer sheets. Pressure drop across the piece can be continuously monitored. The actuating liquid also serves as the environment and, because the system is closed and compact, the entire unit may be loaded with sample and liquid under sterile conditions so that long-term runs in blood may also be accomplished.

Failure in a fatigue test may be defined either as a complete separation of the sample into two parts or as a specific degree of degradation of a particular mechanical property, usually the modulus or compliance. For example, a 20 percent decrease in stiffness might constitute failure for a specific application.

When a material is screened or performance-tested in fatigue, it is sometimes useful to proof-test the material at various points in its lifetime. Sometimes the static mechanical properties such as stiffness and strength will start to degrade long before final failure occurs in fatigue, particularly if the maximum stress level in the fatigue test is significantly below the ultimate static strength of the material. Stopping the fatigue test and running a static stress-strain test (either uniaxial or biaxial) can provide a measure of the material's remaining inherent properties at that point. This type of proof testing is very helpful in evaluating the reliability of a material particularly in its performance configuration.

Properties of Woven Materials

Prosthetic materials, both in clinical use and under development, may consist of woven textile materials. Examples in the cardiovascular system include vascular grafts and heart valve leaflets. It is difficult to characterize these materials mechanically because of their anisotropy and their open porosity. Such materials usually consist of individual filaments or yarns (multiple filaments combined with or without twist, sometimes also called rovings) held relatively straight in the warp or long direction of the loom, and individual filaments or yarns which are woven in the filling direction under and over the warp filaments or yarns. The individual filaments or yarns may also be crimped (bent permanently back and forth like a bellows) in one or both directions to produce a material having totally different properties from uncrimped woven materials.

Normal static mechanical properties such as strength and stiffness as defined earlier for macroscopically continuous materials lose their meaning for woven fabrics because of variable and difficult-to-define cross-sectional areas of solid material through which the load is transmitted. In the case of static tensile properties, load-elongation curves are usually determined in both the warp and filling directions (48). It is also possible to express the load-elongation information as load per filament in either direction. A common way of doing this in the textile fiber industry is to express it as load per denier (1 denier = number of grams per 9,000 meters of single filament).

Although it is not possible to construct a normal stress-strain curve and obtain a classical failure stress or elastic stiffness, it is possible to determine the degree of crimping or percent fabric crimp from the load-elongation information in either major weaving direction (49). The degree of crimping is important because crimping is one method to make a fabric having rather stiff individual filaments or fibers behave like natural tissue in its load-elongation characteristics (low stiffness at low elongations followed by rapidly increasing stiffness as the elongation increases further).

While classical stiffness measurement is not possible, it is possible to obtain a measure of flexural rigidity utilizing a cantilever bending test (50) or a circular bending test (51). The latter is a multidirectional deformation of a fabric in which one face of a flat fabric specimen becomes concave and the other becomes convex.

Two other mechanical properties of interest are tearing strength and abrasion resistance. In the case of tearing strength, the force necessary to initiate and then propagate a rip is recorded using a standard tensile testing machine (52). Abrasion resistance is difficult to measure and requires special testing set-ups. There are two ASTM tests, D4157 which employs an oscillatory cylinder method (53) and D4158 which utilizes a multidirectional rubbing action (54). One can measure how many abrasion cycles it takes to rupture the fabric or, alternatively, the loss of a particular mechanical property such as the fabric breaking load for a given number of abrasion cycles.

Data scatter is a serious problem for all the fabric mechanical characterization tests because of the freedom of the fabric components to move about. Consequently appropriate statistical analysis including standard deviations and t-test applications are especially warranted.

One other useful piece of characterization information concerning fabrics is a pseudo-porosity measurement termed "percent open area." This is obtained from an optical micrograph and is a measure of the looseness of the weave. One simply measures the percent of a fabric plane not occupied by filaments or yarns. Related to this sort of test is an air permeability test (55) in which the cubic feet per minute per square foot of fabric at a stated pressure differential across the fabric is measured. There are also various water retention tests, among them ASTM D461 (56) which is primarily applicable to felts, but which can also be useful for tightly woven fabrics.

For additional information on woven materials, see chapter 6.

Summary

In level II tests of bulk properties, the time-dependent response of the material is probed more deeply in the linearly viscoelastic response regime. It is recommended that at least one transient and one oscillatory type measurement be made, both as functions of temperature. The former is more effective in probing the long-time response while the latter is vital to understanding the high-frequency response. With proper choice of experiments, the combination can be used to give information over 10 or more decades in log time.

This level II information has been used by Smith (14) and by Halpin (57) in studying the ultimate properties of polymer materials. In particular, Smith has shown how a "failure envelope" can be constructed from stress relaxation and creep data.

Fatigue testing is becoming more and more important as polymers and reinforced polymers move into structural, life-supporting roles as prosthetic materials. Qualification for clinical use will depend on a

firm durability data base and measure of the product's mechanical reliability. The characterization of the mechanical properties of woven fabrics is not as exact as that of macroscopically continuous and homogeneous materials. Because of this, it is important to produce a data base amenable to various types of statistical analyses (see chapter 1).

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Note: The complete volumes of ASTM standards (66 volumes) may be found in most technical or engineering laboratories. They may also be purchased individually or as sets from ASTM, 1916 Race Street, Philadelphia, Pennsylvania 19103.

Chapter 5

MATERIAL SURFACE CHARACTERIZATION

RATIONALE

Surface characterization of biomaterials has been considered an important component of biomaterials development, testing, and manufacture for at least the last 20 years. The reasons for this are twofold. First, certain measured surface properties of synthetic materials (e.g., contact angle, X-ray photoelectron spectra) have been shown to strongly correlate with various measures of biological interaction (e.g., protein adsorption, cell and platelet adhesion, thrombosis, and platelet destruction) (1-10). Thus, the implication is that the surface region, for materials which are free of cytoreactive leachable components, controls the nature of the biological response to a material. Second, the surface and the bulk of most materials differ substantially in both composition and morphology. Therefore, more traditional analytical techniques which primarily study the bulk material are ineffective in providing information on chemistry or organization of the surface region.

One can discuss surface characterization of materials both before and after implantation. However, after implantation proteinaceous and cellular films will form on materials. These films can lead to "deceptive" scanning electron micrographs and can "cover over" chemical changes which might have occurred in the surface during implantation. Biologic films can often be partially removed by surfactant-like materials such as sodium dodecyl sulfate (SDS) and Triton X and by enzymes such as pepsin. However, the potential exists that the cleaning agents themselves may alter surface morphology, through extraction and the induction of crazing, and may alter surface chemistry through adsorption (11). Thus, surface characterization experiments on either explanted materials or materials with protein-adsorbed films that are intended to examine the biomaterial itself, and not the biologic film that forms on all surfaces after contact with biologic fluids, are complex and require careful parallel control studies.

Surface characterization of materials for biomedical application should provide the biomaterials investigator or manufacturer with the following:

1. Unknown identification (most commercially available biomaterials and devices are not adequately identified by the manufacturer, particularly with regard to their surface structure).
2. Reproducibility assurance (from experiment-to-experiment and from lot-to-lot).
3. Quality control.

4. Contamination detection.

5. Correlation between measured surface properties and biological response (for fundamental studies or for developing low cost predictive tools for screening biomaterials).

With respect to item 5 above, consider the following: although there has been considerable research activity over the past 20 years on blood-material interactions and some activity attempting to correlate the surface properties of materials with those interactions, the number of noncontroversial correlations which presently exist are very few indeed. Part of the problem can be accounted for by noting that blood-materials work in the past has largely involved the use of materials whose surfaces have not been adequately surface characterized, or, in many cases, not characterized at all. It is generally recognized that the surface of a material is almost inevitably different from its bulk composition, hence the concern over surface characterization. It seems obvious that if one is planning to correlate blood-material interactions with the surface properties of a material, then one must at least be able to measure and characterize those surface properties.

It is the intent of this section to provide a working understanding of what surface characterization is and how it can be performed, either "in house," or using service laboratories. A few review articles have been written which discuss the surface characterization of biomaterials in more detail (12-15).

ORGANIZATION

Three levels of testing are proposed in this chapter. Level I consists of relatively simple test and characterization methodologies which can, in principle, be performed in virtually any laboratory involved with the development of biomaterials. They involve a minimum of apparatus and expense and can be performed with no specialized training. In many cases the physical data required in level I testing will be available from the supplier of the material. The level I testing category has been expanded in this second edition of the guidelines to include certain tests which are particularly information-rich and which can be easily performed either within the average laboratory or through readily accessible service laboratories.

Level II surface characterization assumes that, on the basis of level I testing, a material appears to be of interest for further study or development. Level II bulk characterization tests should already have been completed to provide information on the bulk chemistry and properties of the material. The level II surface characterization will provide relatively detailed information on the surface and interfacial properties of the material. This information can be related to subsequent blood-material interactions. Level II tests often involve sophisticated equipment and, in some cases, equipment that would not be expected to be present in many research laboratories. However, all the

tests suggested at this level can be performed in specialized commercial analytical laboratories and are generally accepted analytical techniques that present few significant problems with interpretation.

Level III tests are generally techniques and methodologies that are still under development or unproven with regard to their significance for biomaterials science. These tests and methods will be important in probing and understanding many of the gap areas discussed in this report. One example is the analysis of surface microheterogeneity of materials, particularly the block copolymer systems that are widely used as biomaterials (e.g., the polyurethanes). The characterization of morphology down to the 25 Å level is a difficult surface analytical problem. It is expected that the level III characterization will only be attempted by those researchers specializing in surface analysis. It is also expected, however, that many of the tests in level III will be refined and developed in the near future for inclusion in the level II category.

In some cases, tests suitable for probing certain properties may be useful only for a specific class of materials. Nevertheless, the types of tests suggested for level I and level II are such that they can be expected to apply across the entire range and breadth of materials normally considered useful and of interest for cardiovascular device development and application.

LEVEL I TESTS

Introduction

Figure 5.1 presents a suggested flow diagram for the level I surface analysis of a typical medical material. The surface of a sample should not be characterized unless some of the bulk characteristics of the sample are already known. Therefore, at least the sample identification aspects of the level I bulk characterization discussed in chapter 4 should be available prior to the level I surface characterization.

Microscopic Examination

All material, as a first step in surface characterization, should be examined optically for surface homogeneity or heterogeneity, the presence of particulate matter, surface defects, microbubbles, scratches, etc. This should be done with a low-power stereo microscope and then at relatively high magnification with a conventional reflected-light optical microscope. For example, a surface roughness of the order of 0.1 µm or greater (16) will make it difficult to interpret any contact angle measurements made on the material. It is reasonable to expect that materials used for blood-material interaction studies would be largely particulate-free and relatively smooth and homogeneous. This means that the materials should be prepared and studied in a dust-free environment, preferably in a laminar flow, filtered air laboratory bench or in a clean room. Low cost bench-top laminar flow hoods are readily

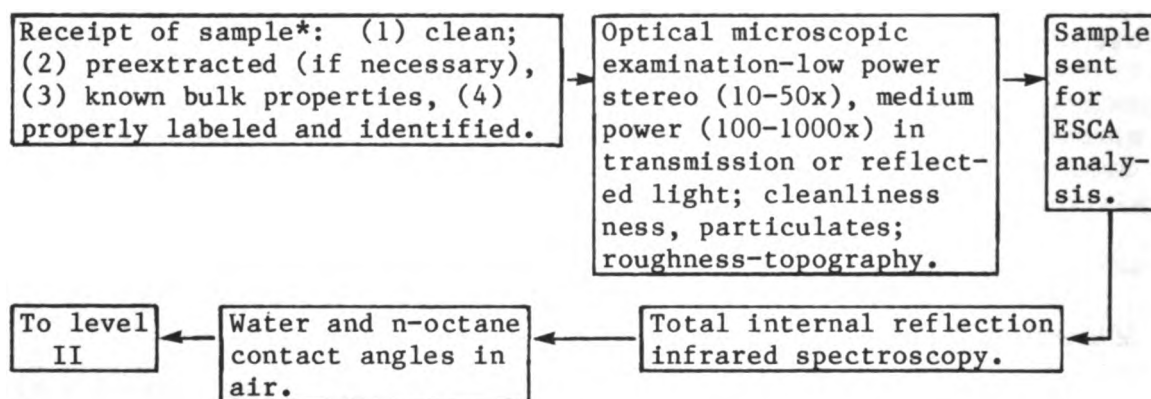


Figure 5.1--Recommended flow plan for level I surface characterization.

*Note: See appendix II prior to initiating surface characterization.

available and are ideal for sample preparation. A detailed discussion of sample handling and particulate contamination is presented in appendix 2.

The microscopic examination should pay careful attention, in addition to particulate matter, to the presence of surface roughness and deposits, including extrusion marks, machining marks, etc. The orientation of such features should be clearly documented. Orientation effects in many polymers can be clearly seen by the use of cross-polarized light microscopy (17).

Certain types of material surface features will be difficult to see because of the optical characteristics of the material, unless one uses interference or phase contrast or scanning electron microscopy (discussed in more detail in level II tests). Depending on the features observed and the nature of the material, it is possible to obtain additional information from optical and scanning microscopy utilizing cross sections of the material. Brittle materials can be fractured and examined edge-on to study roughness and topographic features; materials that are nonbrittle at room temperature can generally be fractured under liquid nitrogen (18).

Although microscopic examination of materials can provide surface information at a low cost, the investigator must be cognizant of the potential of these techniques to introduce artifacts. Such artifacts are particularly prevalent where extensive sample preparation prior to microscopic examination is involved, or where samples are examined in a high vacuum (19-23). Great care should be exercised in interpreting microscopic images and drawing conclusions based upon those images. A variety of optical and electronic distortions can occur. Ideally, features seen in a microscopic image that could be subject to alternate interpretation should be corroborated using a complimentary microscopic or surface characterization technique.

Infrared Spectroscopy

Infrared spectroscopy is recommended for inclusion in both level I and level II bulk characterization protocols. It can be useful in surface analysis (24), particularly for the detection of mold-release agents, surface-active agents, antistatic agents and related additives, surface oxidation, and processing effects. By the utilization of relatively simple total internal reflection accessories, one can obtain infrared spectra of the surface region of nearly all solid polymers (24-26). The mid-infrared region of the spectrum is extremely rich in chemical information, and peak charts are available which allow much of this information to be used in a relatively routine fashion (27).

Attenuated total reflection infrared spectroscopy (ATR-IR) (figure 5.2) allows the recording of the diagnostic infrared "fingerprint" for the surfaces of solids (in the form of films or compressed powders) and liquids. The technique is sensitive enough to detect and identify some films as thin as 10 Å (where absorption bands are sufficiently strong)

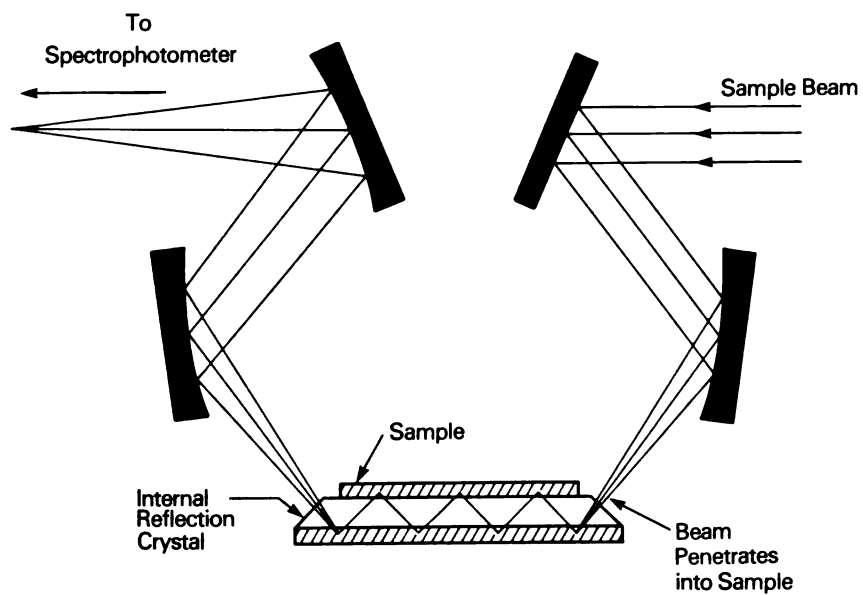


Figure 5.2--Typical geometry for attenuated total reflection spectroscopy.

and to follow changes in the composition, configuration, and bonding within such films. The method is generally nondestructive. However, the pressure required to bring the specimen in intimate contact with the ATR element can distort many samples. In addition, material from certain ATR elements can be transferred to the specimen.

It is important to note that even though the technique can be surface sensitive, the penetration depth of the infrared beam is of the order of the wavelength of the infrared radiation (26), which is in the micron range. Thus, the technique really measures a considerable depth into the sample. In many cases, for surface contaminants that are present only in very small concentrations (portions of a monolayer), for low degrees of surface oxidation, or for surface species with weak IR absorbing bands, ATR-IR may not be adequately sensitive to detect processes occurring at the surface. Also, quantitation of data is more difficult in the ATR mode than by conventional transmission IR. Finally, the energy throughput of most ATR devices is low. Therefore, the signal-to-noise ratio of ATR spectra tends to be lower than for conventional transmission spectra. Fourier transform infrared (FTIR) spectrophotometers readily provide high energy throughput so that excellent spectra can be obtained in ATR mode in a reasonable period of time. Low cost FTIR instruments are now becoming available and are particularly well suited to performing ATR-IR studies on biomaterials.

Nevertheless, IR methods are probably the simplest, least expensive, most widely used, and most easily interpretable methods to obtain chemical information from the surface region of solid organic materials.

Note: The Division of Research Resources, NIH, has funded a National Center for Biomedical Infrared Spectroscopy. It is the intent of this center to provide information, collaboration, and services to those wishing to apply infrared spectroscopic techniques to biomedical problems. For further information, contact:

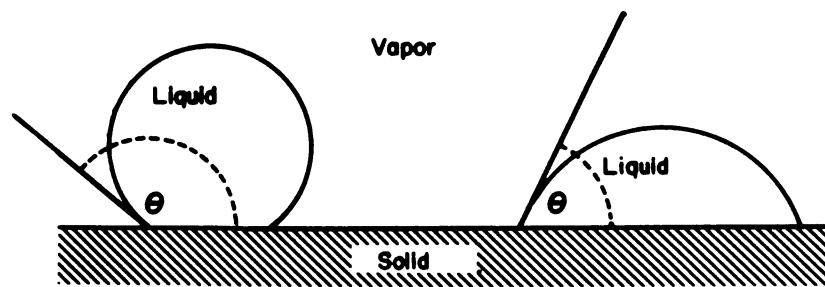
National Center for Biomedical Infrared Spectroscopy
Battelle Columbus Laboratories
505 King Avenue
Columbus, Ohio 43201

Contact Angles (Level I)

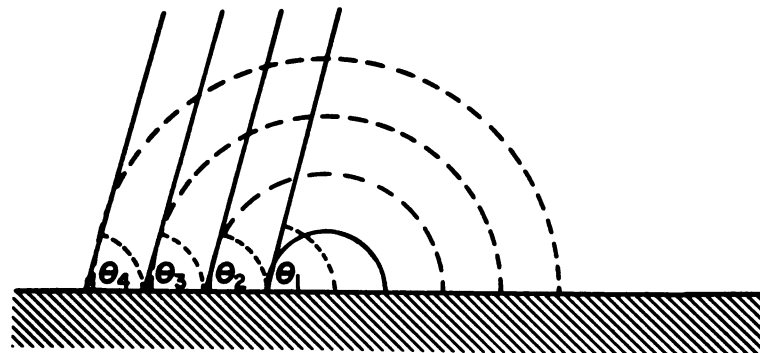
A measure of the surface free energy and/or surface wettability of solid materials is most readily and simply obtained by measuring the contact angle of a diagnostic liquid on the solid surface (figure 5.3). The basic relationship between the properties of the measurement liquid, the properties of the surface and the contact angle is the Young equation:

$$\gamma_{SV} - \gamma_{SL} = \gamma_{LV} \cos \theta$$

where γ_{SV} is the interfacial tension between the solid and the vapor above it, γ_{SL} is the interfacial tension between the solid and the



STATIC CONTACT ANGLES



DYNAMIC CONTACT ANGLES

Figure 5.3--Contact angle measurement on solid surfaces.

liquid, γ_{LV} is the surface tension of the liquid and θ is the contact angle. Measurement of the contact angle is an extremely sensitive technique for identifying changes in surface composition (28,29). In addition, a number of biological responses have been found to be a strong function of the wettability as measured using contact angles (1-3,5-7,30).

Contact angles can be measured in a number of different ways. A listing of the more common of these methods and their classifications within these guidelines into level I or level II tests is presented below:

<u>Contact Angle Methods</u>	<u>Level</u>
Static contact angle measurements	I
Advancing contact angle measurements	I and II
Critical surface tension determination	I and II
Receding contact angle measurement	I and II
Underwater contact angle measurement	II
Wilhelmy plate technique	II

The static or advancing contact angle methods (figure 5.3) are suitable for most of the common rigid polymers or for apolar polymers that cannot rapidly reorient their surfaces in equilibration with their environment. A discussion of the critical surface tension concept for using contact angle data to estimate the surface energy of a solid is given in appendix 5 and reference 28.

The critical surface tension approximation is only one of many ways to relate contact angle information to surface energy or other surface properties. Using various algorithms, the surface energy can be broken down into a dispersion force and a polar force component (30-33). The significance of such an analysis has been questioned in recent years (33,34), and a method involving a breakdown of the total surface energy in components representing dispersion force interactions, (Lewis) acid interactions, and (Lewis) base interactions has been shown to perhaps better explain observed interactions at interfaces (34-38).

Before a contact angle measurement can be meaningful, it is important to guarantee that the probe (i.e., the drop of liquid) has the expected surface tension (is uncontaminated) and is not extracting soluble material from the solid surface (see chapter 4) or, conversely, not swelling or absorbing or adsorbing into the solid. Purity of contact angle measurement liquids has been addressed in some publications (39), and measurement of the liquid contact angle on a clean sheet of Teflon (FEP) can be used to quickly assess the purity of the liquid (see appendix 6 and figure A6.1 in that appendix, or reference 40). Small amounts of residual material which are extractable into the measurement liquid droplet are often surface active and would change the surface tension of the liquid, thus changing the contact angle (see the Young equation) (41). Materials to be examined by contact angle

methods should, if possible, be subjected to extensive preextraction to guarantee that the extractables do not interfere with contact angle studies, subsequent characterization, or subsequent blood-material interaction studies.

For those polymers which have a significant degree of surface mobility and can readjust their surface region in response to their local environment (33,42-45) the advancing contact angle measured in air or vapor is not generally satisfactory. More direct information related to polymer-water interfaces is obtained via comparison of advancing angles to receding angles or using the captive air or captive bubble technique (33,42,44,46,47). These methods are discussed in more detail under level II contact angle methods.

In level I characterization, at the very least, static contact angles, and preferably advancing contact angles, for water and n-octane should be measured (39,48). These measurements can be made with a relatively simple apparatus, a horizontal microscope of about 20-100x magnification. A small drop of liquid is dispensed from a microsyringe or a flamed platinum wire on a test surface. The platinum wire method has the specific advantage that all contamination can be removed from that portion of the wire which enters the test liquid reservoir. Therefore, contamination of both the reservoir and the test drop is avoided. The microsyringe and needle have the potential to introduce such contamination, if not used with care. As mentioned earlier, the purity of the liquids used for contact angle measurements is critically important and should be periodically checked by directly measuring the surface tension of the liquid or by the Teflon test (see appendix 6).

The drop volume is commonly about 1 μ l. The contact angle at the point of tangency can be directly measured with a goniometer eye piece, or the drop dimensions can be measured and related to the contact angle by trigonometric functions (39,46) (also see appendix 7). To simplify contact angle measurement, the drop can also be photographed and the angle measured with a protractor on the photograph. These and other methods are all described in basic textbooks and reviews on surface chemistry (16,39,44,48).

There is usually a significant training period needed to develop the techniques used to deposit the drops and to properly align the crosshair in the goniometer tangent to the edge of the drop. It is suggested that inexperienced operators develop their skills on clean poly(tetrafluoroethylene) (PTFE) and poly(ethylene terephthalate) (Mylar) surfaces. Both these surfaces are easily obtained, readily cleaned, relatively smooth and homogeneous, and easy to work with. Typical value of contact angles for a few liquids on Mylar and Teflon are given in table 5.1. Accepted values for the critical surface tensions of these surfaces would be 19 ± 1.5 ergs/cm² for Teflon and 42 ± 1.5 ergs/cm² for Mylar (see appendix 5).

To measure advancing contact angles, the drop volume is slightly increased with time and the angle of contact measured; the receding angle is measured by slightly decreasing the drop volume with time.

Table 5.1--Typical Values for the Contact Angles (θ)
of a Few Liquids on Mylar and Teflon

<u>Liquid</u>	<u>γ_{lv} at 22°C (dynes/cm).</u>	<u>θ on Mylar</u>	<u>θ on Teflon (PTFE)</u>
Water	77.8	72	102
Glycerol	63.4	57 $\pm$ 6	82
Formamide	58.2	48 $\pm$ 5	--
Thiodiglycol	54.0	37	--
Methylene iodide	50.8	29 $\pm$ 5	71
S-tetrabromo ethane	47.5	12	--
1-Bromanophthalene	44.6	--	63
1-Methylnaphthalene	38.7	--	57 $\pm$ 9
Dicyclohexyl	33.0	--	38
Hexadecane	27.6	--	35
Decane	23.9	0	25 $\pm$ 8

Note: θ values were calculated as the average of the values obtained by two or more workers DVM a 1-year period.

These measurements, particularly receding angle measurements, are more difficult to perform than contact angle measurements with drops of fixed volume. The angles are strong functions of surface roughness and of any heterogeneity in the surface of the material.

Electron Spectroscopy for Chemical Analysis (ESCA)

In the first edition of these guidelines, ESCA (also referred to as X-ray photoelectron spectroscopy, XPS) was listed under level II and III tests. It is suggested that a simple ESCA analysis might be useful as a level I test. The following developments that occurred since the previous edition of the guidelines was published have led to an "upgrading" of this analytical tool:

1. Data obtained with ESCA have been shown to correlate with complex in vivo and in vitro blood interactions (10,49-51).
2. A good fundamental understanding now exists of ESCA as applied to polymers (52-54).
3. Although ESCA instruments are extremely expensive, the technique is readily available through a large number of commercial and academic service laboratories.*

The ESCA technique has a number of advantages for biomedical studies. These include minimal sample preparation (generally no special preparation at all is required), high information content, nondestructive analysis, and the ability to work with hydrated (frozen) specimens (45). ESCA as applied to biomedical problems has been reviewed (14). The following information can be expected from a simple ESCA analysis:

1. Identification and quantitation (± 10 percent) of all elements (except hydrogen and helium) present in top 10-100 Å of the specimen.
2. Information on the organic functional groups present in the surface region.
3. The valance state of any metal or inorganics present.
4. Spatial resolution (analysis within a 150 μm spot) on certain newer instruments.

*The Division of Research Resources, NIH, has established the National ESCA Surface Analysis Center for Biomedical Problems (NESAC/BIO). The purpose of this resource is to provide ESCA information, collaboration, and services to biomedical investigators. For further information, contact:

Dr. Buddy D. Ratner
NESAC/BIO
Department of Chemical Engineering
University of Washington
Seattle, Washington 98195.

ESCA was developed largely through the efforts of Siegbahn and his collaborators in Sweden (55). It is based on the precise measurement of the kinetic energy of electrons ejected from the sample (photoemission) by the action of incident radiation, usually X-rays or UV (basically, the photoelectric effect). The binding energy of the electron prior to ejection (E_B) is obtained from the measured kinetic energy (E_k):

$$E_B = E_{h\nu} - E_k - C$$

where $E_{h\nu}$ is the energy of the monoenergetic excitation radiation (Al K α , 1487 eV in many instruments), and C is an instrument constant which is readily determined experimentally. The high precision of the E_k measurement allows one not only to identify the elements (except for hydrogen and helium) present in the surface, but also to identify their oxidation state. The photoemission energy can be shifted by as much as 10 eV, depending on the oxidation state of the element.

A number of commercial ESCA instruments have been manufactured. If samples are sent out to a service laboratory for analysis, it is worthwhile knowing what type of data might be generated by the specific instrument(s) operated by the laboratory. The following are good performance parameters which might be expected from an ESCA instrument. Performance below the levels suggested in these specifications will result in spectra which are more difficult to interpret (particularly in terms of resolving subpeaks from a complex curve envelope) and will require longer data acquisition times (i.e., the data will cost more).

Good performance standards for ESCA instruments measured using NHLBI reference standard polyethylene sheet are:

1. Cls peak width at half maximum ≤ 1.2 eV. (The peak should be symmetrical and predominantly Gaussian in character.)
2. Signal-to-noise ratio for Cls peak of polyethylene (noise measured in the background): $S/N \geq 150$ after 10 minutes data acquisition time at the above resolution.
3. Total count acquisition rate: 15,000 counts/min. under the Cls polyethylene peak at the above resolution.

It is particularly important for all materials of interest to be examined by ESCA in the wide scan mode (generally 0-1,000 eV), which provides a spectrum that permits one to analyze for any elements present in a semiquantitative way. This is an ideal method of determining the presence of trace impurities and surface contaminants. For example, one often finds fluorocarbon and silicone materials on many commercial polymer surfaces due to mold-release agents. Antistatic agents can also be present. Many commercially prepared polymer surfaces have a small degree of surface oxidation which is not readily detectable by total internal reflection infrared techniques, but is readily detectable with ESCA; the degree of surface oxidation should be measured.

After the total elemental analysis and the assurance that the surface does not contain any undesirable impurities or contaminants, it is important to analyze the ESCA spectrum at higher resolution; in the case of polymers, this generally means looking at the carbon, oxygen, fluorine, silicon, and nitrogen regions. For example, a detailed analysis of the carbon region of the spectrum allows one to deduce the proportion of carbon atoms which are hydrophobic-type carbon (alkyl or aromatic carbons) and those which are hydrophilic (the carbon atoms bonded to electronegative or electropositive groups, i.e., polar groups) by simply comparing the ratios of the appropriate areas of the carbon ESCA spectrum (9). One thus obtains an approximation of the ratio of the hydrophobic to hydrophilic organic functionality of the material in the surface region.

Summary

The level I surface characterization protocol provides information on surface roughness, topography, morphology, and particulate contamination by means of optical microscopy. It provides information on the surface organic chemical functional group character, including the presence of surface contaminants and the presence of surface oxidation by the total internal reflection infrared spectroscopy technique and by ESCA. It also provides information on the surface wettability and the surface energy measured via the contact angle techniques. This information can be obtained in virtually all laboratories since most of the required equipment is either readily available or of low cost. ESCA services (analysis and interpretation) are available through NIH-funded, commercial, and academic laboratories.

LEVEL II TESTS

Introduction

The function of the level II surface characterization protocol is to obtain further information on the surface characteristics of materials that appear to be strong candidates for further development, for further testing, or for cardiovascular device development. It is assumed that level I characterization has already been completed.

The increased sensitivity and new information provided by many of these level II tests (and the level III test described later) relate largely to the sampling depth and area of surface resolution. The "true" surface of a material--the region whose properties differ from those of the bulk material and is important for biological interaction--extends into the material only a few molecular layers. Therefore, techniques which sample much more deeply may obscure surface data. Similarly, highly localized structure, associated with domains or with other surface heterogeneities, can be obscured or overlooked by a sampling probe that has poor surface resolution. Figure 5.4 summarizes level II protocol. Figure 5.5 summarizes the depth and spatial resolution achievable with a number of techniques described in this and the following sections.

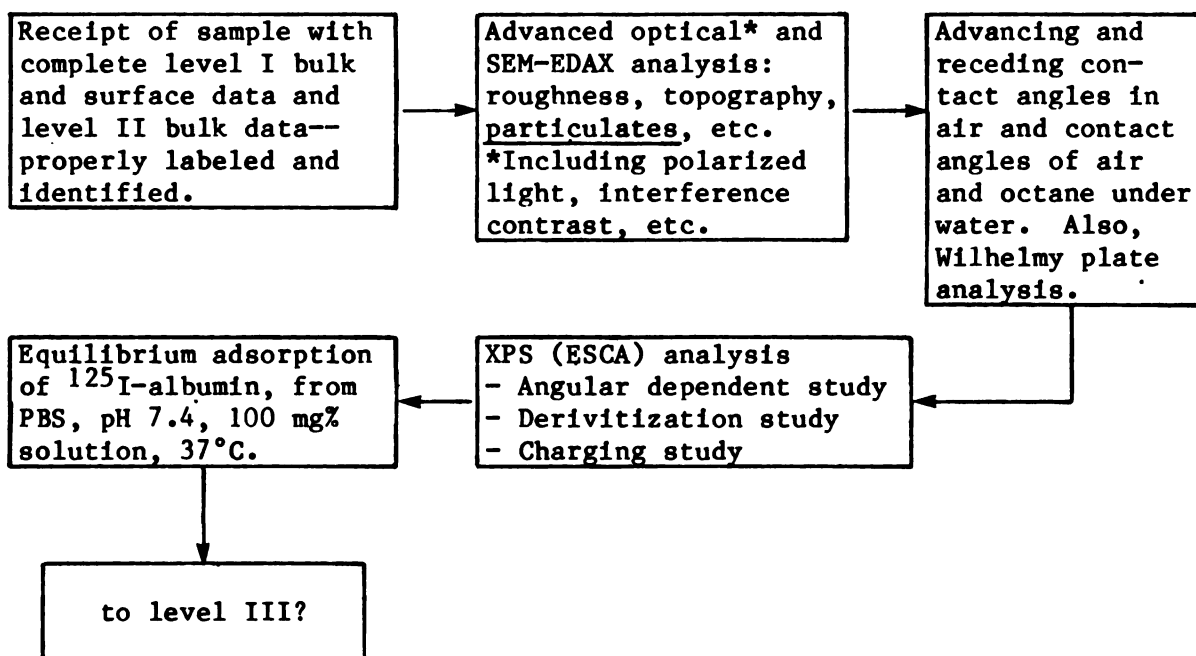


Figure 5.4--Flow-diagram summary of level II characterization.

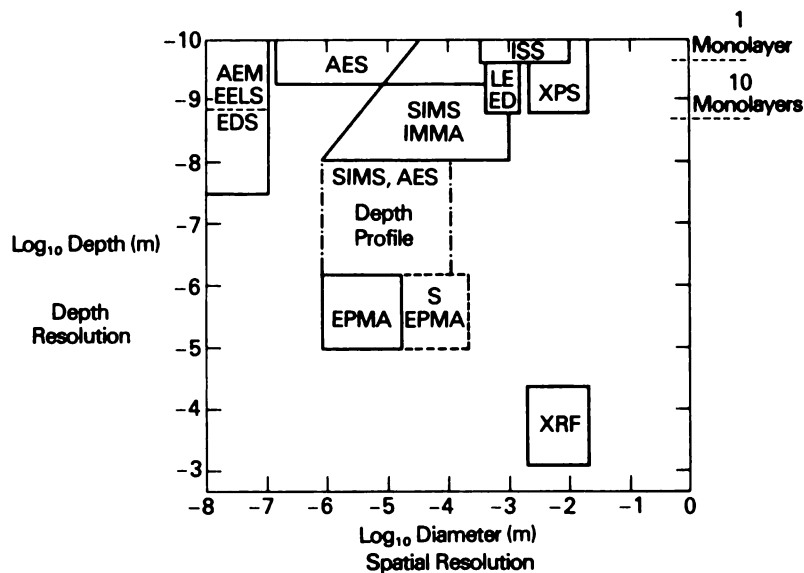


Figure 5.5--Depth and spatial resolution achievable with current analytical techniques, including X-ray fluorescence (XRF), electron spectroscopy for chemical applications (ESCA) [also known as X-ray photoelectron spectroscopy (XPS)], ion scattering spectroscopy (ISS), Auger electron spectroscopy (AES), ion microprobe mass analysis (IMMA), low energy electron diffraction (LEED), secondary ion mass spectroscopy (SIMS), electron probe microanalysis (EPMA) and scanning electron probe microanalysis (SEPMA), and analytical electron microscopy (AEM) including electron energy loss spectroscopy (EELS) and energy dispersive X-ray spectrometry (EDS, commonly referred to as EDAX).

(Note: Contemporary ESCA instrumentation can achieve higher spatial resolution than illustrated above permitting the study of areas as small as 1.5×10^{-4} m.)

Contact Angles (Level II)

Level II contact angle characterization should definitely include measurements with a series of diagnostic liquids as described in the critical surface tension protocol outlined in appendix 5. The liquids to be used must be carefully screened to insure that they do not attack and are not imbibed by the sample material. The protocol should include the measurement of advancing and receding contact angles in order to obtain some measure of surface homogeneity and surface roughness. A good example of the utility of such data for typical commercial polymer surface is given in reference 56. It is also valuable, particularly for surfaces which are partially hydrophilic, which contain two or more constituents distributed in block-like structures, or which contain two or more constituents (one of which may be highly mobile at the interface), that the contact angles be measured under water as well as in air. In materials such as these, the surface region will tend to equilibrate with the environment in question and will appear to be hydrophobic in air and hydrophilic in water (33,42-45). One might conclude from a contact angle measured in air that the surface was hydrophobic when in the environment of major interest for blood-material interactions, in aqueous media, it would be highly hydrophilic. The contact angle tests performed underwater obviate these problems (33,42,44,46,47). A brief description of the underwater contact angle measurement procedure is as follows. The polymer is totally immersed in highly purified water or in a suitable saline solution. An air droplet is then introduced via a curved microcapillary tube and allowed to rise to the polymer-water interface (figure 5.6). The solid-water-air contact angle is then measured. In this case, the polymer is fully equilibrated with water prior to the application of the air bubble, and the polymer-water-air contact angle is a measure of the fully hydrated polymer-water interface. This contact angle value is related to the receding water contact angle in the more common technique which the water droplet would be placed on the polymer in air and gradually reduced in volume by being withdrawn through a microsyringe. One can, in addition, use several water-immiscible liquids as probes to obtain further information about the polymer-water interface. A common probe that has been used for this purpose is n-octane (33,44,47,57). A protocol is given in appendix 7.

An extremely precise measurement of advancing and receding contact angles can be made using a Wilhelmy plate apparatus (39,58-60). With this equipment, the force required to withdraw or immerse a flat plate (often coated with a polymer of interest) into a vessel of highly purified wetting liquid is measured. The relationship between the measured force (F) and the angle of contact (θ) is given by the following equation:

$$F = p\gamma \cos \theta$$

where p is the plate perimeter and γ is the liquid surface tension. Taking into account the buoyancy of the flat plate in the wetting liquid, the following equation can be used to calculate advancing or receding contact angles (59):

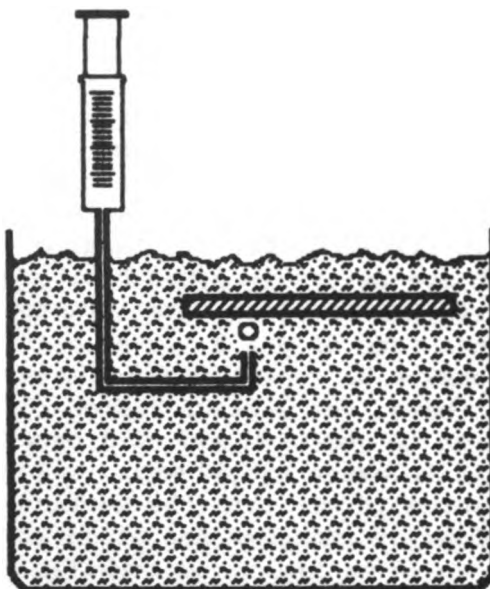


Figure 5.6--Experimental setup for performing underwater contact angle measurements. The bubble can either be air or a hydrocarbon liquid less dense than water.

$$\cos \theta = \frac{mg}{p\gamma} + \frac{V\rho g}{p\gamma}$$

where m is the mass of the flat plate, g is the local gravitational force, V is the volume of the plate immersed under the liquid, and ρ is the density of wetting liquid. A further simplification can be made by extrapolating the advancing and receding slopes to zero depth of immersion, thereby allowing the bouyancy term $V\rho g/p\gamma$ to be eliminated. A diagram of a Wilhelmy plate apparatus is given in figure 5.7. A typical output trace obtained from the insertion (advancing angle) and withdrawal (receding angle) of a Wilhelmy plate is presented in figure 5.8. When the contact angle is zero (i.e., the surface is wet completely), the surface tension of the liquid can be readily calculated. Freshly flamed platinum surfaces, or glass acid-cleaned surfaces immediately after treatment with an argon or helium RF-plasma, will be completely wetted by all common liquids. Therefore, these clean surfaces can be used to insure that the surface tension of the wetting liquid is identical to accepted literature values. This calibration procedure is critically important in Wilhelmy contact angle measurements since a large volume of liquid is repeatedly exposed to large surface area specimens--the potential for contamination of the measurement liquid is high, and the presence of surface active contaminants in this liquid can invalidate the measurement.

Precise measurements of the force of withdrawal or insertion are obtained with a microbalance. A mechanism to achieve constant rates of insertion or withdrawal is also necessary. This has been achieved using hydraulic systems, precise motors, or the crossbeam on a tensile testing instrument. Also important in these measurements are good temperature control and vibration isolation between the moving section of the instrument and the microbalance.

Using the Wilhelmy plate method, contact angles reproducibility to 0.1° can be obtained. Operator bias and extended learning periods are minimized. By using a series of highly purified liquids, a critical surface tension plot can be constructed (see appendix 5 and reference 61). However, as in any type of contact angle measurement, surface roughness and heterogeneity will produce results which are not easily interpreted. Also, strong interaction of the surface under study with the test liquids (e.g., swelling, solubilization) must be avoided.

The Wilhelmy plate technique can contribute to the generation of precise contact angle information. However, the cost of the apparatus is considerably higher than for conventional goniometric contact angle measurement equipment, and great care must be exercised in keeping test liquids pure.

Microscopic Examination

Light microscope techniques used at level II can provide a superior description of the surface at higher resolution than those used for level I characterization. Phase contrast, interference contrast (Nomarski),

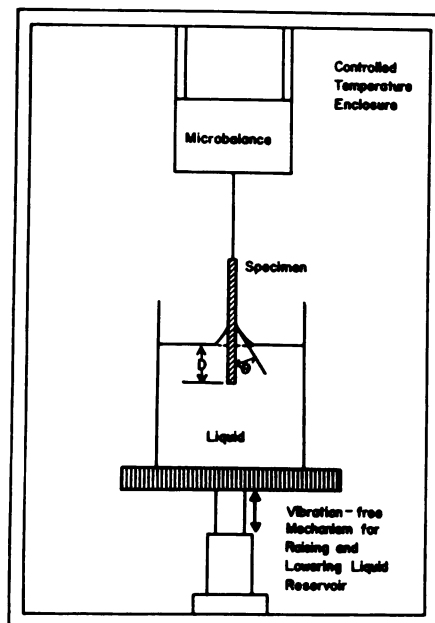


Figure 5.7--A schematic diagram of a Wilhelmy plate apparatus.

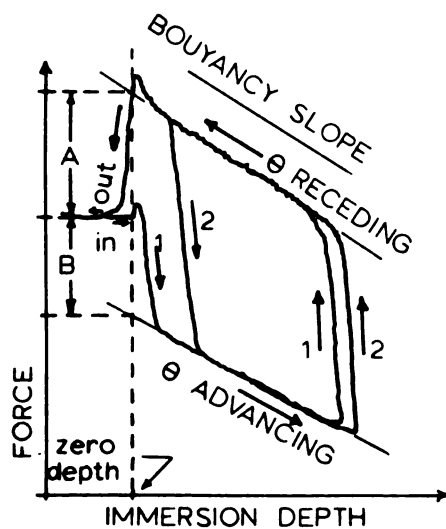


Figure 5.8--An example of a Wilhelmy advancing and receding angle trace (D. Gregonis, et al. reference 60).

and modulation contrast (Hoffman) optics provide a dramatic imaging of surface irregularities and features (18,62). Magnifications up to ~500x are available. Epi illumination (illumination of the specimen through the objective lens used for viewing) can enhance the appearance of surface features in light microscopy. Finally, polarized light methods and interferometric microscopic methods can yield new information on specimen surface structure (18,63).

More sophisticated scanning electron microscope (SEM) methods can also be applied for level II characterization. These include:

1. Energy dispersive X-ray analysis (EDXA) to identify particles and inclusions (generally suitable only for elements with molecular weights higher than aluminum) (64-68). The apparatus to perform these measurements is available on many SEM's through a number of commercial laboratories.
2. Freezing (69) and vapor-metalizing hydrated specimens in situ to maintain structures might be damaged by dehydration (particularly useful for hydrogels).
3. SCRIMAGE to obtain a quantitative assessment of surface roughness in the SEM (Available through the Arvin/Calspan Advanced Technology Center, Buffalo, New York).
4. Stereo pair imaging using the SEM to understand surface topography (67).
5. Electron backscattering imaging and stereo pairs to observe domains near the surface of polymers (70).
6. Etching and selective staining methods to enhance surface and subsurface features (71-74).
7. Auger microprobe to identify the elemental composition of fine particulates (sensitive to all elements except H and He) (75, 76).

Transmission electron microscopic (TEM) methods can also provide a wealth of useful information on the surface and subsurface structure of biomaterials (77-80). In general, TEM sample preparation methods are more complex than those for SEM and consequently are more prone to induce artifacts. In addition, artifacts can be introduced by overfocusing, underfocusing, beam damage, and sample contamination (81-84). The following TEM techniques are available:

1. Cross sections of materials cut with a cryomicrotome can be used to directly observe in profile the surface and subsurface region (85).
2. Replica methods can be utilized to obtain an SEM-like "relief" image for high resolution observation of fine surface features (86).

3. Staining techniques are often necessary to observe domains, surface graft, and other chemically distinctive features (78, 85,87).
4. Gold decoration methods have been used to observe chemically distinct "patches" on the surface of a material (88,89).

Careful control studies and meticulous sample preparation are usually required to minimize the generation of artifactual results.

ESCA

Electron spectroscopy for chemical analysis can be applied to the characterization of biomaterials to give information on surface structure which is considerably more detailed than that obtained in level I analysis. Three ESCA techniques are sufficiently well developed to warrant inclusion under level II. These are angular dependent studies, derivitization, and electrostatic charging experiments. The information obtainable from such studies is extremely valuable and difficult or impossible to acquire by other methods. However, each of these techniques requires great care in performing the experiments and interpreting the results. Collaboration with individuals experienced in ESCA is strongly suggested. Not all ESCA instruments are suitable for all the methods described in this section.

Angular Dependent Studies

Conventional ESCA analysis is generally performed with the position of the sample, the position of the X-ray source, and the position of the analyzer fixed within the instrument. However, by rotating the specimen such that its angle with respect to the horizontal increases while maintaining the position of the analyzer and X-ray source, analysis becomes more surface-localized (90). If one considers that typical ESCA sampling depths are 50-100 Å, effective sampling depths of 10-20 Å might be realized at high glancing angles. This is important since the 50-100 Å "conventional" ESCA sampling depths averages over many molecular layers. From a biomedical perspective, only the uppermost molecular layer (5-10 Å rather than 50-100 Å) might be expected to enter into interactions with proteins, cells, tissues, etc.

The enhanced surface sampling ability at high observation angles can be used in a quantitative sense by applying expanded forms of the following general equations which treat the variation in ESCA signal intensity (I) of core levels i and j an overlayer (l) on a substrate (s) as a function of angle (θ):

$$\text{for the overlayer: } I_i^s = I_i^o N_i (1 - e^{-d/\lambda \cos \theta})$$

$$\text{for the substrate: } I_j^l = I_j^o N_j e^{-d/\lambda \cos \theta}$$

where $N_{i,j}$ = number of atoms/unit area of the element containing core level i or j.

d = thickness of the layer under consideration.

λ = inelastic mean free path for the core level of interest.

I^0 = unattenuated ESCA signal intensity, from core levels i or j for a clean, homogeneous sample.

The use of these equations in the analysis of the surface chemistry and organization of materials has been described in many publications (91-93). Perhaps their greatest value lies in their ability to help discern the nature of the uppermost atomic layer as opposed to an average composition over many atomic layers.

Derivitization Studies

Derivitization studies utilize highly specific organic reactions to place "tag" molecules readily distinguished by ESCA onto organic functional groups which ESCA cannot specifically identify. For example, carbons in ether, peroxy, and hydroxyl functionalities appear in identical positions in carbon 1s ESCA spectra. Organic reactions highly specific for hydroxyl groups can be used to indicate the presence of hydroxyl groups and determine their relative concentrations by placing a tag (e.g., a fluorine-containing organic molecule for a system in which no fluorine was present prior to derivitization) only onto the hydroxyl functionality. A number of papers have been published on this subject (94-98). These all describe the many potential pitfalls of such studies including nonspecific cross-reaction and volatilization of the label in the high vacuum environment. However, if used with care and understanding, derivitization studies in conjunction with ESCA studies can help to better define the chemistry of a surface.

Charging Experiments

When electrons are emitted from a surface during bombardment with X-rays, a positive charge will accumulate on the surface. For an electrically conducting sample in electrical contact with the instrument, electrons can rapidly "bleed" back into the sample from the instrument and from the Earth via the ground connection. For an insulator, this process of replenishing electrons to replace those lost by photoemission is not readily achieved. The extent of charging will be much higher on instruments using monochromatized X-ray sources than on those with nonmonochromatized sources. For most ESCA experiments on insulators, it is of the utmost importance to neutralize this charge since it will lower the kinetic energy of all photoelectrons emitted subsequent to the initial charge formation and can distort the peak shape in the spectra obtained (99). However, one can use this charge buildup as an analytical tool to measure surface conductivity. For conducting specimens, peak shift due to surface charge formation will generally not occur. For nonconducting specimens, massive shifts in the kinetic energy of the emitted photoelectrons are noted. Depending upon the magnitude of the charge buildup, the kinetics of charge buildup and the precise conditions under which the surface charge can be neutralized

(using, for example, a low energy electron flood gun), one can estimate the surface conductivity of the material under study (100,101). This technique is not as quantitative, nor is it as well understood, as some of the other techniques described in this volume for studying surface electrical properties. However, the method is relatively simple, rapid, and intuitively understandable. In addition, data on electrical conductivity for any material can be obtained during conventional ESCA analysis with little or no additional instrument time invested in the analysis.

Protein Adsorption

The adsorption of plasma proteins is the first in a complex series of events that occurs when a synthetic material is placed in the cardiovascular system. It has been shown that adsorption can alter the enzymatic activity of thrombin and prothrombin conversion, modify the subsequent interaction of platelets and red blood cells with surfaces, and passivate the surface if serum albumin dominates the interfacial film. The detailed behavior of proteins adsorbed at the solid/solution interface is currently believed to be one of the keys both to understanding and predicting the behavior of artificial materials placed in the body. In particular, knowledge of time-dependent protein surface composition under conditions of competitive adsorption, the correlations between adsorbed protein conformation and biological properties of the protein, and the effect of the chemical nature of the surface of the material upon protein adsorption are being actively investigated. Review articles on protein adsorption measurement for biomedical studies have been written (102,103).

Unfortunately, the current data on adsorption for a given protein on similar substrates show a wide variation from investigator to investigator. This illustrates the difficulty involved in performing such measurements. The availability of well-characterized reference materials on which the adsorption of a single, well-characterized protein could be studied will remove much of the ambiguity from the results and will ensure meaningful comparisons among investigators.

Protein purity and characterization are critical aspects of these studies. It is difficult to obtain proteins absolutely free of oxidized forms, of other proteins, trace contaminants, or peptide hydrolytic products. To avoid oxidation, denaturation, and growth of bacteria, particular care must be taken in preparation and storage of protein solutions. Moreover, they should be checked for stability and purity by enzymatic, immunological, or other appropriate methods.

Since many studies of protein adsorption involve only measurements of adsorbed amounts, radiotracer techniques are frequently used. Such techniques are conceptually straightforward and involve little interpretation to extract the desired information. However, possible alterations in the protein during labeling and the presence of residual label are complications which must be guarded against. A recommended procedure for carrying out these adsorption studies is the one developed at the

National Bureau of Standards (104). This investigation characterized the effect of stirring, sample withdrawal, rinsing, surface roughness, and protein labeling (preferential adsorption). These factors were found to profoundly modify the adsorption properties reported and must be controlled (105). It is, in general, a good practice to vary the adsorption conditions, the ratio of "cold" to radiolabeled protein (106), and the ratio of solution concentration to surface area to test for complications associated with protein adsorption. Since residual I^- can be particularly insidious in generating artifactual results (e.g., for hydrogels), the tendency of the surface of interest to adsorb non-protein-bound I^- can be separately investigated. Proteins separated from residual I^- by both the usual chromatographic means and by exhaustive dialysis can be used even with difficult surfaces which retain nonprotein I^- .

At present, human serum albumin is probably the most suitable and best characterized protein in terms of adsorption. It should be noted that albumin does have a free sulfhydryl group which is readily oxidizable as evidenced by dimerization and that this protein is the physiologic carrier of free fatty acids and many drugs; it should be noted that even if fatty acids or drugs are present in albumin preparations, they are not detectable by protein chemistry techniques. Whether a locally prepared or commercially labeled protein is used, it should be chromatographically purified on Sephadex G-10 prior to use. For proteins radiolabeled with iodine, the protein preparation should also be exhaustively dialyzed or chromatographed on an ion exchange resin to remove free iodide ion. The purity of the protein and the specific radioactivities should be reported for the protein solution after any purification steps have been completed as well as at the time of each experiment. For proteins with enzymatic activity or other quantifiable functions, this specific activity should also be measured at time of use. Use of these techniques should result in reproducible results for human serum albumin adsorption. However, skilled protein chemistry techniques are required and reproducibility should be validated by studies on reference materials as well as by repeated iodinations of protein from a single lot and from different lots.

Summary

Completion of level II analytical protocol (figure 5.4) yields a detailed measure of surface roughness, texture and, in some cases, morphology which is required prior to any blood-material interaction testing. It also provides enhanced detail concerning the surface chemistry using ESCA. The contact angle evaluation provides a more rigorous measure of the surface hydrophobicity, hydrophilicity, and critical surface tension and, if done underwater, provides a measure of the true hydrophobicity or hydrophilicity of the material-water interface.

All the techniques recommended are readily available, readily interpretable methods, which can be carried out in a fairly straightforward manner. Although Wilhelmy plate measurements, scanning electron microscopy, energy dispersive X-ray analysis, and X-ray photoelectron

spectroscopy may not be directly available in many research laboratories, these are methodologies which are readily available through specialty labs and through commercial analytical laboratories. Thus, level II characterization should not provide a severe burden on any researcher.

LEVEL III TESTS

Introduction

A number of new surface analytical techniques are showing promise for the study of polymers (see figure 5.5 under level II surface characterization tests) (107). The development of many of these techniques has been stimulated by the enthusiastic activity in the areas of surface physics and microelectronics fabrication. For example, in the microelectronics area, the need to perform surface analysis in small areas (i.e., high spatial resolution) has prompted much instrument development. The microelectronics and surface physics literatures are useful for keeping abreast of new developments in surface analysis (see reference 108, for example). However, most of the developments in these areas are not oriented towards polymer systems or biomedical problems. Polymers present unique problems for analysis since they are not electrically conductive and since their surfaces are readily damaged. Therefore, care must be exercised in applying many of the new developments in surface analytical instrumentation to polymer surfaces, the surfaces which are of primary interest for blood-contacting materials. A few newer techniques which show promise for the characterization of polymer surfaces will be briefly reviewed here. Consistent with the philosophy behind level III tests, it is expected that specialists in each of the areas discussed will be involved in such an analysis.

Molecular Surface Characterization

Secondary Ion Mass Spectroscopy (SIMS)

When a surface is bombarded by accelerated ions of energy 1 to 30 KeV, a large number of physical and chemical processes occur at that surface. In one of these processes, molecules at the surface can be fragmented into charged, lower molecular weight species (secondary ions). These can have sufficient energy to eject from the surface into the vacuum environment of a SIMS instrument. A SIMS spectrum consists of a plot of the measured mass of these ejected fragments versus the amount of each molecular weight fragment detected. In many cases, the structure of the ejected fragment can be directly related to the structure of the surface (109,110).

SIMS has a number of unique advantages and characteristics:

1. It observes a surface region of approximately 10-20 Å into the sample. Thus, SIMS is even more surface-localized than ESCA.

2. It is very rich in molecular and chemical information.
3. It is possible to focus the ion beam source, thereby allowing good spatial resolution for SIMS analysis and SIMS imaging.
4. It is extremely sensitive--detection limits are of the order of 10^{-12} g.
5. For some samples, a depth profile can be obtained in a manner analogous to "peeling" layers off an onion.

Difficulty has been experienced in obtaining meaningful SIMS spectra from polymers. This has been because of charging problems and because of the generally destructive nature of the SIMS process. However, only recently, appropriate instrumental parameters have been worked out to allow the interpretable analysis of various biomolecules (111-113) and synthetic polymer surfaces (114-120). In addition, SIMS has been applied in an imaging mode to look at elemental and organic fragment distributions on biological tissue surfaces (121-123) and on polymer surfaces (115). With further developments in theory and instrumentation, SIMS may eventually become a routine tool for studying polymer and biomaterial surfaces. At the present time, it can provide otherwise unobtainable information about polymer surfaces, but only in the hands of operators skilled specifically in SIMS of polymers.

Scanning Auger Microprobe (SAM)

The SAM technique, Auger electron spectroscopy (AES) performed with a focused electron beam to allow good spatial resolution, yields certain information similar to ESCA (75,76). The depth of analysis is almost identical to that observed in ESCA, and, like ESCA, almost all elements can be detected. The primary advantage of this method over ESCA is the excellent spatial resolution (to 500 Å) which can be obtained. However, there are a number of distinct disadvantages to this technique compared to ESCA. Foremost, the focused electron beam used in SAM is so damaging to organic polymers that information obtained by this method is often impossible to interpret. Second, quantitation of SAM data is difficult in most situations. Finally, information on bonding environments or oxidation states can be complex to extract from SAM data. For these reasons, the application of SAM and AES to polymeric biomaterials is generally difficult. However, useful results can be obtained in situations where questions are asked about which elements are present in an inclusion or particle.

Ion Scattering Spectroscopy (ISS)

When an accelerated ion strikes an atom on a surface, it can be deflected away from the surface much like a ball impacting a wall. Some loss of energy occurs due to the collision. This loss of energy is proportional to the mass of the atom with which it collided. By measuring the energy of these scattered ions in a high vacuum environment, the ISS experiment, one can determine which elements are present in a

surface and approximately what their relative concentrations are. Review articles have been published on this method (124), and it has been applied to polymer studies (125).

ISS is extremely surface-sensitive. Only the uppermost molecular layer takes part in the scattering process. It is also sensitive to all elements. However, it gives no information about the molecular environment the elements are in--all carbons, be they from polyurethane or diamond, appear identical. The technique has only been applied to biomaterials studies in a few instances (126). Its potential has not been fully explored in this area, and further investigation would be worthwhile. The ISS technique is often available on SIMS instruments.

Laser Raman Microprobe

The laser-Raman microprobe (also called MOLE, molecular optical laser examiner) can be employed for the characterization of molecular compounds adsorbed on both single particles and large surfaces (127). Detection and measurement are based on the observation of the common Raman effect. In the instrument, a beam of monochromatic, visible laser light is focused onto a small spot (often 1-2 μ) on the sample. The scattered radiation shows Raman shifted frequency differences characteristic of the sample. The Raman pattern therefore represents a molecular fingerprint for identification of surface compounds. If the sample does not adsorb the incident frequency, heating can be kept to a minimum. For adsorbed organic hydrocarbons, a detection limit of 10 to 100 pg has been established.

Depending on the optical properties of the sample and its reflection properties, the spectrum may originate from deep in the sample. This is particularly true for a transparent sample. The technique may be relatively surface sensitive for an opaque or poorly scattering material. Raman spectra, of course, are of considerable interest for organic materials. It must be remembered, however, that the surface sensitivity of this technique and its routine analytical capabilities are still unproven although the technique is developing rapidly and merits careful attention. The potential of the laser-Raman microprobe technique for biomaterials investigations is apparent from studies on silicone polymer inclusions in tissue (128), purinic concretions in histological sections (129), and defects in polymer surfaces (130).

Laser Mass Spectroscopy Microprobe

The new laser mass spectroscopy microprobe utilizes a 1 μ m laser beam which is focused through a microscope onto a particular spot on the sample (131-134). The sample is locally pyrolyzed, and the products fed into a mass spectrometer, producing a mass spectrum of the local sample region. Thus, point-by-point pyrolysis mass spectroscopy of samples can be conducted. Depending on the beam energy, each laser pulse may vaporize a volume of the order of 1 μ or more in depth. That is too deep to be considered a true surface technique, but under

certain conditions it may provide surface information. Few literature references relevant to biomedical studies have appeared on this method (135).

Photoelectron Microscopy

A surface bombarded by X-rays will give off photoelectrons. This is the principle upon which ESCA is based. The energy with which these photoelectrons are emitted is related to the chemistry of the environment from which they emerged. In a photoelectron microscope, a visible light image is generated of the photoelectron emission on a surface which is stimulated by He (I) aUV radiation. Shades of gray in the image could correspond to different photoelectron energies (i.e., surface chemistries) or emission intensities. The potential to achieve compositional imaging at high spatial resolution has been demonstrated, but the instrumentation is presently far from routine. It has been used for studying some biological samples (136,137).

Surface Enhanced Raman Spectroscopy (SERS)

The portion of the scattered light signal used to extract conventional Raman spectroscopic information is generally of low intensity. Consequently, a significant mass of sample has generally been a requirement to obtain high quantity spectra. For this reason, for true surface studies where the mass of sample is extremely small, Raman spectroscopy has not proven of significant value. Recently, it was discovered that when Raman scattering was studied from the surface of certain roughened metals, a 10^6 enhancement in the signal from surface adsorbed species could be observed (138,139). This enhancement, the SERS effect, is sufficient to allow straightforward detection with conventional equipment. The requirement for adsorption to roughened metal surfaces such as silver electrodes limits the application of this method for biomaterials studies. However, some investigators have adopted SERS to systems of relevance for biomaterials studies (140-143). The literature on this technique is extremely active, and it bears watching to see if SERS will develop into a more generally useful method for biomaterials surface characterization.

New Infrared Techniques

The information-rich quality of infrared spectra has already been discussed in this section under level I tests. One of the most demanding aspects of IR technique, particularly with regard to surface studies, has been sample preparation. Two new techniques that require minimal specimen preparation or manipulation are diffuse reflectance IR and photoacoustic spectroscopy (PAS).

In diffuse reflectance IR, the source infrared radiation is focused on the surface of a rough or diffuse specimen. The reflected (scattered) radiation, now missing the portion of the spectrum adsorbed by the sample, is collimated and focused on the detector. Since the intensity

of the scattered radiation is generally low, Fourier transform IR instruments are valuable. The acronym DRIFT (diffuse reflectance infrared Fourier transform) is often used. The surface sensitivity of this method is not clearly established. Also, quantitation is more complex than with conventional transmission IR. However, sample preparation is minimal, and the technique can be used with specimens such as powders and fabrics. The DRIFT method has been of value in studying the functionalization of polymers (144).

In photoacoustic spectroscopy, an infrared source is focused on a specimen. Absorption of the radiation is followed by conversion of this energy to heat. The heat-induced thermal expansion of the specimen results in a pressure change in the adjacent atmosphere in a sealed cell. A transducer (microphone) detects these pressure changes, which are then converted to electrical signals for subsequent spectral processing (145). Fourier transform methods allow significant enhancements in signal to noise. By pulsing or chopping the incident IR radiation, the sampling depth can be varied (146,147). The technique has been applied to depth profiling polymers (147), to proteins (148), and to orientation measurements on polymer surfaces (146).

Surface Charge and Potential

Although there is considerable interest in surface charge, and experiments have been performed suggesting that the surface charge characteristics of materials do correlate with its blood-material interaction behavior (149-154), the field is often controversial and suffers from experimental difficulties. The techniques usually employed to measure surface charge are not applicable to a wide range of materials. Most of the standard electrochemical techniques are meaningful primarily for conductors or semiconductors. Many of the interfacial potential techniques, based on electrokinetic phenomena, are really only applicable to insulators. Surface electrical property measurement is an area in which there is considerable interest and need; the working group considers it to be a problem area where more research is necessary before tests can be recommended for the routine characterization of biomaterials. A discussion of surface electrical properties is given in appendices 9 and 10.

Measurements of surface potential difference are relatively well known and can be obtained accurately, provided samples of appropriate geometry are available. These methods are commonly used for the characterization of polymer electrets (155). Zeta potential techniques, generally based on streaming potential measurements, have been widely used for the characterization of biomaterials (156). While zeta potential measurements are fairly straightforward, the zeta potential is a function of flow conditions and the extrapolation of the value to more realistic device conditions is difficult. The recent development of a flat plate streaming potential apparatus allows the evaluation of a reference material and other materials produced in membrane or sheet form (see appendix 9). However, such an apparatus is not widely available.

Optimum surface charge characterization of conducting materials can be accomplished by standard electrochemical methods, which are discussed briefly in appendix 9, and appropriate text and laboratory books (157,158). Specialized techniques to study electrical charges at polymer surfaces are discussed in chapters 5, 6, and 7 of reference 159.

Surface Area and Roughness

If a material is to be used for studies of protein adsorption and detailed blood-material interactions, it is, of course, important to obtain a measurement of total surface area. This is difficult to do for relatively low surface area materials, such as common sheets and tubes, although it may be possible by the adsorption of radiolabeled molecules from solution provided that adsorption is uniform and a monolayer is formed. A particularly useful technique has been described utilizing both ^{14}C -labeled stearic acid and hexadecane. Overnight adsorption from benzene solution followed by rinsing and counting enabled a comparison of the true surface area of platinum, quartz, and chrome materials. Similarly, Timmons et al. (160) used the technique to compare the effect of polishing iron with different size grits. A direct relationship was found between grit size and the true surface area. The heat of melting adsorbed n-octadecanol has also been used to estimate true surface area (161). By using both polar and nonpolar molecules, the possibility of adsorption artifacts can often be detected. This technique can only be used for materials which do not swell in the solution of interest and for solutes which do not absorb into the material. Moreover, it involves the assumption that the adsorbing molecules deposit in a monolayer. For surfaces with heterogeneous energy distribution, this may not be so. Therefore, there is a need for further work to assure the applicability of the method to various classes of materials.

Closely related to surface area is surface roughness. Surface roughness has been strongly implicated as an initiating factor in triggering blood responses to materials (162-165). One can readily visualize roughness using SEM or light microscopy. However, quantifying the degree of roughness or describing the "quality" of the roughness can be very complex. Still, a meaningful description of the roughness of materials is critical, first to assess the effect of roughness on blood-material interactions and, second, to simplify reproducing specimens for use in later experiments.

A common way to quantitatively evaluate roughness is using an instrument often referred to as a profilometer. This device, incorporating an assembly similar to the stylus on a phonograph, converts the up-down motion of the stylus as it moves across a surface into electrical signals which are plotted against distance traversed. This method has been extensively used in the metallurgy and microelectronics industries (166) and occasionally used in biomaterials research (13,162). It has the potential to scratch soft polymer surfaces, thereby complicating the analysis of results.

Many other methods have been investigated for measuring surface roughness in a nondestructive fashion (167,168). However, none of those methods have been reduced to the level of routine analysis and many are quite complex. The development of good tools for assaying polymer roughness in a quantitative fashion is clearly a goal worthy of further pursuit and have been included in the gap areas section. Appendix 8 presents further considerations on measuring surface roughness and topography.

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Chapter 6

CHARACTERIZATION OF FABRICS AND POROUS MATERIALS

Fabrics and porous materials are commonly used in humans in contact with the blood. The rationale for the use of these materials is based upon the observation that a porous support will form a matrix onto which natural tissue can grow. This porous surface can serve to anchor a device to soft tissue or to allow desired cellularization or passivation to occur on its surface. Blood vessel prostheses are commonly made of porous materials, and their success is attributed to both their ability to anchor to (heal into) the surrounding soft tissue and to the cellularization or passivation observed to occur on their flow surfaces. Blood contact applications for porous materials include patches, insides of blood pumps, flanges on percutaneous access devices, and sewing rings on heart valves, shunts, grafts, and other devices. A number of review articles and books have appeared considering both fibers and fabrics in medicine (1-3) and, more specifically, vascular grafts (4-9). However, because of the importance of this subject, the unique requirements for characterization presented by these porous materials, and because of the relative lack of physical characterization information presented in the published literature, characterization of porous materials will be addressed separately in this chapter of the guidelines.

From the standpoint of the surgeon, a number of exacting technical requirements are placed on porous materials used for cardiovascular applications. They must be easily suturable. The suture must hold well in the porous matrix without pulling out. The cut end of the prosthesis must not fray. The material must be flexible and malleable, and yet should not easily "kink" when bent. Performance requirements are also placed on these devices. In the short term, they must not clot or induce hematoma formation. After long implantation, they must still have a thromboresistant flow surface, must resist aneurism or distension, and must not contribute to the development of anastomotic lesions.

Materials which are prepared in porous form for medical applications include woven or knitted fabrics, nonwoven fabrics (e.g., electrostatically spun porous structures), precipitated polymers, expanded plastics, flocked surfaces, porous materials made by self crystal extraction from polymers, and materials prepared by injection-molding polymers into interconnecting matrices. Each of these types of materials will necessitate unique characterization methods to accurately describe them. This section will concentrate on those methods that are more generally applicable to all porous materials. In particular, the following will be discussed:

1. Visualization
2. Chemical composition
3. Permeability
4. Compliance

5. Strength
6. Thickness
7. Roughness, texture, and porous structure

Visualization is relatively easily accomplished using a light microscope or scanning electron microscope. This provides the simplest and most easily understandable method to appreciate the form, texture, and porosity of porous materials. However, unless coupled with sophisticated electronic image analysis methods, microscopic observation is, at best, only semiquantitative. In figure 6.1, scanning electron microscopic images of many of the types of porous materials discussed above are shown. Further discussion of microscopic characterization methods is presented in chapter 5 of these guidelines. Quantitative analysis can be enhanced by the inclusion of accurate size reference standards such as calibration microspheres in the image. Also, the general appreciation of the nature of the porous material can be enhanced by studying the specimen at a number of magnifications (is there texture on the surface of each fiber in the fabric?), at a few angles, under mechanical stress, and in cross section.

The methods which can be used to identify the composition of a fabric have also been discussed in chapters 4 and 5 in this document. In particular, ESCA and ATR-IR can be helpful. Some of the polymers which have been used to fabricate porous materials for cardiovascular applications include poly(ethylene terephthalate) (Dacron), nylon, Teflon, silicone rubber, polyurethanes, and poly(acrylonitrile). The Association for Advancement of Medical Instrumentation (AAMI) offers a series of helpful material identification techniques in its standards for vascular graft prostheses (10). These are based upon density and solubility. It is also important to use the ESCA and ATR-IR techniques to look for the presence of residual catalyst, oligomeric species related to the monomer(s) used, antioxidants and stabilizers, delusterants, and surface finishes (1). These additives are commonly incorporated into polymers for fabric applications but are of questionable value for biomedical materials.

The permeability (or porosity) of fabrics is generally evaluated by the method described by Wesolowski and developed into an AAMI standard (10) in which the volume of water passing through a fabric under a fixed pressure head in a given period of time is measured. The units for this measurement are $\text{cc H}_2\text{O}/\text{cm}^2/\text{min}$ at 120 mm Hg. Typical values for most graft materials used clinically are in the range of 2,000–3,500 $\text{cc H}_2\text{O}/\text{cm}^2/\text{min}$. However, grafts with higher and lower porosities can be used in special situations.

Although studies have suggested a strong relationship between porosity value and graft implant healing (12–15), controversy continues about the importance of porosity in vascular grafts. Still, porosity measurement is one of the most routine (and, unfortunately, often the only) characterization performed on vascular grafts.

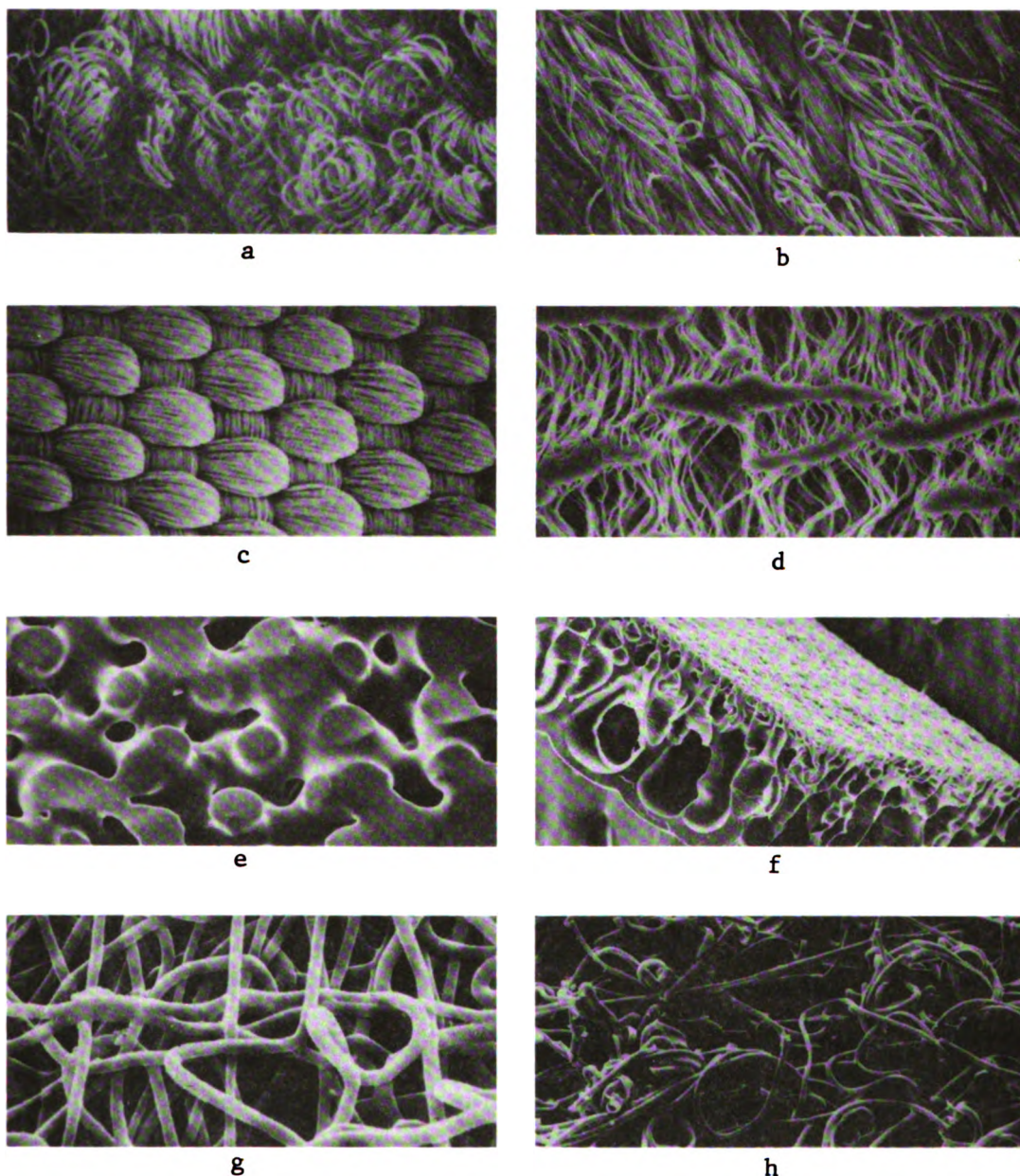


Figure 6.1--Scanning electron micrographs of porous cardiovascular prosthetic materials. (a) Microvel Double Velour Graft (50x), luminal surface (Meadox Medical) (b) Sauvage External Velour Dacron (luminal surface), (C.R. Bard) (c) Cooley Veri-Soft Woven Graft (50x) (Meadox Medical), (d) Goretex (luminal surface) (W. Gore, Inc.), (e) Silicone rubber Replamineform molded vascular prosthesis (R.A. White), (f) Mitrathane porous precipitated polyurethane patch (Mitral Medical) (g) electrostatically spun polyurethane vascular graft (fiber diameter = $1\mu\text{m}$) (D. Annis), (h) Integrally textured polyurethane pump bladder surface formed from negative replica silicone rubber molds (SEM by F. Schoen, see ref. 11).

It has been suggested that the porosity of commercial vascular grafts can vary substantially from point to point along a length. This in turn suggests that routine characterization should include a number of measurements of porosity along the length of a graft and some statistical representation of the degree of variation in porosity in the section of material used in an experimental program.

The compliance of vascular grafts has frequently been suggested as an important factor in the healing of the material, particularly at the anastomoses. Compliance (C), with regard to flexible conduits, has been defined by the following relationship:

$$C = \frac{\Delta D}{D \Delta P}$$

where ΔD = the change in external diameter, D = the diameter in diastole, and ΔP = the pulse pressure (16). Higher compliance values are associated with more flexible conduit walls. The insertion of a rigid or non-compliant vascular graft segment in the compliant arterial tree leads to increased impedance, compliance mismatch, and wave reflection. At the distal anastomosis, between the noncompliant graft and the compliant blood vessel, at certain flow velocities, turbulence can develop. This has been associated with local thrombosis, high frequency wall vibrations, and endothelial cell damage (17). Thus, the hypothesis has been developed that the intimal hyperplasia or lesion formation that is observed to occur with the greatest intensity at the distal anastomosis of an implanted synthetic vascular graft may be related to this compliance mismatch. With this in mind, the compliance, particularly of new vascular graft materials, should be measured. General methods for the measurement of compliance have been discussed in these guidelines and in a number of publications. In addition, for compliance measurement, an AAMI standard (presently under adoption), an ASTM specification for test methods for stretch properties of knitted fabrics (D2594-72, reapproved 1977), and an ASTM specification for tension and elongation of elastic fabrics (D1775) are available.

A number of instances of "ballooning" of the graft wall related to mechanical failure of the synthetic conduit (aneurysm formation) have been reported. This strongly suggests that measurements of the strength of the synthetic conduit before and after implantation are important. Although many evaluation schemes are possible to measure the burst strength of fabrics, clear standardized test procedures set forth in AAMI standards (under adoption), ASTM methods D3786-80a (hydraulic bursting strength of knitted goods and nonwoven fabrics), and ASTM D1682 (breaking strength of uncoated woven fabrics) can be useful.

The thickness of porous vascular prostheses has been cited as important to the end performance of the device. Both the time for tissue ingrowth to occur and the ability of the healed tissue structure to provide adequate vascularization and nutrition to the new tissue will depend upon the thickness of the porous region of the prosthesis. An

ASTM standard method for measuring the thickness of textile materials (D1777-64, reapproved 1975) is available and describes a detailed protocol for performing this measurement in a reproducible fashion. An AAMI standard (under adoption) is also available. SEM imaging of the cut cross section of fabrics can also be used to gain a semiquantitative appreciation of fabric thickness.

The measurement of the texture, roughness, and porosity of porous materials presents special problems and represents one of the gap areas suggested for further development by the guidelines. In order to understand why problems arise in these measurements, it is important to appreciate the complexity of fabrics. When porosity is measured by a permeation test, this only indicates the percent open area available for transport and, to some extent, about the thickness of the fabric. It gives little or no information about the shape of the pores, the geometric relationship of the fibers to each other, the distribution of the fibers across the fabric and through the depth of the fabric, and the geometric arrangement of fibers on the surface. Thus, consider, for example, the levels of organization for a Dacron fabric. The individual fibers (typical diameter for polyester fibers = 11-24 μ) are spun into multiple fiber bundles called yarns, and these yarns can be bundled together into still thicker yarns. These yarns can have different degrees of twist and can also be texturized. The yarns are then combined into fabrics. The variety of fabric types is limitless but usually center around weaving techniques, weft-knit processes, warp-knit processes, and processes which result in nonwoven structures. Naturally, porous materials which do not start with, or incorporate, fibers as their structural unit will have completely different levels of organization.

Precise means to describe this complexity in a quantitative manner do not yet exist. A few guidelines, primarily based upon various ASTM standards, are offered in the AAMI standards. Some level of characterization can be achieved by the use of electronic image analysis. Various distribution functions for surface pore openings, visible fiber lengths, and number of fiber cross-overs can be designed and automatically computed from SEM images. Also, mathematical relationships based upon the weaving pattern used can be constructed (18). Overall, this is an area in need of improvement. When quantitative descriptions of fabrics exist, experimental and clinical reproducibility will be more certain, and formalized quantitative relationships between material performance and porous material structure will be possible.

Further information on woven fabrics testing can be found in chapter 4.

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Chapter 7

BIOMATERIALS TOXICOLOGY

The ultimate safety and performance of materials used in cardiovascular applications depend simultaneously upon satisfactory physical performance and compatibility with the biological system under conditions of actual use. However, some level I tests for toxicity, pyrogenicity, and sterilizability are useful in the early stages of materials development to identify or eliminate materials that may cause problems in the final device. Typically, this kind of testing is best carried out under contract with laboratories that specialize in these procedures.

The toxicological testing of materials has unique constraints which limit the applicability of classical protocols and procedures. Selected reviews of the toxicology of plastics and the testing of materials are listed in references 1 through 6. Classical protocols are almost exclusively designed to obtain quantitative dose-response data on pure chemicals, e.g., drugs and toxicants. In the case of synthetic materials, the complete chemical composition is rarely known; even if it were known, good toxicological information is seldom available on the components. Also, synergistic reactions among the components can cause a toxic reaction even though the individual additives are considered nontoxic.

The toxicity testing of materials is directed towards the assessment of the potential toxic liability of the material under specific conditions, particularly under "end use" conditions. The approach to such evaluation is, therefore, highly empirical. It is in the evaluation of the empirical toxicity data that the large body of fundamental information derived from classical studies comes to bear on the assessment of materials. The relative significance of toxicological responses (acute, subacute, and chronic) to the health of patients has been broadly established for diverse drugs and chemical agents, and it is in that context that the empirical toxicological data generated on materials used in cardiovascular applications must be assessed.

Toxicity testing should take place at logical points in the development of materials and devices. The juncture at which the tests are performed dictates the type of test(s) appropriate. Two logical junctures for toxicity screening are 1) early in the synthesis program, as soon as a material is identified as a candidate material; and 2) as soon as the "end use" configuration of the material is determined and the "device" (or component of a device) is available for testing. Certain tests are appropriate for both situations; however, "end use" testing will carry certain constraints unique to the final configuration and context of application of the device. Sterilization can degrade materials and/or react with nontoxic additives in the material to increase the toxic liability; therefore, all toxicity testing should be done on materials that have been appropriately sterilized.

Acute toxic responses are closely related to the kind and quantity of leachable chemical substances in the materials or devices. While, as a general principle, the absence of such substances is desirable, most materials, when subjected to rigorous extraction procedures, will yield detectable quantities of chemical substances when the extracts are examined by sensitive physical or chemical techniques. It is important, therefore, that an assessment of the potential toxic liability of the material be obtained to guide the synthesis and development program (to preclude the rejection of a material based on chemical analysis alone) and to alert the developers to potential toxic liability of apparently "minor" or undetected substances. Chemical analysis in conjunction with toxicity testing can frequently identify problems at an early juncture and prevent costly developmental work on unsatisfactory materials.

"Subacute" toxicity testing should begin as soon as the final projected formulation of a material or component of a device is known. When a material or component of a device is expected to enter the final device configuration without significant modification, "chronic" toxicity testing should be initiated to avoid long delays after final formulation of the device. Because of the cost and time required for such testing, careful consideration should be given to the most appropriate time and device configuration for testing. In addition, the potential extractables should have been identified with analytical techniques as to type, level, and reproducibility from lot to lot.

Leachable substances may pose a significant problem with respect to aspects of biocompatibility other than acute toxicity, and their presence places greater demands on evaluation at the subacute and chronic levels of testing if the material is to be considered as a serious candidate for use in a cardiovascular device. Therefore, these toxicity tests in no way replace those that are needed for extensive blood-material interaction testing.

The following sections describe tests appropriate for the assessment of acute, subacute, and chronic toxicity. The tests described have been selected for broad applicability to diverse materials and for relative simplicity and ease of execution. However, other tests may be equally pertinent or superior to the proposed tests in specific situations. In any case, judgment is required both in the selection of tests and assessment of test results.

LEVEL I TESTS

These tests comprise a limited series of the more sensitive measures of toxicity. They assess the ability of extractable components to inhibit cell growth, to cause cell death, or to provoke an inflammatory response. While these tests are helpful in screening materials, they are also useful for maintenance of quality control during production or fabrication of a material previously tested. A summary of level I tests is listed in table 7.1.

Table 7.1--A Summary of Level I Biological Tests

Assay	Test Material	Exposure Time	Biological Indicator
Agar overlay	Solid material or extracts applied to cellulose disks	24 hours	Mouse fibroblasts--cell lysis and/or loss of vital dye
Intradermal irritation	Extracts of material in saline, PEG-400, and/or cottonseed oil	4-5 days	Rabbits--intra-cutaneous injection (erythema, edema)
Inhibition of cell growth	Distilled H ₂ O extracts, 121°C 1 hour	72 hours	Mouse fibroblasts--inhibition of cell growth
Hemolysis	5 g solid material in 10 ml normal saline	60 minutes	Rabbit blood

Agar Overlay: Material

Leachable toxic components of materials are detected by observation of the response of a mouse fibroblast monolayer separated from the test material by a thin layer of agar. The magnitude of the response zone and the severity of the response (degree of lysis of the cells) are recorded. Any solid material, e.g., bulk solids, sheets, films, powders, or fibers, that can be placed directly on the agar surface of the test plate (approximately 1 cm² in contact with the agar) can be tested; extracts of materials may be applied to cellulose disks that are placed on the surface in the same manner (see below). Clean but not necessarily sterile samples are required, and positive and negative control materials must be available.

A monolayer of mouse fibroblast cells is propagated on the bottom of a Petri dish, the growth medium is removed, and the monolayer is overlaid with a thin layer of agar containing a minimum concentration of supporting nutrients and the vital stain, neutral red. Duplicate test material samples and positive and negative controls are placed symmetrically on the surface of the agar, two plates per test. The plates are evaluated after 24 hours of incubation, 37°C. Cell death due to toxic substances diffusing from the sample causes loss of neutral red dye from the cells; the test is considered valid if the positive and negative control samples show the anticipated standard response. The response is evaluated grossly and microscopically and graded in terms of the size of the region of dye loss (zone index) and the percent of the cells lysed within the zone regardless of zone size (lysis index) (table 7.2). The test results are reported in terms of a response index (RI), defined as zone index/lysis index, which varies from 0/0 (no detectable response) to 5/5 (zone covering entire plate and over 80 percent of cells lysed).

Positive and Negative Controls

A suitable positive control material will routinely produce a standard response index (i.e., will be toxic) in repeated tests. The magnitude of the RI should be in the range of 2/3 to 3/4 but should not vary with respect to time. Poly(vinyl chloride) sheets formulated with four parts Thermolite (organotin) have been found to be satisfactory and stable for several years. Any material which will routinely produce a response index of 0/0 is suitable for a negative control. High purity polyethylene sheet material has been found to be satisfactory.

Agar Overlay: Material Extracts

The agar overlay tissue culture test can also be used to evaluate the toxicity of extracts of diverse solid materials obtained under selected extraction conditions. Such a modification has the advantage of being independent of the form of the solid material tested and also allows for extraction in polar and nonpolar solvents that may be more effective in removing toxic leachables. This approach is more sensitive than using the solid material directly. The extraction media used include distilled water, saline, polyethylene glycol-400 (PEG-400),

Table 7.2--Agar Overlay Tissue Culture Test

Zone Index	Zone of Vital Dye Loss
0	No detectable zone around or under sample.
1	Zone limited to area under sample.
2	Zone not greater than 0.5 cm in extension from sample.
3	Zone not greater than 1.0 cm in extension from sample.
4	Zone greater than 1.0 cm in extension from sample but not involving entire plate.
5	Zone involving entire plate.
Lysis Index:	Degree of Cell Lysis (microscopic)
0	No observable lysis.
1	Up to 20 percent of the zone lysed.
2	20 to 40 percent of the zone lysed.
3	40 to 60 percent of the zone lysed.
4	60 to 80 percent of the zone lysed.
5	Over 80 percent lysed within the zone.

and cottonseed oil. Water and saline are particularly useful because the aqueous extracts can be combined with other additives to make up a variety of "physiological" solutions. PEG-400 is moderately nonpolar, and extracts must be diluted 1:8.3 (extract:saline) by volume, since undiluted PEG-400 is, itself, toxic. Cottonseed oil is highly nonpolar and extracts the more nonpolar leachables which are often the more toxic. It is advisable, however, to ensure that the oil batch is nontoxic in a control test. For level I testing, three different solvents should be employed.

The extraction protocols are similar to those used for testing plastic containers (USP XX). Standard quantities of materials, based on surface area or weight, are cut to appropriate dimensions and extracted in a sealed tube with the medium of choice for a specified time and at selected temperatures. Two temperatures are recommended: 121°C (autoclave) and 37°C (incubator or water bath). The higher temperature affords more rigorous extraction conditions and is recommended for initial acute toxicity testing. Negative results with such extracts provide evidence for the chemical stability of the test material and the absence of substances soluble in the extraction medium. If extracts obtained at high temperature are toxic, extraction at 37°C should be performed to determine whether thermal degradation products may have contributed to the observed toxicity.

The duration of extraction is important with respect to assessment of leachability. The high temperature (121°C) extraction is usually carried out for 1 hour, following the recommendations in the USP, permitting the evaluation of extracts on the same day as the extraction. Extraction at 37°C is usually carried out for longer periods of time, e.g., 3 days. Under these conditions, an antibiotic, e.g., 100 µg/ml of gentamycin, can be added to aqueous extraction media.

Extracts are imbibed into cellulose assay disks (0.1 ml per disk) and applied to the agar as described above.

Intradermal Irritation Test

The intradermal irritation test is an even more sensitive test of acute toxicity; however, the results of this test correlate with the results of in vitro tests. For this test, extracts of materials or solutions of suspected leachables may be employed. Sterile extracts are desirable, and a small animal facility for the handling and holding of rabbits is necessary. The test is particularly useful in detection of leachable, nonpolar components of materials and takes about 4-5 days. Irritancy, although correlated with acute toxic response, is a qualitatively different response than that assessed in the other acute toxicity tests described and is an important parameter in the evaluation of materials.

Extracts of materials obtained with nonirritating extraction media are suitable for the test. Saline, PEG-400, and cottonseed oil are routinely employed. The extraction medium is employed as a negative control, and 20 percent ethanol in normal saline is a positive control. The extraction

method is as described in the preceding section, and the test procedure is described in USP XX (8). It involves the intracutaneous injection of the extract into rabbits and the assessment of irritant response, with or without the use of vital dyes. The dyes have the advantage that irritant responses can be observed very rapidly, often within the first hour. The standard test without dye requires 72 hours.

Inhibition of Cell Growth

Toxicity may also be observed as inhibition of cell growth under defined conditions. This is one of the more informative tests for leachable substances and is more sensitive than the short-term acute tissue culture tests described above, but does require more time and technical skill. It is also more expensive than the agar overlay method.

Distilled water extracts are prepared as described above in a series of graded weights of material (six weights ranging from 4,000 mg to 1 mg/20 ml). The extracts are incorporated into tissue culture medium and inoculated with a standard quantity of mouse fibroblast cells in tissue culture assay tubes. The tubes are incubated for 72 hours, the medium is decanted, the cell monolayer is gently washed with buffered saline to remove unattached cells, and the extent of cell growth is determined by total protein assay on the individual tubes less the amount of protein in the initial inoculum (zero time). The difference in growth in the presence and absence (negative control) of the extract is expressed as the inhibition of cell growth (ICG). The results are reported for each weight of material extracted and presented as a plot of percent inhibition against weight of material extracted.

Hemolysis

The short-term acute hemolytic activity is significant with respect to potential acute toxic liability of biomaterials. In general, the concentration of toxic or hemolytic substance which produces overt hemolysis is approximately an order of magnitude higher than that required to inhibit cell growth in tissue culture and is correlated with acute toxicity *in vivo*. The detection of overt hemolysis is, therefore, of immediate significance with respect to toxicity. The test is rapid, requires simple equipment, and can be performed in the presence of the material or on the "extract" after removing the material to assess indirectly the contribution of the physical surface of the material to manifest hemolytic activity.

Test samples are cut such that 5 g of material can be submerged in 10 ml of saline. Where this is not possible, the maximum amount of material is added which still allows blood to be freely mixed with the suspension. Clean but not necessarily sterile samples are required.

The test consists of allowing the test material to equilibrate with saline at 37°C for 30 minutes. This is followed by the addition of 0.2 ml of a standardized rabbit blood with gentle mixing, incubation for 60 minutes at 37°C, centrifugation of the suspension, and spectrophotometric determination of the hemoglobin content of the

supernatant saline solution. The results are compared to a totally hemolyzed sample less the amount of hemolysis induced by a negative control (percent hemolysis).

LEVEL II

The level II toxicity tests are basically an expansion of the level I acute toxicity tests with the addition of in vivo tests designed to identify subacute and chronic toxicity. Materials that have successfully passed the level I toxicity screen, have been evaluated for their interactions with blood, and have undergone the appropriate physiochemical and mechanical testing are ready for level II toxicity testing prior to clinical evaluation.

The tests are designed to evaluate both extracts of the material and direct contact with the material. It should be noted that the correlation between in vitro tests and in vivo implantation tests is not high (9). This may suggest that in vitro acute tests measure different aspects of toxicity than the in vivo tests, i.e., leachables (in vitro) versus leachables, physicochemical effects, and biodegradation (in vivo). Different cell lines also vary in their sensitivity to toxic substances (9). In particular, low toxicity to various animal cell lines does not guarantee low toxicity to human cells. At least some consideration should be given to the use of human cell lines as a part of level II toxicity testing.

Table 7.3 lists the recommended level II tests. However, this list is not comprehensive and should not prevent investigators from expanding their toxicity testing program as mentioned above.

Culture Methods

Several tissue culture techniques are available to evaluate both the solid material and extracts of the material (5). Generally, these tests vary in cell type, sensitivity, and the time required to run the test. A large number of cell lines have been evaluated by Johnson et al. (9) regarding sensitivity for toxicity screening. Of the 12 cell lines evaluated, they recommend CCL1 (L-929 mouse fibroblast), CCL74 (raccoon uterus), CCL76 (human skin), and CCL131 (mouse neuroblastoma) as the easiest to culture with optimal sensitivity.

Direct cell contact with the solid material can be done in two ways. The first is to use the material as a substrate for cell culture, and the second method involves placing the test material on a confluent monolayer of cells. The second method is recommended because cell growth may be very sensitive to the nature of the substrate surface. This sensitivity is more a function of attachment (adhesion) of the cells to the surface than to toxicity. A good example of this problem is glow discharge treated polystyrene versus untreated polystyrene. Cells grow very well on the treated polystyrene and do poorly on the untreated polystyrene.

Table 7.3--A Summary of Level II Primary Screening Test for Toxicity

Assay	Test Material	Exposure Time	Biological Indicator
Tissue Culture-- Direct Contact	Solid disk on cell monolayer	24 hours	Mouse fibroblasts confluent monolayer
Erythrocyte Osmotic Fragility	Serial dilutions of material extracts	60 minutes	Rabbit red blood cells (1 percent hematocrit)
Systemic Toxicity	Saline, PEG-400, and cottonseed oil extracts	8 days	Mice injected i.v. (saline) or i.p. (PEG-400, cotton- seed oil).
Intramuscular Implants in Rabbits	Solid material 1 cm rods or strips 1 mm wide	10 days 12 weeks 2 years	Acute Subacute Chronic
<u>In vitro</u> Mutagenicity	Paper disks imbibed with extracts	3 days	<u>Salmonella</u> <u>typhimurium</u> <u>Histodine</u> Auxotrophs

Placing a solid sample on a confluent monolayer of cells can induce cytopathic changes such as cell lysis, granulation, sloughing, or shape changes (10). In addition, investigators should be aware that cell trauma will increase with the density of the test material and the ability of the material to allow adequate gas and nutrient exchange beneath the sample (9).

Erythrocyte Stability (Osmotic Fragility)

This test provides a very sensitive measure of the interaction of leachable substances with the plasma membrane of erythrocytes, reflected as changes in the osmotic fragility of the erythrocytes on short-term exposure of the erythrocytes to material extracts. It can detect leachable components of materials at concentrations one to four orders of magnitude lower than those detectable by tissue culture tests, although the major response occurs at or slightly below the concentration range for which tissue culture systems are sensitive. The quantitation of the fragility response is also more precise and reproducible than that for most tissue culture systems. It is reasonable to assume that leachable substances which interact with the erythrocyte membrane would also interact with other biological membranes.

Any extract which is optically clear and free of particulate matter can be tested with appropriate controls, if it can be adjusted to an osmolarity in the required hypotonic range. Hypotonic saline or distilled water which can be adjusted to the required tonicity is most suitable. Since these extracting media are stable and do not support bacterial growth, a wide range of extraction conditions can be employed. Extracts are adjusted with distilled water or NaCl to form a solution of osmolarity appropriate to hemolyze about 50 percent of the erythrocytes on the basis of hypotonicity alone. Rabbit blood is usually used, diluted appropriately with isotonic saline (about 1 percent hematocrit). One-tenth of a milliliter of this stock erythrocyte suspension is added to 5 ml of extract or a control hypotonic solution, and the surviving cells are counted. The relative hemolysis (number of cells lysed in the extract/number of cells lysed in the control) is reported. It should be noted that greater or less hemolysis with the extract (as compared to the control) indicates membrane interactions. By performing the test with a series of dilutions of the extract, the concentration of extract at which no detectable change occurs can be established and compared with data from other materials or from extracts prepared under different conditions.

Cell size distribution profiles can also be obtained in this test, giving an indication of the degree of swelling or morphologic changes induced in the erythrocyte population. Significant shifts in cell size distribution can be detected at extract concentrations one to two orders of magnitude lower than those producing measurable osmotic fragility changes.

Systemic Toxicity

The systemic toxicity test provides a more direct assessment of neurotoxicity or other adverse systemic effects than do other tests.

This test involves the simple direct intravenous or intraperitoneal injection of extracts of materials into mice. It shows a low frequency of positive responses, but positive responses have been observed in the absence of high positive responses in other acute toxicity tests.

Saline, polyethylene glycol-400, and cottonseed oil extracts of materials are usually employed, and the extraction medium itself serves as a negative control. In brief, mice are injected i.v. (saline) or i.p. (cottonseed oil or PEG) at a dose level of 50 ml/kg. They are observed over a period of 1 week, and deaths or signs of toxicity are recorded. Details of the method are given in USP XX.

Intramuscular Implants

Intramuscular implantation of materials in rabbits provides a direct assessment of acute and long-term tissue reaction to materials and can be performed with simple equipment. The acute test takes about 10 days to complete and it is considered one of the more pertinent and direct tests for leachable components of materials. It is useful in that it provides an assessment of tissue reaction to materials under physiological conditions. Leachable components diffuse rapidly from the implant, and the extent and severity of the tissue reaction is measured. Any solid material which can be cut or formed into small rods of approximately 0.5 to 1.0 mm diameter by 1 cm long is suitable as a sample. Aggregates of fibers or powders can be implanted, but abnormal responses are frequently observed with such samples. The rods should be clean and sterile. The samples are implanted into the paravertebral muscle of rabbits along with positive and negative controls through a trocar needle. For acute studies (primary screen), the rabbits are sacrificed after 10 days, the implant sites with their adjacent tissue are excised, the gross evaluation of the tissue response is recorded, and the implant site is preserved in formalin for histopathological evaluation of the tissue response.

It is possible, however, that time-dependent release of toxic leachable components may extend over longer periods, and in cases where the material degrades in vivo, delayed responses may occur. For these reasons, subacute and chronic extensions of the short-term test are useful.

The subacute test is conducted for 6 to 12 weeks using a sufficient number of rabbits to allow sacrifice and evaluation at weekly intervals. It is designed to determine time dependency of recovery from any observed acute response and to detect delayed response over a moderate period of time. A material showing a positive response in the test, particularly a delayed or protracted response, is probably not a good candidate for long-term implantation.

Chronic intramuscular implantation tests are conducted for periods of 2 years or more and may involve animals other than rabbits. These extensive tests are appropriate for materials which are strong candidates for use in devices intended for permanent replacement or implantation in the body. The tests are designed to provide a preclinical evaluation of any histopathological problems. In view of the variety

of possible end uses of the material and differences in the size and form of any device, the specific animal system, implant location, and experimental protocol must be chosen specifically to fit each particular situation. Every attempt should be made to ensure that the test conditions approximate the end use conditions.

In Vitro Mutagenicity (Modified Ames Test)

Since a relationship is recognized between mutagenicity and carcinogenicity of chemical agents, an assessment of the potential mutagenicity of leachable components is useful in the evaluation of materials and devices. This test provides such an assessment.

Histidine auxotrophs of Salmonella typhimurium will not grow in a minimal medium lacking histidine. Chemical agents, which cause a back mutation (or reversion) of the bacterium to a genetically competent state where the essential amino acid (histidine) is synthesized by the cell, will allow the cell to grow in the histidine-deficient medium. Such back mutations can be detected by the observation of growth of discrete colonies of cells derived from the primary revertant cell. The application of extract impregnated in assay disks to the test plate permits radial diffusion of the soluble components of the extract into the agar seeded with the bacteria. Revertant colonies appear as a radially symmetric zone of growth.

The test takes approximately 3 days to conduct and requires standard bacteriological facilities. Sterile extracts are required, and nontoxic extraction media such as distilled water, saline, or dimethylsulfoxide are routinely used. The extract is imbibed in a cellulose assay disk and placed on the surface of the agar test plate. Testing of both solid and extract is recommended. Several strains of Salmonella typhimurium have been developed for this test. Testing with a minimum of two or three strains is recommended. The protocol follows the qualitative Ames test procedure and involves the preparation of agar plates with minimal medium seeded with standard number of bacteria. The test sample is placed on the surface of the agar, incubated at 34°C for 48 hours, and the presence or absence of revertant colonies in the diffusion zone is recorded. Parallel negative control plates (extraction medium) and positive control plates (known mutagen) are included. A description of the size and visual characteristics of the revertant colony zone is reported.

Although the Ames mutagenicity test has been the standard test for monitoring genetic effects of toxic substances, other useful techniques such as in vitro or in vivo sister chromatid exchange have also become routine (11). Service laboratories specializing in toxicology and microbiology or environmental testing are generally knowledgeable in this area.

STERILIZATION

Sterilization is a necessary and important part of the manufacture of any cardiovascular device. It must also be integrated into the

materials and device development and characterization procedures for two reasons. First, the sterilization process can have a significant effect on the physical, chemical, and toxicological properties of the device and its materials. Second, the device or material, depending on its properties and geometry, may actually protect bacteria or their spores from certain sterilization processes. This is especially true of porous and crystalline materials. Furthermore, following sterilization, it is important to assure that the material's properties have not changed as a result of the process and to assure that the device is, indeed, sterile. The level I testing procedures described in this report will accomplish the necessary physicochemical and toxicological characterization of the device. Guidelines for validation of sterility have been thoroughly addressed by other organizations within government and industry, and therefore only a brief overview of the issues and approaches in sterilization and sterilization assurance is presented in the following paragraphs.

Sterility is defined as an absolute condition of complete destruction or removal of all forms of life; however, because of the practical difficulties involved in the measurement of sterility (13) the best that can be stated about a sterilization process is that it reduces the probability of survival of a single organism in the sterilized object to an acceptably small value. In general, the acceptable probability of microbial contamination of a sterile object is 10^{-6} or less, although there is presently some debate on varying the probability level in accordance with product end use.

Methods of sterilization are classified as either physical or chemical. Physical methods include heat and ionizing radiation. Chemical methods utilize gases and liquids. Most liquids are used as disinfectants (they do not kill spores) rather than sterilants with two notable exceptions, formaldehyde and iodine (especially in combination with anionic detergents). However, liquids should not be used for microporous materials or devices with a complicated geometry since trapped air is difficult to eliminate and acts to protect microbes from the solution. Liquids should only be considered as a last choice and will not be discussed further.

Dry heat sterilization is not considered useful for most polymer materials since temperatures in excess of 140°C are required for long periods of time. Glass and metals can be processed by this method with the recommended times and temperatures listed in Table 7.4 (14). Most plastics soften, deform, or are oxidized at these temperatures. Some commonly used medical materials that can be sterilized by dry heat are PTFE and silicon rubber (15). The advantage of dry heat sterilization, especially for glass or metal, is that endotoxin (pyrogen) is inactivated at temperatures of 180°C for 4 hours (16). The subject of pyrogen removal will be discussed later.

The disadvantages of dry heat are discussed in detail by Ernst (14). Briefly they are: 1) diffusion and penetration of heat is slow in most biomaterials (low thermal conductivity), 2) it requires long sterilizing times, 3) high temperatures are required, 4) many

Table 7.4--Sterilization Conditions for Biomaterials

Methods	Sterilization Conditions	Comments
Dry Heat	170°C, 60 minutes	Can be used for glass, metal, silicon, nylons, PTFE. Longer times above 170°C can effectively inactivate endotoxins.
Steam Under Pressure	121°C (15 psig), 15 min. 126°C (20 psig), 10 min. 134°C (29.5 psig), 3 min.	Generally suitable for plastics. Softening and distortion may occur.
Ionizing Radiation	2.5 M rad	Embrittlement may occur. Can affect PP, PMMA, and PTFE.
Ethylene Oxide	Relative humidity 40 percent; 55°C	Requires degassing.

materials are damaged by heat or oxidation, and 5) temperature zones tend to develop and must be overcome.

Moist heat sterilization (autoclaving) is at least seven times more effective at killing spores than dry heat at the same temperature because of its latent heat of vaporization (14). USP-XX (8) defines steam sterilization as heating in an autoclave with steam-saturated air under pressure (15 psig) for at least 15 minutes at a minimum temperature of 121°C. The 15-minute time period should not start until the material to be sterilized reaches 121°C (8). Shorter times can be used with increased temperatures (see table 7.5).

Many polymers are unsuitable for autoclaving because of heat distortion, loss of mechanical properties, and loss of physical properties. This is especially true if the material is subjected to repeated autoclaving. Elastomers can lose their properties on cooling, and crystalline polymers or polymers with crystalline regions may be drastically changed after autoclaving (17). Table 7.5 lists some common polymers and the effects of sterilization. If one is uncertain about the thermal stability of a particular polymer, the information is available in the Modern Plastics Encyclopedia (18).

Gas sterilization can be extremely effective if certain conditions are met. Although many gases are reportedly germicidal (ethylene oxide, formaldehyde, propylene oxide, beta-propiolactone, ozone, methyl bromide, and others) only ethylene oxide and formaldehyde are considered to be useful sterilants (19).

Formaldehyde gas prepared from paraformaldehyde is reported to be one of the best available methods for sterilization (19). However, it is also reported to be ineffective for porous materials since the gas tends to condense on certain areas (20). Formaldehyde is quite reactive with poly(vinyl alcohol), collagen, and gelatin (17).

Ethylene oxide used as a sterilant requires an adequate understanding of the parameters affecting its activity. Its potential usefulness has been hampered somewhat by misinformation and improper use of this gas (21). It is highly explosive in air in its pure form, but the addition of carbon dioxide or freon reduces this risk. The inhalation toxicity is reported to be mild and nonaccumulative. The maximum safe level in an implanted device is 250 ppm. Bruck has reported that ethylene oxide should not be used to sterilize heparinized surfaces because the gas appears to displace the ionically bonded heparin (20).

The disadvantages of ethylene oxide sterilization are: 1) it is more expensive than heat, 2) more monitoring is required during each cycle, 3) it will not kill spores in crystals, 4) it is necessary to determine if residual ethylene oxide or its reaction products remain in the material, and 5) dirtiness decreases sterilizing efficiency (19,21). Ethylene oxide sterilization is most effective when the relative humidity is controlled at 40 percent and the temperature is maintained at about 50-60°C.

Table 7.5--Suggested Sterilization Methods for Polymers

	Sterilization Procedure		
	Ethylene Oxide	Radiation	Autoclave
Acrylonitrile/ Butadiene/styrene (ABS)	Yes	Yes Single 2.5 M rad dose	No
Butyl rubber	Yes	No	Yes
Cellulose acetate	Yes	No may reduce strength	No softened by heat
Cellulose, regenerated	Yes	No	Yes
Epoxies	Yes	Yes	Yes?
Fluorinated ethylene propylene (FEP)	Yes	No	Yes
Natural rubber	Yes	No	Yes
Polyamides aliphatic	Yes	No Embrittles	No
aromatic	Yes	Yes	Yes
Polycarbonate	Yes	Yes	Yes
Polyester	Yes	Yes	Yes
Polyethylene	Yes	Yes	No Distorted by heat
Poly(hydroxyethyl methacrylate) (HEMA)	Yes	No	Yes
Polymethylmethacrylate	Yes	No	No
Polypropylene	Yes	No Embrittles	Yes
Polysulfone	Yes	Yes	Yes

Table 7.5 (continued)

	Sterilization Procedure		
	Ethylene Oxide	Radiation	Autoclave
Polystyrene	Yes	Yes	No
Polytetrafluorethylene (PTFE)	Yes	No Acids evolve	Yes
Polyurethanes	Yes	Yes	Yes
Polyvinylchloride	Yes	No Acid evolves	No
Silicones	Yes Proper aeration required	Yes	Yes
Styrene/Acrylonitrile (SAN)	Yes	Yes	No Distorted by heat

Ionizing radiation, especially gamma radiation has some advantages over ethylene oxide when used in an industrial setting (22). Fewer controls are required, and continuous processing can be achieved at substantially lower costs. Most plastics are affected in some way by radiation, but dose levels employed in sterilization procedures (2.5 megarads) are usually low enough to avoid any significant changes in most polymers. Oxidation can occur with rubbers, and some polymers become brittle (17, 22). Radiation is especially damaging to PTFE, PMMA, and PP (15). This method is gaining in popularity as manufacturers improve methods to protect polymers from degradation by adding radiation stabilizers. Radiation stability of polymers common to medical applications is presented in table 7.5. This table is a compilation of information found in references 20, 21, 22, and 23.

Sterility Assurance

Methods for the evaluation of sterility are outlined in the USP (8). Two basic methods are described requiring either direct inoculation of the solid materials into culture media or solubilization and filtration followed by incubation of the filter in the culture media. The latter method is reported to be more sensitive by 100- to 1,000-fold (13). Biological indicators consisting of test strips of bacterial spores can be substituted for the device. These test strips are available commercially. After sterilization, the device or test strip is placed in soybean casein digest medium (incubated at 20-25°C) and in fluid thioglycollate medium (at 30-35°) and incubated for 7-14 days (8).

The use of biological indicators is controversial since there appears to be a wide range of variation of commercial test strips sterilized in a steam autoclave at 121°C. For a more detailed discussion of this subject, the reader is referred to reference 13.

It is difficult to assure sterility of a batch of products by endproduct testing only. Therefore, approaches which take sterility system design, product design and manufacture, and process validation into account are used to most effectively and optimally assure the sterility of a product to a given probability level. A systems approach of this type is presented in the Medical Device Sterilization Monographs (24).

After determining which sterilization method is to be used for a given product, the first step in assuring sterility is validation of the process. This focuses on the mechanical equipment specifications and performance of the sterilizing equipment and serves to establish the operational range, parameters, and limitations of the process regardless of the product to be sterilized.

The second aspect of this systems approach is cycle development. At this stage, product characteristics such as chemical properties, configuration, and the packaging system and materials are weighed in the selection of the sterilization cycle. The average bioburden on the items in a batch should be known from past manufacturing history for other lots of the product. The cycle is then developed to establish assurance of sterility to a probability of microbial contamination of

10^{-6} or less. A useful example of this procedure for steam sterilization is available (25). Where gas or liquid sterilization is used, testing for the presence of residual chemicals is also necessary.

The final important aspect of sterility assurance is microbial control in the manufacturing environment. A microbial control program should monitor production environmental conditions and establish good housekeeping through scheduled microbial monitoring. This manufacturing control should be designed to assure that the bioburden (microbial load) of a product remains within limits that maintain acceptable probability of sterilization and to establish procedures for dealing with incidents where bioburden limits are exceeded.

Much of sterilization assurance falls within the realm of good manufacturing practices. However, laboratories which fabricate sterile devices for experimental work should also be available to document and assure the state of sterility of those devices.

PYROGENICITY

Taking great care to ensure the sterility of a device does not ensure that it is pyrogen-free. A pyrogen is any substance that will produce a fever when injected into humans or animals. The most common pyrogens are derived from gram-negative bacteria and are referred to as endotoxins, since they are released when the bacteria are lysed.

Endotoxins (lipid A) are high molecular weight (~106,000 daltons) lipopolysaccharides which are intimately associated with the bacterial cell wall (26). There is good evidence that endotoxin activates complement (26,27), the intrinsic coagulation pathway through activation of factor XII (Hageman factor) (28), aggregates platelets (29), and indirectly promotes generation of thromboplastic activity in leukocytes (30,31) through activation of complement. Leukocytes are thought to play a pivotal role in the febrile response by releasing substances that affect the thermoregulatory function of the hypothalamus.

The most probable primary source for endotoxin contamination of materials is the water used to clean the device. The only way to effectively remove or eliminate endotoxins is to use dry heat at 180°C for 4 hours or 250°C for 30 minutes or to acid treat the material (8,16).

Some polymer materials may selectively adsorb endotoxin. These include polypropylene, polyethylene, and fluorocarbon polymers (32,33). Since extreme dry heat and long-term acid exposure are difficult if not destructive procedures, it is much easier to screen materials for endotoxin rather than attempt to remove it.

Currently, the only official test for endotoxin is outlined in the USP XX (8). This test consists of injecting an eluate or extract from a device into the venous circulation of rabbits and measuring the febrile reaction of the rabbits. A material is considered pyrogen-free only if the rectal temperature of three rabbits injected with the extract does not show an increase of 0.6°C above the respective control temperatures.

The test is time-consuming and expensive, requiring preconditioned rabbits housed under strict temperature control and handled by trained personnel.

More recently, another acceptable test for pyrogenicity has been developed. The Limulus amebocyte lysate (LAL) test (16,34) is based on a distinctive property of the blood clotting mechanism of the horseshoe crab Limulus polyphemus. The test is simple, fast, and relatively inexpensive. However, there are some substances that have an inhibitory effect on this test, such as heparin (35). The Limulus lysate reagent is available commercially. This bacterial endotoxin test (36) is described in detail on pages 888-889 of USP XX, as revised on page 160 of the Second Supplement to USP XX.

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Chapter 8

SPECIES EFFECTS IN TESTING MATERIALS AND CARDIOVASCULAR DEVICES IN EXPERIMENTAL ANIMALS

A large body of information is available pertaining to aspects of hemostasis. However, much of this bibliography is obsolete, nonspecific, and contradictory. Few studies systematically compare platelet function, coagulation, and fibrinolysis among species particularly with reference to these parameters in humans. When species comparisons have been performed, differences in platelet function which extend to platelet release, the thromboxane system, and platelet adhesion to biopolymers have been reported (1-5).

Similarly, with respect to coagulation, species differences begin with those earliest phases of coagulation which are fundamental to surface-plasma interactions, and are no doubt evident throughout the coagulation cascade and extend to the fibrinolytic and complement systems (6-8). Finally, a virtually unexplored area would be the contribution of altered vascular, particularly endothelial, physiology to species variability (9-11).

The summary tables (tables 8.1-8.4) compiled from the literature in the accompanying reference list include reliable available information comparing the relative hemostatic activities of animals and man. However, many published conclusions have been refuted, and much of the literature is contradictory. Examples are the divergent relative hemostatic activities tabulated in earlier reviews (12) and the very low platelet counts reported for goats (13). These discrepancies emphasize the necessity for detailed reporting of experimental conditions and better standardization of experimental design and methodology once the species to be used has been selected.

In vitro and preclinical testing of biomaterials interaction requires the use of experimental animals or animal products. It is essential, however, that the design and execution of studies with animals simulate the human situation so that data can be extrapolated ultimately to man. This requirement cannot always be met because of cost, availability, size, and investigator unfamiliarity with the most suitable species. For example, the hemostatic mechanism of primates parallels but is not identical to that of man, but the expense and current shortage of primates may restrict widespread use of this animal model. Alternative species should thus be considered and used, where appropriate, for less critical in vitro or initial screening evaluations. However, valid quantitative comparisons may still be impossible because many species respond quite differently to anticoagulation both in vitro and in vivo, and the rapid coagulation of most animal bloods often precludes working with native whole blood or blood components (14).

Table 8.1--Coagulation Parameters of Animals Compared to Normal Humans*

Species	Fibrinogen	Prothrombin	Factors										Extrinsic		Intrinsic	
			V	VII	VIII	IX	X	XI	XII	XIII	Clotting	Clotting	Clotting	Fibrinolysis		
Nonhuman Primate	E or I	E or I	I or E	I	I or E	E	E	E	E	I or E	E or D	E or I	E or I	E or I	E or I	
Rabbit	E or I	D	Markedly	D or E	I	I	I	I	I	E	I	E	I	E	E	
Guinea pig	I or E	D	I	D	I	E	D	D	E	E	I	D	E	E	E	
Rat	E	E or I	I	E	I	D or E	D	D or E	D or E	I	E	E or I	I	D, E, or I		
Mouse	E	E or D	I	I	E	E	I	E	E	E	E or I	E or I	E or I	E or I	E or I	
Hamster	E	E		D	I	E	E	E	E	E	E or I	E or I	I	E	E	
Cat	I or E	D or E	Markedly I	E	I	I	Markedly D	I	I	E	E	E	I	E or I	E or I	
Dog	I or E	E or D	Markedly I	I	Markedly I	I	I	I	I	E or I	I	I	I	I	I	
Pig	I or E	E or D	I or E	E or I	I	I	E	E or I	E or I	I	D	E or D	I	E or D	E or D	
Sheep	I	D	I	D	Markedly I	I	D	D	D	I	D	I or E	I	E or D	E or D	
Goat	I or E	E or D	I	E or D	I	E	D or E	D	D	I	E	E	I	E or D	E or D	
Cow	I	D or E	Markedly I	E or D	E	E	E	E or D	E or D	I	Markedly I	D or E	D, E, or I	E or D	E or D	
Horse	I	E or I	I	E	E or D	D or I	I	D	D	D	D	D or E	D	E	E	
Bird	E	D	Markedly D	Markedly D	D or E	Markedly D	E	Markedly D	Markedly D	Absent	E†	E or D	Markedly D	D or absent	D or absent	

E=equivalent; I=increased; D=decreased. Activities as determined in standard coagulation assays.

*Reproduced from Blood-Material Interactions Guidelines, 1st edition.

†With homologous thromboplastin.

Table 8.2--Platelet Aggregation Responses of Various Species^a

Species	Aggregating Agent				
	ADP	Thrombin	Collagen	Serotonin	Adrenaline
Man	++	++	+	+	++
Baboon	++	+	+	+	++
Chimpanzee	++	+	+	+	++
Monkey	+ <u>+</u>	+	+	+	+ <u>+</u>
Rabbit	+	+	+	+	-
Guinea pig	+ <u>+</u>	+	+	-	<u>+</u>
Rat	+	+	+	+	-
Mouse	-	-			-
Hamster	+	+	+		
Cat	++	++		++	++
Dog	+ <u>+</u>	+	+	+	+ <u>+</u> ^b
Pig	+	+			-
Sheep	+	+		+	-
Goat	+	+	+		-
Cow	+	+	+		
Horse	+ <u>+</u>	+	+	+ <u>+</u>	-
Bird and Reptile	-	+	-	-	-

^aCompiled data; ^bhigh doses required.

+, monophasic; ++, biphasic; and ++, occasionally biphasic aggregation; - no response; +, occasional response.

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**Table 8.3--Potentiation of Aggregation and Serotonin (5-HT)
Content of Platelets From Various Species^a**

Species	Potentiated by		Relative 5-HT Content
	5-HT	Adrenaline	
Man	+	+	Low
Baboon	+	+	Low
Chimpanzee	+	+	
Monkey	+	+	Low
Rabbit	+	+	Very high ^b
Guinea pig	<u>+</u>	-	Low
Rat	+	-	Low
Mouse	+		
Cat	+	+	High
Dog	+	+	Medium
Pig	+	-	
Horse	+	-	Medium ^c

^aCompiled data; ^b20-25 times human; ^ctwice human.

+ potentiated

- no potentiation

+ equivocal reaction

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Table 8.4--Platelet Membrane Proteins in Various Species^a

Species	Major Glycoproteins ^b			Major Membrane Protein ^c
	I	II	III	
Man	+	+	+	+
Baboon	+	+	+	+
Gibbon	+	+	+	+
Monkey	+	+	+	+
Rabbit	+	+	+	+
Rat	+	+	+	+
Cat	-	+	+	+
Dog	+	+	+	+
Pig	+	+	+	+
Cow	+	+	+	+
Deer	+	+	+	+
Lion	-	+	+	+
Puma	-	+	+	+
Leopard	-	+	+	+
Bear	+	+	+	+
Wart hog	+	+	+	+

^aCompiled data. Molecular weights: ^b95,000 to 240,000
^c40,000 to 50,000.

+ present
- absent

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Although it is desirable to study blood-biomaterial interactions in the biological normocalcemic environment, most investigations are carried out on anticoagulated blood or in anticoagulated animals. However, as noted above, the response of different animal species to anticoagulants such as heparin and citrate varies widely (14). For example, various heparins cause platelet aggregation or thrombocytopenia in many species including man. Rabbits, dogs, cats, and guinea pigs are the most susceptible to the in vitro and in vivo effects of the anticoagulant. By contrast, mouse, rat, and hamster platelets are relatively nonreactive to heparin. Problems related to species also exist with citrate, as the optimum concentration required to anticoagulate blood for platelet and coagulation studies varies. Citrate concentration is known to affect platelet reactivity, and most animal species require a high final concentration (0.32 to 0.38 percent is compatible with all species).

One of the most important considerations when working with animal blood is the method of collecting and processing samples. Proper technique is critical to minimize the activation or contamination of samples with thromboplastic materials. Thus, cardiac puncture is not a suitable alternative to a careful, direct venipuncture or arterial puncture. Any anesthetization or tranquilization procedures should be noted and controlled as these drugs may themselves alter the behavior of blood components. In small rodents, blood is best collected by abdominal sampling from the vena cava or aorta of the anesthetized subject. When native blood is to be used, it is essential that collection techniques minimize tissue trauma. The preferred way to obtain anticoagulated specimens involves first placing required amounts of anticoagulant into the syringe and then withdrawing the appropriate volume of blood from the animal. Rapid blood flow is desirable as is immediate mixing after collection.

All of the problems alluded to become compounded if the testing of platelet-active drugs is a contemplated part of the experimental protocol. The bioactivity, vasoactivity, metabolism, and antiplatelet properties of any agent would clearly vary from individual to individual and species to species (15-20, table 8.5). Unexpected biotransformation must be considered as well as the stability of the agent under study. Furthermore, many drugs are pH dependent in their activity, and anticoagulated animal blood tends to become more alkaline (pH 8) on standing than does human blood. Thus, it is often wise to compare the effects of any platelet-active drug on human and animal platelets to guarantee that, at least with respect to platelets, its properties are not species dependent.

In summary, it is extraordinarily difficult, indeed impossible, to list unequivocal guidelines which would permit selection of species for biomaterial research. It is perhaps wiser to have well-defined goals and then choose a species which seems particularly suitable. Dog (see below for additional details), sheep, and calf, for example, have the advantage of size, while cat and rabbit are economical. In attempting to study in vivo platelet activation and thrombosis, it is well to keep

Table 8.5--Inhibition of Platelet Aggregation in Various Species^a

Species	Inhibitor					
	Adenosine	AMP ^b	Dipyridamole	PGE ₁ ^c	PGD ₂ ^d	PGI ₂ ^e
Man	+	+	+	+	+	+
Baboon	-		-			
Chimpanzee			+			
Monkey			<u>+</u>			
Rabbit	+	+	+	+	+	+
Guinea pig	-	-	-			
Rat	<u>-</u> ^f	<u>+</u>	-	+	-	+
Mouse			-			
Hamster		+		+		
Cat			-	+		
Dog	+	+	+	+		
Pig	+	+	+	+		
Horse	-	-	-		+	+
Sheep					+	+
Bird and reptile	+					

^aCompiled data; ^badenosine monophosphate; ^cprostaglandin E₁; ^dprostaglandin D₂; ^eprostacyclin; ^fadenosine receptor present but has no measurable effect.

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in mind that cat platelets are very sensitive (table 8.2) and tend to aggregate spontaneously. Thus one can take advantage of this property in model design. Rabbit platelets have a high concentration of serotonin which could be exploited for certain projects. Sheep have a very high coincidence of chronic lung disease which might preclude their use for certain cardiopulmonary work and yet have favorable anatomy (accessible thoracic duct) which might make them indispensable. It is often difficult to obtain peripheral blood from the dog without activating the extrinsic pathway of coagulation (table 8.1), a problem not often encountered in calves. Dogs simply are not affected by vascular disease similar to that seen in humans and consequently might not be an optimal species if vascular considerations are paramount. The rabbit is often quite acceptable with regard to its hemostatic system, yet peripheral blood leucocytes are often very difficult to isolate. In the final analysis, there is probably no substitute for consulting a laboratory with experience with a particular species once the research goals and the experimental design have been defined.

RECOMMENDATIONS

The following can be considered as minimum standards for animal usage in studies of biomaterials interaction:

1. Definition and characteristics of species

- a. age, sex, lifespan.
- b. source, strain (outbred, inbred, random bred, closed colony).
- c. health and nutritional status (viral profile for small laboratory animals, free of bacterial and parasitic diseases, incidence and type of diseases of aging, and tumor incidence).
- d. acclimated and/or conditioned laboratory animals versus farm, field, raw, or unconditioned animals.
- e. environmental factors (housing and temperature requirements).
- f. cost of procurement and maintenance including reutilization.
- g. variability between and within stocks and strains.

2. Preferred species

- a. Human blood is preferred for most in vitro testing. Animal blood is useful, however, for some specific, well-defined test systems and conditions.
- b. Animal blood is used for preclinical testing including extracorporeal and in vivo tests. In general, the species selected should be relatively large (nonhuman primate, dog, calf, sheep, goat, or pig) so that devices can

eventually be tested in their configurational (intended use) form. Smaller laboratory animal species (rabbits, cats, guinea pigs, rats) are useful under special circumstances.

- c. Despite extensive use of the dog for both in vitro and in vivo testing of biomaterials and cardiovascular devices, the dog may not be the species of choice for the following reasons: Low susceptibility to cardiovascular disease and pulmonary embolism; very active hemostatic and fibrinolytic mechanisms; a hyperplastic vascular response to incitants causing rapid occlusion of vessels; and the response of canine platelets to heparin, rendering this anticoagulant undesirable for comparative in vitro and in vivo studies. On the other hand, the considerable literature available and experience with the dog as a test species argue persuasively for its continued use (see references).

4. Selection of test system

- a. The aim in selecting species, methods, and test interaction should be to simulate performance in humans or with human blood.
- b. Species which differ significantly from man in a certain aspect of biocompatibility should not be selected to test this aspect (e.g., variation in species response to citrate versus heparin anticoagulants, and to the amount of ADP required to elicit platelet aggregation and the release reaction).

5. Handling and sampling techniques

- a. Presence or absence of anticoagulation; type of anticoagulant required or recommended; sample collection methods and site, including syringes, needles, collecting and storage containers, any ancillary items; ease of handling the species for collection of samples (size, temperament); restrictions on sample volume by size of animal must all be noted.

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Chapter 9

EVALUATION OF BLOOD-MATERIAL INTERACTIONS IN VITRO, EX VIVO, AND IN VIVO

Perhaps the most controversial and misunderstood concept in bio-materials research is that of blood compatibility or hemocompatibility. Hemocompatibility is ultimately defined by the nature of the end-use application of the material. Frequently, investigators do not have a specific end-use in mind but would like data from a blood-material interactions protocol to provide information concerning possible end-uses for the material.

Many different test systems have been devised to assess BMI's, but in only a few instances have well-characterized surfaces been tested in vitro and the results correlated with in vivo data (1-4). As discussed in earlier chapters, the materials must be well characterized before reproducible BMI can be expected.

The in vitro and ex vivo test systems discussed below should be regarded as simple screening tests; the results from these tests may offer guidance as to which of the more sophisticated in vivo tests to pursue in animals. The simpler tests can also be used to improve in vivo protocols in order to produce the most useful and reliable information. The possibility of false negative and false positive results must be realized when using in vitro tests; for this reason, many investigators do not use in vitro procedures for evaluating materials but screen new materials using animal testing methods. Unfortunately, false positive or false negative results can occur in vivo as well. Changes in design and/or flow conditions can improve the apparent in vivo hemocompatibility of a mediocre material. Likewise, a material with superior blood compatibility may appear to be extremely thrombogenic if the in vivo test system is inappropriate.

In vitro testing implies that blood or fractions of blood are isolated from a subject and then exposed to different materials to observe changes in either the material or the blood. Ex vivo tests refer to exposure of the blood to the test material directly from the animal. The test chamber is located outside the body but the test protocol can be either single pass, i.e., blood passes directly from the animal into a test chamber into a test tube; or recirculating, as in the case of a shunt model. In vivo testing involves implanting the material into the animal. Patches, vascular grafts, rings, etc., are examples of typical configurations used in in vivo testing. In vivo tests are designed to examine long-term hemocompatibility, whereas in vitro and ex vivo tests evaluate acute blood interactions.

Much of the confusion concerning blood compatibility of materials is due to the lack of materials characterization before blood tests are performed and to the fact that most studies only rely on one blood test to determine blood compatibility. The conclusion that hydrogels are hemocompatible based solely on the nonadherence of platelets to hydrogels

is a good example of the single parameter "trap." The importance of multiparameter testing cannot be overemphasized. This is true for in vitro, ex vivo, and in vivo approaches to materials evaluation (5,6). Materials to be used in a low flow (venous) environment may respond quite differently when used in high flow (arterial) situations (1).

The tests recommended in the previous NHLBI Guidelines for Blood Material Interaction (17) are listed in table 9.1. These tests are still useful but there are improved assays for most of the coagulation factors (chromogenic and radioimmunoassays) (7-9) and platelet activation factors (radioimmunoassays) (10,11). More recently, immunoassays are commercially available for human blood but are not generally available for other species. Table 9.2 lists currently available assays. The human test kits usually do not cross react with other species except for some nonhuman primates. Care should be taken when designing in vitro test systems to ensure that one is actually measuring activation due to the test material and not an artifact of the system.

Wilson et al. have recently reviewed in vitro, ex vivo, and in vivo test methods for assessing blood materials interactions (6). Their compilations of static in vitro tests and dynamic in vitro tests are given in tables 9.3 and 9.4 respectively. Tables on ex vivo and in vivo tests may be found in reference 6.

IN VITRO TESTS

There is general agreement that in vitro tests are most useful in the detection of artificial surfaces that are highly reactive with blood. It is also agreed that human blood should be used for in vitro tests whenever possible. However, if one desires to correlate in vitro results with in vivo studies then in vitro studies should also include experiments using blood from the species selected for in vivo analysis.

Variables that must be considered when utilizing in vitro test methods include anticoagulants, sample collection, sample age, aeration and pH, temperature, sequence of test versus control studies, and surface-to-volume ratio and fluid dynamic conditions (especially wall shear rate). If possible a statistician should be consulted at the early stages of experimental design.

All materials under investigation should be characterized as to their bulk and surface properties as noted in chapters 4 and 5, respectively. Ideally, the material should be sterilized since sterilization can alter both surface and bulk properties of the polymer.

As mentioned earlier, data from in vitro tests are useful before expanding to more expensive in vivo testing. However, in vitro tests are very useful for quality-control purposes as well. New materials for devices that are used to handle or process blood such as syringes, blood bags, tubing, test tubes, etc., are more appropriately tested using in vitro methods. Also, pharmacologically active polymers can be well characterized using in vitro blood tests.

Table 9.1--Recommended Tests for Blood-Material Interactions
After Blood Exposure (17)

	Level I	Level II
<u>In Vitro</u>	Examination of Test Material	
	Gravimetric Analysis (thrombus weight)	Scanning Electron Microscopy
	Light Microscopy (adhered platelets, leukocytes, aggregates, etc.)	Transmission Electron Microscopy
	Examination of Anticoagulated Blood	
	Whole Blood Smear	Recalcified Whole Blood
	Platelet Count	Clotting Time
	Leukocyte Count	Specific Coagulation
	Erythrocyte Count and Morphology	Factor Assays
	Partial Thromboplastin Time	Euglobulin Lysis Time
	Prothrombin Time	Fibrin Degradation Products
	Thrombin Time	Fibrinopeptides
	Plasma Fibrinogen	Platelet Aggregation
	Plasma Free Hemoglobin	Platelet Research Reaction
		Leukocyte Mobility and Chemotaxis
		Leukocyte Phagocytosis
		Erythrocyte Fragility
		Lactate Dehydrogenase
		Complement Factors C3 and C5
		Characterization of Eluted Proteins
<u>Ex Vivo</u>	Acute (minutes to hours)	
	Cannulae or Shunts (platelet adhesion and activation, coagulation activation)	Stagnation Point Flow Experiments (platelet and leukocyte adhesion) Annular Axial Tests (thrombus deposition)
	Chronic (days to weeks)	
	Arteriovenous Shunt (platelet consumption, platelet activation, coagulation activation)	Arteriovenous Shunt (radiographic analysis, gamma imaging)

Table 9.1 (continued)

	Level I	Level II
<u>In Vivo</u>	Acute (hours to days)	
	Intravascular Catheters	Vena Cava Ring Test
	Chronic (days to months)	
	None Recommended	Vena Cava Ring Test Renal Embolus Test

Table 9.2--Commercially Available Assays for Platelet and Coagulation Factors
(From References 7 and 9)

<u>Factor</u>	<u>Types of Assay</u>
Plasminogen	Colorimetric Fluorogenic
Antithrombin III	Chromogenic Fluorogenic
Factor X	Chromogenic
Factor Xa	Fluorogenic
Antiplasmin	Chromogenic
Prekallikrein	Chromogenic
Kallikrein	Fluorogenic
Platelet Factor 4	Radioimmunoassay Chromogenic
β -thromboglobulin	Radioimmunoassay
Factor VIII	Chromogenic
Factor IXa	Fluorogenic
Factor IX	Chromogenic
Factor XIIa	Fluorogenic
Factor XII	Chromogenic
Fibrinopeptide A	Radioimmunoassay

Table 9.3--Static In Vitro Test Methods
(Modified From Reference 6)

Configuration	Species	Anticoagulant	Exposure Time (Min.)	Measured Variable	Reference
Test Tube	Human, Dog	None	--	Clotting Time	38
Test Tube	Dog	None	--	Lee-Howell-White Clotting Time	39
Test Tube	Rabbit, Calf	None	--	Lee-Howell-White Clotting Time	5
Sheet or Film	Human	Acid Citrate	1	Platelet Adhesion	22
Sheet or Film	Human	None	2-20	ST, PTT, Hb, ATP	40
Sheet or Film	Human	None	7	Platelet Adhesion	41
Sheet or Film	Human	None	--	Lindholm Clotting Time	42
Membrane	Dog	None	20-120	Thrombus Weight PC, Hb	43 44
Sheet or Film	Human	None	5	PTT, ST, LC, PC HB, Hct	45 46
Sheet or Film	Rabbit	None	15	ST, PTT, PT	5
Closed Tubing	Human	None	1-120	PTT, ST, LC, PC Hb, Hct	47 46
Sheet or Film	Human	None	1/60-2	Platelet Adhesion	48
Membrane	Human	Citrate Citrate/Heparin	7	Platelet Retention	50
Sheet or Film	Human	Heparin	1/60-10	Platelet Adhesion	51-52

Table 9.4--Dynamic In Vitro Test Methods
(Modified From Reference 6)

Configuration	Species	Anticoagulant or Plt. Soln.	Shear Rate (Sec-1)	Exposure Time (Min.)	Measured Variable	Reference
Closed Loop	Human	None	150	—	Thrombus Growth Time and Morphology	53
Closed Loop	Human	None Heparin	—	10	PTT, ST, LC, PC, Hct Platelet Morphology	54
Closed Loop	Dog	Citrate	N.R.	90-360	PTT, PT, PC, ST	55
Annular Axial Rod	Rabbit	Citrate	12- 16,500	2-40	Platelet Adhesion Thrombus Deposition	56 57-58
Annular Axial Rod	Human	Citrate	50- 10,000	1-40	Platelet Adhesion Thrombus Deposition	59
Annular Axial Rod	Human	Citrate	650- 3,300	2-7	Platelet Adhesion Thrombus Deposition	60
Annular Axial Rod	Human	PRP	800	3-5	Radiolabeled Platelet Adhesion and Morphology	61
Hemodialysis Unit	Human	Citrate Hep./Cit.	N.R.	30	PTT, ST, LC, PC, Hct, Hb	62
Elliptical Cell	Human	Heparin	N.R.	15-60	Platelet Adhesion	63
Flat Plate Flow Chamber	Human, Dog	Citrate Heparin	400- 1,000	10-20	Platelet Adhesion	64
Flat Plate Flow Chamber	Human, Dog, Rabbit, Pig, Sheep, Calf, Macaque, Baboon	Heparin Citrate	400- 1,000	5-180	Platelet Adhesion	13,65
Flat Plate Flow Chamber	Dog	Heparin	740	5-10	Radiolabeled Platelet Fibrinogen Deposition	66
Series Shunt	Rabbit, Baboon	Citrate, ACD	500- 1,000	2-120	Platelet Adhesion	67

Investigators should realize that in vitro tests are designed to detect activation of the blood components responsible for coagulation or thrombus formation as discussed in chapter 2. Tests which expose blood to the test material and then attempt to evaluate something like whole blood clotting time or activated partial thromboplastin time should realize that changes in levels of some coagulation factors must be reduced as much as 95 percent to affect the test.

PLATELET ASSAYS

Adhesion of platelets to a material has been a longstanding measure of hemocompatibility (12,13). However, it is quite clear that activation of platelets is a much more important parameter to monitor than adhesion alone. Activation can occur without adhesion and transient adhesion of platelets does not necessarily result in platelet activation (14). In fact, in vivo studies have demonstrated that platelets can release alpha-granules without releasing dense granules (15). For a complete review of platelet interactions with biomaterials, see reference (16) by Anderson and Marchant, particularly for the details of platelet activation as well as coagulation activation induced by biomaterials.

Procedures to evaluate platelet function in vitro include static tests, bead-column methods, nonrecirculating flow systems, closed-loop flow systems, rotating devices, and aggregation methods. All of these methods have been reviewed (16,17).

Whichever test system is used it is strongly recommended that blood collection techniques also be reviewed and standardized (18). Platelets can become hypersensitive under a variety of conditions, not the least of which is improper blood collection (19). It is also recommended that more than one parameter be monitored during the experiment, e.g., adhesion and platelet factor 4 release.

COAGULATION ASSAYS

A variety of coagulation methods are available for evaluating BMI's. Test systems can use native whole blood, anticoagulated whole blood (usually citrated), platelet-rich plasma, or platelet-poor plasma. In all cases the test material should be used as the activator in the assay.

Since most of the standard coagulation assays are designed to detect clinical coagulation disorders which result in prolonged clotting or excessive bleeding, the protocols should be modified to appropriately evaluate coagulation induced by biomaterials. Reagents commercially available for such tests as partial thromboplastin time usually have an activator added to the solution. Kaolin, celite, and ellagic acid are some examples of activators used in common coagulation assays. Reagents with activators should be avoided except when testing biomaterials designed to release pharmacologically active agents such as heparin.

Blood is exposed to test materials either in a static chamber such as the parallel plate cell (21,22) or in a closed-loop system where the inner surface of the tubing is the test material (23,24). Blood must be anticoagulated in the closed-loop system, however, nonanticoagulated blood is more useful in the static system. After a predetermined contact time with the test surface, the unclotted blood is drained (into citrate for the static test system) and coagulation assays performed on the blood to determine the extent of intrinsic activation. The most frequently used clotting assays include the partial thromboplastin time, thrombin time, prothrombin times, and plasma fibrinogen (6).

The commercial development of nonclotting assays to determine levels of specific factors, fibrinopeptides and platelet release factors has made it easier for investigators to evaluate these parameters on a routine basis. However, most of these tests are only available for human blood. Radioimmunoassays are available for platelet factor 4, beta-thromboglobulin, fibrinopeptide A, and various prostaglandins (9). Chromogenic assay kits are available for antithrombin-III, antiplasmin, plasminogen, prekallikrein, heparin, factor X, factor Xa, and factor II (9). Tests for factors VIII, VII, and XII are under development and should be commercially available soon.

Substrates are also commercially available for fluorogenic assays of antithrombin III, X, Xa, plasminogen, antiplasmin, plasmin and plasminogen activators, kallikrein and prekallikrein, IXa and XIIa (9). Investigators are starting to use the substrate assays for biomaterials evaluations, however a much larger data base is necessary to determine the value of these assays for hemocompatibility testing (20). The caution should again be restated that the material should serve as the activator and proper controls should be run with each test.

EX VIVO TESTS

The term ex vivo is usually applied to test systems that shunt blood directly from the test animal into a test chamber. The blood may be shunted directly into the animal (recirculating) or collected into test tubes for evaluation (single pass). In either case, the test chamber is located outside the body.

Advantages of ex vivo tests are that flowing native blood is used, several materials can be evaluated since the chambers can be changed, and it is possible to monitor some events in real time. Some disadvantages included variability in blood flow from one experiment to another, and usually, only short or intermediate time intervals can be evaluated. In this sense ex vivo systems, like in vitro systems, are useful models for screening biomaterials but are not considered to be accurate predictors of long-term (months to years) hemocompatibility (4).

Ex vivo test systems have been reviewed in Wilson et al. (6). A majority of these test methods examine platelet adhesion exclusively. However, methods are available to monitor emboli generation (25), fibrinogen deposition (26), thrombus weight, bleeding time (27), white cell adhesion (28), platelet consumption (29), and platelet activation.

Table 9.1 lists the recommended tests as described in the previous Guidelines (17).

Many of the ex vivo test systems utilize radiolabeled blood components to monitor hemocompatibility of materials. Radiolabeled platelets and fibrinogen are the most commonly labeled components of blood (30). Investigators should realize that labeling procedures for platelets may alter platelet reactivity. Tests to verify normal platelet reactivity are usually done using an aggregometer. Unfortunately, unreactive platelet preparations are easily detected using this method, but hyper-reactive platelets are not detected. Therefore, an effort should be made to determine if platelets become hyperreactive, and if possible, the ratio of labeled to nonlabeled adherent platelets should be determined.

IN VIVO TEST SYSTEMS

Implantation of materials or devices that do not allow blood to flow outside the body are considered in vivo experiments and have been reviewed by Wilson et al. (6).

The current general opinion of biomaterials researchers is that in vitro and ex vivo blood material interactions do not predict the long-term in vivo response (4). Each animal model offers advantages and disadvantages with no single species acceptable to all investigators. Baboons are the closest model to humans; however, the dog is considered a "conservative" model with the advantage of a large data base. Many differences are discussed in chapter 8.

Patency is the most common measure of success or failure for most in vivo experiments. The percent occlusion and thrombus weight are monitored after the device is removed and embolus generation by the material is evaluated by the examination of downstream organs. The renal embolus test is an example of this approach (31).

Methods to evaluate in vivo performance without terminating the experiment have been reported. Arteriograms are used to determine graft patency or thrombus deposition on devices (32). Radioimaging has been used to monitor platelet deposition at various time periods in vivo (33) and platelet consumption can be used as an indicator of hemocompatibility and passivation due to neointima formation (34,35).

It is important to emphasize that in some in vivo test systems the material's properties may not ultimately determine the blood material interactions. Flow parameters, compliance, porosity, and implant design may be much more important than blood compatibility of the material itself. Low flow systems have been shown to give substantially different results when compared to the same material evaluated in a high flow system (36,37,29).

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Chapter 10

EVALUATION OF CARDIOVASCULAR DEVICES AND PROSTHESES DURING IN VIVO FUNCTION

Procedures used to evaluate cardiovascular devices and prostheses in animals are essentially the same as those employed in the clinical setting. However, animal models permit continuous device monitoring, and systematic controlled study of important variables. Procedures will be considered for use in conjunction with animal as well as clinical evaluations and will be classified separately for each specific device.

The test protocols recommended in this section will follow certain general guidelines. Thrombosis and thromboembolism are emphasized as the major deterrents to the use and further development of advanced cardiovascular prostheses. For devices employed or implanted for short periods, important measurements are related to the acute extent of variation of hematologic, hemodynamic, and performance variables, gross thrombus formation, and possible embolism. With chronic implants, emphasis is placed on serial measurement techniques that may yield information regarding the time course of thrombosis and thromboembolism, the consumption of circulating blood components, and other device failure modes. Serial measurements are emphasized. Thrombus formation may be greatly influenced by surgical technique, variable time-dependent thrombolytic and embolic phenomena, superimposed device infections, and possible alterations in exposed surfaces, e.g., endothelialization. Many of the recommended tests may also be useful in other settings or for the evaluation of prostheses not considered here.

As noted earlier, animal models are useful in the evaluation of materials and prosthetic cardiovascular devices. However, such testing is not without limitations. The choice of an animal model may be restricted by the availability of certain species and the costs of adequate maintenance. Animal size relative to that of a particular prosthesis type may also present difficulties; for example, studies of a totally implantable artificial heart obviously necessitate the use of larger animals. It is critical that the investigators be mindful of the physiological differences and similarities of the species chosen with those of man, particularly those relating to hematological parameters such as coagulation, and the response to pharmacological agents such as anticoagulants and antiplatelet agents. Suitable control experiments must be designed and carried out. The local and systemic effects of surgical procedures, anesthesia, and "routine" maintenance treatments may be difficult to quantify. Despite the fact that infection is a serious problem, results may be modified by prophylactic antibiotic treatment. It is also important to consider that many humans and experimental animals with prosthetic devices are on regimens of anticoagulation, antiplatelet therapy, or thrombolytic therapy (1-6).

GENERAL CONSIDERATIONS

Primary Tests

Primary tests assess device function and alterations in the level and function of circulating blood constituents.

Secondary Tests

Secondary tests are recommended also, recognizing either that they are difficult to perform or that at the present time these tests are of limited or unproven relevance. It is speculated that various tests to be described may be positive in the absence of gross thrombus formation, or that results may correlate with results obtained from primary tests. Platelet survival times (7,8) and plasma levels of the platelet-specific proteins PF-4 (9) and beta thromboglobulin (9,10) may reflect the extent of platelet activation in vivo (and perhaps risk of thromboembolism) even in the absence of significantly elevated rates of platelet consumption. Platelet size distribution abnormalities may also reflect their injury (11). Measurements of fibrinogen peptide A (11) or fibrinogen/fibrin complexes (12) may reflect activation of intrinsic coagulation. Localization of thrombotic material by radionuclide imaging techniques of radiolabeled platelets (13-17) has been demonstrated in studies of vascular grafts and mitral valve prostheses, or localization of coagulation proteins may be useful (18). These studies may also yield information regarding the survival and turnover of radiotagged blood constituents. The use of noninvasive radioisotopic imaging of visceral thromboemboli with 111-In labeled platelets, recently demonstrated in dogs with heart valve prostheses, represents an especially attractive possibility (15). Other secondary tests will be considered in relation to particular devices or prostheses.

A survey of procedures which have been used to assess BMI in association with cardiovascular devices and prostheses is presented below. This literature review is carefully selected from available medical literature and is intended not to be exhaustive, but rather to emphasize recent developments and trends. Based upon this review, test procedures of documented or potential usefulness are identified. The following applications have been considered: 1) cannulae, 2) catheters, 3) extracorporeal oxygenators, 4) heart assist devices, 5) heart prosthesis, 6) heart valves, 7) hemodialyzers, and 8) vascular grafts.

Cannulae

Cannulae are typically inserted into one or more major blood vessels to provide repeated blood access. They may be studied acutely or chronically and are commonly studied as arteriovenous (AV) shunts. The use of cannulae appears to induce little alteration in the levels of circulating blood cells or clotting factors. Hemolysis appears minimal. It is recommended that platelet and leukocyte counts be done before and after cannula placement to document the absence of a serious challenge to these systems. Clinically, vascular access devices may in some respects represent a more severe environment than vascular substitutes of the same materials (19).

The survival and turnover of radiolabeled (^{51}Cr or ^{111}In) blood platelets may be altered significantly in animals with chronic AV can-
nulae. The rates of utilization of radiolabeled platelets and radio-
labeled fibrinogen can be measured simultaneously, and this approach
may be particularly useful to assess the relative participation of
platelets and the coagulation component of the hemostatic mechanism.
Control studies with labeled blood constituents should be performed in
normal animals to ensure that 1) the label is specific for the intended
blood species, and 2) the disappearance rate of labeled species has not
been artifactually shortened as a result of blood collection or label-
ing procedures.

Cannula blood flow rates should be measured noninvasively with
either Doppler or electromagnetic flow probes. These measurements may
be of considerable significance, since hemodynamic effects may largely
govern the nature and extent of thrombotic events including consumption,
thrombus deposition, and embolization. Alterations in flow rates may
be a useful indication of the extent of thrombotic occlusion. If pos-
sible, materials should be compared under uniform flow conditions.

Catheters and Guidewires

Most of the tests considered under cannulae may be relevant to the
study of catheters (20-22). The location or placement of catheters in
the arterial or venous system may be of critical importance. Variable
flow effects over the length of the catheter are probably unavoidable.
Whenever possible, simultaneous control studies should be performed
using the contralateral arteries or veins. Significant differences in
thrombus accumulation have been observed with different catheters, but
care must be taken not to strip off thrombus upon catheter withdrawal.
In situ evaluation may permit assessment of the extent to which intimal
or entrance site injury contributed to the thrombotic process. Kinetic
studies with radiotagged blood constituents (see Cannulae) are recom-
mended only with chronic catheters, but may be useful for imaging
thrombus accumulation in vivo. Angiography may also be useful.

Hemodialyzers and Extracorporeal Oxygenators

The hemostatic response to cardiopulmonary bypass may be signifi-
cant and acute (23-25). A thorough coagulation screen is recommended
before, during, and after bypass, including factor assays, cell counts,
measures of hemolysis, clotting times, and fibrinogen/fibrin degradation
products. Primary tests recommended are the prothrombin time and par-
tial thromboplastin time tests. Many variables influence test values
such as use of blood suction, composition of blood pump priming fluid,
blood contact with air, and others. The standard template bleeding
time is also recommended (26). Previously discussed in vitro assays,
considered potentially useful, may be particularly useful here. Con-
siderably elevated levels of PF-4 have been demonstrated in patients
undergoing bypass, and similar results in animal experiments seem
likely. Emboli in outflow lines may be detected by screen filtration
pressure measurements in vivo, the periodic placement of blood filters
ex vivo, or the use of ultrasound or other noninvasive techniques.
Thrombus accumulation can be indirectly assessed during bypass by moni-

toring performance factors such as pressure drop across the oxygenator and oxygen transfer rate.

Although there is not a large body of literature concerning animal evaluations of BMI with hemodialyzers, they may induce considerable blood damage and are subject to thrombotic occlusion.

There has been recognition recently of complement activation caused by both hemodialyzers (28-31) and cardiopulmonary bypass apparatus (32-35). Clinically significant pulmonary leukostasis and lung injury with dysfunction may result (36,37). It is also important to recognize that embolization of biomaterials from these devices does occur and may have clinically significant consequences (38,39).

Heart Assist Devices and Total Artificial Hearts

These devices may induce considerable alteration in various components of the hemostatic system. The acute response to such prostheses should be assessed as previously described for extracorporeal oxygenators (40-42). Chronic studies should include measurements of hemolysis and red blood cell survival (43,44), platelet and fibrinogen concentration and turnover (see Cannulae), and close monitoring of liver, renal, pulmonary, and CNS function.

Heart Valve Prostheses

The frequencies of the specific sources of clinically encountered valve-related mortality and morbidity have been well described (45-47). Previous studies in the area of development and evaluation of heart valve replacement have dealt almost exclusively with measurements of patency, animal survival rates, and gross thrombus formation. Although these evaluations remain of primary importance in pathologic analysis, more sophisticated tests are available for periodic monitoring during chronic function of heart valve prostheses.

Technologies most useful in evaluating prostheses and associated myocardial function include fluoroscopy, phonocardiography, echocardiography, radionuclide ventriculography, and cardiac catheterization. Fluoroscopy (real-time radiography) is of particular benefit in the evaluation of mechanical valves with radiopaque occluders (48-51). Normal bioprosthetic valve leaflets are generally not visualized with fluoroscopy; however, calcified bioprostheses may become visible. Fluoroscopy of the sewing ring may be of value in suggesting dehiscence with a paravalvular leak if excessive mobility is noted or changes in poppet motion may denote dysfunction (52). Phonocardiography (electronic recording of heart sounds) also is of particular value in evaluating mechanical prostheses. The timing, intensity, and acoustic spectrum of the opening and closing sounds may be determined (53-55). Early degeneration of tissue valves with stiffening of the leaflets often leads to a change in the acoustic frequency of the heart sounds which may be detected by spectral analysis (56).

Echocardiography is the single most useful noninvasive diagnostic modality available for evaluating the function of prosthetic valves.

Echocardiography uses pulsed high-frequency (2-5 MHz) sound waves (ultrasound) to image cardiac structures. Reflections occur when the incident sound waves strike an interface composed of tissues of materials of differing acoustic impedance. Mechanical prostheses emit strong echo signals, and the movement of the occluder can usually be clearly imaged (57-59). Any abnormality in the pattern of motion of the poppet, particularly if the timing of motion is altered, is usually easily determined (51,60,61). Echocardiography may also be useful in the assessment of function of tissue-derived valve prostheses (62-64). Either M-mode or two-dimensional (2-D) echocardiography, or both, may be useful. In many patients with mechanical valve or bioprosthetic dysfunction, the echocardiogram can furnish sufficiently diagnostic information concerning structural abnormalities that surgery for replacement or revision may be performed without cardiac catheterization. This may be of major importance in patients with marked functional impairment in whom diagnostic delay may be disastrous. On the other hand, a valve may be abnormal without eliciting definitive echocardiographically determined abnormalities. Doppler ultrasound may add a new dimension to the detection of hemodynamic abnormalities where structural changes are subtle (65). Using this method, quantification of pressure gradients (66) and degree of insufficiency (67) is feasible in native valve lesions, and bioprosthetic stenosis, not apparent on 2-D echocardiography, has been diagnosed (68).

Radionuclide ventriculography uses a radioactive tracer to label blood flow (by labeling the red blood cells with technetium-99m) and a gamma scintillation camera to image the tracer within the heart (69). Since the detected counts are proportional to ventricular volume, a time/activity curve may be generated, which is analogous to a time/volume curve. From this curve, the efficiency of ejection of blood from the left and right ventricles may be assessed with high accuracy and reproducibility. Radionuclide ventriculography is used principally to assess left ventricular size and systolic myocardial performance. Although providing no direct information about prosthetic valve function, it assesses global cardiac performance in these patients (70), thereby helping to distinguish myocardial from valvular causes for hemodynamic deterioration.

Cardiac catheterization is an invasive procedure used to measure intracardiac pressures and flow and to inject contrast material into coronary arteries and cardiac chambers (71). It is the most definitive of the available diagnostic procedures with a low, but finite, rate of complication.

Measurements of platelet survival, turnover, and deposition as well as other blood parameters are recommended (see introductory portions of this chapter).

Vascular Grafts

Both porous and nonporous materials can be implanted at various locations in the arterial or venous system. The choice of implantation site should be determined largely by the intended usage for the prosthesis, but must represent a sufficiently severe test of function so that

achieved patency justifies further consideration of the implant configuration. For example, patency of a given graft is enhanced by larger diameter and shorter length (72). An acceptable rule of thumb for grafts less than 4 mm ID is that the length exceed the diameter by a factor of 10 (i.e., 40 mm for a 4 mm graft) for a valid model. Patency can be documented by palpation of distal pulses in some locations and by periodic angiography. Results of serial platelet kinetic studies correlate with the area of nonendothelialized graft surface in baboons (73). In dogs, platelet biochemical and functional alterations relate to their interaction with fabric vascular prostheses (74,75). It is important to realize that species variation in these effects may be significant. Specific differences are as yet undetermined, so that caution should be observed in making absolute extrapolations to the human from experimental data. As previously discussed, radiolabeled platelets facilitate noninvasive imaging of mural thrombotic accumulations. Serial measurements of platelet count, platelet release constituents, and fibrinogen/fibrin degradation products also are recommended.

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Chapter 11

PATHOLOGY EVALUATION OF EXPLANTED CARDIOVASCULAR BIOMATERIALS AND PROSTHESES

Procedures used to evaluate cardiovascular devices and prostheses after implantation in animals and humans are essentially the same. However, animal models permit in situ evaluation of fresh implants after desired time intervals. Elective sacrifice is highly recommended, so that specimens from experimental animals can be obtained immediately, avoiding autolytic changes which often impair the ability to perform some sophisticated analyses on human-derived material or improperly handled experimental specimens. In vivo perfusion fixation at physiologic pressure may be used to obtain optimal conditions for electron microscopy.

It is imperative that the planning stage of implant experiments involve the detailed preparation of protocols for ultimate specimen analysis in order to maximize information gain. Where possible, division of specimens into frozen and multiply fixed portions allows the completion of analyses which may initially seem mutually exclusive. Formulation of such a protocol ideally has contributions from all study collaborators, including a pathologist familiar with the pertinent issues, who will ultimately be responsible for ensuring that the relevant questions have been addressed by analysis. It is important to emphasize that the guidelines on specimen analysis included herein are recommendations aimed at completeness of investigation. Deviations from these procedures may, and should, occur in order to approach specific details and answer selected questions. In general, however, it is recommended that procedural alterations be reserved for specific indications. Procedures considered for use in conjunction with animal as well as clinical evaluations will be classified as primary or secondary (table 11.1). For all analyses, efforts toward systematic quantitative data gathering and presentation are strongly encouraged.

GENERAL CONSIDERATIONS

Level I Tests

These include assessment of 1) the extent and mechanisms of device dysfunction should it exist, and host/device interactions, 2) the extent and type of gross thrombus formation, and 3) the frequency of emboli and consequent embolic damage to distal organs and tissues. These tests are the most important and readily performed. Procedures specific to particular devices and prostheses will be recommended below. In all cases, generous use of gross photography is encouraged; intermittent use during the dissection is often useful.

In general, in order to eliminate artifactual thrombosis, experimental subjects should be therapeutically anticoagulated with heparin,

**Table 11.1--Techniques in Evaluation of Explanted
Cardiovascular Devices and Prosthesis**

Primary

**Animal necropsy or human postmortem examination (when
applicable)**

Gross inspection of device

Microbiologic cultures

Secondary

Light-level histology (required in all but screening studies)

Transmission electron microscopy (TEM)

Scanning electron microscopy (SEM)

Immunohistochemical studies

Radiographic examination

Biochemical and spectroscopic analysis

Functional studies

immediately prior to sacrifice and specimen harvest. Appropriate heparin dosages have not been published for most experimental species, but it is recommended that doses of approximately 3U/ml blood volume or 250 U/kg body weight be given for primates (equivalent to human dose), and that the dose for dogs and other quadrupeds be double this. Caution is urged against massive overdosing, since heparin in extremely high concentrations may promote lytic action, thereby potentially causing artifactual removal of thrombotic deposits.

For all types of cardiovascular devices incorporating biomaterials, various specific modes of dysfunction including thrombus formation itself must be anticipated for each device. The observed location of thrombus should be specifically related to areas of flow separation or stasis, junctions of valve components, etc. It is mandatory that a careful gross examination of implant and associated host tissues (both local and distant) be performed at animal necropsy after sacrifice, during human postmortem examination, or receipt of a specimen taken at reoperation. This should be done in situ wherever possible because removal of the implant from its tissue attachments may impair detection of critical pathologic findings. For example, removal prior to careful in situ examination of a catheter may artifactually dislodge significant thrombotic deposits. In all but the most acute experiments, relevant injury due to emboli is primarily demonstrated by the presence of organ infarcts in the distal circulation, whose age and location are often consistent with the duration of the implant. However, since not all emboli produce infarcts, careful gross dissection of major arterial beds is often required for their documentation. Histologic examination (see below) provides confirmation of gross observations of emboli and resultant infarcts, as well as documentation of microemboli (thrombotic, septic, or biomaterial fragments). Since intravascular infection may be either a cause or a consequence of thrombus formation, properly taken microbiologic cultures of the device, attached thrombus and the subject's blood must be used to establish whether infectious organisms are present and perhaps causal to any device failure apparently due to thrombosis.

Level II Tests

Tests in this setting either elucidate specific complications of cardiovascular devices or represent research approaches to the detailed pathophysiology of blood/surface interaction in unique clinical and experimental applications.

Gross evaluation of extent of endothelial coverage of native vessel luminal surface or of prosthetic blood contacting surface may be obtained by preterminal injections of Evans blue dye, which stains de-endothelialized surfaces. Endothelialized surfaces are not stained with Evans blue (1) and relative extent of areas of "neo" and "pseudo" intima may be assessed by morphometry of gross specimens which have been stained by this technique.

Light microscopy (LM), transmission electron microscopy or scanning electron microscopic examination are used to supplement gross inspection and provide details of the morphology of BMI in specific situations.

Carefully done light-level histologic studies provide near-mandatory documentation in all but the most superficial investigations. Depending on the information desired from the analysis, the method of handling the specimen may vary. Tissue may be quick frozen and cryosections cut for immunofluorescent staining of adherent proteins such as immunoglobulins or coagulation factors or histochemical staining for enzymes. If the specimen is to be used for routine light microscopy or scanning or transmission electron microscopy, in general it must be fixed in solutions which denature proteins by cross-linking. The standard fixative used in gross and light microscopic pathology is 10 percent neutral buffered formalin (a 4 percent formaldehyde solution). Although formalin penetrates tissue thoroughly and rapidly, despite relatively large specimen size (up to 1 cm thick), its use generally precludes study by electron microscopy, since preservation of subcellular structures is poor. Glutaraldehyde solutions are the preferred fixatives for electron microscopy. Frequently, 3 percent glutaraldehyde is used. Although glutaraldehyde preserves ultrastructural details extremely well, solutions containing glutaraldehyde alone do not penetrate tissue well and thus the tissue must be less than 0.1 cm in maximum thickness for good glutaraldehyde fixation. Glutaraldehyde-fixed specimens must generally be partially processed within a short time after fixation, to remove excess glutaraldehyde, and arrest fixation.

In recent years, fixatives combining the best features of both glutaraldehyde and formaldehyde fixation have been developed. Two such solutions are Karnovsky's (3) and McDowell-Trump's (4). Fixation of specimens that have only thin tissue layers benefit from fixation in either of these media since they are adequate for both light- and electron-microscopy use, and thus apportionment of fragments for these various techniques may occur subsequent to fixation.

Specimens for light microscopy are generally embedded in paraffin and sectioned to 5-6 μ m thick using conventional microtomy with metallic knives. For work where a tissue/biomaterial interface is to be maintained intact, it is useful to embed specimens with glycolmethacrylate (JB-4 medium, Polysciences, Warrington, PA). Using a special microtome and glass knives, sections may be cut at 2-3 μ m, overall tissue morphology is generally superior in glycolmethacrylate. However, with plastic technique, specimen size is limited approximately 1.0 x 0.5 cm face (compared with 2 x 3 cm with paraffin), most special stains (e.g., for collagen, elastin, fibrin) are only currently being modified for this application, and cost, including materials and technician effort, are greater.

Immunofluorescence or immunoperoxidase staining for endothelial cells (factor VIII/von Willebrand's factor antigen) and various serum proteins, including immunoglobulins from surface-accumulated material may help to assess sequential development of mature pseudo- or neo-intima (5-7). Some analyses are possible on formalin-fixed, paraffin-embedded tissue. While identification of antigens using immunoperoxidase (particularly for factor VIII/VWF) techniques has been achieved on formalin-fixed specimens, the effect of glutaraldehyde on these reactions is largely unknown. Previously, autoradiography and peroxidase labeling

techniques were mandatory yet cumbersome and often imprecise for ultra-structural localization of macromolecules. Recently, low-temperature embedding colloidal gold immunochemical methods have become available and provide a high resolution, reliable and technically facile alternative to the established techniques (8-10). In general, where advanced planning is possible and specimens may be easily divided, material for immunohistochemical procedures should be separated, fresh frozen and stored for possible use at a later time. Special histochemical stains for neutrophils (chloroacetate esterase) and mononuclear phagocytes (alphanaphthyl esterase) may allow absolute identification of these cells in thrombotic deposits (11), using frozen or formalin-fixed paraffin-embedded specimens. Enzyme histochemistry may be useful for certain studies (12).

Radiographic analysis may be appropriate in some cases, especially where calcification is at issue (see chapter 14). A useful device for specimen radiographs is the Faxitron (Hewlett-Packard, McMinnville, Oregon); exposure of 35kV for 1.0 min is suitable for most studies. In vitro postimplant functional studies (e.g., on heart valve prostheses) may add important information in particular situations. In some instances, biochemical studies, with or without spectroscopic analysis, may be used to investigate uptake of blood-borne compounds (such as lipids), polymer degradation effects or deposition of minerals.

Since blood contacting surfaces of fibrillar biomaterials such as vascular conduits occasionally become collagenized, and since collagens may be differentially thromboresistant (13,14), it may become useful to determine the collagen types in a pseudointimal membrane (15). Studies describing collagen types in biomaterials-associated tissues have been reported (16,17).

The relationship of surface defects in biomaterials to microthrombi or calcification may be assessed by careful dissolution of surface protein layers or mineral deposits using a dilute solution of proteolytic enzyme (e.g., trypsin) or decalcification agent (e.g., RDO Rapid Bone Decalcifier (Dupage Kinetic Laboratories, Naperville, IL) or 5-percent formic acid), respectively, to expose the underlying surface. It is critical, however, to be extremely careful in assigning cause and effect relationship if a defect is revealed, as well as in ruling out the possibility that the dissolution treatment has, in itself, damaged the surface, since proteolytic enzymes may have polymer degradative properties under some circumstances.

It is important to keep in mind that in some cases the studies of pertinent pathology distant from the implant may be informative. For example, histologic examination of thromboemboli may allow determination of their age, the presence of renal hemosiderosis may be indicative of hemolysis (18), and visceral biomaterials emboli may be indicative of device dysfunction (19).

SPECIFIC DEVICE CONFIGURATIONS

This literature review is highly selected from available medical literature and is not intended to be exhaustive. Based on this review, test procedures of documented or potential usefulness are identified. The following devices are considered: hemodialyzers and extracorporeal oxygenators, cardiac assist devices, cardiac valve substitutes, and vascular grafts.

Hemodialyzers and Extracorporeal Oxygenators

Following bypass or dialysis, the system should be flushed with physiologic buffer and disassembled. Flushing prior to disassembly with appropriate fixatives may enhance microscopic analyses. The location of thrombi should be noted with particular reference to areas of flow separation or stasis, junctions, connectors, etc. Thrombus composition may be determined by histological studies or by analysis of relative deposition of radiolabeled species. The various materials incorporated into the system should be considered also. Control studies with the pumping system, inflow and outflow tubing, etc., are recommended.

Heart Valve Prostheses

Clinical cardiac valvular replacement has been widely available for over two decades. It is estimated that almost 1 million individuals have received substitute heart valves and that approximately 40,000 such devices are currently implanted each year in the United States. Most patients so treated have benefited by enhanced functional capacity and prolonged survival. Despite considerable evolution of surgical techniques and prosthetic design over the years, failures are commonly associated with the prostheses available (20-24). The definition of valve failure provided by the Stanford group (22,23) is widely accepted. Valve failure includes any of the following events, if they lead to reoperation and/or cause death: 1) anticoagulant-related hemorrhage, 2) prosthetic valve occlusion (thrombotic or pannus ingrowth), 3) thromboembolism, 4) prosthetic valve endocarditis, 5) hemodynamic prosthetic dysfunction (including structural failure of any prosthetic component), 6) reoperation for any other reason (e.g., hemolysis, noise, incidental), and 7) bland periprosthetic leak.

Four major designs of cardiac valvular substitutes have been widely used: caged-ball, caged-disc, tilting-disc, and tissue valves (usually the bioprosthesis, a tanned porcine (pig) aortic valve or bovine (cow) pericardial tissue mounted on a prosthetic stent). The first three are mechanical prostheses, composed entirely of nonphysiological materials. The pathology of cardiac valve replacement has been recently reviewed (25-28). The most frequent clinically encountered serious mechanical valve-related complications are thromboembolic problems: thrombosis, thromboembolism, and anticoagulation-related hemorrhage (22-24). Thrombosis may lead to local occlusion of the prosthesis or thromboemboli which may cause critical obstruction to blood flow in distant organs. Valvular malfunction may result from a small thrombus or tissue fragment

impairing movement of the occluder, or from materials degradation. Partial dehiscence (separation) of the suture line anchoring the valve occasionally occurs, leading to a paravalvular leak (regurgitation of blood around the prosthesis). Destruction of blood components on their passage through the prosthesis, especially through a degraded valve, or around the device when there is a paravalvular leak, leads to destruction of erythrocytes (hemolysis) and may cause anemia. Infective endocarditis (infection on the valve prosthesis) is a devastating and often fatal complication. Durability problems caused by materials degeneration vary widely with design and materials. Problems with valve structural deterioration have been responsible for most failures of bioprostheses (28). In addition, all valves are, to some extent, intrinsically obstructive to forward flow (26).

Each prosthesis configurational type is subject to the various complications listed above, but differences in materials of construction and mechanisms of action of particular prostheses lead to marked differences in the frequency and nature of problems due to thrombogenesis, degradation, and hemodynamic abnormalities. In general, thromboemboli and durability considerations are both materials- and design-dependent, while valve hemodynamics and hemolysis are primarily design- rather than materials-dependent.

The recommended protocol for the dissection of the heart with a valve prosthesis is modified from that described by Silver (27,29). The general schema is summarized in figure 11.1. The chambers proximal and distal to the prosthesis should be opened, but the incisions should not extend to any valve annulus containing a prosthesis. For each specimen the type (and size, if available) of prosthetic valve must be described. Keys for identification of clinically used previously and presently available heart valve substitutes have been published (27,30,31). This information is essential for understanding the types of complications which may arise. The valves should be inspected for evidence of thrombosis, infection or dysfunction. Microbiologic cultures should be taken of pertinent areas. The size and location of any dehiscence or other defect at the valve perimeter should be noted and probed to determine extent. The lining of the chambers both proximal and distal to the prosthesis may reveal alterations ("jet lesions" or generalized endocardial fibrosis) produced by turbulent blood flow. Whether from autopsy or surgery, the prosthesis should be photographed from both inflow and outflow aspects, prior to its removal from the tissue bed, if possible. Adherent material should be submitted for histologic examination to rule out a possible contribution by infection. Evidence of exuberant tissue overgrowth should be recorded.

Detailed analysis of prosthetic heart valves varies according to type and design. For mechanical prostheses, cloth wear, poppet wear (variance or abrasive wear), cage wear or misalignment, free movement of the poppet and competence of valve closure should be noted and recorded. This analysis is summarized in table 11.2. Conversely, for a tissue-derived prosthesis, the leaflets should be examined for flexibility, intactness, tears or calcification. Proper examination of clinical and experimental bioprostheses includes radiography (for

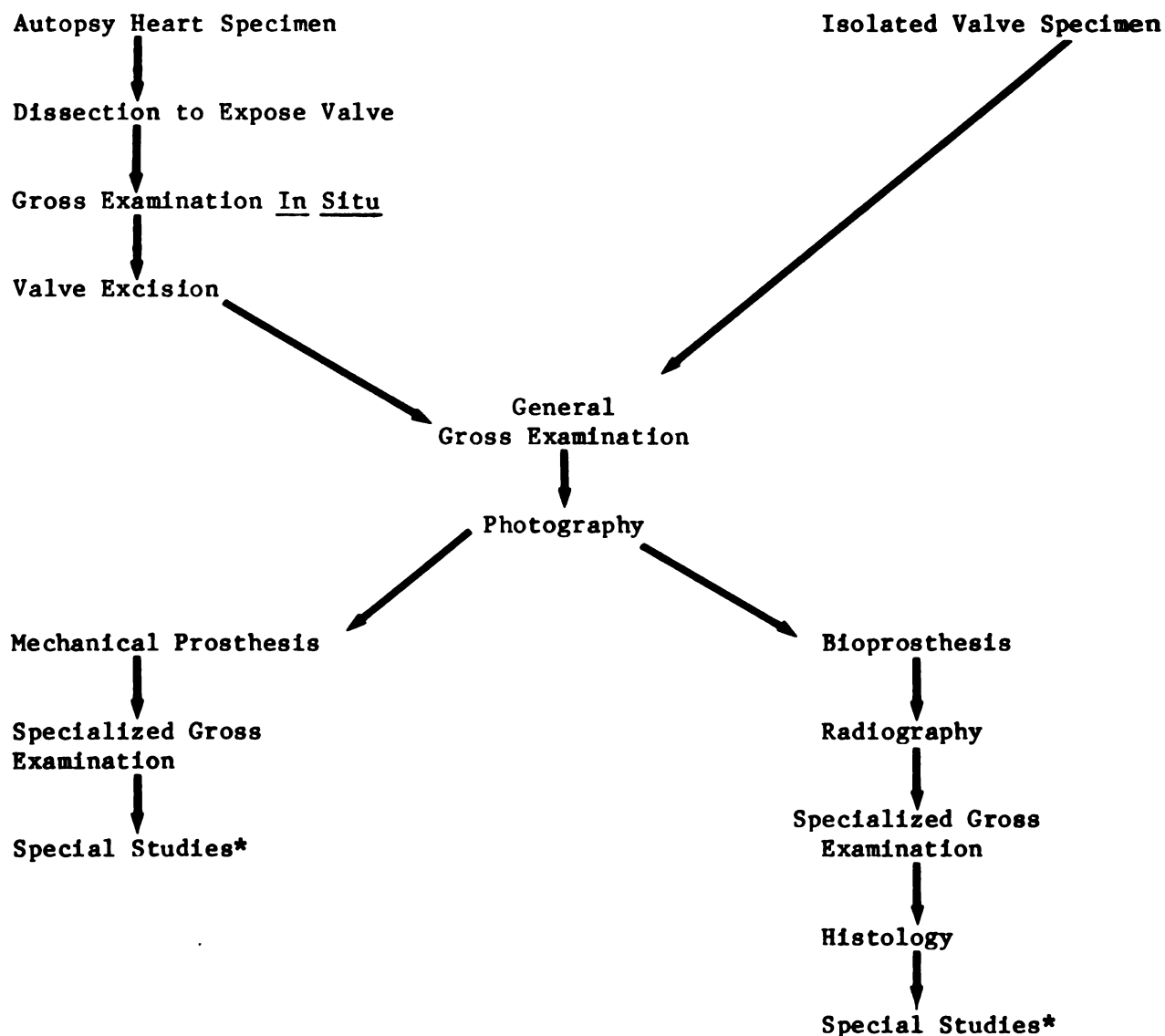


Figure 11.1--Explant analysis of cardiac valve prostheses.

***Recommended for specific indication only.**

Table 11.2--Detailed Analysis of Mechanical Prostheses

<u>Gross Examination</u>	<u>Special Studies*</u>
Poppet excursion	Poppet
Thrombi	Dimensions and weight
Vegetations	Surface roughness
Component defects and fractures	Ultrasound
Symmetry	Lipid uptake analysis
Adherent tissue	Strut/cage
	Cloth/tissue complex histology
	Scanning electron microscopy

***Recommended for specific indication only**

location and extent of calcific deposits) and light microscopy of representative histologic sections (32-34). Specific features to be assessed are outlined in table 11.3. These may be graded semiquantitatively for comparative studies. For specific valve types, special analyses and approaches may be appropriate such as lipid uptake studies on silicone poppet ball valves (35), SEM of critical wear areas (36), histology of tissue ingrowth into cloth-covered struts (37), or determination of tissue calcium concentrations in failed bioprostheses (38).

Cardiac (Ventricular) Assist Devices (VAD's) and Artificial Hearts

In the early experimental total artificial heart implantations, many animals (generally calves) died from technical and mechanical failures. As design and techniques improved, survivals were prolonged and the causes of death were usually related to pulmonary insufficiency, low cardiac output and thromboembolism (39-42). Biomaterials emboli were occasionally noted (39). Although clinical utilization of left ventricular assist devices has been reported by several investigators, there is little description of relevant pathology and patient/device interactions (43-45).

The structure of the pump bladder pseudointima (PI) has received considerably more attention in the long-term experimental studies of both smooth and rough (fibrillar) blood pump surfaces (46-53). Surface accumulations after short-term (days) assist with fibrillar bladders appear similar in man and experimental animals (43,54,55). The bladders accumulate a uniform thin time-dependent fibrin/platelet/leukocyte thrombotic deposit which is stable and nonfriable in the thickness achieved after up to 1 month in animals and up to 8 days pumping in humans. Emboli and other pump-related complications are unusual during temporary VAD support in man, with pumps utilizing either smooth or fibrillar bladders, but continued close surveillance is vital. It is probable that the pseudointimal layer will continue to thicken and may undergo some organization after prolonged clinical pumping (months) using long-term devices; it is possible that calcification, observed in long-term implants of this configuration in animals, will occur after extended duration in humans (37).

The general schema for analysis of a cardiac assist device is summarized in figure 11.2. At device removal, the pump should be rinsed with normal saline, disassembled while in saline, and the pump diaphragm and valves removed. Consideration should be given to setting aside a portion of the pump bladder, which will be fresh-frozen rather than fixed. This will allow immunohistochemical or biochemical studies to be done, if necessary. Following careful examination of the specimen grossly and under a dissecting microscope, and subsequent macroscopic photographic documentation, the pump bladder and valves are radiographed to document calcification, should it exist. It is important to be mindful of the fact that specimens may be undergoing autolytic degradative changes during all prefixation manipulation and documentation; thus, it is important to either freeze or place specimens in fixative as soon as possible after explantation. Furthermore freeze/thaw cycles lead to considerable deterioration in tissue morphology due to ice

Table 11.3--Detailed Analysis of Bioprostheses

Gross Examination

Cuspal stiffness

Tissue overgrowth

Thrombi

Cuspal hematomas

Calcification

Vegetations

Fenestrations and tears

Strut relationships

Radiography

Calcification

Degree

Localization

Cuspal body

Commissures

Basal attachments

Free edge

Histology

Vegetations

Thrombi

Host cell interactions

Calcification

Degree

Morphology

Location

Surface

Spongiosa

Fibrosa

Ventricularis

Atrial muscle shelf

Aortic wall

Miscellaneous Analyses*

Calcium/phosphorous determination

GLA-proteins

Lipids

Immunohistochemistry

Transmission electron microscopy

Scanning electron microscopy +/-EDAX

***Recommended for specific indication only**

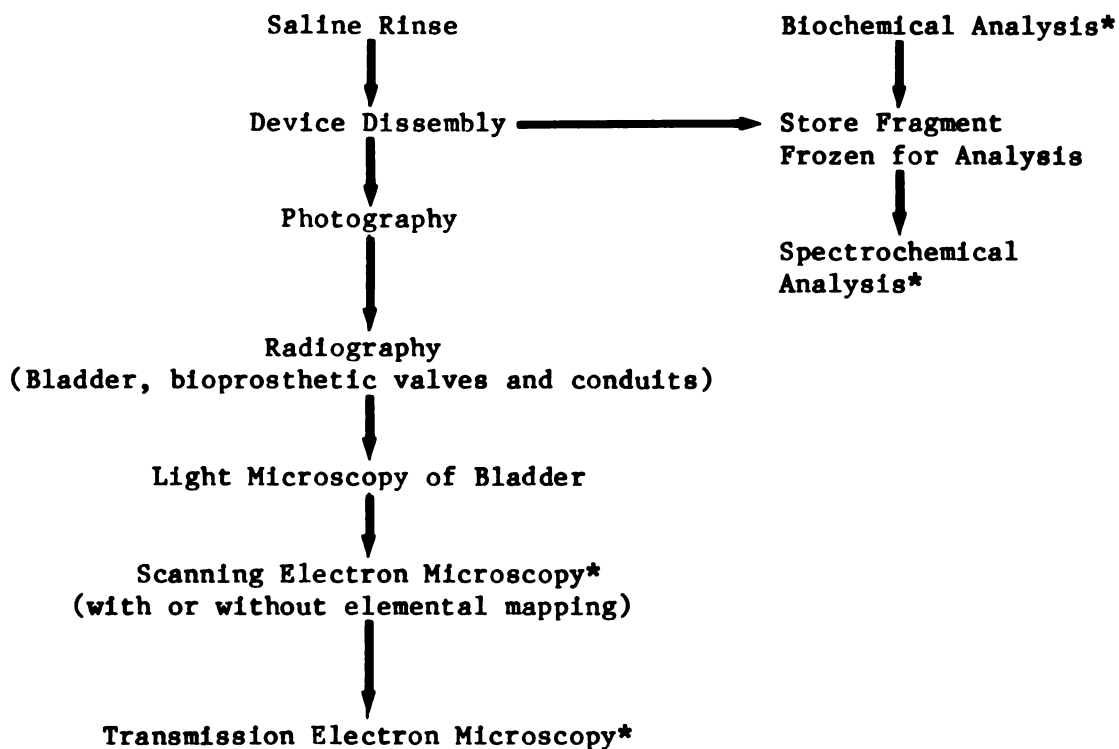


Figure 11.2—Explant analysis of cardiac assist device or artificial heart.

*Recommended for specific indication only.

crystal formation, so that previously frozen tissue is generally not suitable for good morphologic studies. Sections for LM and SEM should be selected from the inflow, intermediate and outflow portions of the pump bladder and valves, as well as any other pertinent areas, and subsequently prepared according to standard procedures for LM and SEM. LM of selected paraffin-embedded and stained cross sections may be used to map pseudointimal thickness variations along the pumping bladder surface and to identify cellular and noncellular components such as platelets, leukocytes, fibroblasts, endothelial cell, fibrin, collagen and calcium deposits. SEM may document surface characteristics of the PI such as smoothness, uniformity and the presence of specific cells and thrombi. Sections of bladder with associated surface deposits should be examined by LM in cross section preferably using glycol methacrylate-embedded sections to facilitate maintaining intact the tissue/biomaterials interface during sectioning. Where the PI is sufficiently thick, it may be lifted from the underlying polymer by sharp dissection and the fragments processed in paraffin.

Vascular Grafts

As discussed by Silver and Wilson (27), the tissue processes that occur when a vascular prosthesis is implanted must be understood if complications are to be appreciated. The experimental investigations and human autopsy studies done over the past three decades by Szilagyi et al. (56), Sauvage et al. (57), Wesolowski (58,59) and others have provided much knowledge in this area. As long as healing of a vascular graft remains incomplete, and the bloodstream interface consists of layers of platelet-fibrin aggregate, the risk remains of distal embolization or thrombotic stenosis with ultimate occlusion. Only when the entire graft is incorporated in a fibrous tissue matrix, and its luminal surface is covered by endothelium, can healing be considered complete.

Man's ability to heal prostheses in the cardiovascular system may be limited in degree and rapidity when compared with that of animals such as the dog, pig, calf, and baboon (57). Nevertheless, focally complete healing, including surface endothelialization, of arterial prostheses may be feasible (60). There are three potential sources of endothelium in graft healing: 1) ingrowth across anastomotic sites from the host vessel; 2) differentiation of cells growing through the interstices of the graft wall; and 3) differentiation of pluripotent mononuclear cells deposited from the flowing blood (27,61,62). While the first mechanism is of undisputed relevance, the role of the other two is controversial. (In humans, endothelial cells typically extend 10 to 15 mm across an anastomosis.) This allows man to heal fabric patches placed in the heart as well as the sewing rings of heart valve prostheses, and to coat the cloth surface of the various metal rings, etc., used to reconstruct diseased heart valves. However, it is generally insufficient to heal long, synthetic vascular grafts.

At animal sacrifice, or human post mortem examination, the location, length and diameter of the graft should be noted, as well as thrombus formation and coverage by endothelial or other cell types with particular reference to the anastomoses. Cell type should be documented.

The extent of surface coverage, cell morphology, and the thickness of ingrown or deposited layers should be assessed. Analysis of vascular graft neointima using a multifaceted histomorphologic and immunocytochemical approach has been described (63). Consideration should be given to the possibility of fixation by physiologic pressure perfusion in vivo with glutaraldehyde solutions. While this is generally the most favorable procedure for preservation of fine structural detail for transmission electron microscopy, it clearly precludes the selection of unfixed specimens for immunohistochemical or biochemical analysis. This again emphasizes the need for a carefully designed protocol for specimen harvest and analysis, in order to most efficiently address major issues.

When vascular grafts are examined at autopsy, incomplete healing may be obvious with large exposed areas of graft pseudointima. However, it is easy to mistake a smooth, yellow-white, glistening surface of unorganized, compacted fibrin for an endothelialized intima, so that before making statements regarding endothelialization it is necessary to examine the flow surface in detail, by Evans blue staining, by LM of conventional paraffin-embedded sections, and/or by gross Evans blue staining characteristics (1,2), and by SEM which allows detailed examination of large areas of the graft surface. The extent of anastomotic and other fibrous neointimal hyperplasia should be documented. Furthermore, any tissue reaction or perigraft hematoma formation should be noted. Distal organs must be examined carefully at autopsy for embolic damage.

Measurements of prostacyclin and thromboxane B₂ production may be useful to further characterize neointimal function (64).

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BIOLOGICALLY DERIVED SURFACES

FRESH AUTOLOGOUS VESSEL

The preferred graft for coronary artery bypass or for lower leg arterial reconstruction is a freshly dissected, reversed, autologous saphenous vein (1,2). Coronary artery bypass with this graft is performed with low morbidity and mortality and high early patency rates and is generally more successful than lower leg artery replacement. However, medial fibrosis and intimal hyperplasia often superimposed with atherosclerosis and/or thrombosis are long-term problems which can lead to late graft occlusion in either location.

To qualify for use as an arterial graft, it is essential that the vein be of good quality. The quality of the vein is determined by its diameter (preferably more than 4 mm at the smaller end), its wall structure, and the presence or absence of varices, fibrotic valves, and phleboscclerosis (3). In 10-25 percent of cases, the vein may be unavailable or of inadequate quality by the above criteria. Szilagyi (4), for lower limb arterial grafts, found a 48-month cumulative patency rate (5) of 62 percent for veins of good quality versus 28 percent for those of poor quality.

The most important variables in preparation of a vein graft appear to be: Distension pressure, components of the perfusate, and temperature. The optimum combination of variables (table 12.1) should prevent vein contraction (spasm) and thus prevent endothelial cell protrusion into the lumen, which leads to endothelial sloughing (6,7). The pressure to which the vessel should be distended is somewhat controversial. Pressures of less than 100 mm Hg are recommended by most studies (6,8). However, Bonchek points out that leaks are sometimes not discovered at pressures less than 300 mm Hg (9). Bonchek notes that pressures above 500 mm Hg can injure the vessel and that these pressures are often produced by irrigating the vein with a manually controlled syringe. Contraction of the smooth muscle cells can be controlled by perfusing the vessel during dissection with a saline solution containing papaverine, a vasodilator. The distension pressure becomes even more critical in papaverine-treated veins, as the veins can easily be overstretched such that gaps between the endothelial cells are created. The level of papaverine must also be carefully monitored, although the dosage used locally in experimental animals has no systemic effects (7). The perfusion components favored by some investigators (6,7) are nutritive saline solutions such as tissue culture medium with added papaverine, while others favor blood (8,9). Using blood as a storage medium may not be optimal because it contains vasoconstricting substances, and platelets tend to adhere to the luminal surface (6). Temperature of perfusion and storage is another significant variable (10). For distention and perfusion, 37°C is recommended. Haudenschield (7) recommends 37°C for storage while others recommend 10°C (6) or 4°C (8).

Table 12.1--Recommended Variables for Fresh Vessel Preparation

<u>Distension Pressure</u>	<u>Vessel</u>	<u>Storage Temperature</u>	<u>Components of Perfusate</u>	<u>Reference</u>
100-500 mm Hg	Rabbit common carotid (n = 20 rabbits)	37°C	0.9% NaCl or tissue culture medium (heparinized + papaverine)	7,10
<100 mm Hg	Dog cephalic veins (n = 10 dogs)	10°C	Plasmalyte ^R (heparinized + papaverine)	6
<100 mm Hg	Human saphenous vein (n = 20 veins)	4°C	Heparinized blood	8

ISOLATED LIVING CELLS

The addition of living cells to the blood-contacting surfaces of cardiovascular prostheses can increase their long-term thromboresistance. Living cells have been incorporated into prostheses by 1) seeding them onto the graft in a preclotting procedure immediately after harvest and immediately before implantation; 2) harvesting the cells, growing them in tissue culture, and then seeding the prosthesis and implanting; and 3) growing cultured cells directly onto the prosthesis. These methods appear to convey the same degree of thromboresistance, and their advantages and disadvantages are listed in table 12.2. Note that pros and cons of "prosthesis culture" include those of "cell culture," because usually cell culture precedes prosthesis culture. One drawback to the implantation of living cells is the questionable immunologic status of any implanted cells that were not derived from the individual receiving the implant. Results of implantation of isolated living cells are summarized in table 12.3 and discussed as follows.

Xenografts

Living cells or tissue from one species implanted into a different species constitutes a xenograft (heterograft, xenotransplant). Immunologic rejection invariably occurs with implants (11), consisting of leukocyte infiltration, presence of giant cells, fibrinoid necrosis of the foreign cells and stroma, and local edema. The rate and extent of the pathologic reaction varies with the phylogenetic distance between the species and other factors such as tissue vascularity and immunocompetency of the host. For example, human fibroblasts cultured on Dacron velour and implanted as blood-contacting surfaces in dogs were more thrombogenic than controls and were found to be immunologically rejected at necropsy after 21 days (12).

Allografts

Living cells or tissue implanted in one individual from another of the same species is an allograft (allotransplant, homotransplant, homograft, allogeneic transplant, or homologous transplant). Allografts are subject to immunologic rejection by the host, but not as rapidly or with as extensive a reaction as xenografts.

Allogeneic fibroblasts cultured on microfabric substrates (13) and implanted into young calves as aortic grafts had more than twice the depth of thrombus buildup as controls (1-16 mm) (14).

Fetal allogeneic fibroblasts as linings for left ventricular assist devices (LVAD's) in calves have proved more successful. Fibrous linings of these bladders after implants of 11 months were rarely greater than 1.0 mm in depth whereas control linings varied from 5-8 mm (15). Occasional sites of calcification were present in both but were diminished in seeded implants. Radioautography on portions of the lining at autopsy allowed identification of the cultured fibroblasts, which had been labeled with tritiated thymidine prior to implantation. Instron peel tests showed fibroblast-seeded pseudointimas were more tightly bound to the polymer substrate than control linings on polyester

Table 12.2--Advantages and Disadvantages of "Cell Seeding," "Cell Culture," and "Prosthesis Culture"

	<u>Cell Seeding</u>	<u>Cell Culture Prior to Seeding</u>	<u>Prosthesis Culture</u>
Advantages	Requires no special tissue culture facility	Controlled <u>in vitro</u> testing of cells can be done	Completely endothelialized grafts can be achieved
	One surgical procedure (one or two sites)	More cells can be obtained and therefore larger areas endothelialized	Improves small artery patency without pharmacologic intervention
	Pure culture not required	Two surgical procedures (one site each)	Fragile monolayer may be disrupted during implantation or after implant due to blood flow
	Immediate availability of cells	Pure culture may be required	Special medium perfusion may be required through long tubular prostheses during culture
Disadvantages	Rapid procedure		
	No characterization of cells prior to implant	Possible cell transformation in culture	Longer period of culture needed to achieve coverage
	Potential for early thrombosis without pharmacological intervention	Possible loss of cells from bacterial contamination or unknown causes	Other disadvantages of cell culture prior to seeding
	Atherogenic potential of seeded smooth muscle contaminants	Requires special tissue culture facilities	
	Probably not applicable to small artery or venous prostheses	Not available for immediate applications	
	Requires more donor vein to seed an equal graft area		

Table 12.3-Implantation Results

Type of Transplant	Donor/Recipient	Cells	Substrate	Cultured Prior to Seeding (C/S), or Seeding (S), or Cultured (C)		Implant Procedure	Result	Reference
Xenograft	Human/dog 2 exptl/ 2 control	Skin fibroblast	Dacron velour 1608 (USCI)	C		Atrial implant	Acute immunorejection of cell lined materials	12
Allograft	Calf 20 exptl/ 8 control	Fetal fibroblast (ligamentum nuchae)	Fibrillar polyurethane	C/S		LVAD 30-335 days	Cell lined LVAD's were thinner and more collagenous than surfaces of unseeded devices	15, 16
Autograft	Calf	Granulation tissue fibroblasts	Microfabric B (Union Carbide) polyester micro-fabric (Abcor)	C		Aortic grafts 1-6 months	Cell lined grafts developed more thrombus than controls	14
Autograft	Calf	Carotid artery EC ¹	Microfabric B (Union Carbide) polyester micro-fabric (Abcor)	C		Aortic graft 1,4,8 weeks	Microfabric B grafts-cell lined and nontrombogenic polyester grafts-nontrombogenic where cell lining was intact	19
Autograft	Calf 7 exptl/ 4 control	Aortic EC	Microfabric B (Union Carbide)	C		Atrial patch 1 week	EC lined areas were thrombus-free	20
Autograft	Calf 6 exptl	Carotid artery EC	Microfabric B (Union Carbide)	C		LVAD 8 hours-4 weeks	Loss of EC lining between 8 and 16 hours after implant	21
Autograft	Dog 14 exptl	Saphenous vein EC	14 different materials	S		Aortic graft 2 weeks	Best endo-thelialized surfaces-Weft knit Dacron materials	30

Table 12.3 (continued)

Type of Transplant	Donor/ Recipient	Cells	Substrate	Cultured Prior to Seeding (C/S), or Seeding (S), or Cultured (C)		Implant Procedure	Result	Reference
				C/S	S			
Autograft	Dog 7 exptl/ 6 control	Jugular vein EC	Dacron Double velour	C/S		Thoracoab- dominal (TA) Bypass 1 day- 4 weeks	All patent-- Controls 10% EC lined, exptls 60-70% lined at 2 weeks 80% lined at 4 weeks	24
Autograft	Dog 14 exptl/ 14 control	Jugular vein EC	Dacron double velour	C/S	S	TA Bypass 1 day- 4 weeks	All patent-- 1-7 days: C 81% clot free E 95% clot free 2-4 weeks: C 40% clot free E 88% clot free	25
Autograft	Dog 8 exptl/ 5 control	Jugular vein EC	Expanded PTFE (Goretex)	C/S		TA Bypass 2,4 weeks	All patent-- 64% EC covered exptls 2 weeks 91% at 4 weeks None of controls EC covered Controls-progressive accumulation of thrombus	31
Autograft	Dog 14 exptl/ 14 control	Jugular vein EC	Weft knitted noncrimped filamentous Dacron velour, externally sup- ported (USCI)	C/S		Iliofemoral bypass 2,4 weeks 2,4 months	2 weeks--all patent cum. patency rate from 4-16 weeks= 27% controls, 73%exptls exptls EC lined by 4 weeks	26
Autograft	Dog 8 exptl/ 9 control	Jugular vein EC	Double velour knitted Dacron (Meadox)	S		TA bypass 7-8 weeks	All patent--platelet PST returns to normal sooner in seeded grafts. More EC lining exptls than controls	29

Type of Transplant	Donor/ Recipient	Cells	Substrate	Cultured Prior to Seeding (C/S), or Seeding (S), or Cultured (C)	Implant Procedure	Result	Reference
Autograft	Human 18 seeded 20 unseeded	Convenient Superficial vein exclusive of saphenous	Knitted Dacron	S	"Extra-anatomic" bypass	Seeded insignificantly better patency than unseeded	56
	Human 18 seeded 17	Saphenous vein rejected for graft material	Knitted Dacron	S	Femoral-popliteal bypass	Seeded patency > unseeded in non-smokers Unseeded patency > seeded in smokers Lipid vacuoles in smokers' smooth muscle cells	
Autograft	Dog 23 seeded	Saphenous vein EC	Knitted Dacron 3-4 mm ID	S	17 femoral artery 7 femoral AV fistula 3 iliac AV fistula 5 carotid artery up to 1 month	100% occlusion	55
	Dog 67 cultured	Jugular vein EC	54 ePTFE 10 Dacron 3 tanned umbilical vein	C	21 carotid artery up to 1 month	6 patent (without anticoagulant) lining stable 1 month	
				9 occluded < 10%		coverage 1 occluded-mixed culture 5 occluded	

flocked and fibrillar polyurethane bladders. Histologic and biochemical data showed that the linings were largely types I and III collagen.

Fetal fibroblasts are obtained from ligamentum nuchae, excised under sterile conditions from 2-4 months gestation bovine fetuses (15,17). The tissue is rinsed in sterile saline, minced and explanted (positioned to allow adherence to the bottom of the flask), then fed with a complete medium. When confluent, these cells are passed three to six times, then frozen in liquid nitrogen for storage (18). Several days prior to implant, the cells are defrosted and grown in monolayer culture, then seeded onto the pump surface 3 hours before surgery.

In order to test the immunologic compatibility of the allogeneic cells, Holstein fibroblasts were implanted into Hereford recipients (15). Histocompatibility differences between these breeds were detected by mixed lymphocyte culture and fibroblast lysis as produced by in vitro sensitized lymphocytes. Thus, the explanation of the lack of rejection of cell-lined LVAD bladders is not clear.

Autografts

Tissue or cells moved from one location to another in the same individual is an autograft (autotransplant; cells are autologous or autogenous). No immunologic reaction occurs, but these tissues may elicit an inflammatory reaction if they become necrotic before or after transplant.

Autologous fibroblasts seeded onto microfabric substrates and implanted as aortic grafts in calves accumulated more thrombus than unlined control surfaces, whereas autologous endothelial cells prevented thrombus formation (19). Endothelial cells were deposited on "microfabric" substrates of polyester or polypropylene fibrils (2 μ m in diameter, 25 μ m thick) extruded onto a nonporous backing (13). These materials without endothelial cell linings were very thrombogenic; and, although intact endothelialized surfaces were produced in vitro, they were too fragile and became disrupted after implantation. As linings for aortic grafts, which are hemodynamically simple, these surfaces performed well (19). In geometrically complex devices with compliant walls or moving parts, the cells detached (20,21). However, the cell linings were thromboresistant when they remained intact.

Endothelial Seeding

Thromboresistance of synthetic vascular grafts was increased by including freshly harvested autologous endothelial cells in blood used to preclot the grafts (22). This technique has since been further investigated by others (23,26). Endothelial cells were harvested from 12 cm segments of canine external jugular veins and cultured for 2 weeks. The cultured cells were mixed with blood, and 6 mm thoracoabdominal grafts were preclotted (27) and implanted. Although all

grafts were patent 2 and 4 weeks after implantation, seeded grafts were 60-70 percent covered with endothelial cells [as assessed by factor VIII immunofluorescence (28)] at 2 weeks and at least 80 percent covered at 4 weeks. Control grafts, preclotted with blood alone, were not endothelialized. The thickness of the pseudointima developed was 30-50 μm on the controls and 10-20 μm on the seeded grafts.

A parallel investigation was carried out on freshly harvested cells which were not cultured prior to preclotting. Autopsies on days 1 through 8 showed all grafts were patent. On days 1 through 4 the surfaces of control vs. seeded grafts were histologically indistinguishable. This means the endothelial cells were not "filtered out" at the luminal surface during preclotting. The 2- and 4-week grafts were 88 percent thrombus-free compared to 40 percent of the controls, confirming the finding with the grafts seeded with cultured cells.

The sequence of healing in the endothelialized grafts is distinguished by clear areas where fibrin has been lysed beneath the endothelial cells as they migrate to the lumen. Perigraft ingrowth of host connective tissue occurs as in control grafts. However, the connective tissue cells do not produce the many layers of "intimal thickening" luminal to the Dacron velour which is seen in the controls (23).

Substrates

The techniques of producing precultured endothelialized surfaces, and seeding material with blood containing endothelial cells, may require different substrates. Choice of substrate is just as critical as choice of cell line in determining the fate of the implant, if the cells are cultured directly on the substrate. If the cells are seeded in a clot, substrate may be less important, as the cells do not contact it directly.

The endothelial seeding and fibroblast seeding techniques are being used for different prostheses. In porous grafts, in which perigraft healing along the length of the graft occurs, endothelial seeding increases thromboresistance. The effect of fibroblast seeding is to limit neointimal growth on the surface of nonporous blood pumps in which healing must occur (in the absence of seeding) through chance deposition of cells from the flowing blood. Whether the substrate is porous or nonporous dictates the healing process.

In arterial replacement, the biomaterials in current use, knitted Dacron velour and expanded PTFE, have been seeded with endothelial cells for implantation. The results were variable (30) and emphasize the importance of material properties (e.g., porosity, fiber size) in the ability of endothelial cell seeding to provide a thromboresistant graft. Efforts to seed expanded PTFE with endothelial cells have been less successful, in terms of efficiency of seeding or cell retention, than Dacron materials which are preclotted. However, the seeded PTFE grafts seem to be as thromboresistant as the Dacron grafts (31).

Identification of Cultured Vascular Cells

One problem in determining the identity of any cell in culture is that it has been removed from the tissue of origin and adapted to a foreign environment. Thus, structure and function may change. Sometimes cells alter drastically such that little expression of the original cell type is evident; that is, the cells dedifferentiate. Markers of function unique to the particular cell type should be monitored, along with morphological properties, to assure maintenance of the differentiated state in vitro.

The best way to be sure of the identity of cultured cells is to harvest them from tissues which contain only one cell type. In the case of the endothelial cells, this is almost impossible. Freshly harvested endothelial cells are almost always contaminated with underlying smooth muscle cells. The extent of this contamination varies with the structure of the vessel from which the cells are harvested, e.g., dog carotid artery endothelial cells are rarely subject to smooth muscle contamination in contrast to bovine aortic endothelial cells. It is important to be able to harvest endothelial cells in as pure a state as possible because blood contacting smooth muscle cells may prove atherogenic after implantation, and because smooth muscle cells tend to overgrow endothelial cells in culture. This holds true for bovine aortic and human umbilical vein endothelial cells (the most widely used and well characterized cultured endothelial cells).

The most obvious property of a cultured cell is its morphology. There are several commonly used terms to describe cells in culture at the light microscope level. Polygonal, or epithelioid cells are rounded or circular, with no major axis or processes (endothelial cells, figures 12.1 and 12.2). Spindle-shaped cells are more irregular and elongated, usually somewhat round in cross section (smooth muscle cells, figure 12.3). Fibroblastic cells are elongated but may have pointed processes and are flatter in cross section (fibroblasts, figure 12.4). These terms are very subjective and relative, but are commonly used.

1. Endothelial Cells (figures 12.1 and 12.2)

Under the light microscope, endothelial cells (EC) in culture form a "cobblestone" monolayer of polygonal cells which are contact inhibited. Contact-inhibited cells do not grow on top of each other, and mitosis slows to a minimal rate once the substrate is covered with a monolayer of cells (i.e., confluence has been reached).

This growth pattern is unique among the vascular cell types and is the most obvious hallmark of EC. However, the culture must be confluent before it is judged by this criterion to be free of other cell types. Freshly harvested EC attach as clumps of polygonal cells, but subcultured cells that are dispersed into single cells and replated attach singly and assume spindle shapes. Some EC tend more toward a spindle

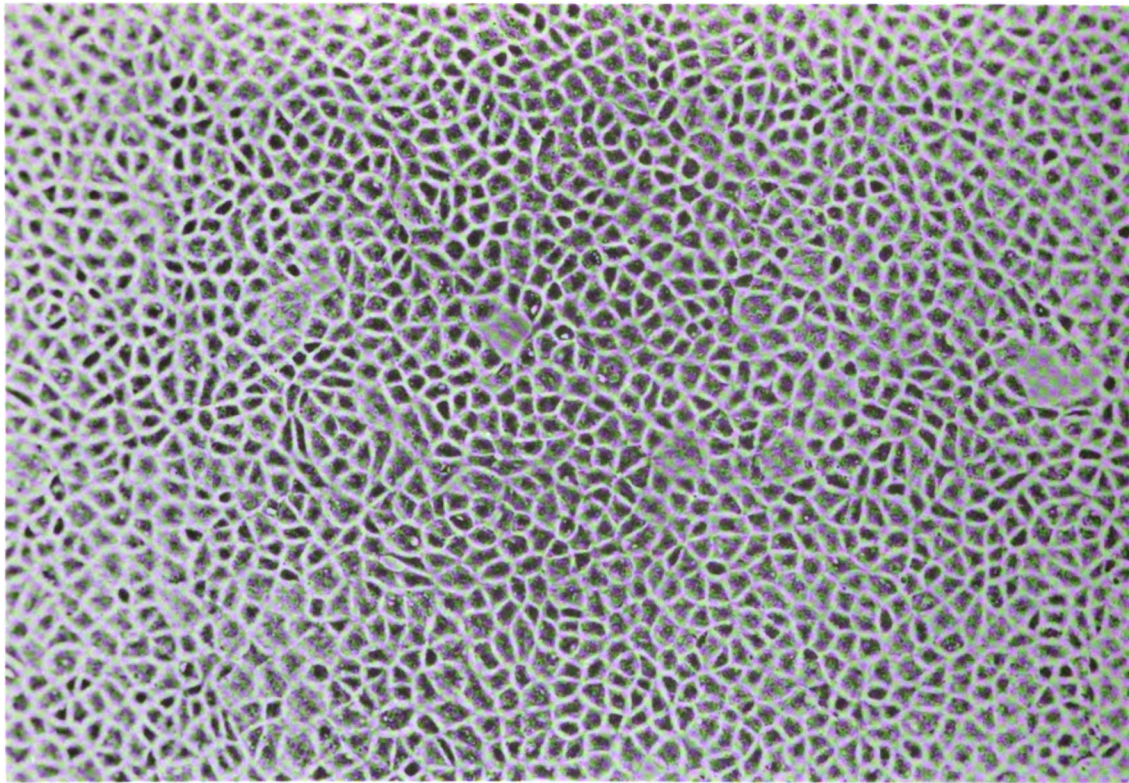


Figure 12.1--Bovine aortic endothelial cells, sixth passage, x100.

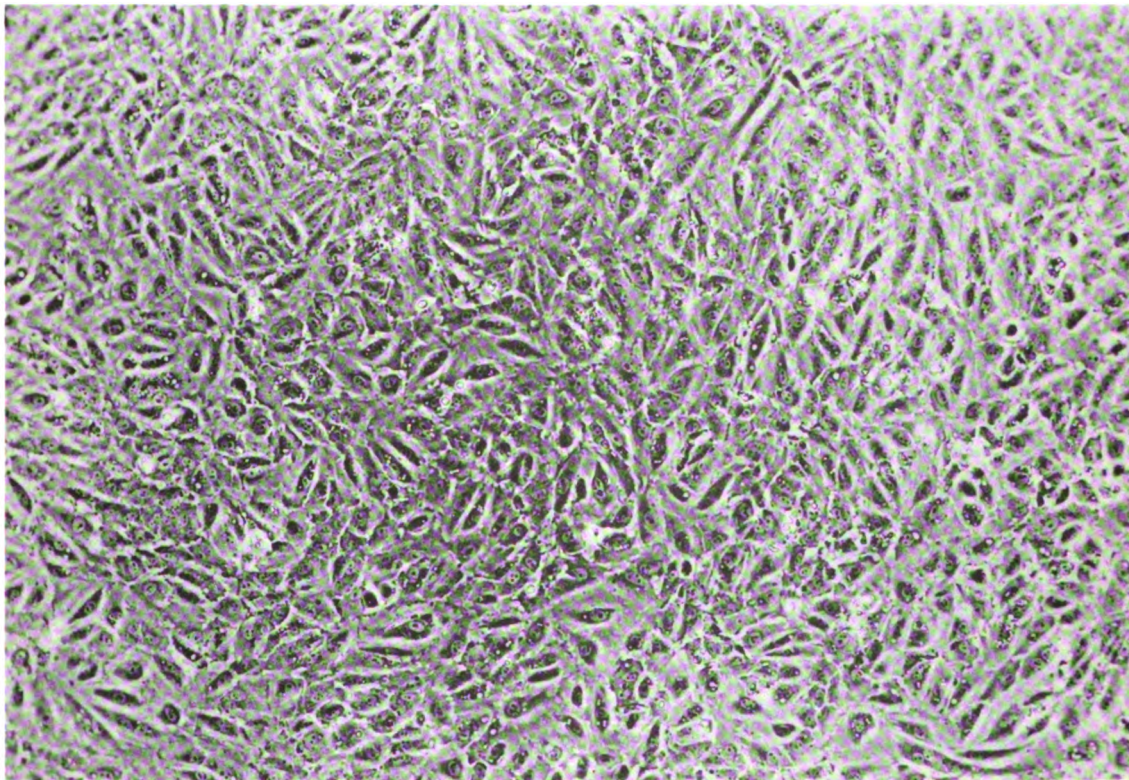


Figure 12.2--Human umbilical vein endothelial cells, primary culture, x100.

than a polygonal shape, and the monolayer they form tends to be less the classical "cobblestone" pattern, e.g., human umbilical vein EC.

Ultrastructurally the EC have prominent nucleoli, many mitochondria, extensive endoplasmic reticulum and Golgi, and many pinocytotic vesicles. The peripheral cytoplasm usually has microfilaments. The edges of EC form overlapping articulations with neighboring cells and have gap junctions continuous with tight junctions (32).

The following properties are sufficiently unique to EC to be considered markers or allow them to be distinguished from other vascular wall cells.

- a. presence of Weibel-Palade bodies in some cells (notably human umbilical vein EC (33)).
- b. presence of vWF (von Willebrand's factor (34)).*
- c. absence of platelet adherence in vitro (35).
- d. presence of angiotensin converting enzyme (36).
- e. presence of blood group antigens in some cell lines (37).

The Weibel-Palade bodies, organelles unique to EC are found in about 30 percent of the EC depending on vessels and species (33). However, they are rarely seen in bovine or canine aortic EC or canine jugular vein EC. Their presence does not indicate culture homogeneity; it simply means that one particular cell is an EC. vWF is found associated exclusively with platelets, megakaryocytes, and EC. Its presence is demonstrated by staining histochemically with antibody to vWF (34). The staining pattern is punctate in the cytoplasm. Care must be exercised in interpretation of data. Antibodies can be contaminated with nonspecific material (e.g., commercially available factor VIII-related antigen has been found to be contaminated with fibronectin antibody). Also, interpreting the brightness of a reaction is a subjective judgment which is difficult to quantitate. However, this problem is minimized by use of the unlabeled antibody peroxidase-antiperoxidase (PAP) method (38). vWF is species specific in many cases, e.g., human anti-vWF does not react with canine antigen. Finally, as with Weibel-Palade bodies, the presence of vWF positive cells does not insure the purity of a population. Another property of EC which could be quantitated is their resistance to platelet adherence in vitro (35). Platelets adhere to smooth muscle cells and fibroblasts in vitro. "Passivated" surfaces which appear to be modified smooth

*vWF refers to von Willebrand's factor, that factor referred to in the literature as factor VIII-related antigen.

muscle cells and develop on prostheses in vivo (neointimas) are resistant to platelet adherence.

Human vascular endothelial cells have the major blood group (ABH or ABO) antigens of the donor, whereas vascular smooth muscle cells and fibroblasts do not (37). H-antigens were present in 20-60 percent of endothelial cells cultured from human umbilical veins. A- and B-antigens were less frequently observed. To demonstrate the ABH antigens, mixed cell agglutination reactions were used, in which cultured cells were presented with anti-A serum (for example), then washed, and reacted with A-erythrocytes. Mixed cell agglutination is a more sensitive assay than immunofluorescent techniques, which failed to demonstrate ABH antigens on endothelial cells.

The HLA (human leukocyte antigens), a separate histocompatibility system from the ABH system in humans, evoke strong allograft immunity. When tissue-typing is carried out for organ transplant, it is the HLA which are characterized. The HLA are found on the surface of lymphocytes and macrophages. Human umbilical vein endothelial cells also have some antigens of the HLA complex (39). Human endothelial cells may have additional antigens, not present on the autologous lymphocytes (40).

PGI₂, plasminogen activator, and tissue factor are important endothelial cell products for inhibiting thrombogenesis (41), promoting fibrinolysis (42), and promoting thrombogenesis (43), respectively. However, they are also produced by vascular smooth muscle cells; thus, they are not markers for endothelial cells.

Recent reviews (44) and symposia (45,47) on endothelial cell function are available.

2. Smooth Muscle Cells (figure 12.3)

Under the light microscope, these are spindle-shaped cells when sparse. As they multiply they form whorls and tend to pile up forming "hills and valleys" (48,49). The "hills" are formed by 5-10 cells in which cells are randomly arranged. The "valleys" are comprised of monolayers of cells.

Smooth muscle cells (SMC) can assume either a "synthetic" phenotype or a "contractile" phenotype in vitro as well as in vivo (49,50). The synthetic phenotype is characteristic of proliferating cells and resembles a fibroblast. The contractile phenotype is a differentiated cell characteristic of the confluent culture, in which the cells have the same ultrastructural properties as medical SMC in situ and are theoretically capable of contraction in vitro. The synthetic phenotype results from in vitro modulation of the differentiated SMC. Modulation refers to reversible dedifferentiation (49), or the loss of differentiated properties

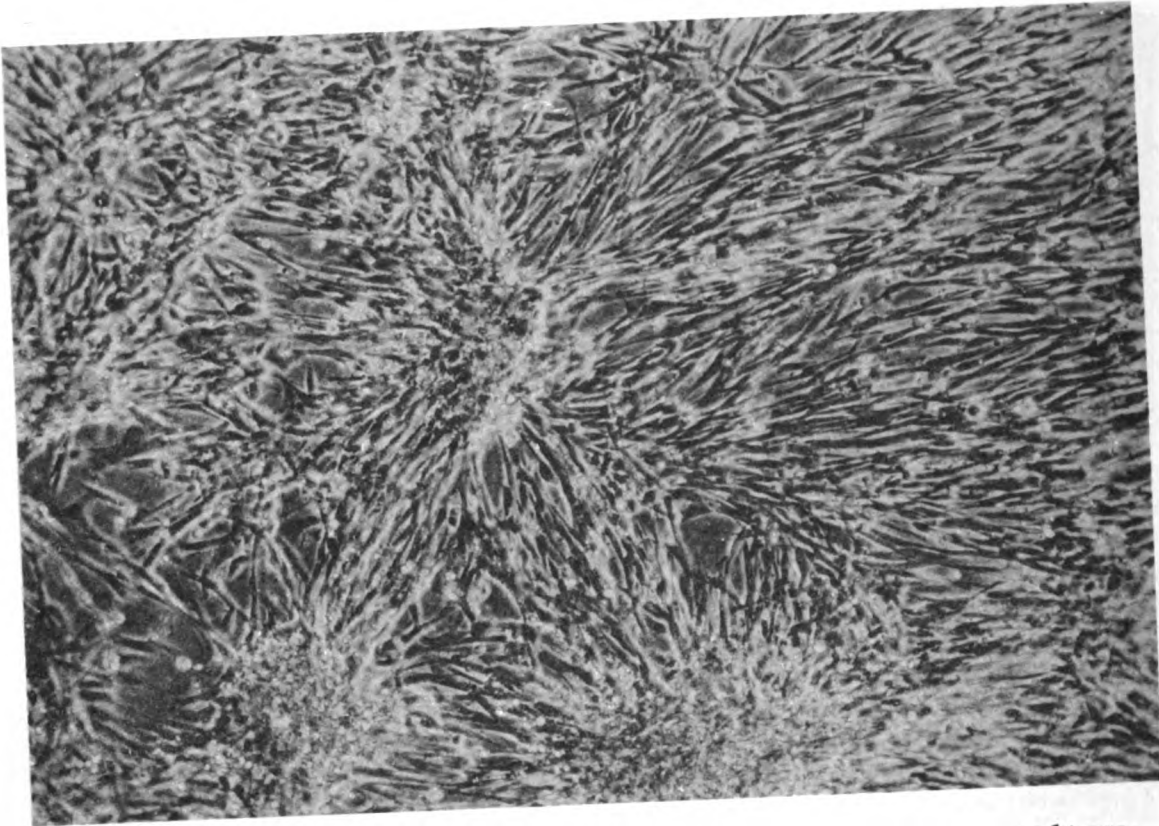


Figure 12.3--Dog saphenous vein smooth muscle cells, primary culture, x100.

which may be regained in time. How long a population of SMC remains in the synthetic phenotype depends on the method of harvesting, the density of seeding, and the time in culture. They may never regain the contractile phenotype. Contraction, per se, has rarely been demonstrated in vitro in vascular SMC, although it would be an excellent criterion by which to distinguish SMC from fibroblasts (51).

SMC of the "contractile" phenotype are characterized by an abundance of cytoplasmic microfilaments containing dense bodies, fewer mitochondria, and less endoplasmic reticulum than fibroblasts, and these organelles are concentrated around the nucleus. The most diagnostic ultrastructural feature is a basal lamina around the cell membrane; but it is not always present in vitro.

SMC have been positively identified in the synthetic phenotype by demonstrating the presence of SMC myosin, SMC actin, tropomyosin, and intermediate filament protein, with fluorescein-conjugated antibodies against each protein (52). It is necessary to obtain antibodies produced against these proteins in vascular smooth muscle cell', as they do not cross react with these proteins in other smooth muscle types. Such antibodies are not widely available, and their use and interpretation requires extensive experience.

In short, it is difficult to positively identify vascular SMC in culture, as there are not consistently reliable properties which distinguish them from fibroblasts.

3. Fibroblasts (figure 12.4)

Fibroblasts, the cells which make up the adventitial layer of the vessel wall, are elongated cells (fibers) which grow in concentric whorls in culture, in which the cells are parallel to each other (53). When sparse they cannot be distinguished from SMC. The function of these cells is to synthesize and secrete extracellular connective tissue proteins, principally collagen and elastin; and these proteins are ultrastructurally and biochemically evident in cultures after confluence is reached.

Fibroblasts are rich in cytoplasmic organelles associated with synthetic activities. They also contain actin filaments, although not as many as smooth muscle cells.

Conclusions

The clinical use of isolated living cells as nonthrombogenic linings for cardiovascular prostheses merits further investigation. Some considerations are:

1. Whether or not the cells must be autologous.

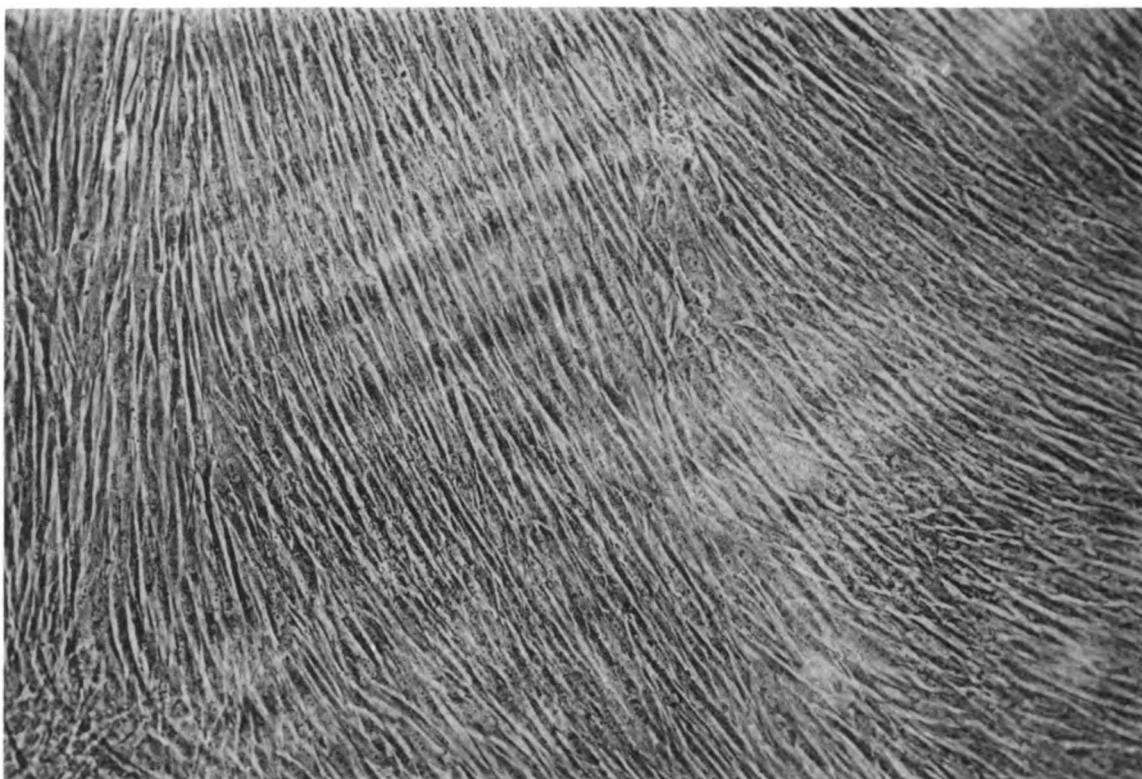


Figure 12.4--Human foreskin fibroblasts, fourth passage, x100.

2. Whether culturing the cells prior to seeding is advantageous.
3. Whether the cells adhere to the graft material and remain adherent after implantation.
4. Whether or not the cells must be endothelial.

The use of homologous over autologous cells has advantages in that a well-characterized supply of cells could always be available. Thus, although endothelial cells have many cell surface antigens (37,40,54), the potential use of homologous cells should not be dismissed. As with homologous blood, careful matching procedures should be developed. It would then be possible to establish a frozen cell bank of well-characterized cells using, for example, human umbilical vein endothelial cells.

Whether immediate one-stage seeding of grafts, culturing cells prior to graft seeding, or preculture of the graft itself will prove best is unknown. Findings indicated the one-stage seeding results provided the same thromboresistance as cultured cells (see table 12.2). The promising results on implanting isolated endothelial cells were obtained on large to medium diameter grafts with high flow rates. These grafts probably would have remained patent without cell linings as evidenced by patent controls.

More severe tests of cell linings were recently reported (55). Single-staged (immediate) seeding of autologous endothelium failed to confer patency on long 3-4 mm diameter grafts in the canine carotid or femoral arteries. Nevertheless, when e-PTFE grafts were successfully endothelialized in culture first, grafts of similar dimensions remained patent for 30 days without the use of platelet inhibitory drugs. Histologically, the endothelial linings were stable and nearly devoid of smooth muscle cells or fibroblasts.

A randomized clinical study of endothelial seeding was reported recently using mechanically derived cells seeded into Dacron grafts. The patients that did not smoke demonstrated excellent patency of femoral popliteal bypass grafts and a significant improvement over nonsmokers receiving unseeded control grafts. Smokers had significantly worse patency with seeded than unseeded grafts. Heavy lipid vacuolization of the smooth muscle cells were identified in the graft lining of one of the smoking patients, suggesting that the contaminating smooth muscle cells may be the target of continued atherogenic stimulæ (56).

The graft material on which the cells are seeded or cultured is critical. Dacron velour has proved a good "scaffold" for endothelial cell function, but results become less favorable as the graft diameter decreases. Expanded PTFE has proved difficult to seed and culture probably because it is extremely hydrophobic. The earlier microfabrics, developed for this application (13), proved too difficult and expensive to fabricate and failed mechanically (21).

Whether the cells must be endothelial or not is questionable. The fetal fibroblast has proved a beneficial addition to the specialized fibrillar polyurethane bladders (15,17) in that more stable pseudointimas

are produced. These linings consist of more collagen and thus are more stable than the fibrous linings on unseeded bladders, which are less well organized. The fibroblast in general is less antigenic than the endothelial cell (37). However, the use of autologous fibroblasts has proved thrombogenic (12). Fibroblasts may prove the cell of choice for nonporous, large flexing prostheses on which little natural healing would occur. But for increasing the thromboresistance of small to medium diameter grafts, the endothelial cell would seem to be preferable.

Although there is no evidence for such an occurrence, the potential danger that cultured cells may become changed in vitro and cause problems upon implantation must be considered. The most obvious possibility is that the cells may be transformed to a carcinogenic form in vitro, either virally, chemically, or spontaneously. Transformation of cultured cells is accompanied by drastic changes which are readily observed (although chromosomal differences may precede the obvious changes). Change in cell shape and behavior (loss of contact inhibition) and a sharp increase in rate of growth occurs. Spontaneous transformation rarely occurs in human cells in vitro. In fact, they are difficult to transform with viruses or chemicals (57). SV40 is the only oncogenic virus reported to infect endothelial cells, and these transformed cells are not positive for vWF (58). Monitoring of the karyology of cells that have been maintained in culture for many passages can be used to prove that a cell strain has not transformed. Although karyological typing is recommended for cultured cells destined for implantation, since these cells are implanted in early passages (less than 10 subcultures), transformation is highly unlikely.

If autologous cells are harvested and seeded onto a prosthesis and implanted immediately, the precautions should be the same as for a freshly harvested autologous saphenous vein. That is, cells should be kept sterile and not subjected to any toxic substances. If the cells are to be cultured prior to implantation, it may be necessary that they be harvested as free of other contaminating cell types as possible (meaning fewer cells are harvested in the case of endothelial cells), and the cells must be free of bacterial or fungal contamination. They must be kept alive throughout the culture period using the proper nutrients. The enzymes, nutrients, and antibiotics necessary to maintain cells in culture are of little danger to the recipient individual. Any residual enzymes would be rapidly inactivated by the recipient's serum. Other residual culture medium components would serve as nutrients for the recipient. Cells are much more vulnerable to destruction in vitro than they are in vivo, where the defenses of the organism are available for their maintenance. Also, implantation of isolated cells is a procedure in which extremely small amounts of living material are added to the prosthesis, orders of magnitude smaller than a recipient would receive in a blood transfusion.

There are problems related to these grafts that endothelialization is unlikely to solve. Insufficient arteries downstream of the graft may limit blood flow rate and cause thrombosis (poor runoff). Compliance or diameter mismatch may create hemodynamic conditions that predispose a graft to thrombosis and intimal hyperplasia.

Nevertheless, there are applications in vascular surgery for which cell seeding could provide some needed thromboresistance. These are the small diameter grafts (coronary artery bypass and lower limb reconstruction), arteriovenous fistulas for hemodialysis, left ventricular assist devices, and any artificial organ which interfaces with the blood, e.g., kidney, lung, or liver. For grafts, it may prove helpful to culture the graft itself, probably under conditions of in vivo shear stress. Then an endothelialized surface able to withstand the physical forces encountered in vivo would be available from the onset of implantation.

FIXED TISSUE (BIOPROSTHESES)

Bioprosthetic materials are derived from living tissues that have been fixed or otherwise preserved by protein cross-linking to provide a nonliving protein (predominantly collagen) framework which is structurally stable over extended periods and not immunologically reactive. The stimulus for development of such materials was the suboptimal performance of synthetic biomaterials as valve replacements and as arterial grafts in locations other than where the flow rate is high, flow pattern is laminar, and the artery is large. In contrast to fresh autologous grafts in which living tissue is preserved, the aim in bioprostheses is to provide a nonliving structure which does not encourage host tissue ingrowth.

Bioprostheses are used clinically in cardiac valve replacement, in small-to-medium caliber arterial reconstruction, and in arteriovenous fistulae for dialysis. Less commonly used clinically are large vascular conduits and patches for pericardial closure.

Valve Replacements

Of the bioprosthetic valves in use clinically, the glutaraldehyde-preserved porcine aortic valve and valves constructed from preserved bovine pericardium have been the most successful. The advantages of these bioprostheses over mechanical valves are their low thrombogenicity and fluid dynamic properties. Anticoagulants are generally used for a short time postoperatively (2 months) in all patients but thereafter may be discontinued in most. In contrast, all patients with mechanical valves require lifelong, carefully monitored anticoagulation.

Bioprosthetic valves are most advantageous in patients in whom anticoagulation is a risk (chronic hemodialysis patients, patients with bleeding disorders, children or other patients in whom close anticoagulant monitoring would be difficult). The disadvantages of bioprosthetic valves relate primarily to their eventual degeneration (59), especially calcification of the fixed tissue (see chapter 14) which makes them less desirable to use in patients under age 35 (as calcification proceeds faster in young persons). Also, success rate with bioprostheses is related to experience in handling (60). Whether the bioprosthetic valves are as durable as the mechanical valves over the long term remains to be evaluated. Ten-year followup data are

just beginning to accumulate (61,62). The potential modes of failure must be considered independently of their frequency in that failure of the bioprostheses is often sufficiently slow that replacement may be accomplished. However, failure modes of mechanical prostheses are often catastrophic.

The most important factor determining the durability of bioprostheses is their preparation (63). The tissue is dissected from the donor, rinsed in sterile saline, fixed as soon as possible, generally in glutaraldehyde. Glutaraldehyde fixation prevents degradative enzymes from breaking down the tissue after implantation by cross-linking the tissue protein, most of which is collagen. The fixation procedure is critical, as it must permit the glutaraldehyde to penetrate the tissue prior to the onset of cross-linking. This may be accomplished by soaking the tissue at a low pH so that glutaraldehyde penetrates but does not cross-link; then raising the pH to achieve cross-linking. Glutaraldehyde reduces the antigenicity as well as increases the stability of the tissue. Further preparative procedures are aimed at reducing soluble antigenic components. A detailed preparation procedure is given by Carpentier (64). Since tissue valves are now commercially produced, newer preparative techniques are proprietary. Woodroof (65) discusses the chemistry and biology of these valves.

Recently, Carpentier et al. (60) have noted that tissue failure in bioprosthetic valves after implant can often be traced to long intervals (10-120 hours) between harvesting and glutaraldehyde fixation, high pressure used during fixation, and inadequate preservation during shipping. They recommend rinsing hearts twice after excision from the pig in cold buffered saline, a maximum shipping period of 72 hours, and a fixation pressure of 4 mm Hg in glutaraldehyde. These techniques would seem adaptable to the handling of pericardial tissue also.

Commercially available valves are mounted on stents (often flexible) after fixation. The stent is then cloth covered, and a cloth sewing ring is added. Stents consist of a corrosion-resistant alloy of nickel and cobalt (Elgiloy^R), polypropylene with a stellite ring to stabilize the annulus, or titanium. Stents are covered with knitted PTFE or knitted Dacron.

Before the commercial valves are sold, they are tested for leaks under physiological closing pressures, tested for sterility, and the flow characteristics are determined (66). Hemodynamic testing includes assurance of an adequate orifice when the valve is open, a small pressure drop across the valve, and simultaneous cusp opening.

Histologic and ultrastructural differences between the porcine valves and the bovine pericardial valves have been described (67). The collagen fibers are oriented differently in pericardial cusps than in the natural valve cusps, and the pericardial tissue has 25 percent more collagen than the natural valve. Furthermore, while pericardium is almost completely type I collagen, heart valve has been shown to be a mixture of types I and III (68). The spatial distribution of

collagen types in the valves is unknown but may have implications in surface thrombogenicity (69,70).

Valves fabricated from human dura mater have been used extensively, particularly in South America where they were developed (71). The results are not as successful, particularly in the aortic position, as the previously discussed bioprosthetic valves, as judged from 6- to 8-year followup data (72). Factors which may contribute to the propensity for late dysfunction in this valve are the use of glycerol for preservation and storage rather than glutaraldehyde. Glycerol does not sterilize completely and leads to postoperative infections (73). Thus, some degree of biologic remodeling occurs in these valves in addition to calcification. Macrophages and fibroblasts were found in the tissue after 23-108 months of implant (74). These valves may be more immunologically reactive and more subject to host tissue invasion (due to the preparative procedures) than glutaraldehyde fixed bioprosthetic valves. Furthermore, the use of a nonflexible stent might have contributed to tearing of cusps. More recently, a flexible stent has been used.

Thorough discussions of contemporary bioprosthetic valves and advanced design concepts are found in Cardiac Bioprostheses (75) and Tissue Heart Valves (76). Key references have recently been compiled by Schoen and Braunwald (77).

Small-to-Medium Diameter Grafts

One of the greatest clinical needs in cardiovascular surgery is a successful small diameter artery replacement (about 4 mm) for use in the lower legs and for coronary artery bypass. The reversed autologous saphenous vein is the present replacement of choice, but there are many cases where it is unavailable or inadequate (3). The success of a small diameter graft is in large part determined by the length of the graft (resistance to flow is proportional to length) and the size and number of arteries downstream of the graft (runoff patency). Alternative grafts for medium diameter artery prostheses when the autologous saphenous vein is not available have included:

1. Glutaraldehyde fixed homograft (human umbilical vein) externally reinforced with Dacron mesh.
2. Dialdehyde starch tanned heterograft (bovine carotid artery).
3. Negatively charged glutaraldehyde tanned (NCGT) graft (bovine carotid artery).
4. Mandril-grown conduits.

The glutaraldehyde-processed human umbilical vein has been developed for this purpose (78). The vein is fixed in glutaraldehyde and is reinforced with a very loose weave Dacron mesh wrapped around the adventitial surface. It is used clinically for femoral-popliteal and femoral-tibial reconstruction. Dardik has reported cumulative patency rates equivalent to those obtained with autologous saphenous veins for arterial replacements in the lower leg (79) (63 percent after 36 months

in the femoral popliteal position, 54 percent after 36 months in the distal tibial). However, others (1,80) have found the autologous saphenous vein clearly superior. Sauvage et al. (81), in comparing this graft to grafts of Dacron and PTFE in the dog carotid and aorto-coronary position, found the synthetic materials to be superior.

As in the case with valve bioprosthesis, the proper surgical techniques for handling the glutaraldehyde fixed vein are critical.

The modified bovine graft is an adult bovine carotid artery digested enzymatically with ficin resulting in a nonliving, nonantigenic conduit of collagen, which is then strengthened by dialdehyde starch tanning. This graft was used clinically for femoropopliteal and iliac artery bypasses as well as for arteriovenous shunts for hemodialysis. Because of the incidence of late aneurysm in the lower extremity grafts, their high cost, and poor response to infection, their use has been discontinued (82). As dialysis fistulas, they have been more successful, although expanded PTFE is the replacement of choice presently (autologous saphenous vein is rarely sacrificed for fistulas). Sawyer et al. (83,84) have obtained promising results with negatively charged glutaraldehyde-tanned collagen prosthesis based on the bovine carotid artery.

Besides natural veins, silicone mandril-grown connective tissue conduits have been used successfully as vascular prostheses (85). Perloff et al. (86) report high patency rates both clinically and experimentally for glutaraldehyde-tanned polyester mesh supported conduits grown subcutaneously in sheep. An obvious problem in extension of this concept to the clinical arena is the local inadequacy of the circulation in most patients which might limit the ability to "grow" a graft in the sites in which they are needed.

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PHARMACOLOGICALLY ACTIVE MATERIALS

A popular technique for imparting thromboresistance to biomaterials has involved the incorporation of heparin (1,2). More recently, other pharmacologically active agents such as prostacyclin or urokinase have been used to enhance the thromboresistance of the underlying material. These agents have been either chemically bonded or physically blended with the substrate. Both covalent and ionic immobilization procedures have been used with consequent differences in the degree of permanence of the chemical bond. Regardless of the method used, it is useful to distinguish between materials which rely on the immobilized species for their functional properties and those for which the pharmacologically active agent is released into the blood. These two classes are distinguished primarily by the release or leakage rate* of the active species. Since heparin is the active species most studied, it will be discussed in detail.

Materials which selectively absorb or adsorb a particular species from blood (e.g., albumin (3) or antithrombin III (75)) can be considered to be pharmacologically active in vivo although in terms of characterization they are more like inert materials. Thus alkyl-derivatized surfaces (3) and heparin-like surfaces (75) are not specifically considered further.

REVIEW OF EXISTING TECHNIQUES

Controlled Release Systems

Selected techniques used to produce materials which are considered to release heparin at biologically significant rates are summarized in table 13.1. Some chemical bonding techniques (e.g., ionic heparinization) have been included here because of their known or suspected release properties. Techniques for other agents are listed in table 13.3. Readers are directed to a recent review which provides further details (1).

The rate of release in blended materials is determined by the permeability of the active agent in the polymer and its solubility in blood. The relationship between these parameters and the geometry of the device are discussed extensively in various reviews of controlled release systems (e.g., 4,5). A particular feature of such diffusion-limited systems is that the release rate decreases with time, unless relatively unique features are incorporated to correct for or eliminate this effect. In the other systems, the rate of release is limited generally by the rate of reaction of a particular chemical bond. This would normally result in a constant rate of release unless a diffusion-limiting situation was superimposed; e.g., the

*Release and leakage refer to the same phenomenon; the nomenclature reflects the difference between the planned and inadvertent appearance of the active species in the liquid medium.

Table 13.1--Systems Based on Controlled Release of Heparin

Binding Agent or Key Reagent	Specific Substrate*	Reference
A. Blends		
-----	silicone rubber	6
-----	polyhydroxyethyl methacrylate gel	88
B. Quaternary ammonium salts		
benzalkonium chloride	graphite	7-12
benzalkonium chloride		13
tridodecylmethyl ammonium chloride (TDMAC)		14
cetyltrimethyl ammonium bromide	collodion	20
cetyltrimethyl ammonium bromide	polyethylene, polypropylene	22
cetylpyridinium chloride	polyethylene, polyvinyl chloride	34
methyltricapryl ammonium chloride	--	35
trimethyl dodecyl ammonium chloride	polyethylene, polyurethane	36
C. Glutaraldehyde stabilization		
quaternary ammonium salt	--	23, 37
alkylamine	polyethylene	23-39
acid treated tertiary amine copolymer	polyamidoamine	38, 39
D. Quaternized polymers		
various methods	polystyrene, cellophane, butadiene/vinyl pyridine copolymer	15, 17
ethylene imine dimethylaminoethyl methacrylate (DMAEM) copolymers	cellulose membrane	40-43
quaternized	--	15, 17
quaternized	copolymer with methoxypolyethylene glycol methacrylate	49-53
acid treated	copolymer with acrylonitrile	54
other acid treated amine	diamine cured polyurethane	56
other acid treated amine	amine cured epoxy	57
other acid treated amine	diethylaminoethyl cellulose	55, 58
other acid treated amine	tertiary aminated glycol-polyurethane	59
	protamine/glutaraldehyde treated vascular graft	60
E. Cyanogen bromide		
no spacer	sepharose	61, 63
no spacer	regenerated cellulose	34
no spacer	polyvinyl alcohol, polyhydroxyethyl methacrylate (poly HEMA)	63
with/without ϵ -aminocaproic acid or putrescine spacer	agarose	32
	poly HEMA/N-vinyl pyrrolidone copolymer	33
F. Other		
anion exchange	silastic	64
γ -aminopropyltriethoxy silane	glass	65
γ -aminopropyltriethoxy silane	silica/silicone rubber	67
γ -aminopropyltriethoxy silane	dacron	66
γ -irradiation	diethylaminoethyl cellulose acetate	68
chitosan	polyethylene	69

*Substrate as specified in references. This list should not be considered a list of materials to which method can be applied since designated substrates can frequently be used as coatings on other materials. (-) implies no specific substrate.

active species was at some distance from the surface, necessitating its diffusion to the surface and/or diffusion from the surface of the reagent causing disruption of the bond.

Immobilized Systems

Table 13.2 summarizes selected covalent immobilization schemes for heparin which result in stable systems; i.e. they do not appear to leak at appreciable rates. A review is available for further details (1). Table 13.3 summarizes systems involving other agents.

CHARACTERIZATION STRATEGY

Since controlled release and immobilized systems function differently, correct classification of a particular material has scientific and practical significance. The mechanism and duration of action, the fate of the agent, and presumably the regulatory approach will be different for each class of material. Thus determination of the release or leakage rate and the associated chemical characterizations are extremely important. Similarly, the biological activity of the incorporated species must be determined with the appropriate controls to demonstrate the utility of the modified material. Furthermore, it may be helpful to conduct specific biochemical assays which serve to detail the local mechanism of action of the immobilized or released species. Finally, attention must be given to the potential adverse effects of the pharmacologically active species on nontargeted sites of action (e.g., heparin on platelets, vasodilation induced by PGI₂).

Proper attribution of a measured effect to the active agent necessitates consideration of nonspecific factors such as surface charge or energy. Thus, both pharmacologically active materials and the nonactive controls should still be subject to the standard array of surface characterization assays. Furthermore, a conservative therapeutic approach implies presumably that only the minimum amount of active agent needed to impart thromboresistance should be used. Accordingly it is expected that the agents will be incorporated into the best available (i.e., the most thromboresistant) well-characterized substrate materials. Thus the procedures outlined below should be considered supplementary to those surface characterization procedures described elsewhere in this volume.

CHARACTERIZATION OF HEPARINIZED MATERIALS

Heparin Characterization

Heparin is a heterogeneous molecule, which should be characterized itself, prior to incorporation into a material. The molecular weight (110), antithrombin III affinity (111), and biological activity (112) distribution of heparin are well known (table 13.4). Accordingly, the biological activity of different commercial or fractionated heparins varies widely in both amount and kind. Low-molecular-weight

Table 13.2--Systems Based on Immobilized Heparin

Heparin Functional Group	Binding Agents or Key Reagent	Specific Substrate*	Reference
- OH	isocyanate	poly p-isocyanatostyrene	70
	cyanuric chloride	aminated polystyrene, aminated silicon/silicone rubber	18
	cerium (IV) initiator	polymethyl methacrylate	71-73
	cerium (IV) initiator	polymethacrylic acid	74
- COOH	carbodiimide	aminated polymethyl acrylate (PMA)	45
	carbodiimide	aminoalkyl Sepharose	61,62,84
	carbodiimide	albumin	87
	carbodiimide	albuminated PMA	46
	carbodiimide/ 4-nitrophenethyl amine	diazotized aminophenethyl agarose	92
	Woodward's reagent	diaminoalkylated Sepharose	85,86
- CHO	nitrous acid	polyethylene imine	30
- NH ₂	glutaraldehyde/H ₂ SO ₄	polyvinyl alcohol (PVA)	44
	glutaraldehyde/H ₂ SO ₄	PVA/N-vinyl pyrrolidone copolymer	93
	glutaraldehyde/MgCl ₂	polyvinyl alcohol	94-100
	glutaraldehyde	diaminoalkylated polymethyl acrylate (PMA)	45,46
	glutaraldehyde	albuminated PMA	46
	carbodiimide activated carboxylic acid	hydrolysed PMA	45
Heparin-like materials	sulfonate/ α -amino acid	polystyrene	75-77,79-82
	sulfamide derivatives	carboxymethylated Sephadex	78
	derivitized cis 1,4-polyisoprene		102,103
Other	NaBH ₃ CN	NH ₂ -Sepharose	84
	radiation grafting	acrylic acid-g-Dacron	83

*Substrate as specified in reference. This is not a list of materials to which method can be applied since designated substances can frequently be used as coatings on other materials.

Table 13.3--Other Pharmacologically Active Surfaces

Active Agent	Substrate	Binding Agent or Key Reagent	Reference
A. Blend			
ditazole	cellulose acetate	--	104
PGE ₁	Biomer, polyvinyl chloride	--	89
PGE ₁ , PGD ₂ , PGI ₂	Biomer, Tygon	--	90
B. Reacted systems			
urokinase-heparin	graphite	Benzalkonium heparin	105
urokinase	surface hydrolysed nylon	carbodiimide	106
	surface hydrolysed nylon	polyethylene imine,	107,108
	polyester, silicone,	carbodiimide, maleic	
	polyvinyl chloride,	anhydride copolymer	
	polyurethane		
	Sepharose	cyanogen bromide	109
	derivatized Sepharose	carbodiimide	109
	derivatized Sepharose	N-hydroxysuccinamide	109
ditazole	surface hydrolysed nylon	urethane linkage	104
	surface hydrolysed nylon	carbodiimide	104
	cellulose	triethylamine	104
dipyridamole	glutaryl chloride	triethylamine	104
	activated cellulose		
prostacyclin/PGF _{2a}	diaminodecyl polystyrene	carbodiimide	91

Table 13.4--Summary of Affinity Fractionation of Heparin Components
(From Reference 113, With Permission)

Heparin Species	Relative Abundance (%)	Specific Anticoagulant Activity (units/mg)
Molecular weight ~7,000 daltons*		
Highly active	7.9	360
Active	25.1	267
Relatively inactive	67.0	4
Molecular weight ~20,000 daltons†		
Highly active	10.0	738
Active	30.0	400
Relatively inactive	60.0	20

*The specific anticoagulant activity of the starting pool averaged 125 units/mg.

†The specific anticoagulant activity of the starting pool averaged 250 units/mg.

heparin typically has a higher antifactor Xa activity and partial thromboplastin time (APTT) activity than high-molecular-weight heparin, while mucosal heparin has a higher anti-Xa activity than lung heparin of the same APTT or USP (United States Pharmacopeia) activity (114). Heparin's enhancement of ADP-induced platelet aggregation has been found to increase with increasing molecular weight, while there was a roughly reciprocal relationship between platelet aggregation activity and antithrombin binding affinity (115). The therapeutic benefit of low-molecular-weight heparin is currently subject to a large-scale clinical trial (116).

Because of these variations, it is desirable to know (and report) the nature of the starting heparin so that appropriate comparisons can be made among different heparinized materials. The commercial supplier, tissue source, USP activity or equivalent, and the generic type (i.e., low or normal molecular weight, fractionated or not) of heparin should be readily available and reported. The average molecular weight, anti-factor Xa, APTT (or equivalent) activities should be determined if not available from the commercial supplier. If effects on platelets are being studied, it would be desirable to quantify the effect of the heparin on ADP-induced platelet aggregation in an aggregometer.

Other features of the starting heparin are of interest because of their impact on the chemistry of the bound heparin (see next section). Thus it may be desirable to determine, for example, the serine, total amino acid, free amine (de-N-sulfate), or total sulfate content of the starting heparin to aid in the chemical characterization of the final material.

Chemical Features

Site of Binding*

The heterogeneous nature of heparin makes delineation of the site of binding of heparin both difficult and important. In addition to the distributions outlined in the previous section, there are numerous chemical heterogeneities which may or may not be associated with these distributions. Such differences in chemical structure may result in the fractionation of heparin upon incorporation into a material, so that only those molecules with a particular chemical composition are bound while the remainder are not (96). If the biological activity and chemical composition do not parallel each other, the bound heparin may have a different activity than the unreacted material (ignoring the question of inactivation due to chemical modification). Similarly, some of the heparin may be bound more tightly than the remainder, so that simple relationships between release rate and duration of biological effectiveness may not be applicable.

Although infrared spectroscopy (117) and ^{13}C -NMR (118) should be useful in confirming that the intended site of chemical reaction

*Blended systems require little or no characterization as to site or amount of binding.

is the actual site, in practice the complexities of the spectra of pure heparin make unambiguous structural assignments rather difficult. ESCA has been used to confirm the presence of a Schiff base in a glutaraldehyde-stabilized ionically heparinized material (31) and as a means of characterizing a heparin-like material (79) and a heparinized cuprophane (85). Unfortunately the poor ESCA sensitivity for sulfur, the low sulfur and nitrogen contents of heparin, and the need to deconvolute the spectrum makes this an expensive and difficult procedure.

Alternatively, a probable site of binding may be inferred by preparing heparins with different chemical structures (96) or using heparins from different commercial sources (95). Heparin can be prepared with a radiolabel at its reducing end to distinguish the chains with and without a terminal serine (119). Free amine groups may be N-sulfated (120) or N-acetylated (121), while de-N-sulfation may be used to increase the number of free amines along the chain (121). The effectiveness of such modifications must, of course, be examined by the appropriate methods (122). Extensive dialysis of heparin results in the removal of contaminant amino acids that may also be a site of binding or may interfere with other structural determinations; on the other hand, amino acid-heparin complexes may be formed by simple incubation (123) to increase the potential for binding through these contaminants. Other modification techniques are available in the heparin literature. Heparins from different suppliers or animal sources have different composition (e.g., terminal serine content) so that the heparin of one supplier may bind preferentially relative to another heparin; structural information can be inferred from these observations if detailed composition information is available.

It is also prudent to examine the chemical structure (and the biological activity) of the unbound heparin to determine the extent of structural alteration (e.g., de-N-sulfation, depolymerization) that occurs during heparinization. However, it may not be possible to distinguish between adverse effects of the binding procedure and the consequences of the heterogeneous nature of heparin, without a clear picture of the site of binding.

Amount Bound*

The quantity of bound heparin may be determined by direct sampling of the heparinized material or indirectly by mass balance from the amount of unreacted and released heparin and the amount used initially. Release measurements are described in the next section.

Although the sensitivity of the techniques are limited, in principle at least, bound heparin contents can be determined by elemental analysis for sulfur (50,70) or nitrogen (if there is no nitrogen in the base polymer or reactants). ESCA, for example, may be used to confirm the presence of heparin on the surface, although

*Blended systems require little or no characterization as to site or amount of binding.

quantitation in terms of a surface concentration (i.e., $\mu\text{g}/\text{cm}^2$) is difficult unless appropriate calibration standards are used. Other techniques which have been used include ellipsometry (29), autoradiography (20), Auger electron spectroscopy (26), and X-ray dispersion analysis EDAX (15,50,52). EDAX determination of sulfur in a scanning electron microscope is particularly useful since it can provide an approximate picture of the uniformity of distribution of heparin in the sample. Transmission electron microscopy with gold deposition has also been used for a similar purpose (124).

It has been most common to use radiolabeled heparins (either ^{35}S - or ^3H -labeled) to determine the amount bound and the subsequent release rate. However, the labeled heparin may have a different chemical composition than the unlabeled heparin, even if it is prepared from the same lot of heparin. The chemical steps required to incorporate the radiolabel (e.g., de-N-sulfation or N-acetylation) are similar to those suggested above for binding site characterization with consequent differences in the amount bound. For example, heparin tritiated at the terminal xylose did not bind to a polyvinyl alcohol gel, whereas the heparin without a label, terminating in a serine residue, did bind (96). Because of the need for terminal serine for binding in this particular material, ^{35}S -heparin significantly underestimated the amount of heparin bound (95). Regardless of these specific concerns, the lability of the radiolabel should also be considered in interpreting the results.

Liquid scintillation counting of the solid or aqueous solution or both may be used, although the counting efficiency in the solid may be difficult to determine with as great a precision as that for the solutions. The sensitivity of the determination in solution depends on the specific activity of the heparin used, but appears to be on the order of $2 \mu\text{g}/\text{mL}$ for most reports in the literature.

Similar sensitivity ($2 \mu\text{g}/\text{mL}$) can be obtained for the indirect chemical assay of heparin in distilled water or buffered saline using toluidine blue (95) or other dyes (e.g., azure A) which undergo a metachromatic shift in the presence of heparin (125) (see figure 13.1). Azure A recently has been used for determining heparin in plasma without a prior extraction (125). A direct method to determine the amount of heparin bound to an agarose gel using toluidine blue is available (127). Absorption of toluidine blue by the heparinized gel is quantified by comparing the depleted and starting solutions and is related by a standard curve to the amount of heparin. This technique may need to be modified for different materials.

A problem with the use of dyes or some other chemical methods to detect heparin is that, in theory, the heparin used to calibrate the assay should be identical with the heparin being measured (generally the released heparin). A difference between the released heparin and the starting heparin (that normally used for calibration), due, for example, to a chemical modification during immobilization, would yield inaccurate results. Such a problem has been noted (96) with the toluidine blue assay because of the dependence of the metachromatic shift on N-sulfate content (125). Thus, it is useful to use the

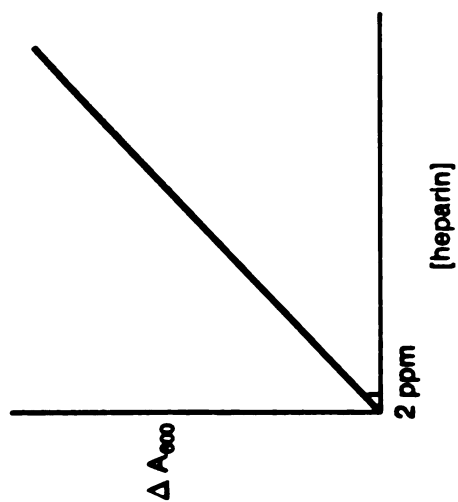
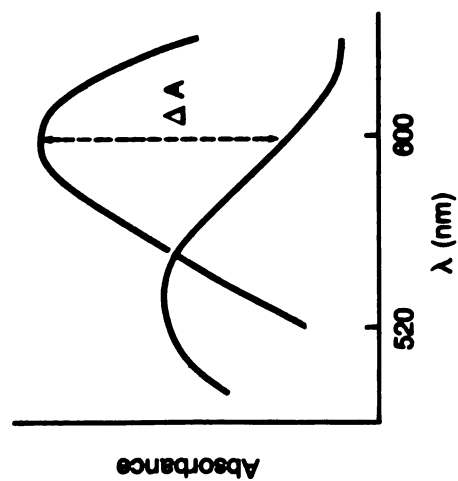
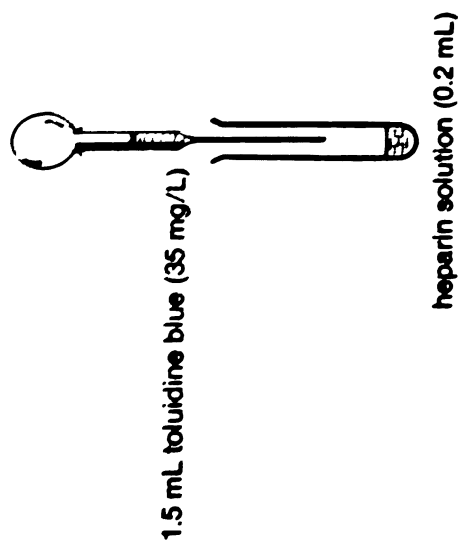
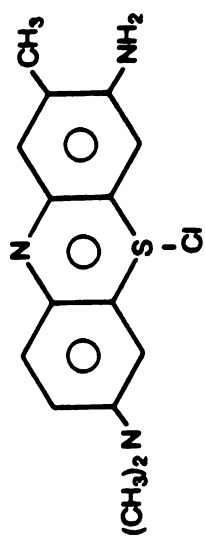


Figure 13.1--Toluidine blue assay for released heparin.

chemically altered heparin if available to calibrate the assay. Alternatively, the heparin concentration can be detected in at least one or two samples by one of the numerous sugar assays that are available (e.g., glucosamine by indole assay (128), uronic acid by carbazole method (129)). This more accurate determination can be used to provide a correction factor for the toluidine blue or other simpler and more generally useful assays.

Release Rate (or Leakage Rate)

Failure to properly determine the release rate has been at the core of most discussions regarding the adequacy of heparinized materials for long-term applications. The release rate is necessary to distinguish between controlled release systems (even if prepared by chemical reaction) and immobilized systems; only the latter presumably can be considered generally useful for long-term application. In the past, heparinized materials have been claimed (wrongly) to be "stable" or "permanently immobilized" on the basis of release tests with inadequate sensitivity. Release rates of $4 \times 10^{-2} \mu\text{g}/\text{cm}^2/\text{min}$ from a heparinized catheter in the inferior vena cava of dogs has been reported to be the minimum release rate required to impart thromboresistance in that configuration (40,50). The general validity of this value and the relationship among release rate, surface concentration, and a heparin microenvironment at the blood/material interface has been demonstrated recently (101).

Release rate can be determined using one of the chemical assays for heparin in solution as above (e.g., toluidine blue, radiolabeled heparin) or indirectly by a clotting or other biological activity assay. The latter assays are described further in the next section. Some of the assays that have been used are listed in table 13.5. Since the concern with release ultimately arises from the interpretation of a biological assay and since these assays are generally more sensitive than the chemical assays (e.g., detection limit in thrombin time test is $0.2 \mu\text{g}/\text{mL}$), it is prudent to test for release by both methods.

Release rate is normally determined by incubating a known quantity of heparinized material (with known surface area) in an aqueous medium (e.g., phosphate buffered saline, citrated plasma, serum) at 37°C or room temperature. Samples are taken at regular intervals and the concentration of heparin determined by chemical or biological assay. High sensitivity requires the use of long incubation times (1-2 weeks or more, with daily sampling), small volumes of liquid, and large surface areas of material. For example, a release rate test using the toluidine blue assay to determine the concentration of heparin in samples taken once a day from 1 L of saline in which were incubated 30 g of heparinized beads ($\sim 10^4 \text{ cm}^2$) has a detection limit of less than $10^{-4} \mu\text{g}/\text{cm}^2/\text{min}$ (95). Smaller volumes of liquid could be used, provided the wash solution was changed frequently so that an equilibrium between immobilized and released heparin is not established. With plasma as the released fluid, incubation time is limited to 1 hour, although longer times can be used if the deterioration of plasma is not a concern. High sensitivity can be obtained with plasma, if a large surface area of material is avail-

Table 13.5--Heparin Release Rate Assays

Analytical Procedure	Medium	Reference
³⁵ S-heparin		
direct	tap water	56
	unlabeled heparin solution	56
	plasma, blood (<u>in vitro</u>)	24
	blood (<u>in vivo</u>)	10,56
indirect	distilled water	16
	saline	16,97
	3M NaCl	94
	plasma	18,24,97
	plasma proteins/saline	21
	blood	24
³ H-heparin		
direct	saline, plasma	11
	blood (<u>in vivo</u>)	11,19
indirect	saline	14,22
	blood (<u>in vitro</u>)	22
¹⁴ C-heparin		
direct	plasma	132
^{99m} Tc-heparin		
direct/indirect	saline	55
toluidine blue		
indirect	phosphate buffered saline	95
thrombin time	saline	41,72
	plasma	41,49,50,95
thrombin activity/chromogenic substrate	plasma	47
	blood (<u>in vivo</u>)	34
Factor Xa inhibition/chromogenic substrate	saline	69
	plasma	27,30
radioimmunoassay for thrombin/antithrombin III complex	plasma	48
detection of heparin in blood/plasma after clotting test by:		
whole blood clotting time		13,20,23,41,65
thrombin time		17,40,42,44,72,75
partial thromboplastin time		17,18
³⁵ S-heparin		44
³ H-heparin		85

able (e.g., beads) and if a biological assay is used to determine heparin concentration. Frequent changes of the plasma do not result in an improved sensitivity although a single batch of material can be followed for a longer period of time.

There is little advantage to using 3M NaCl or other high ionic strength solution for these tests unless a particular mechanism of release is being investigated. Some differences in release rate have been noted for ionically heparinized materials in different media and between in vivo and in vitro tests (56). Although this latter difference was attributed to nonspecific factors, the more recent discovery of endoglycosidases in various tissues (130) including platelets (131) that can depolymerize heparin poses additional, albeit unresolvable, questions regarding the in vivo stability of the bound or otherwise incorporated heparin. However, none of these enzymes have been detected in plasma as yet, and since the platelet heparitinase, for example, is localized within its lysosome (131), the relevance of these enzymes to the in vivo fate of the heparinized material is unknown.

The release rate is determined directly from the slope of a plot of concentration against time (assuming linearity, at least over a part of the release curve). Sufficient samples should be determined with replication to calculate the 95 percent confidence interval of the slope or to apply the F-test to determine if the slope is significantly different from zero. Even if no statistically significant release is detected, it is imperative that the detection limit of the assay system be reported; i.e., the release rate should be reported as less than the detection limit and not simply as "zero" or undetectable.

In some cases it is useful to quantify the leakage rate of the binding agent (e.g., quaternary ammonium salt) into the blood, particularly if this agent may have adverse effects. The determination of the release of ^{14}C -TDMAC from a TDMAC-heparin preparation is one example (14).

Blended Systems

Only measurement of release rate has any particular significance for these systems. It is generally assumed that the amount incorporated is equal to the amount of heparin added to the blend during its preparation. If loss of heparin results during fabrication, then the appropriate determinations similar to those outlined previously need to be made. For these systems, sensitivity is not an issue since release is generally high enough that a microenvironment is created at the interface. On the other hand, the order of release (i.e., zeroth or first) is of considerable interest; reviews on controlled released systems should be consulted for further details (4,5).

Biological Activity

In Vitro Tests

Because of the special relevance of heparin to the intrinsic coagulation system, clotting tests remain important for demonstrating

the biological activity of the heparinized material. Recalcified plasma or whole blood clotting times and partial thromboplastin times as well as more specific clotting tests (e.g., thrombin time, factor Xa inhibition assays) are useful. The latter are particularly helpful in assessing the effectiveness of the heparinized material at specific steps in the coagulation cascade. Table 13.6 lists some of the tests that have been used.

Tests can be done with the material as tubes or in the form of beads with clumping or adhesion of the beads to the tube wall taken as the "clotting time"(95). In the latter case, the tube wall presents a different surface than the beads so that coagulation may proceed at different rates in different regions of the tube; this complicates interpretation of the results unless the nonheparinized beads are used as controls.

For immobilized systems, in addition to using the nonheparinized material as a control, it is necessary to check for release of heparin by incubating the plasma or blood with the heparinized material at the same temperature as the test, for times equal to or exceeding the total test time (clotting time + preincubation time). Prolongation of the clotting time in a clean glass tube after separation from the heparinized material must be considered evidence of significant release of heparin unless evidence is presented to show that the incubated plasma or blood is hypocoagulable due to significant depletion, by adsorption, of key coagulation factors.

Results may be reported directly as clotting times or in terms of units of biological activity. Units of heparin per unit area or mass may be determined by comparison of the clotting time obtained with the heparinized material with the clotting time obtained in equivalent fashion with heparin solutions of different concentration. Similarly, the number of thrombin units inactivated per unit mass or area may be determined by comparison with standard curves relating the clotting time to thrombin concentration (75,76).

Instead of using the appearance of polymerized fibrin as the end point of the assay, the amount of active thrombin or factor Xa (135) may be determined using a chromogenic substrate. After incubating the heparinized material (e.g., beads) with enzyme and a source of anti-thrombin III (e.g., defibrinated plasma), the residual enzyme activity (adsorbed and in solution) is determined spectrophotometrically (color yield) after the addition of the appropriate specific chromogenic substrate (63,80). The color yield is in turn related to a standard curve prepared with solution of different enzyme concentration but without beads. Separation of the material from the incubating medium prior to exposure of the material to the chromogenic substrate (as tubing (28) or a column of beads (97)), results in determination of the adsorbed enzyme activity. Addition of chromogenic substrate to the supernatant demonstrates the depletion of enzyme, but whether the enzyme has been inactivated or simply adsorbed cannot be distinguished. The unheparinized material is useful as a control as in a clotting test. Effects of released heparin on the enzyme/chromogen reaction (136) should be considered but are not expected to prove a problem. A clotting test using

Table 13.6--In Vitro Assays for Heparinized Materials

Test	References	
	Heparinized Tube	Heparinized Beads in Tube
A. Clotting tests		
whole blood clotting time	7,16,17,23,42,44	70
plasma recalcification time	20,63	71,95
partial thromboplastin time	16,44,94	
thrombin time	44,132	20,71,75,76,78,95
B. Enzymes		
thrombin/chromogenic substrate	28	45,63,74,80
factor X _a /chromogenic substrate		63,80
bound thrombin activity/chromogenic substrate	28,30	97
bound factor X _a /chromogenic substrate	28	
radioimmunoassay for inactive complex		48
C. Protein adsorption		
total protein		24,41,52,58,85
radiolabeled protein components		15,37,43,73,126
albumin/fibrinogen (electrophoresis)		58
fibrinogen (fibrinopeptide A)		27
antithrombin III	32,46,73,74,80,82,92,102	
thrombin/factor X _a		30,32,80,82
thrombin/antithrombin III affinity chromatography		61,84
thrombin/antithrombin III displacement		98,99
D. Clotting factor alterations (generally depletion)		
after exposure to material*		
use of clotting factor deficient plasmas		44,62
specific clotting factor assays		73,80
clotting time tests	14,41,44,58,85	
thromboelastography		132,133
E. Platelet interactions		
platelet count differences		43,44,54,58
⁵¹ Cr platelet adhesion		25
platelet retention in bead column		45,46,47,85
microscopy of adherent platelets		53,100
ATP level (bioluminescence) of adherent platelets		25,27,30
platelet factor 4 release		85
acid phosphatase/stypven time assays		14
aggregometer behavior after exposure to material		14,18,53
F. Other		
hemolysis		17,58,105,134
lipid/lipoprotein depletion		14,50
fibrinolytic activity		105,132
kallikrein generation		102

*Release of heparin may interfere with interpretation.

purified fibrinogen has also been used to measure the binding of enzyme to a heparinized material (63).

Since the chromogen is much smaller than fibrinogen as a substrate of the coagulation enzymes, it is conceivable that diffusion limitations may add to the apparent delay in conversion of fibrinogen to fibrin (longer clotting time) but not affect the cleavage of the chromogen; a difference in the activity of heparin immobilized to a gel measured by clotting and chromogenic substrate assays has been observed by one group (63). Chromogenic substrate assays, however, have the advantage of being more precise than clotting tests since detecting the appearance of a "clot" is somewhat subjective.

A radioimmunoassay for inactive thrombin-antithrombin III complex (137) has also been used to determine the activity of certain immobilized heparins (48). Heparinized beads were incubated in plasma, thrombin was added, and 15 seconds later the reaction was stopped by adding hirudin. The amount of complex generated was then quantified using an antibody with 3,000:1 specificity for the complex compared with either thrombin or antithrombin III alone and can detect heparin levels as low as 6.7 ng/mL (48). However failure to release the complex into the medium after inactivation of the thrombin may result in artificially low immobilized heparin activities; this was observed for heparin-sepharose (48).

The sequence of exposure of the heparinized material to thrombin and antithrombin III has been found to be important, in one system at least, in demonstrating the inactivation of thrombin (97). This observation was related to the mechanism of the inactivation reaction, which remains a subject of controversy (130). It is presumed that the sequence of exposure of the heparinized materials to the reagents may be significant in systems in which each reagent is separated from the material prior to the addition of the next one. Altering the sequence of exposure may only give slight differences in other systems in which the material is exposed to all reagents together. Because of the mechanism question and to avoid rejecting an active material due to a flaw in the assay, it is wise to alter the sequence of exposure to examine its effect in at least a few cases.

Ex Vivo or In Vivo Tests

Most ex vivo or in vivo tests of blood material interaction can be applied to heparinized materials without special modification, provided of course that heparin is not added as an anticoagulant. Table 13.7 lists many of the test systems used for heparinized materials. Comparison of the heparinized material with the same material without heparin as a control provides a measure of the biological activity of the heparin under the particular test conditions. Fibrinopeptide A assays (138) are particularly relevant to demonstrating the anticoagulant activity of the incorporated heparin. Since some of these tests measure aspects of BMI that may not be directly related to heparin activity (e.g., platelet or fibrinogen adhesion), observed differences between the heparinized and nonheparinized material may be due to nonspecific effects. Under

Table 13.7--In Vivo or Ex Vivo Investigations* of
Heparinized Materials

A. In Vivo

Gott inferior vena cava ring	7,9,14,49,57,67
atrial implant	6,9,10,11
vein implant	
patency/histology	42
adherent thrombus at explanation	68
vascular graft	
patency/histology	56,60,66
catheter	
adherent thrombus at explanation	13,49.

B. Ex Vivo

arteriovenous shunt	
patency	23,42,51,56
patency (pig)	100
platelet count differences	43
platelet deposition (^{51}Cr or ^{111}In)	27,67,68,81
other cell deposition	51,81
fibrinogen deposition (^{125}I or ^{131}I)	69,81
fibrinopeptide A generation	27
adsorbed thrombin/factor X_a activity	28
Dudley test	36
bypass circuit	35,37

*In dogs unless specified otherwise.

these cases, the appropriate control, if the effect of heparin itself is to be distinguished, may be an inactivated heparin, another glycosaminoglycan (e.g., chondroitin sulfate (46)), or other sulfated or nonsulfated polyanions.

In releasing systems, attention presumably needs to be given to the fate of the released heparin, which is by its nature concentrated at the interface. The released heparin may, depending on the fluid mechanics of the situation, interfere with the BMI in part of the test circuit distal to the heparinized material. This is presumed to be especially relevant when two different materials (one of them heparinized) are tested simultaneously.

Under some test conditions, no effect of the incorporated heparin may be noted, presumably because of the relative insensitivity of the test to surface chemistry (139). Since heparinized materials are generally under consideration for low flow situations, and heparin has more effect on the intrinsic coagulation system, it is speculated that high flow tests may have only limited relevance. A chronic A-V shunt system with 3 percent of the flow bypassed through a small diameter (1 mm) heparinized tube is being developed to precisely and inexpensively examine heparinized materials under low flow conditions (140).

Controls

Three types of control materials may be used in the biological characterization of heparinized materials:

1. The unmodified substrate.
2. The substrate modified by the chemical steps used in immobilization, but in the absence of heparin.
3. The identically modified substrate containing equivalent quantities of an immobilized, yet biologically inactive, analog molecule.

These controls supplement the use of reference materials and laboratory controls required to calibrate the specific test of BMI. Each control is used to test a different aspect of heparin's effectiveness. The unmodified substrate provides an overall measure of the benefits (or adverse effect) of the bound/released heparin, when compared with the heparinized material. The chemically modified substrate, without heparin, enables the effect of heparin alone to be distinguished, without the complicating influence of the chemical alterations which may or may not be beneficial on their own. The analog molecule (e.g., chondroitin sulfate, partially de-N-sulfated heparin), without significant antithrombin activity is particularly useful if the mechanism of heparin's effect is under investigation. Such analogue molecules, either bound or released, enable a distinction to be made between specific biochemical activity (enhancement of rate of thrombin inactivation) and nonspecific factors such as surface charge or surface energy. Examples of the latter two types of controls are the use of a

polyvinyl alcohol hydrogel without heparin as a control for the same hydrogel but with heparin (95) and the use of chondroitin sulfate bound to polymethyl acrylate (46).

Other Tests

Understanding the mechanism of action of heparinized materials and the effect of heparin on other systems requires additional testing.

Platelet interaction can be quantified using the same tests described in earlier chapters, provided of course that heparin is not used as an anticoagulant in preparing the platelets. Investigators with releasing systems should be aware of the fate of the released heparin in their test systems and correct for artifacts. Controls are as before (the nonheparinized material) and for immobilized systems a sample of platelets that have been incubated with and then separated from the heparinized material to check for biologically significant heparin release.

Protein adsorption studies are also of interest, particularly those using antithrombin III, thrombin, or other serine proteases of the coagulation cascade. Both adsorption and desorption profiles have been determined; the latter is of interest in demonstrating that the inactive thrombin-antithrombin complex can be readily displaced from the surface (98,99). Methods for purifying thrombin and antithrombin III are available (97). Radiolabeling thrombin (using immobilized enzymes) appears to result in unaltered affinity to sepharose-heparin while radiolabeling antithrombin III (at least by the same method) results in significant deterioration in its affinity for sepharose heparin (99). The utility of other radiolabeling techniques for antithrombin is unknown.

Characterization of the interaction with other biological systems as described in earlier chapters is also relevant to heparinized materials. Complement and leukocytes may prove to be of special importance.

OTHER PHARMACOLOGICALLY ACTIVE AGENTS

The strategy for materials with other active agents is similar, although the actual tests, whether they be chemical or biological, will be different for each agent (table 13.8). Since in most cases these agents are more homogeneous than heparin, much of the problem associated with the chemical characterization of heparin can be avoided. Release rate again remains a critical test since it distinguishes between controlled release and immobilized systems. As with heparinized materials, a detection limit should be reported if no release is observed. Since some of these agents (e.g., prostacyclin) are highly labile, standard assays may need to be modified to shorten the incubation time if activity is to be detected. Also shelf life is presumed to be a concern with these compounds, necessitating additional tests (90) to demonstrate the absence of chemical or biological change over a period of weeks or months in the final form.

Table 13.8—Assays for Other Pharmacologically Active Materials

Agent	Assay	Reference
urokinase	amino acid content (amount bound)	109
	fibrinolytic activity	
	clot adherence	105
	fibrin plate	105,108
	caseinolytic activity	109
	scanning electron microscopy	108
ditazole, dipyridamole	protein adsorption	104
	examination of implants for thrombi	104
prostaglandin	release	
	tritiated PGE ₁	109
	radioimmunoassay for 6 keto PGF _{1α}	91
	platelet adhesion	90,91,109
	ADP induced aggregation	90,91,109

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CALCIFICATION OF CARDIOVASCULAR BIOMATERIALS

This section describes an approach to the evaluation of pathologic calcification of biomaterials, a special problem in the interaction of blood with nonphysiologic materials. This complication, which may cause device failure under certain circumstances, is considered separately from other manifestations of blood/surface interactions in the other portions of these guidelines. It is, at present, an area of intense investigation, whose analysis goes beyond the usual limits of blood/surface interaction studies.

For the purposes of this document, calcification is defined as the deposition of crystalline calcium phosphate mineral, consisting primarily of hydroxyapatite. Pathologic calcification is that occurring in an ectopic site (1). Calcification related to cardiovascular biomaterials will be considered from two broad perspectives. First, the various in vivo systems in which calcification occurs will be outlined. Cardiovascular, as well as noncardiovascular, and physiologic as well as pathologic situations are described. Second, the detailed analysis of specimens for investigation of calcification will be outlined. The objectives of this section are to illustrate the similarities and differences in calcific degeneration of cardiovascular devices in different settings, to summarize available understanding from these studies and from other studies of pathologic and normal mineralization, and to suggest how to document and investigate the occurrence of pathologic calcification in blood-material interaction studies. The potential prevention of medical device-associated calcification is considered as a gap area.

CALCIFICATION SYSTEMS

Cardiovascular Biomaterials

Calcification in the cardiovascular system has been observed clinically related to prosthetic heart valves and vascular grafts and experimentally in heart valve substitutes and ventricular assist devices.

Early trileaflet polymeric cardiac valve prostheses were found to calcify heavily when used clinically (2); this problem has persisted in recent experimental attempts to further develop this design concept in animals (3).

Glutaraldehyde-preserved stent-mounted porcine aortic valve bioprotheses, implanted in several hundred thousand patients since 1972, were an important development in cardiac valve replacement. However, many of these valve substitutes have failed with clinically significant tears or stiffening due to calcification (4-20). Calcification occurs both in orthotopic valve implants and in valve implants in prosthetic right ventricular-to-pulmonary artery conduits. Calcification has also occurred clinically in bioprotheses fabricated from fascia lata (21,22),

glutaraldehyde-preserved dura mater (22,23), and pericardium (24,25) as well as formalin-preserved porcine and human cadaveric valves and other tissue valve prostheses (22,26). Large animal (calves and sheep) experimental models have simulated the pathology of bioprosthetic valve calcification in orthotopic valve implants (27-30), valves implanted in left ventricle-to-aortic conduits (31,32) and valves used in conjunction with cannulas for blood pump systems (33). While clinically important calcification of glutaraldehyde-preserved porcine valves implanted in adults generally occurs only in some patients and only after 5 years of function, adolescents and preadolescent children with orthotopic or conduit-situated bioprostheses commonly have considerable calcification with valve failure in less than 4 years (13,19). Experimentally induced calcific degeneration of bioprostheses implanted in calves or sheep is sufficiently accelerated so that after 6 months it is almost ubiquitously seen, to the extent seen in clinical failures (28-30,32).

Calcification of bioprostheses occurs predominantly by deposition of calcium phosphate crystal, predominantly deep in the spongiosa of the residual valve tissue and is enhanced at the highly stressed areas of the cuspal attachments, especially at the commissures (4,8,34). Calcific deposits involve residual cells and collagen of the porcine valve (4,35). Calcific deposits may also form rapidly in the necrotic inflammatory cells of superficial thrombotic accumulations or the vegetations of infective endocarditis (4,7,11). This process is clearly distinct from the slowly developing intrinsic cuspal mineralization described above as emphasized by recent reports of a calcified vegetation on a porcine bioprosthesis 18 days postoperatively (36) and a calcified thrombus associated with a pericardial bioprosthesis 3 days after implantation in a man with acute renal failure (37). Renal failure and other disorders of calcium metabolism are considered to be predisposing conditions in all forms of cardiovascular calcification (38). Similarly, rapid calcium metabolism associated with somatic growth is thought to be largely contributory to the accelerated failure encountered in younger patients.

There has recently been interest in studying calcification of implants in small laboratory animals. Calcification of glutaraldehyde-preserved porcine aortic valve cusps occurs after subcutaneous implantation in rabbits, rats, and mice in relatively short periods of time: 6 months for rabbits (39) and 3 weeks for rats and mice (30,40-42). The morphology and biochemistry of mineralization are similar to that which occurs in the cardiovascular system in clinical implants and in large animal models (30,40). These models, despite the lack of dynamic mechanical stress characteristic of the circulatory environment, reproduce the basic metabolic processes of mineralization, and hence have been used to investigate various host and implant factors in the pathophysiology of degenerative calcification of bioprosthetic heart valves.

Specifically, porcine valves cusps implanted in 3-week-old rats calcify progressively with mineral deposition comparable to that noted in clinical retrievals and that noted in experimental calf implants. The levels of calcification achieved in rat subcutaneous implants and clinical large animal circulatory implants are comparable (20,30). The

time course is markedly accelerated with rat implants with diffuse calcification beginning within 48 hours (30,40). Deposits are principally related to devitalized porcine cells, especially their cell membranes, and collagen. They increasingly predominate in the valvar spongiosa as diffuse microcrystals (30,40). The process of calcification in porcine aortic valve cusps is similar to that occurring naturally in the aortic valves of aged individuals (43,44), in other pathological calcification (1), and in physiological bone mineralization (1,45). Like these calcifications, mineral deposits in bioprosthetic valves are predominantly hydroxyapatite (40). There is late enhancement of calcific deposits to confluent nodules at the spongiosa-fibrosa junction, and at static mechanical strain foci, such as folds (40).

While the calcium-binding protein gamma-carboxyglutamic acid (46) is accumulated in bioprosthetic valves proportional to the calcium uptake (32,40,47), its role as a pathogenetic or regulatory factor is unclear. Nonpreserved porcine aortic leaflet implants do not calcify, indicating that glutaraldehyde-induced cross-linking of proteins is a prerequisite for cuspal calcification (40). Cusps implanted for 21 days in 8-month-old rats calcify less than cusps implanted in 3-week-old rats, thereby simulating the clinically observed age effect (40). Glutaraldehyde-preserved bovine pericardium calcifies with similar kinetics and morphology (30). Neither anticoagulation nor immunological factors appear to play a role (32,40,41).

The subcutaneous model is rapid, convenient, and economical, and the results may be quantified. Specimens are amenable to detailed and sophisticated biochemical and morphological study. It is logical to use rat subcutaneous implants to investigate the metabolic pathophysiology of bioprosthetic valve calcification and to screen specific hypotheses for its inhibition. The most promising approaches may then be investigated in the circulatory environment. Nevertheless, the validity of this approach has been recently questioned (48).

Calcification is a common occurrence associated with the flexing polymeric pump bladders of experimental cardiac assist devices after extended implantation (48-55). On smooth-surfaced bladders, calcific deposits appear to be associated with surface defects originating during bladder fabrication (52,54). A pathogenetic relationship between thrombotic deposits and calcific nodules in this situation is not apparent. Calcification associated with the pseudointima of flexing bladders with rough (fibrillar) surfaces also originates at the polymeric surface/tissue junction. It appears to be a consequence of excessive growth of PI to a thickness exceeding the diffusion path of nutrients from the lumen. Thus, the deeper portions of the tissue at the interface of PI and solid bladder can lose viability, and dystrophic calcification may result. In flexing assist device bladders, calcification is clearly enhanced at points of greatest dynamic mechanical stress (50,52,56), and its site-specific intensity and extent may be semiquantitatively described (57).

The term "biolized" describes a surface preparation technique utilizing aldehyde-treated proteins, such as gelatin (58). Calcification has been seen in a "biolized" ventricular assist pump bladder surface (59) and in fibroblast-generated collagenized surfaces (60).

Studies have attempted to modify calcification of assist device bladders by pharmacological modulation of the host (51,61).

Calcification after long-term implantation has also been noted deep in the tissue ingrowth into fabric-covered struts of cloth-covered caged-ball heart valve prostheses (62) and in association with clinical vascular grafts (63).

Other Nonosseous Biomaterials Calcification

In vivo calcification has been observed in association with contact lenses (64,65), intrauterine contraceptive devices (66), and other implanted biomaterials and prostheses (67).

Calcification in Native Hard Tissues

Calcification occurs naturally in bones, cartilage, dentin, and enamel. Although the extent of the similarity between physiologic and pathologic mineralization has not been elucidated, it is important that researchers investigating calcification in the cardiovascular system be mindful that extensive work has been done toward understanding the mechanisms of calcification of native tissues (45,68,69). Postulated pathogenetic factors in native tissue mineralization include the vitamin K-dependent calcium-binding proteins with gamma-carboxyglutamic acid residues (46,70,71), matrix vesicles (1,72,73,76), collagen (75), phosphoproteins (76), lipids/glycolipids (68,77), osteo- and chondro-nectin, proteoglycans and mitochondrial calcium regulation, and various enzymes and hormones.

METHODS FOR INVESTIGATION OF CALCIFICATION PROCESSES

The various methods for evaluation of physiologic and dystrophic calcification are described below, despite the fact that only some have been applied to studies of prosthesis-associated calcific degeneration.

Gross and Microscopic Analysis

Specimens should be fixed, described grossly, photographed from various pertinent aspects, and radiographed (for example, 40 kV X 0.8 min) (Faxitron Model 43805, Hewlett-Packard, McMinnville, Oregon). Multiple portions of the main specimen should be dehydrated according to standard procedures and embedded in either paraffin or glycolmethacrylate medium, or both, sectioned at 6 microns (paraffin) or 1-3 microns (plastic), and stained with hematoxylin-eosin. Selected sections should be stained by von Kossa's method (for calcium salts) and by alizarin red for calcium (78-81). Additional specimens may be decalcified, embedded in paraffin, sectioned, and stained.

Analysis of frozen tissue may be useful (82,83). Additional microscopic histochemical staining and analysis of frozen sections for enzymes such as alkaline phosphatase may be useful in localizing sites of active mineralization. These techniques have been applied to studies

of physiologic mineralization (79). A recently developed glyoxyl-bis-(2-hydroxyanil) staining technique is potentially capable of detecting ionized calcium, and soluble and insoluble calcium salts in tissue (84,85).

Ultrastructure and Crystallography

Scanning electron microscopy with energy dispersive X-ray analysis allows the quantitative mapping of calcium ions and the major anions of their salts. This allows determination both of concentration and localization of mineral. Resistance of the deposits to microincineration verifies their inorganic nature.

Crystal structures of mineral phases are determined by X-ray diffraction on powdered material or electron diffraction of ultra-thin sections in the transmission electron microscope (40). High resolution electron probe X-ray microanalysis may be performed in a scanning electron microscope capable of scanning transmission operation (30). Minute amounts of powdered materials can be analyzed in capillary tubes using a Debye-Scherrer camera. Comparisons with hydroxapatite standards quickly confirm the presence of apatitic materials.

Conventional transmission electron microscopy and scanning electron microscopy may be extremely useful in investigating the morphology of calcific deposits and the relationship of mineral-to-substrate tissue structural features. In general, specific preparation and analysis use conventional methodology.

Potassium pyroantimonate has been used as a histochemical stain for calcium localization at the ultrastructural level (86-90). Although the technique has been applied in several detailed studies, and it is considered to be useful by investigators mindful of its limitations (91), this technique has been criticized recently for its lack of ion specificity (92).

In any study of calcification at the ultrastructural level, consideration should be given to the fact that the use of aqueous fixatives and solvents, especially electron-microscopic stains, may introduce significant changes in both the location and nature of mineral phases (92). These changes have likewise been demonstrated in a light microscope study (93). The extent of such artefacts in specific systems is not known. Nevertheless, two available experimental approaches which may circumvent the difficulties created by loss, migration and redistribution of transformation of ionic or elemental species or other tissue components are nonaqueous (ethylene glycol or alcohol) fixation and/or storage and manipulation of specimens (94-97), and ultra-cryomicrotomy (92). The details of these techniques have been described in studies of physiological mineralization.

The organic/inorganic relationship at sites of biologic calcification may be indicated by the presence of a phosphotungstic acid positive-staining matrix which reflects the presence of proteoglycan (98).

Chemical and Biochemical Analysis

Tissue to be studied for calcification has to be properly prepared in a uniform manner for analysis. This typically involves copious saline rinses to remove adherent blood and serum. The material is then best freeze-dried. The tissue can then be finely minced, or ideally nitrogen milled to a uniform powder (40,47). Powdered specimens for analysis should be stored in a vacuum dessicator and dried to a constant weight when sampling—a process that involves replicate weighings over a 5- to 7-day period.

Mineral analysis of calcified material should include at least ash content with analysis for calcium and phosphorous. At best, total mineral content includes analysis for magnesium and carbonate levels and X-ray diffraction to establish the degree of crystallinity. There are a number of valid approaches which can be best tailored to suit the particular experimental situation. Ashing of hard tissue can be performed by a dry procedure in a furnace oven at 600°C overnight for samples over 50 mg or by wet ashing in concentrated nitric-perchloric acid (4:1) solution (75). The dried ashes are then reconstituted for appropriate mineral analysis. Three- to 5-day extraction in 0.5M HCl at room temperature effectively decalcifies mineralized soft tissues and is a procedure commonly used in atherosclerosis research (99). Six-N HCl protein hydrolysis conditions (100°C for 24 hours) will effectively decalcify all hard tissues and provide the option of performing simultaneous aminoacid analysis procedures on aliquots of the same material analyzed for calcium and phosphorus content (47).

Elemental analyses for calcium is most reliably performed using atomic absorption spectroscopy. Aliquots for analysis are derived from the decalcification procedures discussed above. A reaction solvent of acidified lanthanide is necessary to eliminate phosphate interference with the detection of calcium in a quantitative manner. Phosphorus analyses are most reliably performed using classic procedures (100) or the modern modification of these procedures such as the methodology of Chen (101). Great care must be taken to use acid-cleaned glassware in all aspects of phosphorus analysis, since the opportunity for contamination is otherwise great. While extremely important and useful for comparative studies, such information averages mineral over an entire specimen and does not usually provide details of site specificity. The presence of hydroxyapatite is either inferred by determination of calcium/phosphorus ratio in the appropriate range of 1.6 to 2.0 or by independent determination of crystal structure by X-ray diffraction (75).

Analysis for proteins associated with tissue mineralization may be useful in some studies. A specific class of protein, the vitamin K-dependent calcium binding proteins containing gamma-carboxyglutamic acid have been found in physiologic bone mineralization as well as several dystrophic calcification situations including clinical and experimental porcine bioprosthetic heart valves (47) and experimental long-term ventricular assist devices (54). Specialized alkaline hydrolysis procedures to avoid inadvertent decarboxylation are necessary

for the preparation of samples for analysis (102). Specific programming of an automated amino acid analyzer for this purpose can be done (102). Gla can also be quantitated by HPLC procedures and anion exchange chromatography (103).

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Chapter 15

GAP AREAS

This chapter attempts to identify problem areas which the working group felt are very important. In general, there are two types of gap areas: first are problems which have been recognized since the beginning of the biomaterials program at the NIH, but for which no really satisfactory solution or resolution has been attained. Two examples are the validity of animal models for human device testing and the validity of in vitro testing for predicting in vivo biomaterial response. The second type of gap area is one where progress in the field has brought to the forefront a new previously unrecognized (or undefined) problem. To illustrate, calcification of flexing blood contacting biomaterial surfaces was a gap area created by the success of long-term animal support with artificial hearts and the extensive use in humans of porcine aortic valve prostheses. This problem is now under extensive scientific investigation, as reviewed in chapter 14. The ability to noninvasively evaluate calcification is still a gap area.

Our hope is that by identifying present problem areas, we will stimulate debate and interest. The goal, of course, is to find solutions for or at least generate a better understanding of these problems. The list is not intended to be all inclusive nor does the order imply priority.

LIMITED INTERPRETATION OF IN VITRO AND ACUTE IN VIVO TESTING

A number of relatively simple screening tests have been used for the evaluation of blood-material interactions. Although such testing may permit rapid and economical evaluation of a large number of materials, serious limitations exist. The clinical relevance of most in vitro tests remains undefined. Thus the usefulness of acute testing in vivo for predicting long-term thrombotic and thromboembolic events remains to be established. Although it has not been shown that initial blood-material interactions are predictive of subsequent sequelae such as gross thrombus formation and thromboembolism, it is recognized that knowledge regarding the full-time course of events following the exposure of surfaces to blood is of fundamental importance.

In addition, testing which involves the use of idealized flow geometries may be of uncertain relevance to the performance of actual cardiovascular devices and prostheses, where the fluid mechanics may be quite complex. Extracorporeal circuits, incorporating a variety of flow geometries, have been widely utilized to study initial interactions between blood and artificial surfaces. Most studies have involved the evaluation of materials per se rather than the behavior of devices in their actual usage configuration. With these systems, exposure times rarely exceed several hours. The assessment of blood-material interactions provided by these tests is thus limited to the acute deposition of hemostatic elements, and particularly platelets. Results with these test systems should be interpreted with caution since studies with in

vitro as well as ex vivo systems have shown that initial thrombotic events may be strongly influenced by species-related and fluid mechanical effects. The presence or absence of such effects may depend on the material being tested.

Thus, despite extensive study of biomaterials, the application of artificial cardiovascular devices remains limited, in large measure due to the unsolved problems arising from their persistent thrombogenicity. Since no acute in vitro testing has yet been shown to predict chronic in vivo blood-material compatibility, reliable testing to establish predictive capability in the experimental animal and clinical setting is critical in the integration of developmental strategies.

ANTITHROMBOTIC THERAPY

Since the use of small caliber prosthetic grafts is precluded by the high frequency of occlusion, studies are needed to characterize the relative importance and time course for the antithrombotic agents from all three classes (antiplatelet, anticoagulant, and fibrinolytic) singly or in combination with respect to endothelialization and anastomotic intimal lesion formation. This is an important, but complex, area that requires definition to establish the therapeutic usefulness of these agents. Some of the questions to be addressed include: What fibrinolytic agents and regimens can restore graft patency without surgical intervention? What is the role of antiplatelet therapy in the prevention of acute graft occlusion? What is the role of combining anticoagulant therapy with other acute antithrombotic measures in preventing occlusion of those small caliber grafts subject to periods of low flow? How can small caliber graft patency be maintained following endothelial seeding during the period required for graft endothelialization to be completed?

THE ROLE OF BIOMATERIAL-PROTEIN INTERACTION IN DETERMINING THROMBUS FORMATION

Based on the current literature, several conclusions can be drawn. Patterns of protein adsorption have been observed to depend on material type and may determine also the extent and type of subsequent platelet, white cell, and red cell interactions. However, the complex process of competitive adsorption and desorption of proteins in vivo, apparently involving a large number of blood proteins, has not been defined. At the present time, protein-surface interactions are not predictable from considerations of material properties. Moreover, no general relationship has been established between the time course of protein-surface interactions in vivo and subsequent thrombotic events.

With the present availability of monoclonal antibodies and the capacity to label proteins covalently with imageable isotopes together with our current understanding of the identity of the attaching proteins, studies need to be performed to define the role of protein-biomaterial interaction in predicting subsequent thrombus deposition.

THE ROLE OF WHITE CELLS IN BIOMATERIAL THROMBOSIS

Leukocytes and monocytes are involved in the blood response to biomaterials. The role of these cellular reactions involving white cells in the process of biomaterial thrombogenesis requires definition.

INTIMAL AND ANASTOMOTIC NEOINTIMAL FIBROUS HYPERPLASIA

Late failure of clinical and experimental vascular grafts has frequently been observed to be associated with a diffuse thickening of the neointima because of fibrous hyperplasia (1). In some grafts, late failure is associated with a significant hyperplasia at the anastomosis without excessive thickening of the remainder of the neointima. This anastomotic neointimal fibrous hyperplasia (ANFH) appears to be most compromising in grafts less than 6-8 mm in diameter used to bypass arteries. ANFH has also been observed in endarterectomized arteries, in radial arteries used as aortocoronary bypass grafts, and in autogenous veins in the femoropopliteal or aortocoronary position (1). It has been observed in failed Dacron grafts, bovine heterografts, and most recently in expanded PTFE grafts (2).

Myointimal hyperplasia in vein grafts appears to be a result of the intimal migration and proliferation of the medial smooth muscle cells which functionally have been transformed into multipotential mesenchymal cells (3). A recent detailed morphologic study demonstrated, beneath the endothelium, collagenous ground substance and myofibroblasts mixed with smooth muscle cells (4). Two forms of smooth muscle cell pathomorphogenesis were recognized. Formation of myofibroblasts induced medial fibroplasia, whereas degeneration of muscle cells progressed to medial necrosis. Intimal hyperplasia has not been observed to occur in the absence of endothelial cell injury, such as that which occurs during the harvesting and preparation of a vein graft or that which is caused at the anastomosis. Although the injury is initiated at this time, it is promoted by ongoing hemodynamic alterations, elastic compliance disparity, and other contributing factors.

Among the possible causes of ANFH, we can include one or more of the following: 1) operative trauma, 2) graft and host vessel surface thrombogenesis, 3) delayed or incomplete endothelialization of the graft and adjacent vessel, 4) hemodynamic factors associated with the anastomosis 5) mechanical mismatch between the graft material and host artery, and 6) local alterations in the wall stress of the artery due to geometrical factors at the anastomosis.

Early experience with vascular grafts indicated that operative trauma could result in thrombus formation at anastomotic suture lines and that anastomotic intimal thrombus may contribute to acute graft thrombosis (5). However, no direct evidence is available that non-occluding thrombus in the early postoperative period results in ANFH or causes late occlusion of grafts.

In general, the more thrombogenic materials develop the most severe ANFH. This suggests that thrombogenicity or some other characteristic of the graft material used may be related to the development of ANFH. A role of platelet-derived growth factor, a potent mitogen for smooth muscle cells which possibly contributes to the formation of atherosclerotic plaque (6), is not established. Intimal hyperplasia is stimulated after endothelial denudation experimentally in normal arteries (7,8) and clinically in arterial damage with medial exposure and in reversed autogeneous vein grafts (9,10). Furthermore, the role of vascularization of vein grafts is not clear (11,12).

Hemodynamic and mechanical factors probably play additional albeit unclear roles (10). Mechanical mismatch has been postulated to explain why different graft materials might produce different degrees of endothelial trauma, platelet adherence to subendothelium, stimulation of fibrous tissue and smooth muscle cells, and subsequently ANFH. Materials with different mechanical properties that are joined together and placed in a cyclic stress system such as arterial pulsations could result in bending or buckling at the site of the anastomosis and may result in a chronic recurring injury to the vessel wall. Evidence has been presented that elasticity of grafts as measured by their compliance can influence patency (13,14). In one animal model, the compliance of the human femoral artery was compared with that of grafts currently in use: human saphenous vein, knitted Dacron, glutaraldehyde-treated umbilical cord vein, bovine heterograft, and expanded polytetrafluoroethylene (14). Increased clinical patency in the femoropopliteal position correlated with a decreasing disparity between host artery and graft compliance. Cyclical stress is capable of causing hyperplasia of arterial smooth muscle cells, and vascular wall tension in vivo correlates with fibrous tissue content and wall thickness (15).

Although matching elastic properties of graft and host vessel is of undisputed benefit, the diameter ratio of graft to artery may be a far more significant determinant of impedance matching, both for blood flow and pulse waves (16). The typical "cobra-head" configuration of an end-to-side anastomosis greatly enlarges the overall diameter of the artery in the region of an anastomosis. A 200-percent to 300-percent increase in wall tension in a typical end-to-side anastomosis, using a small artery, is derived from LaPlace's law (tension = pressure x radius of curvature). This stress level far exceeds any possible stresses from mismatched impedance and would be constant, unlike the cyclical stress from mismatched impedance.

To inhibit ANFH, design of a practical small caliber prosthesis must be predicated upon assuming physiologic stress levels at all points in the distal artery, normal shear stress at the endothelial surface and minimal blood activation by the prosthesis. No available graft has been constructed which completely meets these criteria. Therefore reduction in failure rates could be achieved by limiting endothelial damage by better vein preparation techniques, by altering local mechanical factors, or by treatment with antiplatelet and other pharmacological agents. If platelets are indeed contributory to ANFH, then it is logical to assume that antiplatelet drugs might be effective

in preventing it. In canine experimental vascular grafts, patency was enhanced and ANFH was diminished relative to controls in animals receiving aspirin and dipyridamole (17). Nevertheless, clinical studies of the use of platelet inhibitory drugs in patients undergoing aortocoronary bypass grafts have conflicting results (5,18,19). Experimental inhibition of ANFH by immunosuppression (azathioprine) but not by steroids (methylprednisolone) has been also demonstrated (20).

ANIMAL MODELS

In order to assess suitability for clinical use, animals are necessary for evaluating materials and prosthetic cardiovascular devices. However, such testing is not without limitations. The relative suitability of different animal species for the evaluation of devices or materials intended for particular applications has not been systematically investigated.

The choice of an animal model may be restricted by the availability of certain species and the costs of adequate maintenance. Animal size and possible growth relative to a particular prosthesis may also present difficult problems in system design and hydrodynamic scaling. Testing of certain prostheses such as a totally implantable artificial heart obviously necessitates the use of larger animals. The extent to which the biologic system is stressed may be reflected in rates of organ dysfunction and morbidity and will obviously govern overall alterations in the levels and function of circulating blood constituents. The effects of surgical procedures may be difficult to quantify. Infection is a serious problem, and results may be modified by antibiotic, anticoagulant, or other therapies. The impact of a prosthesis on a biologic system may thus range from minimal (not measurable) to disastrous.

The hemostatic mechanism of most if not all animals differs significantly from that of man in a number of respects (see chapter 8). While these differences may be critical, the relative suitability of various species for the evaluation of a particular device or implant has not been fully elucidated. Animal studies may, however, be useful since they permit systematic evaluation of the relative importance of in vivo variables, material properties, and system design. Prudent development in biomaterials, because of the problems of thrombogenesis, demands having a suitable animal model relevant to man.

PREVENTION OF CALCIFICATION

Since calcification in the cardiovascular system is generally deleterious, there is considerable interest in eliminating or reducing it. Early studies by McLean and Urist indicated that mineralization may be controlled through the use of calcium-binding dyes and other compounds (21,22). More recently, considerable attention has been focused on the use of detergents (surfactants) and diphosphonates (23-25). Surfactants have been widely investigated in vitro (23) and in vivo (26,27). Although calcification is clearly reduced by the detergent sodium dodecyl sulfate in noncirculatory model systems (i.e., subcutaneous implants)

(26,27), there are contradictions concerning the efficacy of such compounds with experimental bioprosthetic heart valve calcification in orthotopic implants in calves (26) and sheep (28). Furthermore, there may be differential effects on porcine aortic valve and bovine pericardial bioprostheses, the use of surfactants being less efficacious in the latter (28).

Surfactants probably change the nature of the lipid micelle and the charge and conformational properties of the lipid molecules in early mineral nucleation (23). Thus, the reaccumulation from the blood of lipid constituents may obviate the effect of surfactants in actual valve implants. Diphosphonates may act not only as surfactants, but also as hydroxyapatite crystal poisons, thereby preventing growth of mineral deposits (23). These compounds are widely used clinically in the treatment of patients with diseases of bone and calcium metabolism (24). Research has begun to elucidate the mechanisms of action of diphosphonates and related compounds (25,29). One such diphosphonate, 1-hydroxyethane 1, 1-diphosphonate (EHDP), has recently been demonstrated to prevent calcification of subcutaneously implanted porcine bioprosthetic heart valve leaflets in rats (27,30). While systemic therapy is potentially applicable to clinical cardiovascular calcification, caution is indicated since these compounds may have a severe growth retardation effect when given systemically for long periods. Nevertheless, administration through either controlled, localized drug delivery systems at the site of potential calcification or systemic pulse therapy may be beneficial. Experimental suppression of atherosclerosis by the calcium antagonist lanthanum has also been reported (31).

NONINVASIVE EVALUATION OF CALCIFICATION

In the past, evaluation of calcification associated with cardiovascular devices has been done either post mortem or after removal of the device due to diminished or lost function. Special histochemical stains for calcium and the use of microradiography, X-ray microanalysis, backscattered electron imaging, and atomic adsorption analysis all require retrieval of the implant. The only noninvasive methods currently available are based on radionuclide imaging using technetium coupled diphosphonate or pyrophosphate or regular X-ray diagnostic techniques. However, these approaches are not sensitive enough to detect early mineral growth and, as a result, do not provide much additional information relative to a posteriori tests.

Research efforts in the immediate future should be directed towards noninvasive diagnostic methods to identify mineralization at early stages and monitor the effectiveness of therapeutic measures to retard or reverse the mineralization process.

Possible methods to diagnose calcification of pneumatically driven blood pumps includes analysis of the air pressure wave forms using a cardiac output monitor and diagnostic unit (COMDU), the use of sophisticated computer aided gamma imaging techniques, or ultrasonic or thermographic imaging techniques. At present, none have the required sensitivity to detect early mineralization of valves, vascular grafts, or

blood pumps, although many of these techniques are clinically useful in other situations.

QUANTITATION IN IMPLANT PATHOLOGY

Pathology analysis of cardiovascular explants is hampered in general by limitation of data and approaches relevant to two areas: 1) lack of methods to quantitate gross radiographic and histologic features of morphology sufficient to characterize and provide comparative analyses and 2) lack of data relating to the number of clinical devices of a given configuration that have been implanted.

BMI data have been reported as, for example, weight or thickness of thrombotic accumulations or semiquantitative assessments of light histologic features of pathology. There has been, in general, considerable difficulty in describing implant-related pathology with the degree of sophistication and quantitation characteristic of other research in the biological and physical sciences. Quantitation allows the use of formal statistical analyses in the comparison of experimental and control data. Data collection should be aided by progress in the use of computer-aided image analysis, in conjunction with stereologic methods, where possible. Standard protocols including data collection forms, complete yet realistic, are needed.

Many pathology analyses of clinical explants--for example, heart valve prostheses and vascular grafts--suffer from the lack of knowledge of the population of surgical implants from which specific pathologies are derived. While implant registries are beginning to accrue in certain areas, there is, for the most part, little knowledge of the frequency of specific implant pathologic features.

CONTROL OF INFECTION OF IMPLANTED DEVICES

Infection and resulting vegetative thrombus is a common problem with many types of cardiovascular devices. Even using rigorous sterile techniques, implants such as heart valves and blood pumps can still provide a protective environment for bacteria that normally enter the bloodstream through breaks in the skin, after dental work, or other types of exposure to pathological organisms. After the bacteria have become established in seams or small defects in the device, antibiotic treatment is not very effective in eliminating the organism completely.

Prophylactic antibiotics can be started if, for example, the patient knows dental work is required. However, this does not protect against the unexpected introduction or exposure to invasive organisms. Incorporation of antibiotics into the sewing ring of artificial valves has shown some reduction in vegetative thrombus but is not considered a useful approach because it does not protect against resistant organisms, and the drug will eventually diffuse out.

Elimination of percutaneous leads in existing blood pump designs should greatly reduce the incidence of infection associated with these

devices. Other design improvements should include the elimination of gaps where two parts are joined together and a reduction in microscopic defects that can provide a protective environment for bacteria.

CELLULAR COVERAGE OF GRAFTS

During the healing process, vascular grafts become lined with endothelial cells in all experimental animals in which they have been implanted. The rate of endothelial coverage varies with species and age. This is not the case in man, where very few cases of patchy endothelial coverage have been reported. In man, the blood contact surface is usually compacted fibrin which gives way during healing to connective tissue proteins (collagen and elastin). This surface becomes "passivated" to thrombus formation (platelet adhesion) relative to the freshly implanted surface. How the thrombogenicity of the "passivated" surface compares to an endothelialized surface is unknown. However, it is characterized by progressive thickening (presumably from stimulation of smooth muscle cells by platelet-derived growth factor) which can embolize if the underlying layers become necrotic, or can build up and stenose the graft. An endothelialized graft is not subject to progressive thickening of the pseudointima or subendothelial layers.

In animals endothelial coverage proceeds from the adjoining vessels by ingrowth of cells across the anastomoses (pannus ingrowth). Whether or not "fallout" seeding of the graft surface occurs from the flowing blood is unknown. Whether perigraft ingrowth can provide endothelialization from vasa vasorum is also unknown.

Questions suggested by the differences in endothelialization observed between man and experimental animals are: How do the human endothelial cells differ from those of experimental animals? Are there factors other than the endothelial cells in man which inhibit endothelialization? How does a "passivated" surface resist thrombogenesis? Does "fallout" seeding of endothelial cells from the blood occur?

POROUS VERSUS NONPOROUS SURFACES

At present all vascular grafts and cardiovascular implants that are intracorporeal are porous. Porous Dacron prostheses are filled with thrombus by preclotting procedures (to prevent excessive bleeding through the material) prior to implantation. Less porous materials made from PTFE need not be preclotted, but immediately post implantation some thrombus forms on and within the graft. The thrombus formed initially is then organized by connective tissue ingrowth through the interstices along the length of the graft.

Nonporous materials are required in applications where all or part of the prosthesis is extracorporeal or must have an air interface—for example, the left ventricular assist device. Nonporous materials do not have the benefit of connective tissue ingrowth across the entire surface of the prosthesis. They must depend on pannus ingrowth from

the nearest anastomosis with the vasculature which may be quite distant, or from "fallout" seeding of endothelial cells from the blood, if it occurs. Alternatively, these materials must resist healing and the thrombus formation that precedes it. The materials which most closely meet these requirements, presently used for these devices, are polyurethanes and the "biolized" surfaces of Nose, et al.

During the course of long-term (more than 1 month) implantation in experimental animals, ingrowth of tissue on these nonporous surfaces does occur. It is subject to excess thickening no matter what the material is. Such thickening is more critical on nonporous than porous surfaces, because no abluminal exchange is possible; thus necrosis is inevitable when the lining thickens beyond 50 microns, the limit for the supply of nutrients by diffusion from blood. Necrosis is followed by embolization and/or calcification of the lining, particularly if the surface is flexing continuously. Prostheses including nonporous blood contacting surfaces are more complex than porous grafts or patches. There is more surface area of biomaterial, relative to natural endothelium, that the blood must contact. They also normally have complex geometries and moving parts. The thrombogenicity of these factors, together with the difficulty of healing the nonporous surfaces, make prostheses such as left ventricular assist devices formidable challenges in blood-material interactions.

METHODS FOR IDENTIFICATION OF CELLS ON MATERIAL SURFACES

The identity of a given vascular cell as endothelial, smooth muscle or fibroblast, derives from the location of the cell in the intact vessel wall. Living cells that have been isolated from their tissue of origin are difficult to identify. Endothelial cells are the easiest vascular cell type to identify because there are concrete markers for them (see chapter 12). To positively establish the identity of an isolated population of smooth muscle cells or fibroblasts would require extensive expertise and techniques found only in a few laboratories. If these cells are cultured to confluence on a typical culture surface, the pattern of growth in the population indicates cell type, but does not provide positive proof of identity. The identity of isolated vascular cells seeded onto or growing on biomaterials becomes even more difficult because of the interference of the biomaterial (e.g., in complicating sectioning and complex topography). If the cells are mixed with preclotting blood, because of poor optical transmittance, discriminating between blood cells and rounded-up vascular cells becomes problematic.

Even if a pure cell type is seeded on a biomaterial to be implanted, assessing seeding efficiency is a problem. Techniques using cells prelabeled with high energy isotopes (e.g., Indium-111-oxine) which can be noninvasively imaged hold promise for estimating seeding efficiency. Otherwise cells must be radiolabeled, seeded onto the biomaterial, then fixed, sectioned, and subjected to radioautography. Determining whether a healed prosthetic surface is made up of pre-seeded cells or cells which migrated in after implantation likewise

requires prelabeling of cells and radioautography after harvesting the prostheses.

VALIDITY OF SHORT-TERM TESTING FOR PREDICTING BIOMATERIAL PERFORMANCE

One of the major unresolved questions in biomaterials science, and particularly in blood-material interaction research, is what is the relationship between short-term or initial biointeraction events and the long-term performance of implanted biomaterials and devices. It is almost universally accepted that protein films formed on the surface of an implanted material during the first few seconds after contact with biological fluids will direct subsequent events in blood reaction and healing and will influence how a device will perform years "down the road." However, no one can relate the nature of the protein film formed in this initial period to that long-term response. Certainly there is strong evidence that adsorbed protein films can influence complex biological events. Also, the concept of the protein film influencing healing is intellectually satisfying since we know that the protein film, and not the surface of the implant, forms the interface between the implant and the body. Why, then, can we not predict how a device will perform based on the nature of the protein film formed? The future systematic exploration of this idea should be encouraged so that this hypothesis can either be laid to rest or can form the basis for new tools useful for understanding the performance of implants.

These relationships between short-term and long-term events are not limited to a consideration of adsorbed protein films. Measurements of platelet adhesion, white cell interaction, PT times, and PTT times are used to assess the performance of materials intended to contact the bloodstream. How well do these measurements actually predict the performance of implanted devices? In order to attribute any levels of significance to presently used blood-material interaction tests, research must be dedicated specifically towards answering these difficult questions.

SURFACE TOPOGRAPHY

Surface topography characterization (roughness, surface domain structure, etc.) and the related problems of fabric characterization remain major areas clearly in need of further study and standardization. To achieve reproducibility in experiments and in production of devices used in the clinic we must be able to express quantitatively not only a magnitude of roughness, but a "quality" of roughness. Most frequently, the only roughness characterization performed is microscopic imaging. This is, at best, only semiquantitative and is generally inadequate for precisely expressing the nature of the surface used. Similarly, for fabrics, one can measure a porosity in terms of liquid permeation, but this does not express the difference between two materials of equal water porosity, one with straight, parallel pores, and the other with tortuous, sponge-like pores. Attempts at more precise characterizations of rough surfaces are being made in the microelectronics area and in the technology related to precision machining and fabrication. In particular, lasers,

profilometers, image analysis and mathematical functions (correlation methods) are being used in attempts to describe complex surface topographies in a more quantitative manner. These tools should be applied to biomaterials as part of the routine characterization of materials. However, some exploratory studies are suggested in order to determine the applicability of the many new methods available to cardiovascular biomaterials and to assess the significance of each particular type of measurement with regard to its influence on biological reactions to the material.

REFERENCE MATERIALS

Reference materials are essential for the calibration of experiments and for performing meaningful interlaboratory comparisons. Much controversy in the literature can potentially be traced to research groups referring to materials by generic names. Although two groups could be using a material called "polyethylene," the two lots of material could, in fact, be quite different. An excellent start towards achieving some level of uniformity, consistency, and standardization has been made by the NHLBI with its introduction of poly(dimethyl siloxane) and polyethylene reference materials. However, both materials are relatively hydrophobic, and they are available only in one tube diameter and in sheet form. Thus, the needs of a large portion of the biomaterials research community may not be satisfied by these standards. The solution to this problem should be to have many more standard materials in many more sizes and geometries developed for biomedical researchers. This admirable course must be tempered with the realization that the development of reproducible, stable reference materials is technologically complex, and the costs associated with large quantities of materials in many geometries are high. Private industry may be helpful in resolving these problems. Great expertise often exists in companies dealing with specific classes of materials. With some developmental subsidy in exchange for reasonable pricing, the biomaterials community could make available to itself a range of appropriate reference materials.

TRAINING OF POLYMER SCIENTISTS AND ENGINEERS WITH INTERESTS IN BIOMATERIALS

The field of biomaterials is intrinsically a composite one. Traditionally, the field had its roots in the medical community. However, as materials, computer, and electronics technologies exploded, it became impossible for any one person in any one field to keep abreast of developments, let alone single-handedly apply these technologies to the problems in medicine. Now, developments in biomaterials depend on combined efforts among materials scientists, engineers, chemists, physicists, and physicians. Generally people with classical training in any one of the fields and a common interest in solving a biomaterial problem band together, teach each other the jargons of the various fields, and successfully, but sometimes painfully, make strides forward in the field. The "pain" usually comes in organizing and leading the group and in maintaining fruitful interaction among its members. Patience is a key virtue which must be exercised by all in the group

primarily because everyone in the group must go through a learning curve. From an educational point of view the question of shortening or eliminating this learning curve needs to be addressed.

Several universities currently offer a graduate curriculum in biomedical engineering. These curricula usually reside in portions of classical engineering and science departments as interdepartmental programs or in some cases in independent departments of biomedical engineering. The area of biomaterials is sometimes covered in these curricula, but more often it is not dealt with on a formal basis. Thus the question remains--what sort of formal training would be advantageous to a person, trained at the B.S. level as a classical scientist or engineer, who wants to do graduate research or prepare for a career in research and development in the biomaterials area? Put differently, what formal courses should be added to existing curricula to provide a good background for biomaterials research?

The scope of biomaterials does not yet warrant designing an entire graduate curriculum focused on biomaterials, but key areas for formal course requirements can be identified. By far the most prevalent types of biomaterials used today are polymers, both synthetic and natural. Four key courses, which usually already exist in polymer science and material science curricula, are organic synthetic polymer chemistry, polymer physical chemistry, mechanical properties of Polymers, and rheology and processing of polymers.

Interactions of materials with the body fluids and tissues are of course important. Desirable courses from this point of view would be cell biology, general biochemistry, and general physiology.

Lastly, it is important to have at least a qualitative knowledge of the structural engineering aspects of these materials when they must take on specific geometries and support specific loads and strains under specific environments for specific lengths of time. Toward this end, courses in strength of materials, statics, dynamics, viscoelasticity, and anisotropic mechanics would be useful.

It is clear that no one graduate program would likely require all of these courses, primarily because any one classical undergraduate curriculum does not adequately prepare a student to do graduate course work in all these areas. However, one or two survey courses in each area would be greatly beneficial. Training in the biomaterials area, whether it be formal course work or on-the-job exposure, is greatly augmented by people in one area patiently interacting with people in another area on a one-to-one basis. Chemists should learn some basic mechanics. Mechanical engineers should learn some basic polymer chemistry. Both should learn some basic physiology. Much of this exchange can take place informally on a day-to-day basis through seminars, casual dialogues, and recommended reading lists.

There is a shortage in industry and in academia of trained biomaterials scientists. This is particularly true for individuals with expertise in the areas of polymers for medical applications. As stated above, the number of institutions offering specific programs dealing both

with materials science and medicine is relatively small, and only a tiny fraction of these programs emphasize polymers. Three issues must be addressed. First, how many trained biomaterials scientists are needed? It is estimated that there are 8,000 companies in the U.S. developing and/or manufacturing devices utilizing biomaterials for implantation or other clinical application. If each company needed one trained biomaterials scientist on its staff, how many years would it take to graduate 8,000 individuals? Second, should there be a degree program and a specialty area called "biomaterials science?" Can we define a meaningful, rigorous, and unique biomaterials curriculum, or can our needs be met by traditional academic units? A number of committees are addressing these issues (e.g., the U.S. Society for Biomaterials Professional Development and Education Committee), and their deliberations over the next few years should be studied with care since the course of biomaterials training in the immediate future may well be set by their reports. Finally, how can we attract a larger number of motivated, intelligent individuals into the biomaterials field? Scholarships or postdoctoral fellowships offered by Federal agencies may be one means to provide to the health care industry and to academia individuals knowledgeable in biomaterials science. In general, the consideration of biomaterials science as a unique professional specialty has been given relatively little thought or study and represents a distinct gap area.

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Appendix 1

BLOOD HANDLING

The reliable laboratory evaluation of all components participating in the initiation and regulation of hemostatic mechanisms is highly dependent upon the appropriate collection and processing of the blood samples. While there are general principles regarding the handling of blood, those tests measuring activated platelets or activated plasma hemostatic proteins usually have specific requirements regarding sample preparation that necessitate rigorous adherence to protocols.

PLATELET EVALUATION

Platelet Counting

Blood for platelet counting should be collected in EDTA (lavender top tube) and mixed thoroughly. Platelet enumeration remains reliable for up to 18 hours if the sample is kept unagitated at room temperature. The concentration of platelets can be estimated by chamber (hand) counting using phase contrast microscopy or by electronic particle counting. Electronic particle enumeration increases precision owing to increased number of particles detected in short counting period. Most laboratories obtain platelet counts on whole blood, using electronic impedance or laser light scattering that are simple, rapid and also provide size distribution curves. For automatic procedures, results of individual determinations have a variance of + 2 to 3 percent. Reliable platelet counts will not be obtained below 50,000 per μ l without special care in handling particle-free material and using larger volumes of plasma to reduce dilution error. Platelet counts are unreliable when white cell counts are greatly elevated ($>100,000$ per μ l). Severe red cell fragmentation may produce an artificially increased count. Low counts may be more reliably assessed by the hand (phase) technique.

Platelet Aggregation

Blood may be collected from a forearm vein; the occlusion time of the vein should be minimized so that excessive stasis is avoided and the oxygenation of the blood and the pH are not significantly affected. In addition, care should be taken to avoid undue stress in the donor since it has been demonstrated frequently that platelets prepared from the blood of stressed individuals are hypersensitive to aggregating and release-inducing stimuli.

Blood may be collected by free flow into a tube containing an anticoagulant or directly into anticoagulant contained in a syringe. Either of these methods can be used successfully provided the blood is gently mixed with the anticoagulant as soon as possible. The blood should be mixed in a way that avoids frothing. Blood is kept at room temperature for aggregation studies. Depending upon the type of experiment to be done, blood can be collected into a chelating agent such as trisodium citrate or into heparin or hirudin.

Trisodium Citrate

One part of a 3.2 percent solution of trisodium citrate dihydrate is required for each 9 parts of blood. The trisodium citrate dihydrate solution is prepared by dissolving 3.8 g of trisodium citrate:2H₂O in 100 ml of distilled water.

Since platelet aggregation is dependent upon the Ca²⁺ concentration, it is important to maintain the citrate concentration constant. To correct for the variable plasma volumes, the normal 9:1 blood to citrate ratio should be modified by adding more or less anticoagulant as the hct deviates from 45 (table A1.1):

Table A1.1

Hematocrit	mls of 3.2% Sodium Citrate for 9 mls Blood
55	0.74
50	0.82
45	0.90
40	0.98
34	1.06
30	1.15
25	1.23
20	1.31
15	1.39

Heparin

Heparin (5-10 U/ml) can be used as an anticoagulant. Heparin is prepared by dissolving the powder in 0.154 mol/l (0.9 percent) saline. Since different batches of heparin and heparin from different sources tend to behave differently, preparations should be screened to determine whether they induce platelet activation in the absence of other stimuli. This can be examined by viewing a drop of blood beneath the microscope or by determining whether the heparin causes the platelet release reaction (this can be tested by adding heparin to gel filtered or washed platelets prelabeled with ¹⁴C-serotonin and measuring the amount of ¹⁴C that appears in the platelet suspending medium in the presence of heparin and comparing it with the amount of ¹⁴C-serotonin in the suspending medium in the absence of heparin).

Hirudin

The amount of hirudin used to anticoagulate blood varies among different preparations of hirudin. A range of 5-10 units per ml is usually sufficient to inhibit thrombin generation, although occasionally higher concentrations are required.

Acid-Citrate-Dextrose (ACD)

For the preparation of washed platelets, blood can be collected into a acid-citrate dextrose anticoagulant. This anticoagulant not only chelates calcium ions in the plasma but also serves to prevent platelet aggregation because the final pH of the whole blood is 6.5; platelets do not aggregate at this pH. Acid-citrate-dextrose is prepared by dissolving 25 gm of trisodium citrate:2H₂O, 15 gm of citric acid:H₂O, and 20 gm of dextrose in 1 liter of distilled water. The pH of this solution is approximately 4.5 and the osmolarity about 450 mosmol/l. By routinely measuring the pH of the solution and its osmolarity it may be possible to detect errors in the preparation of the solution. Use one part of ACD solution for six parts of blood.

Blood should be collected and stored in polyethylene or polycarbonate test tubes or siliconized glass tubes. There are several forms of silicone available for use in preparing siliconized glass surfaces. To avoid small micelles of silicone from becoming associated with the platelets it is preferable to use a preparation of silicone that binds irreversibly to the surface of the glass tube or container.

Platelet-rich plasma (PRP) can be prepared by two methods: 1) the red cells can be allowed to settle under the influence of gravity and the platelet-rich plasma can be transferred to a fresh, siliconized glass tube with a siliconized glass Pasteur pipette (again taking care to avoid frothing the plasma); or 2) the whole blood can be centrifuged (preferably in a centrifuge with a swing-out head) at 190 x g for 15 minutes for blood collected into trisodium citrate, acid citrate dextrose or hirudin. When blood is collected into heparin it should be centrifuged at a slightly lower g (100 x g) since a number of investigators have demonstrated that the platelet count of plasma obtained from blood centrifuged at higher g tends to be low and that platelets tend to be lost not only in the buffy coat, but also along the walls of the centrifuge tube, even when the tube has been freshly siliconized. The reason for this is not clear, but could be related to the properties of different batches of heparin, or to changes in the surface of the platelets.

If the platelet-rich plasma is to be used immediately for tests of platelet function, no special precautions need to be taken. However, for long experiments, or if the study cannot be undertaken immediately, special steps are necessary to maintain the pH of the plasma. This can be done by storing the platelet-rich plasma at 37°C in a sealed container under an atmosphere of 5 percent carbon dioxide.

Release of Platelet α -Granule Constituents

Platelet-specific α -granule proteins, generally PF4 and β TG, are readily released from platelets, and changes in their plasma levels can be sensitively detected by radioimmunoassay. Unfortunately, artifactual increases in plasma levels occur readily due to release during sample collection and processing. Thus, blood collection and processing require rigorous attention to detail in order to minimize the in vitro release of α -granule contents.

On the basis of a series of studies reliable results are obtained when blood is withdrawn through a 20-gauge "butterfly" into a 5-ml syringe precooled on ice containing 1 ml acid-citrate-dextrose (ACD, NIH Formula A), 30 mM acetylsalicylic acid (ASA), and 1 μ M prostaglandin E_1 (PGE_1) final concentration. The ASA is prepared in ethanol (180 mg/ml), and 80 μ l is added to the syringe immediately before drawing the blood. The PGE_1 is also prepared in ethanol (100 μ g/ml), and 10 μ l is added per ml of ACD immediately before drawing. The samples are mixed and maintained on ice for a minimum of 5 min and a maximum of 30 min prior to centrifugation at 48,000 g for 20 min at 4°C. One milliliter of plasma is immediately collected from below the lipid interface and above the cell pellet and stored frozen at -20°C until assayed. To remove adherent plasma proteins, the tubes to be used for centrifugation of the blood sample are cleaned by soaking them overnight in 2 M NaCl and a wetting agent (BRIT -35), brushed clean, and rinsed six times with tap water and six times with distilled water.

Blood samples in clinical studies are drawn in duplicate into pre-cooled syringes containing ACD-ASA- PGE_1 as described. If variable results are obtained between duplicates, the samples are drawn again and measurements repeated. Additionally, it is important, if multiple samples are to be collected from a patient, to collect the sample for platelet protein and fibrinopeptide determination first, because there is evidence that thrombin is generated at the site of an indwelling needle and this can cause artifactual increases in the levels of secreted proteins.

Other anticoagulant-antiplatelet mixtures for blood collection have been used successfully, e.g., chilled evacuated siliconized collection tubes to which has been added 1/10 volume of anticoagulant containing 1400 U/ml heparin (to inhibit thrombin), 1000 U/ml Trasylol (to inhibit plasmin), 0.01 mol/l adenosine and 0.02 mol/l theophylline (to inhibit platelet release) in 0.05 mol/l Hepes, 0.1 mol/l NaCl, 0.02% Na azide, pH 7.4. This anticoagulant mixture is suitable for measurement of platelet factor 4, β -thromboglobulin, fibrinopeptide A (the first product of thrombin cleavage for fibrinogen) and thrombin-increasable fibrinopeptide B (an early product of plasmin cleavage of fibrin I). After collection the blood tube is placed immediately on ice and maintained there for up to one hour before centrifugation.

Platelet Morphology

The following procedure is used to fix platelets for study in the electron microscope. Ordinarily the number of platelets present in 1 milliliter of platelet rich plasma is sufficient to make a good preparation.

Blood is collected by venipuncture as described previously into syringe containing acid citrate dextrose anticoagulant (nine parts blood to one part ACD) and maintained at room temperature. ACD is prepared according to the table.

Anticoagulant

1. 0.1 mol/l citric acid 10.508 g to 500 ml with distilled H₂O
2. 0.1 mol/l sodium citrate 2H₂O 14.706 g to 500 ml with distilled H₂O

To make the reagent:

35 ml 0.1 mol/l citric acid
465 ml 0.1 mol/l sodium citrate
12.25 grams dextrose (0.1 mol/l)
Add 5+ drops 10N NaOH (0.1 mol/l) to pH 6.5
Refrigerate

Platelets should be collected and separated from whole blood at room temperature (190 x g for 15 minutes) and maintained at 23°C. Initial fixation in glutaraldehyde should be carried out in this same temperature range.

Combine samples of PRP with an equal volume of 0.1 percent glutaraldehyde in saline buffer for 15 minutes at 37°C, centrifuge the sample to a pellet at room temperature, discard the supernatant and replace it with 3 percent glutaraldehyde in saline buffer.

Fixatives

1. 0.1% glutaraldehyde in saline buffer:
8.9 ml distilled H₂O
0.5 ml solution A
0.5 ml solution B
0.1 ml 10% glutaraldehyde
2. 3% glutaraldehyde in saline buffer:
6 ml distilled H₂O
0.5 ml solution A
0.5 ml solution B
3 ml 10% glutaraldehyde

Stock Solutions for Saline Buffer

Solution A.

14.0 g NaCl
0.75 g KCl
0.55 g MgSO₄
1.5 g Ca(NO₃)₂·4H₂O
Add distilled H₂O to 100 ml
Refrigerate

Solution B

1.1 g NaHCO_3
0.22 g $\text{Na}_2\text{HPO}_4 \cdot 7\text{H}_2\text{O}$
0.052 g KH_2PO_4 anhydrous
0.01 g phenol red
Add distilled H_2O to 100 ml
When dissolved at room temperature bubble CO_2 in for
about 15 seconds, if necessary. Mix well and pH to 7.4
Refrigerate

Fixation in 3 percent glutaraldehyde is continued either at room temperature or 4°C for 40-60 minutes. The supernatant is then discarded and the cells washed once with saline buffer and then combined with 1 percent osmic acid in saline buffer for 1 hour.

1% OsO_4 Prepare and keep in a dark brown glass-topped bottle
2.5 ml 4% OsO_4 aqueous
2.0 ml saline buffer
0.68 ml stock salt solution
2.0 ml 0.1N-HCl (0.1 mol/l)
2.8 ml distilled H_2O
Keeps one week in refrigerator

After exposure to osmic acid fixation at 4°C for one hour, the cells are dehydrated in a graded series of alcohol and embedded in Epon 812. Contrast of thin sections cut with a diamond knife on an ultra-microtome is usually enhanced with uranyl acetate and lead citrate.

EVALUATION OF COAGULATION AND FIBRINOLYSIS

Screening Tests

Anticoagulant

The standard anticoagulant used for screening tests of coagulation and fibrinolysis and for specific factor activity assays is 3.8 percent trisodium citrate (0.13M). Note that 3.2 percent trisodium citrate is used for platelet aggregation studies. Nine volumes of blood are added to one volume citrate. The citrate complexes the calcium ion by lowering the free Ca^{++} concentration to the point where clotting is interrupted. There are a number of citrate anticoagulant preparations available in the commercially prepared vacuum tubes (usually blue-topped). Care should be taken to use the same anticoagulant as in the normal control plasma against which the screening tests are standardized. EDTA is too strong a Ca^{++} chelator for the clotting screening tests and causes spurious results. In some situations, heparin inadvertently contaminates the sample. Reliable results from coagulation tests may still be obtained, however, by using a simple column or tablet adsorption to remove the heparin.

There is an optimal Ca^{++} concentration for these tests, and problems with excess or too little anticoagulant occur when the hematocrit deviates from the normal of 45, as with polycythemia or more severe anemia, respectively. To correct for the variable plasma volumes, the normal 9:1 blood to citrate ratio should be modified by adding more or less anticoagulant according to the hematocrit as described in the Table presented in the section related to platelet aggregation, except that the citrate concentration for the clotting screens is 3.8 percent (rather than the 3.2 percent used for platelet aggregation tests).

Collection

Blood for coagulation studies must be collected with particular care to avoid denaturation or activation, by drawing the blood gently into a plastic syringe from a clean venipuncture with minimal stasis. If there is difficulty with the venipuncture, blood flow, or improper mixing with the anticoagulant, the sample may be partially clotted, resulting in unreliable and misleading results. For vacuum collection tubes, the high flow-rate from filling may cause foaming and denaturation; blood should be transferred from a syringe to the tube by allowing the blood to run gently down the side of the tube. If the vacuum seal has leaked, the ratio of blood to anticoagulant can be altered.

The citrated sample is immediately placed on ice.

Processing Procedures

Citrated whole blood centrifuged at 4°C at 3,000 to 4,000 rpm for 10 minutes will yield platelet-poor plasma. This plasma is satisfactory for coagulation studies run within 4 hours. A higher centrifugal force (e.g., 40,000 g for 30 minutes) is required to remove enough of the remaining platelets prior to freezing the plasma so that it can be stored at -80°C before testing.

A variety of manual, semiautomated, and automated techniques are used to assess clot formation visually or by changes in electric conduction or optical density. Commercial reagents vary considerably, especially for the prothrombin and partial thromboplastin times.

Factor Assays

A deficiency of procoagulant or profibrinolytic factor or inhibitor protein activity may occur either because of impaired synthesis or because a variant of a molecule is synthesized deficient in biological activity. For example, in assays based on clotting activity, both types of abnormality will be evident as deficiency; dysfunctional proteins must be detected immunologically. Unfortunately, traditional activity assays are subject to many inaccuracies, since factor activity is estimated approximately by adding dilutions of citrated test plasma to known factor-deficient plasma and comparing the results with those obtained by the addition of normal citrated plasma. In such assays, plasma is prepared as for the screening clotting tests. In the future, activity measurement will probably be quantitatively determined using specific chromogenic substrates or similar indicators.

Activated Factor Products

In the detection of activated species, the technique of blood collection and processing is critical to avoid artifacts due to continued activation or nonspecific proteolysis. The published assay procedures usually involve a different inhibitor cocktail for each assay. For example, to measure fibrinopeptide A the plasma should be prepared from blood containing 140 U/ml heparin and 100 U/ml aprotinin, while other assay protocols recommend the addition of a different serine protease inhibitor phenylmethanesulfonyl fluoride (PMSF) 2mM. No universal inhibitory cocktail has yet been documented that is suitable for all assays.

Appendix 2

RECEIPT, PREPARATION, CARE, AND HANDLING OF SPECIMENS

INTRODUCTION

It has been well documented that the surface character of a material can strongly influence the course of biologic interactions with that material. However, as an unneeded complication for biomaterials investigations, it has been demonstrated that surfaces are also easily contaminated or altered. Therefore, in order to perform meaningful, reproducible blood-material interaction experiments on materials received for evaluation or prepared "in house," certain strict protocols should be adhered to to prevent contamination or alteration of the surface intended for evaluation. This section will briefly consider how specimens are fabricated, suggest protocols for handling and cleaning, and offer advice on maintaining biomaterials in a condition conducive to meaningful experimentation. Although the surface of a specimen is most subject to contamination, these protocols apply equally to maintaining the bulk of the specimen in an uncontaminated condition.

Three sets of protocols must be considered: specimen receipt and preparation for evaluation, specimen storage, and specimen handling. The general principles are relatively simple: open specimen containers containing clean specimens only in a dust-free (laminar flow hood) environment, clean specimens using well-thought-out protocols, handle specimens only with clean implements, store specimens in clean environments and, finally, utilize characterization tools described in other sections of this report to monitor specimen quality and handling protocols. In practice, the implementation of these ideas can be far from straightforward. This appendix is intended to offer suggestions which can make the appropriate handling of specimens simpler, and reduce the possibility of introducing contamination or alteration.

SPECIMEN FABRICATION

Many specimens for biomaterials investigations, as opposed to being received from other suppliers, are fabricated "in house" into a form appropriate for the experiments planned. The advantages of such "in house" fabrication include the ability to maintain total control of the composition and quality of the test specimen, and, in most cases, substantially reduced cost. The disadvantages stem from the realization that eventually dawns on most fabricators--that there is relatively little science and much art in material fabrication. Every materials system seems to have its own set of fabrication "tricks" that must be learned to achieve good quality specimens. This fabrication information is not easily ferreted from the vast scientific literature. The best source of information on each specimen preparation (both fabrication method and type of material must be specified) is often an individual experienced in that particular material fabrication. Although the subject of fabrication is simply too voluminous to even begin a useful discussion, it is

perhaps helpful just to list some of the commonly used techniques and annotate this list with comments specifically of importance to this appendix. This list is included as table A2.1.

UPON RECEIVING A SPECIMEN

Upon receipt of new specimens destined for blood or biologic interaction studies from an outside supplier, the following are important:

- Specimens should be removed for examination from sealed packaging only in a laminar flow hood or clean room environment.
- Adequate labeling and documentation should have been provided. Information to be requested from the supplier includes lot numbers, fabrication method, the nature of the materials used for fabrication, the presence of additives, method of sterilization if supplied sterile, and characterization data on the physical, chemical, biological and surface properties obtained by the manufacturer.
- Record a superficial description of the specimen upon receipt (e.g., "looks clean and homogeneous," "obvious particles on surface," "spotty, white areas on surface," etc.).

PARTICULATE CONTAMINATION

One of the most common types of contamination that can unintentionally be introduced at the time of initial receipt of a specimen is that represented by particulates becoming imbedded in, or strongly adherent to, surfaces. To minimize the incorporation of particulate impurities into surfaces, specimens should be prepared in a "clean" environment or subenvironment, regardless of the type of specimen involved or the type of blood interaction test intended. As a minimum precaution, the use of a standard laminar flow bench is recommended. This type of facility, if maintained properly, can provide a subenvironment having an airborne particle count on the order of $100/\text{cm}^3$. To achieve this low particulate level the laminar flow bench should be positioned in an area that has minimum amount of foot traffic and is sheltered from drafts and air currents. A "sticky" doormat of the type manufactured for clean rooms can be positioned at the entrance to the room in which the laminar flow hood is kept. These mats are disposable and low cost and have the potential to reduce one of the major sources of airborne particulates--particles from shoes. For a few key works on particulate control and detection see references 8-13 at the end of this section.

Particle contamination of surfaces can also occur by deposition from wash and rinse liquids. Liquids to be used for cleaning or hydrating biomaterials should be filtered through $0.2\ \mu\text{m}$ Teflon filters into bottles that have been washed with particle-free water and dried in a dust-free environment. The microelectronics industry has dealt with liquid-borne particles, and articles discussing particle filtration from liquids have been written (14,15).

SPECIMEN HANDLING

If specimens for blood and biological interaction studies must be handled within the clean subenvironment, their blood contact surfaces should be protected. This requires that these surfaces not be touched by bare hands. A fingerprint can easily be detected on a clean polyethylene surface by means of attenuated total reflectance infrared spectroscopy. Blood proteins can be expected to react differently with this oily contamination than with the surface itself. Thus, personnel performing work in a laminar flow bench should wear gloves of a type that have been shown to be free of plasticizer and have been rinsed externally to remove antistick powder often applied to their inner and outer surface by manufacturers. Although the airflow in the laminar flow bench is relatively particle-free, due to electrostatic charge buildup, particle transfer to specimens can easily occur from unrinsed gloves.

The rinsing of gloves can be accomplished with a 1:1 methanol/water mixture. This methanol/water mixture should be prepared and stored in a vessel that has never had contact with silicone oils or grease. In laboratories where concern for the nature of surface is not a serious consideration, it is common to find silicone lubricant contamination of glassware. The transfer of silicone monolayers will spread from item to item unless the silicone-contaminated glassware is quarantined.

The blood contact surface of specimens may be touched, if necessary, with forceps provided the force applied is controlled so that no imprints remain. Microimprints and microscratches are of potentially critical dimensions, i.e., the order of magnitude of the size of a platelet. Upon exposure to blood flow, platelets accumulate in the valleys and crevices, followed, more often than not, by aggregation.

The cleanliness of forceps made of even the most inert metals can not be taken for granted, especially when their tips are grooved. Traces of metal oxide particles located within the grooves can be transferred resulting in "hot spots" on the test surface. Thus, it is prudent practice to use either Teflon-tipped forceps, or to cover the tips of forceps with a heat-shrinkable Teflon tubing that, after being shrunk to size, is subjected to extraction in lukewarm 95 percent ethanol for several hours to remove migratable additives.

Consistent with the practices taken to minimize surface contamination related to particulates during specimen receipt, appropriate measures should be implemented in all further operations until a specimen is placed in contact with blood. Thus, consider that specimens made of polymeric materials are likely to develop surface charges. These charges can attract particulate contamination from the air and render it strongly adherent. This is the case in blood interaction tests performed in vivo when implantable specimens are exposed in a possibly sterile but not particulate-free atmosphere (e.g., to the surgeon's gloves). Since sterility and freedom from particulate contaminants are entirely different conditions, in implantation procedures, the same precautionary measures should be taken to prevent particle transfer from surgical gloves as those exercised during specimen preparation. The

gloves should be rinsed free of antistick powder with a 1:1 methanol/water mixture. A low-cost antistatic device emitting a low flux of ionizing radiation may also be useful in reducing static charge on specimens. Such devices are available from many laboratory supply houses.

Prehydration, prior to blood-material interaction experiments, can be important in many experiments. A disequilibrium state may be induced upon insertion of a dry surface into aqueous biologic media which could make reproducible results in the biointeraction studies more difficult to achieve. A 24 hour soak, for nonhydrogel materials, might be a reasonable starting point for developing a prehydration protocol. Hydrogels are highly hydrophilic materials, and, depending upon their thickness and chemistry, may need longer prehydration and equilibration periods. Appropriate prehydration and rehydration solutions should be sterile, isotonic and physiologically well tolerated. Prehydration can be performed at refrigerator (4°C) temperatures. The investigator should always be cognizant of the fact that ions or other substances in the prehydration solution may, themselves, induce artifacts. Control experiments are called for.

Depending on the nature of preparative procedures involved and the experiments to be performed, it may become necessary to subject the blood contact surface of test specimens to a superficial rinsing with a liquid that may be water, an aqueous solution, or an organic solvent such as alcohol. It is often a common practice to rinse off specimens prior to implantation. As a general rule, all rinsing liquids should be free of particulates. Saline used in preimplantation rinsings, should not only be sterile but also pyrogen and particulate free. Rinsing solutions free of particulate contaminants can be prepared in quantity by passing the solution through an appropriate filter having a mean nominal pore size of 0.5 μm into a clean bottle or vessel, with the filtration carried out within a clean subenvironment. Filters themselves are sometimes treated with surfactants. Extracts from filters can be examined for such contaminants. Teflon filters, though more expensive, can be thoroughly extracted and cleaned.

A simple method for delivering particle-free rinsing liquids utilizes plastic syringes fitted with self contained filter units. The filter unit attaches to the syringe with a standard Luer-lock fitting. Prior to the use of any brand of plastic syringe, determine whether the plunger elastomer or plunger lubricant is of the silicone type. The presence of silicones may result in the deposition of silicone monolayers similar to the cross-contamination phenomena discussed earlier. Syringes constructed only of polyethylene and polypropylene are now available. Clean glass syringes are also excellent.

Thorough washing of surfaces can introduce complex problems. Washing can, by itself, alter the chemistry of a surface by extracting surface-localized material and altering ionic distributions. Any washing (as opposed to more superficial rinsing or to prehydration) should be followed up with surface analysis and control studies to assess what changes, if any, might have been induced in the surface. Some washing or cleaning procedures which have been shown to be effective for certain

classes of materials are described below. For many of the procedures involving liquids, the effectiveness and/or speed of the cleaning can be enhanced by performing the cleaning procedures in an ultrasonic cleaner. Soxhlet and other extraction techniques can be used for removing leachable materials from the bulk of polymeric specimens. Such extraction can induce extensive morphological and chemical changes, however, and the material under treatment should be thoroughly recharacterized after extraction.

Glass and Ceramic Surfaces

Cleaning by the UV-ozone method (16) is perhaps the most effective method if the initial surface is relatively clean to start with. Soaking in a hot, concentrated nitric-sulfuric acid mixture followed by thorough rinsing can also be effective in cleaning surfaces of this type, although the ionic distribution at the surface can be altered by the treatment. Etching in an argon gas RF plasma discharge can be an excellent method to clean glass.

Metal Surfaces

The UV-ozone method (16) can be effective for cleaning metal surfaces although oxidation may be induced. RF plasma discharge methods can also be useful. Other cleaning procedures should be tailored to the specific metal.

Nonporous, Nonhydrogel Polymers

Sonication in a 0.1 percent solution of Ivory flakes followed by copious pure water rinses can be effective. Many other cleaning agents have the potential to contaminate surfaces or induce chemical or morphological changes in them. Solvents can be retained in the bulk of these materials or cause surface changes (e.g., crazing).

Porous Polymers or Hydrogels

These specimens can be most easily contaminated. Sonication in a pure, volatile organic solvent (e.g., methanol, ethanol or acetone) can be effective if it is established that these solvents do not damage the specimen.

These are only preliminary guidelines for specimen washing. Any protocol must be backed up with appropriate bulk and surface characterization to assay for induced surface changes, entrained solvent, or other contamination inadvertently introduced during the attempt at removing contamination.

The potential for entrained solvent is high for any polymeric or porous system which has been fabricated, rinsed or washed with organic solvents. Two methods might be effective in removing such solvent. Leaching the entrained solvent into a large volume of water or isotonic saline solution in which the specimen is soaked can be highly effective for solvent removal. Placing the specimen in a vacuum environment can eventually remove volatile, entrained solvent. For some polymer-solvent

combinations, however, this process can be slow. Using either technique, solvent removal rate will depend upon temperature, solvent type, material type, the size and geometry of the specimen, agitation and flow and possibly other factors. Thus, to err on the side of "excessively" extreme conditions for solvent removal is not unreasonable. UV measurement of leaching solutions is a good method for monitoring the transfer of UV-absorbing solvents into the aqueous phase. Other analytical protocols may have to be developed for non-UV absorbing solvents.

Reuse of specimens can result in special problems. Exposure to blood is known to leave an adsorbed layer of a mixture of plasma proteins on any surface. Some of these proteins may undergo partial elution, but most of them tenaciously resist quantitative removal. Polymeric specimens would be difficult or impossible to clean properly. Even if a particular type of specimen is suitable for glow-discharge treatment that might be used to etch off proteinaceous residues, the results obtained with this technique remain questionable, since the treatment may well induce other alterations in surface composition. Control experiments must be performed to ascertain the efficiency of protein removal techniques. If possible, avoid reusing devices which have contacted blood.

SPECIMEN STORAGE

Storage of specimens often presents difficult problems. Storage under vacuum in a clean chamber is generally a good technique for lengthy storage. A well-maintained backstreaming filter on rotary vacuum pumps will prevent migration of pump oils back to the specimen surface. Silicone vacuum grease should be scrupulously avoided in all parts of the vacuum storage system. Storage containers suitable for use in air or under vacuum and which have been found to be noncontaminating include small polyethylene envelopes sold for electron microscope applications, tissue culture dishes and tubes, and polypropylene cases designed for the microelectronics industry to store silicone wafers. Other containers may also be suitable, but appropriate experiments should be performed to assess the potential of the containers to induce contamination. Specimens stored wet have the greatest potential for contamination. Bacterial and fungal growth can be particularly insidious under wet storage conditions.

EVALUATION OF SPECIMEN HANDLING PROTOCOLS

In order to evaluate protocols for material handling, cleaning or storage, the following general procedure can be useful. Prepare a gold surface on a glass coverslip by vacuum deposition (most electron microscope facilities can readily fabricate this surface). Clean this surface using UV-ozone (16), hot acid or argon RF plasma immediately prior to use. Deposit on this surface extracts or solvents to be evaluated for potential to contaminate, or store the gold surface under the conditions planned for storage. Compare a "control" gold surface (e.g., stored in an acid-cleaned glass petri dish kept in the laboratory environment) with the specimen stored or treated under the conditions

of interest. An ESCA study of the control and experimental gold surfaces will indicate differences between the two. Differences greater than 10 percent or the addition to the experimental sample of any elements not present on the control are indicative of potentially undesirable changes or contamination.

CONCLUSION

The discussion presented above concerning specimen- and procedure-related factors capable of affecting blood-material interaction test results is only illustrative rather than complete. Depending on the nature of test specimens involved, the precautionary measures that are to be taken to protect the integrity of the blood contact surface can vary over a broad range. For example, there are procedural controls that are applicable to smooth-surfaced materials but may not be automatically transposed to the handling of textured surfaces or hydrogels. The illustrative enumeration of specimen-related factors given above suggest that even the simplest procedure may induce artifacts. However, the reproducibility of results from biological assays requires that attention be paid to a number of seemingly insignificant, often cumbersome details to minimize variation in the nature of the surface under study. Consultation with individuals familiar with each specific class of material is advised due to large variation in specific requirements for each type of material. For a few general references on specimen cleaning and contamination, see 2, 9, 15, and 17-25.

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Table A2.1--Commonly Used Fabrication Techniques for Preparing Specimens for Blood-Material Interaction Studies

<u>Method</u>	<u>Comments</u>
Extrusion	Well suited for fabricating many thermoplastic polymers into tubing, rods, or sheet, particularly in large quantity. Additives are sometimes needed in the polymers to allow easy extrusion. Surface oxidation can occur due to the high temperatures required.
Injection Molding	Excellent for preparing small complex parts, particularly in large quantity. Additives are often used in the polymers to assist with the fabrication. Mold release agents that can contaminate surfaces are also often used.
Machining and Polishing	Bulk materials can often be machined and polished to produce required configurations for testing. Machine oil contamination, abrasive contamination and surface oxidation are the major concerns.
Polymerization Molding	A useful method for making sheets or small parts from specialty monomers, particularly those that can be polymerized by free radical method or reaction injection molding (RIM) methods. Removal from the mold, optical quality and general homogeneity can present difficulties.
Solvent Casting or Coating	Excellent for fabrication where only small amounts of the desired surface material are available. Many problems can be encountered in achieving good coating quality (uniformity, appearance) and in adhesion to the casting substrate.
Surface Grafting and Immobilization	Can be used to prepare unique surfaces unobtainable by any other technique. Much technology and skill is usually required for such fabrication.
Vapor Deposition	Metals can be deposited as thin films on many surfaces using this technology. Coatings are not contiguous for very thin films. Thicker films can delaminate.

Appendix 3

REFERENCE MATERIALS

OBJECTIVES

The objective of the reference material program is to provide to the community of researchers investigating blood-material interactions with stable, consistent and well characterized materials which can be used for the following purposes:

1. To provide a basis for interlaboratory comparison of biological interaction results.
2. To calibrate new blood compatibility assays with older, more accepted assays.
3. To draw conclusions concerning the blood compatibility of polymer specimens.
4. To correlate surface energy or chemistry with various biological interactions.
5. To calibrate secondary standards prepared "in house."

It is accepted that the number of potential variables in biological interaction experiments is large. Using well designed reference materials, one of these variables (the nature of the surface) can, perhaps be held constant.

DESCRIPTION

The ideal reference material should have the following characteristics:

1. Identical to, or related to, commonly used biomaterials
2. Stable (surface and bulk)
3. Well-characterized (level I and level II tests)
4. Smooth
5. Available in many forms (sheet, tube, microsphere) to allow incorporation into a number of test systems
6. Available in large supply
7. Sterile or sterilizable

THE REFERENCE MATERIAL PROGRAM

The NHLBI has released two reference materials: low density polyethylene and unfilled poly(dimethyl siloxane). Both these polymers are available as sterile tubing (~3mm i.d.) or in sheet form. All materials have been exhaustively characterized and are periodically monitored to check stability. They are packaged to maintain purity, sterility, and surface quality and to provide unique code numbers for identifying individual pieces of material. A program has been set up to compile and coordinate research results obtained by biomedical investigators with these polymers. An effort to develop perfluorinated ethylene-propylene polymers as a primary reference material has not met with success to date.

The following characterization has been performed on NHLBI Reference polyethylene and poly(dimethyl siloxane):

1. Differential Scanning Calorimetry
2. Thermogravimetric analysis
3. Equilibrium swelling
4. Wettability (γ_c)
5. ESCA
6. Gel permeation chromatography (for the polyethylene)
7. Scanning electron microscopy
8. Infrared analysis

NHLBI reference polyethylene and poly(dimethylsiloxane) can be obtained by qualified biomedical researchers by writing to:

Reference Material Program Coordinator
Devices and Technology Branch
National Heart, Lung and Blood Institute
Federal Building, Room 312
Bethesda, Maryland 20205

RECOMMENDATIONS

In 1982, the Devices and Technology Branch of the NHLBI solicited suggestions for new primary reference materials. Table A3.1 lists many of these suggested materials and comments on their potential to meet the criteria suggested as desirable for primary reference materials.

Based upon the considerations in table A3.1, the following materials deserve additional consideration:

Polystyrene

There is strong biomedical interest in poly(styrene). It is consistent, easy to fabricate, and inexpensive. It is probable that this material could be produced rapidly in sheet and tube form and quickly gotten into the hands of interested investigators. Bead configuration would require considerably more work, but would still have a high probability of success.

Quartz

The high purity and availability of quartz are advantageous for reference material purposes. Fabrication would have to be entrusted to one of a limited number of companies and research groups that could handle this material. The cost might be high. Surface contamination is likely to be the most complex problem. Some contamination will always be present (1,2). Achieving consistent levels of the normal hydrocarbon contamination found on quartz surfaces may be difficult and would require further investigation.

Poly(Methyl Methacrylate)

Poly(methyl methacrylate) has had extensive implant application, particularly in the orthopedic and ophthalmologic areas. It is simple to work with and inexpensive. There are some potential problems concerning the true surface structure of this polymer (3), but these issues are, as yet, unresolved, and most experts would consider the surface structure relatively straightforward. Fabrication should be simple and rapid, except for bead form where a major development effort would have to be launched to insure surface consistency and purity.

Poly (HEMA)

There is extensive biomedical interest in poly(2-hydroxyethyl methacrylate) (Poly HEMA) and much controversy in the literature over its actual blood compatibility. It can be made in a highly pure form, but there is evidence that its surface structure changes in a "chameleon-like" manner in response to its environment (4-6). These surface structure rearrangements could make characterization more difficult. It can become readily contaminated from substances diffusing into its polymer matrix. Fabrication in the form of sheets is simple. Tubes are more difficult. Pure microspheres may be very difficult. Widespread biomedical application coupled with contradictory data on biological performance raises this material to a high priority for developing a pure, consistent, well characterized supply for the biomedical community to investigate. The extensive experience in fabrication, storage and sterilization accumulated by the contact lens industry might be put to good use for developing P(HEMA) as a reference material.

Poly(Ethylene Terephthalate)

The biomedical interest in poly(ethylene terephthalate) (Dacron fabric or Mylar sheet) is obvious since it is widely used in vascular prostheses. Surface structure as examined by ESCA for both fabric and

film, is remarkably consistent, but polymer chain orientation effects may be troublesome. The presence of low-molecular-weight oligomers, and processing, stabilization and appearance-enhancement additives may also present difficulties. This material could be rapidly developed as a standard in sheet form. Tubes should be significantly more difficult. Beads may be extremely difficult.

PTFE Teflon

Poly(tetrafluoroethylene) (PTFE) is used extensively in contact with blood, particularly in the form of an expanded, porous material (vascular graft). With proper handling and fabrication, nonporous PTFE could make a good standard. Achieving a smooth surface and a reproducible morphology may present the major problems for the fabrication of sheets and tubes. However, one can anticipate a stable, well-defined surface chemistry. Beads may be very difficult to prepare.

Polyurethanes

Although the complications involved in the development of reproducible polyurethane standards are great, high interest in this polymer as a blood contact material may stimulate reference material development. Since "polyurethane" is not the name for a single material but rather a broad class of materials with widely varying physical and chemical properties, more than one polyurethane standard might be considered. Polyurethanes containing polyether blocks would be most appropriate as most polyurethanes used in medicine are of this type. Particular problems for developing reference materials might arise due to low molecular weight polymer fractions, surface and bulk phase segregation, fabrication sensitivity, and contaminants introduced because of side-reactions (7). Hard segment-rich model polyurethanes, and polyurethanes purified by precipitation or extraction may be viable approaches to developing reproducible, stable polyurethane reference materials.

Table A3.1--Potential NHLBI Reference Materials

	<u>Purity</u>	<u>Consistency*</u>	<u>Availability</u>	<u>Fabrication</u>	<u>Steriliz.</u>	<u>Stability</u>
Polystyrene	Excellent	Good	Excellent	Simple	Simple	Excellent
Glass	Poor	Poor	Excellent	Complex	Simple	Fair
Quartz	Excellent	Fair	Excellent	Complex	Simple	Fair
PMMA	Excellent	Fair	Excellent	Simple	Simple	Excellent
P(HEMA)	Excellent	Fair-Poor	Excellent	Potential Problems	Simple	Good-Excellent
Graphite Carbon	Excellent	Good	Fair	Complex	Simple	Good
Stainless Steel	Fair-Poor	Poor	Excellent	Complex	Simple	Poor
Custom-Synthesized Polyurethanes	Fair	Poor	Excellent	Potential Problems	Potential Problems	Fair
Poly(Ethylene Terephthalate)	Fair	Good	Excellent	Potential Problems	Simple	Good
PTFE Teflon	Excellent	Excellent	Excellent	Complex	Simple	Excellent
Acrylic Copolymers	Good	Fair	Custom Synthesis	Potential Problems	Simple	Excellent
Poly Carbonate	Fair	Good	Excellent	Potential Problems	Simple	Good
Polyolefin Rubber	Fair	Good	Fair	Potential Problems	Simple	Good
Collagen	Variable	Poor	Good	Fair	Potential Problems	Fair

*Batch-to-batch consistency.

Appendix 4

TISSUE CULTURE METHODS FOR ENDOTHELIAL CELLS

In carrying out culture of any cell type it is essential to practice meticulous sterile technique. Working surfaces, supplies, and solutions must all be sterile. Details may be found in a tissue culture text (1).

1. The vessel to be used is carefully dissected from its surroundings - attempting to avoid stretching or clamping it. Also, avoid contact with external skin or hair of the donor. The vessel lumen is gently rinsed free of blood using sterile saline such as Dulbecco's phosphate buffered saline (PBS) (GIBCO). Antibiotics may be added to the PBS at twice the concentration used in complete medium (see below). The vessel is transported to a laboratory containing a sterile tissue culture area, preferably a laminar flow hood.
2. The vessel is dissected free of adherent tissue, branches are tied off and, depending on the wall thickness, it is used as a sack or everted over a rod. Proteolytic enzymes (collagenase and/or trypsin) are used to cleave the endothelial cells from the vessel wall. Type CLS collagenase (Worthington Diagnostic Systems, Inc.) at a concentration of 0.05-0.1 percent (depending on the units/mg) in PBS is commonly used. The time of incubation varies with temperature (room or 37°C) and batch. Ideally enzyme treatment is varied to achieve a maximum endothelial cell number with a minimum of smooth muscle cell contaminants. The cells can be scraped off the vessel wall mechanically, although most investigators believe this leads to more endothelial cell damage and contamination by underlying smooth muscle cells than does enzymatic extraction.
3. The harvested cells are transferred to a centrifuge tube and gently spun down (100 x g).
4. For endothelial seeding in prosthetic vessels (2), the cell pellet is dispersed into aliquots of the recipients' blood which is then used to preclot a graft using the four step method (3).
5. For endothelial cell culture (4), the cells are dispersed in complete tissue culture medium (e.g., Dulbecco's modification of Eagles' medium (GIBCO), with 10 percent fetal bovine serum (Sterile Systems), 15 mM Hepes buffer and 100 U/ml penicillin, 100 mcg/ml streptomycin and 50 U/ml Fungizone) and seeded into small (25 cm²) polystyrene tissue culture flasks or petri dishes. The cultures are incubated in a humidified atmosphere of 5 percent CO₂ in air at 37°C, and left undisturbed for at least 6 hours. The cells should begin to attach immediately, spread within 48 hours, and begin to divide by the third day

in culture, at which time they should be fed with fresh medium. When the cells are confluent, they are subcultured (removing them with 0.5 percent trypsin in 1:5000 EDTA, and diluting the suspended cells with fresh medium) into new flasks or on-to the prosthetic surface. They may then be implanted directly or allowed to grow and cover the surface in vitro prior to implant.

The above procedure is appropriate for endothelial cells which are relatively easy to maintain in culture--for example, bovine aortic endothelial cells. Most human endothelial cells do better with the addition of growth factors and/or heparin (5), and 20 percent fetal bovine serum, or possibly pooled human serum. The number of cells seeded in primary culture is a significant variable in determining the growth rate and success of the culture. The culture vessel should be varied accordingly. For example, a human saphenous vein 2 cm long is going to provide only a few cells and would best be seeded into one well of a Multiwell tissue culture plate (Falcon 3008), or a 35 mm petri dish.

Early consultation and support from investigators already expert in tissue culture of endothelial cells will greatly accelerate the learning process inherent in successfully applying the deceptively simple procedure given above.

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Appendix 5

CRITICAL SURFACE TENSION

An important requirement in biomaterials research is for an extremely sensitive and reliable method of monitoring the surface properties of coatings and substrates. Perhaps the fastest and most sensitive technique known for identifying changes in surface constitution of any material is based upon contact angle measurements. Using air-dried specimens, critical surface tensions are measured from the slowly advancing contact angles of drops of pure liquids of known surface tension.

Contact angle methods were developed extensively over the past four decades by Zisman and coworkers at the U.S. Naval Research Laboratory. A large body of reliable data was accumulated and a vast literature exists correlating contact angle data with surface properties of liquids and solids. The critical surface tension, γ_c , is a useful parameter which can be obtained by measurement of the contact angle (figure A5.3) of each of a series of homologous liquids on the test material and a graphical treatment of the contact angle data (figure A5.1). For most liquids, the values fall along the same straight line or cluster closely around it in a narrow rectilinear band. Other interpretations and treatments of contact angle data are being developed and used.

A most important consideration for screening materials for potential contact with blood or tissue is that the critical surface tension values are empirically related to the surface constitution. Even small changes in the outermost atomic layer are reflected in a change of critical surface tension, while other properties might remain essentially unchanged. For example, a simple hydrocarbon surface (like that of polyethylene) exhibits contact angles leading to a critical surface tension of about 31 dynes/cm. Gradual replacement of hydrogen atoms in the surface by fluorine atoms gradually decreases the critical values to 19 dynes/cm (as observed with polytetrafluoroethylene). Conversely, gradual replacement of surface hydrogen with chlorine atoms leads to an increased critical surface tension gradually approaching 41 dynes/cm (as with polyvinylchlorides).

Figure A5.3 defines the contact angle of a liquid on a solid surface. The angle is measured through the liquid phase so that a contact angle of zero degree ($\theta=0^\circ$) indicates that spontaneous spreading of the liquid takes place; larger equilibrium contact angles ($0^\circ<\theta<180^\circ$) indicate poorer, or incomplete, surface wetting by the liquid phase.

The critical surface tension, γ_c , of a polymer can be deduced from a Zisman plot, illustrated in figure A5.1. In a Zisman plot the cosines of the contact angles for a variety of pure liquids on a given solid are plotted against the liquid/vapor surface tensions of the liquids. It has been learned empirically that linear fit to the data points is most often obtained. Since the intercept at the $\cos \theta = 1$ axis is different for different surface, the intercept surface tension value is recorded as the critical surface tension for the solid in question.

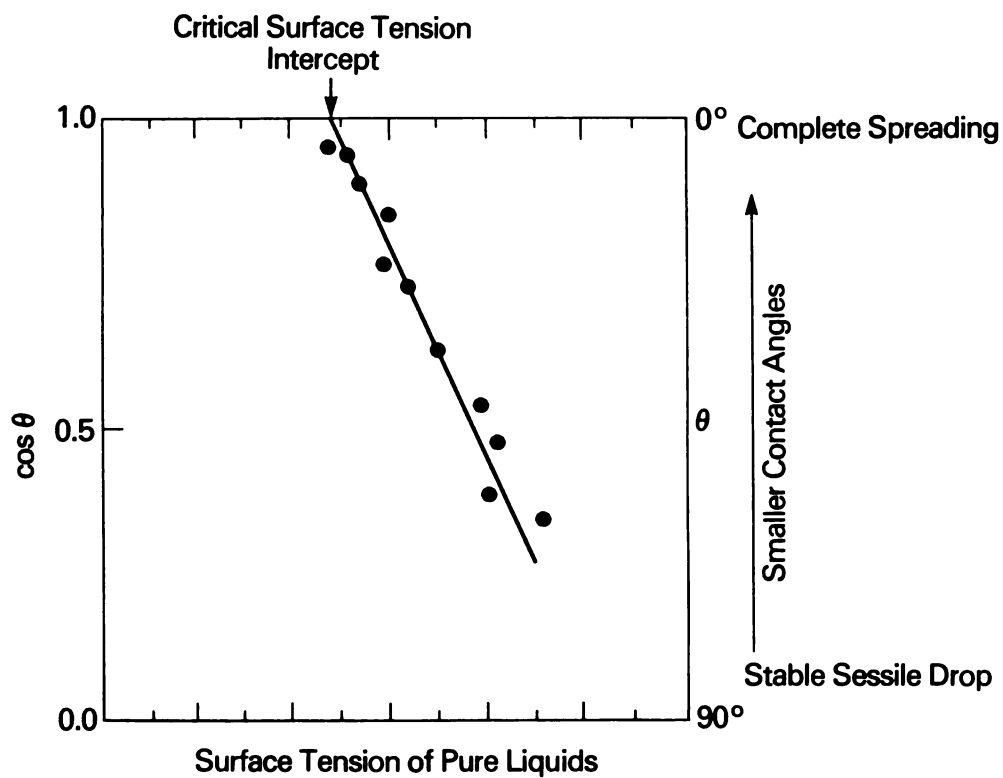


Figure A5.1--A representative Zisman plot.

Appendix 6

THE TEFLON TEST

Contact angle methods can also provide rapid and inexpensive answers to questions of surface cleanliness and stability. A simple, rapid method (the teflon test) can be used to detect the surface activity of minute amounts of organic contaminants, often encountered as coating compounds.

The following diagram (figure A6.1) is a standard plot of the contact angles obtained for a series of pure wetting liquids on Teflon. To perform the test for surface contaminants, a drop of pure liquid—for example, water—is placed on smooth Teflon FEP film; its contact angle is recorded—for water this would be about 110° ; the material having unknown surface properties is touched to the drop; if there are no transferable surface active constituents, the contact angle of the drop remains unchanged—a good test of the surface cleanliness is thereby completed; if surface active agents are present, the drop profile changes instantaneously to exhibit a new equilibrium contact angle reflecting its new surface tension; the surface tension which corresponds to the new contact angle can be deduced from the standard Teflon plot; the difference between the surface tension of the pure liquid and the surface tension of the contaminated liquid represents the surface activity (i.e., the spreading pressure) of the agent.

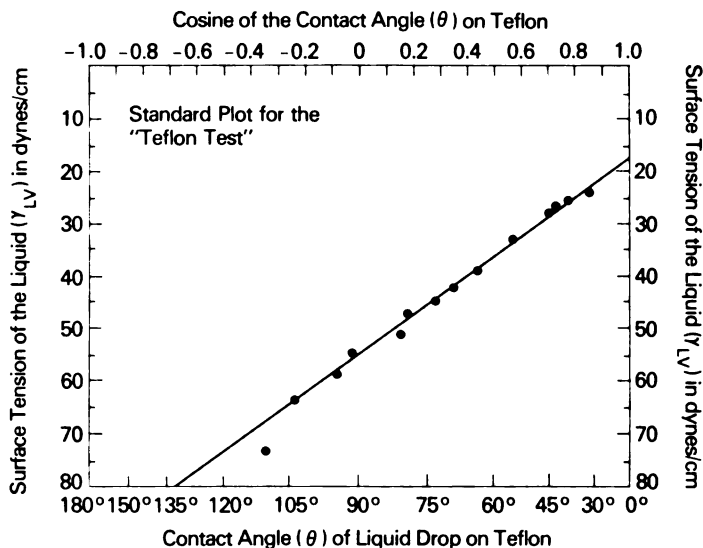


Figure A6.1--A standard plot for the Teflon test. The contact angles of Teflon with each of a series of pure liquids is presented. A change of the contact angle of Teflon with one of these liquids after the liquid has been exposed to a test material is indicative of the presence of leachable components in the test material.

Appendix 7

PROTOCOL FOR CONTACT ANGLE MEASUREMENTS AT THE POLYMER-WATER INTERFACE

Contact angles at the polymer-water interface are determined by immersing the sample in double distilled water and measuring the height and diameter of both air and octane bubbles in water (figure A7.1). In addition, dodecane is often used as a third liquid as a correlative check on the octane data.

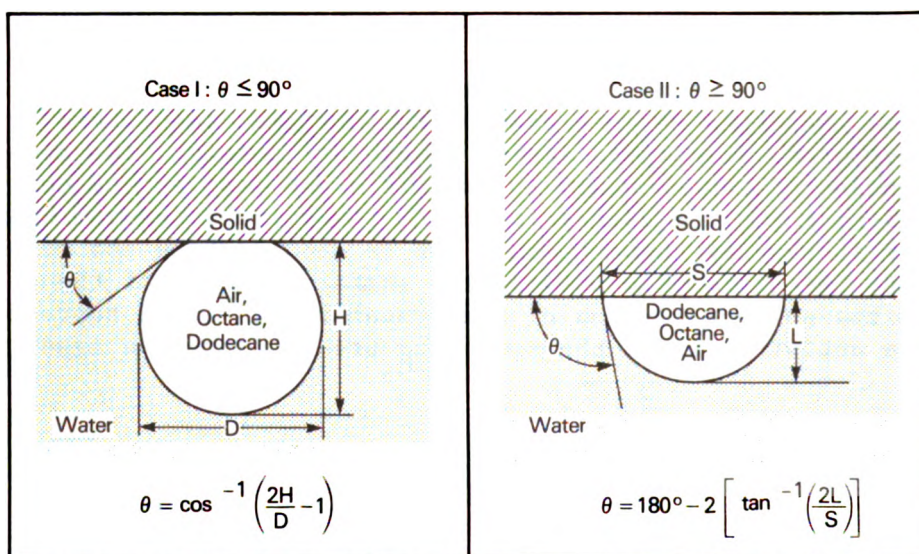


Figure A7.1--Measurement of contact angles at the polymer-water interface.

The bubbles of air or liquid are introduced to the immersed sample surface using a micro liter syringe equipped with a U-shaped needle. The highest purity octane and dodecane available (99 percent or better) should be used. The height and diameter are measured by use of micrometers which manipulate a stage holding the sample immersed in water. The bubbles are observed through a microscope using a 20x objective with a long working distance and a 15x eye-piece equipped with a fine crosshair reticle. The bubble is manipulated across the crosshairs and the dimensions are read directly from the micrometers. The sample box is back-illuminated by a variable light source. The bubble volume is approximately 0.1 to 0.2 μl . The bubble volume is minimized in order to avoid buoyancy effects. The bubbles are applied to the surface by forming a bubble at the tip of a microsyringe and then tapping the syringe body with the index finger to allow the bubble to dislodge and float from the needle tip up to the water-sample interface. Figure A7.1

schematically illustrates the geometry of the contact angle measurement. The contact angles are calculated using the equations.

$$\theta = 180^\circ - 2[\tan^{-1}(\frac{2L}{S})]$$

for angles greater than 90° and

$$\theta = \cos^{-1}(\frac{2H-1}{D})$$

for angles less than 90°

Appendix 8

SURFACE TEXTURE

The surface profile of a synthetic or engineering material can be described with three major components (see figure A8.1):

- The roughness or primary texture.
- The waviness or secondary texture.
- Longer term (lower frequency) errors of form.

Generally speaking, most surface metrification and observation methods readily identify the roughness and perhaps the waviness of a surface under study. The lowest frequency component, error of form, may have a wavelength so long that observing it makes roughness and perhaps waviness disappear by loss of resolution. The converse is, of course, equally true. Observations made using light microscopy and the scanning electron microscope suffer from this particular shortcoming. Thus, SEM and light microscope observations, while adequate (for some materials) in characterizing roughness and perhaps waviness, probably do not adequately address errors of form.

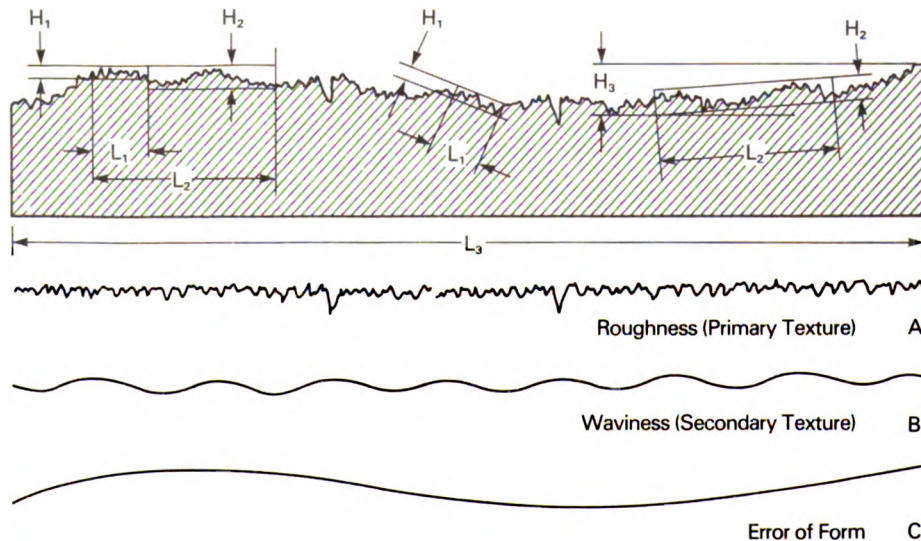


Figure A8.1--Major surface texture components.

Other important moments of surfaces are skewness and kurtosis, as discussed below and illustrated in figures A8.2 and A8.3. Since the relationship of roughness, waviness and error of form is complex, filtering operations must be used to separate these components one from another.

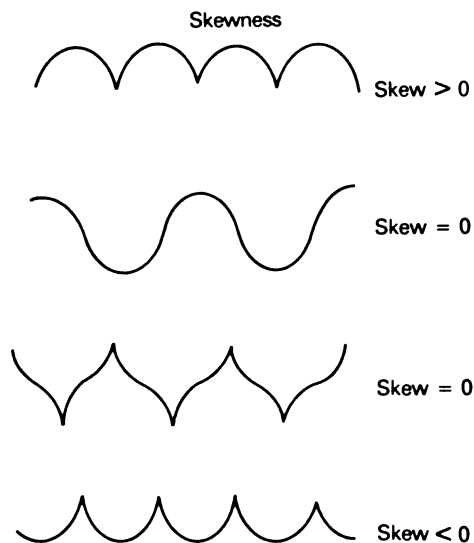


Figure A8.2--Skewness

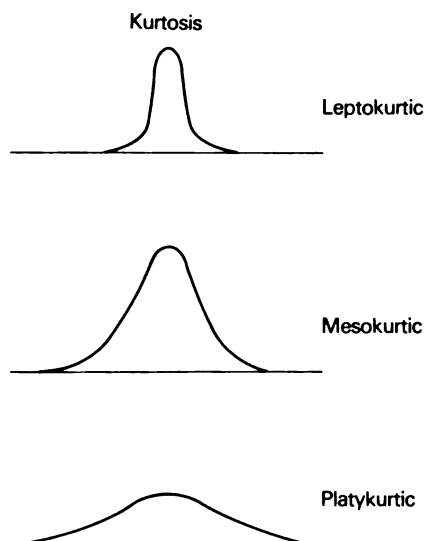


Figure A8.3--Kurtosis

It is obvious that sampling or cutoff lengths required to quantify roughness, waviness and error of form are vastly different, varying in direct proportion with some multiple of the component's frequency. While some of these components may be meaningful in characterizing a surface for some applications, it is not expected that error of form or perhaps waviness (if long-term) will be directly meaningful for application in characterizing the effect of surface roughness on blood interactions. The effect of surface roughness on thrombus formation is expected to be independent of error of form and, perhaps, waviness.

Thus, it is probably most useful to concentrate on moments and derivatives used to characterize the roughness of a specimen surface. Elements of roughness considered at this point to be potentially important to thrombus formation include frequency, lay, skewness and kurtosis, and maximum and minimum height value.

The frequency of the roughness is the number of peaks per unit length of the surface. Lay is the directionality of a surface with repeating textural units. Skewness is a measure of the lack of symmetry about the mean of the distribution of heights as illustrated in figure A8.2. Kurtosis, illustrated in figure A8.3, is an indication of the peakedness of a distribution. Relative to surface measurement, leptokurtic distributions are characteristic of surfaces in which heights tend to cluster near a mean level with only occasional extreme excursions. In platykurtic distributions, there is more of a tendency toward frequent extreme swings from high to low elevations, with relatively little "dwell" near the center line.

Various measures and derivatives thereof are used to characterize surface geometries. There are currently 10 common parameters used in characterizing a material's roughness:

1. Arithmetic average roughness.
2. Root-mean-square roughness.
3. Maximum peak-valley roughness.
4. Ten-point height.
5. Average peak-valley roughness.
6. Average spacing of roughness peaks.
7. Swedish height of irregularities.
8. Bearing length ratio.
9. Leveling depth.
10. Waviness height.

The above listing includes most parameters used in industrialized nations to describe surface textures.

Arithmetic average roughness (AA) is computed using the formula

$$AA \text{ or } R_a = \left(\frac{1}{N} \sum_{i=1}^N |y_i - \bar{y}| \right) / N$$

where y_i is the i^{th} height value, N is the total number of data points, and

$$\bar{y} = \left(\frac{1}{N} \sum_{i=1}^N y_i \right) / N, \text{ the mean.}$$

Root-mean-square roughness (RMS) may be defined thus:

$$RMS = \left[\left(\frac{1}{N} \sum_{i=1}^N (y_i - \bar{y})^2 \right) / N \right]^{1/2}$$

The frequency of roughness elements or peaks may be defined by computing the average spacing (A_r) of roughness elements or peaks. If A_{rn+1} is the spacing between peak n and peak $n+1$, then

$$A_r = \left(\sum_{i=1}^n A_{ri} \right) / N$$

yields the average distance between the peaks or the frequency.

The moment of a distribution are defined as expected values weighted according to the probability of encounter in a random process. The first moment of a distribution is the mean ($\bar{y}$, see above). The second moment is the variance and measures the dispersion of a distribution. Skewness and kurtosis are two other important functions or moments. Skewness may be expressed as the nondimensional ratio.

$$\text{Skewness} = \frac{M_3}{M_2^{3/2}}$$

where

$$M_3 \text{ (the third moment)} = \frac{1}{N} \sum_{i=1}^N (y_i - \bar{y})^3$$

and

$$M_2 \text{ (the second moment)} = \frac{1}{N} \sum_{i=1}^N (y_i - \bar{y})^2$$

Kurtosis is expressed by

$$\text{Kurtosis} = \frac{M_4}{M_2^2}$$

where

$$M_4 \text{ (the fourth moment)} = \frac{1}{N} \sum_{i=1}^N (y_i - \bar{y})^4$$

and M_2 is the variance or second moment.

Thus, from simple measurements amenable to digitization, a number of parameters may be defined that are potentially useful in describing the texture-based contribution to blood-material interactions.

Reference Standards

When there are needs to investigate a material's surface textural contribution to thrombus formation under varying conditions of flow, experiments designed to accomplish this task should involve at first a screening set to determine whether differences can be detected when blood or blood fractions encounter grossly differing surface roughnesses. More subtle effects, springing from differences in lay, skewness and kurtosis, should be reserved for later study in experiments using closely characterized surfaces. In addition to differences in skewness, inversion of skewness should be examined by using mirror image replicas.

To aid in such work, there are readily available replicas of surface standards in a variety of roughness. In the surface metrology terms introduced in the preceding pages, the available surface standards may be described as

honed--linear lay; skewness >0 ; leptokurtic;
ground--linear lay; skewness ~ 0 ; meso- to platykurtic;
Blanchard--lay is angular in both directions;
 skewness ~ 0 ; meso- to platykurtic;
shaped (turned)--highly directional lay; skewness = 0;
 leptokurtic;
profiled--lay is angular in both directions; skewness ~ 0 ;
 lepto- to mesokurtic;
side or end milled--circular lay; skewness ~ 0 : lepto- to
 mesokurtic;
periphery milled--linear lay; skewness ~ 0 ; lepto- to
 mesokurtic;
cast finishes--no lay; skewness ~ 0 ; lepto- to mesokurtic;
grit blasted surfaces--no lay; skewness ~ 0 ; lepto- to
 mesokurtic;
shot blasted surfaces--no lay; skewness ~ 0 ; lepto- to
 mesokurtic; and
EDM surfaces--no lay; skewness >0 ; meso- to platykurtic.

Surface roughness standards are commercially available. The standards from one manufacturer are electroformed nickel replicas of original manufacturing surfaces. Typical replicas supplied are

Machined surfaces including honed, lapped or polished, ground with periphery of wheel, ground with flat side of wheel (Blanchard), shaped or turned (these are equivalent), side milled, end milled, or profiled, and milled with periphery of cutter. These specimens include a high degree of lay and hence represent a repeating surface.

Cast finishes in roughnesses equivalent to die castings ($\sim 20 \mu$ in. RMS) through normal ferrous green sand castings ($560\text{--}920 \mu$ in. RMS).

Grit blasted surfaces ($32\text{--}1,000 \mu$ in.).

Shot blasted surfaces ($32\text{--}1,000 \mu$ in.).

Electrical discharge machined surfaces (16-250 μ in.).

The above standards except the machined surfaces, do not have lay and therefore are completely random.

Appendix 9

MEASUREMENTS OF SURFACE ELECTRICAL PROPERTIES

Electrical parameters of solid surfaces may eventually be correlated with the earliest phases of blood-material interactions. The electrical parameters that are described here are

- 1) Surface charge
- 2) Surface potential difference
- 3) Rest potential
- 4) Zeta potential (streaming potential).

Most solid surfaces under study allow these electrical parameters to be measured before and after exposure to blood or other biological phases.

SURFACE CHARGE

Isolated surfaces can be electrostatically charged by a number of methods. Electrification by contact, rubbing, or through permanently charged groups are all possible. Although the origin of this charge is of interest, this question will not be further addressed here.

In the case of conductors, this charge is easily dissipated. On the other hand, insulating surfaces remain charged for long periods of time. A simple apparatus can be fabricated to determine the magnitude of the charge on various materials. The required apparatus is essentially a Faraday cup which is electrically isolated from a metal container in which it is centrally located. A sketch of one such device is shown (figure A9.1).

The inner cup shown is connected electrically to an electrometer using a very short piece of coaxial cable. The shielded part of the cable connects to the outer shield of the assembly and to the ground post of the electrometer.

When a charged insulator (the specimen) is placed on the hook attached to the inner cover and the specimen is lowered into the Faraday cup (this cup being previously grounded using a grounding button on the assembly) it will charge the inner cup by induction. On the other hand, if the specimen is a conductor, it will charge the Faraday cup by contact. The charge transferred, or induced on the inner cup, results in a voltage, V , which is measured with the electrometer.

The expression relating the charge to the voltage is $Q = CV$ where

Q is the charge in coulombs on the specimen
 C is the total capacitance in Farads
 V is the measured voltage in volts.

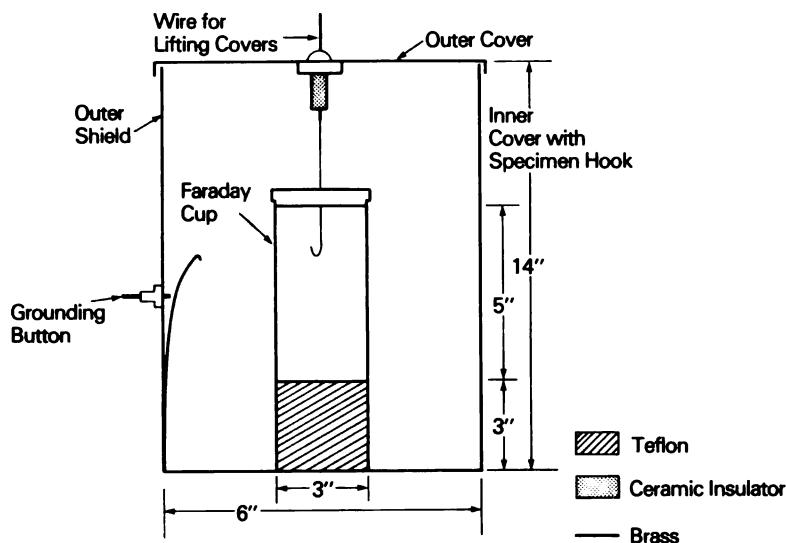


Figure A9.1--Faraday cup for measurement of surface charge.

SURFACE POTENTIAL DIFFERENCE

The key to understanding the information which is obtained from a measurement of the surface potential difference is in the word, difference. This means that the voltage between two phases is measured and then, after introducing a change to one of the phases (e.g., adsorption of a protein), this voltage is remeasured and the difference between this and the first value is recorded. Since the measured voltage is developed entirely in the space between two phases, none of the uncertainties of the phase boundary potentials is involved.

Consider the surface potential between two clean dissimilar metal surfaces as determined using an ionizing electrode technique described below. The voltage recorded in this case is due to the difference in work function between the two metal surfaces. If a film is placed on one of the metals, it will result in a different measured potential. The difference in potential between the clean and the film-coated metals has been related to the electrostatic dipole moments (both permanent and induced) in the film. It has been found that even nonpolar materials exhibit a surface potential difference due to induced polarization of the film by the electrostatic field generated by the metal on which the film is placed. In the case of the surface potential between a metal and a liquid, a similar potential is developed when a film is spread on the liquid. In this case, there may be a contribution to the surface potential difference due to a reorientation of the liquid phase dipoles as a result of an interaction between the film and liquid molecules. Although surface potential measurements have often been used in the investigation of the orientation of molecules spread on a liquid by relating the measured electrostatic dipole moment and with computed values deduced from molecular structure, the unknown contribution of the restructured liquid phase to the surface potential difference makes the results difficult to interpret.

Consider the surface potential difference, V , to result from the displacement of charge as in an electric dipole, μ , leads to the equation attributed to Helmholtz, which is

$$\Delta V = 4\pi\mu n$$

where

ΔV is the surface potential difference in volts absolute

n is the number of molecules per cm^2

μ is the electrostatic dipole in Debyes.

This equation can be modified to the more convenient

ΔV in millivolts

A as the area per molecule in $\text{\AA}^2/\text{molecule}$

μ as the electrostatic dipole in milli-Debyes, to give

$$\Delta V = 12\pi\mu/A$$

Actually, μ is the only component of the electrostatic dipole perpendicular to the surface. It can be resolved into two parts, $\mu_p + \mu_i$, in which μ_p is the permanent part of the dipole and μ_i is the induced part. Using this expression for μ in the above equation gives

$$\Delta V = 12\pi(\mu_p + \mu_i)/A$$

In this expression, the two parts of the electrostatic dipole term correspond to their components perpendicular to the surface of the film. Hence, a dipole orientation term (i.e., the angle of the dipoles relative to the surface) must be known in order to use ΔV for a determination of overall molecular orientation.

Up to this point, the film of material either at an air/liquid interface or adsorbed on a metal has been considered electrically neutral. This is not always the case, since some films contain ionized groups which will also contribute to the measured surface potential difference. This has been included in the theory by adding to the above equation a term, Ψ , which represents the potential difference across the film because of the double layer for surfaces, films on solutions, and ionized groups within the films. With this term incorporated, the expression for ΔV is

$$\Delta V = 12\pi(\mu_p + \mu_i)/A + \Psi$$

This equation, in principle, can be used to gain information on the orientation of the molecules within a film, provided the kind and strength of the electrostatic dipoles present and the packing density of the molecules are known, and some estimate of Ψ can be made.

A typical instrument used to determine the surface potential of a metallic plate utilizes a sealed, gold-coated alpha particle radiation

source to reduce the impedance between the test material and the measuring electrode. Therefore, the surface potential recorded is the potential of the metal with respect to the gold surface of the radiation electrode. When the ionizing electrode used is an American 241 sealed source with a mean alpha particle range of 2 cm in air, supplied as a thin foil soldered to the head of a #6-32 bolt, the sensitive area for surface potential measurements is approximately the full area of the bolt head, or 0.635 cm diameter.

The measuring apparatus typically utilizes a null detector scheme to determine the surface potential. This is done by applying a voltage opposite in sign but equal in magnitude to the surface potential to be determined. Under this condition, a high impedance voltmeter connected between the radiation source and the specimen will be at null. An instrument to accomplish this measurement is illustrated schematically in figure A9.2. With this instrument the surface potential measurement is fairly insensitive to small changes in the separation between the radiation electrode and the specimen. The reading on the digital voltmeter is equal and opposite in sign to the surface potential between the specimen and the radiation electrode.

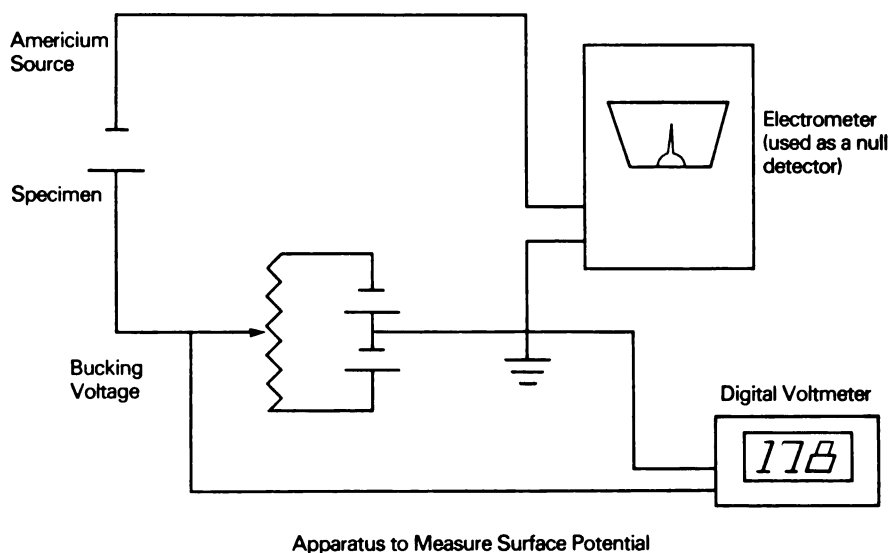


Figure A9.2--Apparatus to measure surface potential.

REST POTENTIAL

Any material in contact with an electrolyte has at its solid/liquid interface a potential determining layer (made up of desolvated ions chemically adsorbed to the surface) and a double layer which consists of a dense part (Helmholtz or Stern layer) and a diffuse part (Gouy layer). These separate regions of charge are indicated in figure A9.3 for a surface placed in an electrolyte.

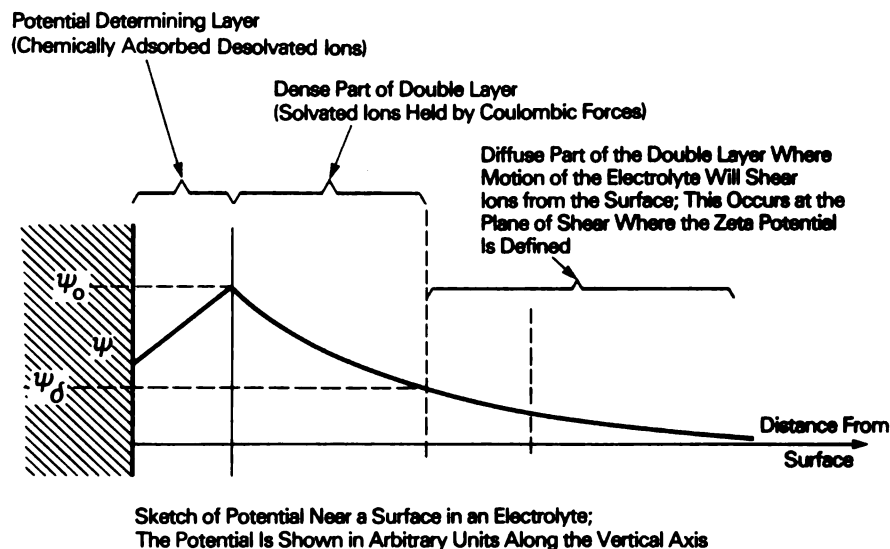
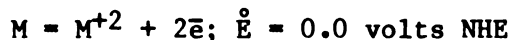


Figure A9.3--Changes in potential near the surface of an electrolyte.

The potential-determining layer can make the potential, ψ_0 , positive or negative and depends on the specific potential-determining ions present. The case where the potential is increased is shown in figure A9.3. The electrical potential outside the layer of potential-determining ions always decreases. The ions in this layer are believed to be electrostatically adsorbed and only partially solvated. This reduces the potential ψ_0 to ψ_δ . Outside of this layer is the diffuse part of the double layer consisting of positive and negative ions with a greater number having a sign opposite to the potential ψ_δ . The plane of shear within the diffuse part of the double layer is where the zeta potential is defined.

Rest potential as discussed here is the potential of an electrochemical cell consisting of the material of interest in contact with an electrolyte, referenced to a Ag/AgCl electrode. The rest potential results from a corrosion reaction of the test material in the electrolyte. In many cases, the reaction is not reversible and there is a loss of material from the test material's surface. The discussion which follows gives a measure of the magnitude of the effect which would be observed under equilibrium conditions. Any electrochemical cell under study can usually be divided conceptually into two half cells. The first half cell may be considered to be the test material electrode in contact with the electrolyte. The second half cell is then the Ag/AgCl electrode in contact with the same electrolyte. Where the electrolyte contacts the test material's surface, an ionic double layer is set up from the material's surface and extends into the electrolyte. If the material is negatively charged, as proposed in many biomedical development efforts, cations from the solution are attracted to its surface. The concentration of these cations near the surface (called the Helmholtz layer) differs from that in the bulk solution. Outside of this layer is the diffuse layer which usually contains the probable plane of shear. The material surface in contact with the electrolyte, then, is considered to undergo an anodic oxidation which releases electrons to the bulk material. These electrons give rise to

the potential measured as the rest potential. The half cell reaction for a semiconductor in salt solution, for example, might be



where $\overset{\circ}{E}$ is the standard electrode potential measured with respect to the normal hydrogen electrode, NHE.

The concentration of metal ions in solution as a result of this anodic oxidation can be calculated using the familiar Nernst equation. This equation relates the electrode potential E to the equilibrium concentration of a metal ion M^{+z} in solution. This equation is

$$E_{M^{+z}} = \overset{\circ}{E}_{M^{+z}} + \frac{.06}{z} \log M^{+z}$$

where z is the number of electrons exchanged. $E_{M^{+z}}$ is the standard electrode potential. The half cell potential for the Ag/AgCl measuring electrode can also be calculated from the Nernst equation. In this case, the cell potential $E_{Ag/AgCl}$ is equal to its standard electrode potential $\overset{\circ}{E}_{Ag/AgCl}$ if the reference electrode is saturated with Cl ions. The cell potential (or the rest potential) developed between the test material and the Ag/AgCl measuring electrode, when both are in contact with the same buffered salt solution, is given by the difference between these half cell potentials, i.e.,

$$E_{\text{cell}} = \overset{\circ}{E}_{Ag/AgCl} - [\overset{\circ}{E}_{M^{+z}} + \frac{.06}{z} \log M^{+z}].$$

It is often desired to relate the change in the rest potential, or corrosion potential, of a solid surface to the adsorption of protein from solution on that surface or to other basic blood-material interactions. The measurement of the rest potential is very simple and was one of the first techniques used to correlate the position of a metal in the electromotive series with its apparent blood reactivity. Some investigators have suggested that metals more negative than +300 mv with respect to the normal hydrogen electrode (wrt NHE) would be thromboresistant when implanted in the blood stream, but others that the rest potential of several metals does not appear to correlate with their blood compatibility.

Rest potentials are easily measured in normal saline or other electrolytes by making the specimens one electrode in a Josephson bridge and with a saturated calomel electrode as the other (reference) electrode as shown in figure A9.4.

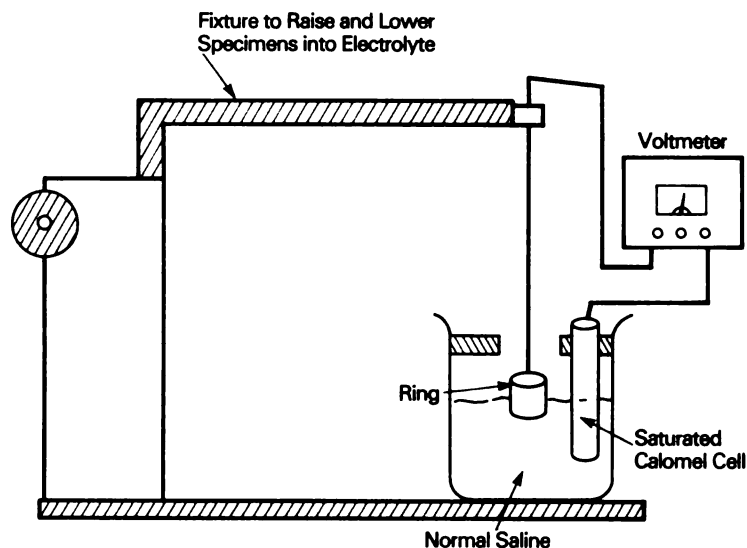


Figure A9.4--Apparatus for measurement of rest potential.

The rest potential of a material (surface) is best measured using a unity gain high impedance amplifier connected to a digital voltmeter. This electrical arrangement should give an input impedance of $10^{12} \Omega$ for voltage measurements. The unity gain high impedance amplifier thus assures that the electrochemical potential under study is not disturbed electrically, since the measurement current is extremely small.

The rest potential should be recorded immediately after submersion of the specimen in the saline solution and then again about 10 minutes later. It is interesting to note that the initial rest potential, in most instances, will be different from the value obtained after 10 minutes. In fact, the rest potential of many material fails to level off to a stable value even after 60 minutes in contact with a conductive solution. No evolution of gas should be noted during any of the measurements.

ZETA POTENTIAL (STREAMING POTENTIAL)

In order to determine the zeta potentials of materials, it is possible to either make electrophoretic mobility measurements on small particles ($2 \mu\text{m}$ diameter) or to flow an electrolyte solution through a capillary made of the same material. The first technique measures the movement of the particles in an electric field and the second measures either the streaming potential or streaming current to determine the zeta potential of the surface. Neither the use of small particles nor capillaries is suitable for the complete list of materials of interest, but rectangular flow cells can be assembled using two flat plates of such materials substituting for the capillary geometry.

In streaming potential experiments, the plane of shear locates the region from which ions are displaced by the flowing electrolyte. The distance of this plane from the layer of potential-determining ions is given by the Debye-Huckel theory, which expresses this distance (called

the Debye length, D) as the effective thickness of the diffuse double layer. The expression (requiring rationalized MKS units for its evaluation) for the Debye length relates the term, K^2 , where $D = K/2$, and other physical parameters describing the electrolyte contacting the surface:

$$K^2 = \frac{e^2}{\epsilon KT} \sum_{i=1}^n n_i z_i^2$$

where e is the electron charge = 1.6×10^{-19} coulomb
 k is the Boltzman constant = 1.38×10^{-23} joule/ K°
 T is the absolute temperature
 ϵ is the permittivity for the solution
 n_i is the number ions of the i th type per unit volume
 z_i is the valence of the i th ion.

This expression can be rewritten in terms of the ionic strength of the electrolyte, c . During this and evaluating the constant terms in this expression at $21^\circ C$ gives

$$K = 0.322 \times 10^{10} \sqrt{c} / m^{-1}$$

Converting to angstroms and taking the reciprocal gives the Debye length,

$$D = \frac{30}{\sqrt{c}}$$

It is usually assumed that the surface in contact with an electrolyte influences which ions are released from the plane of shear due to electrolyte motion near the surface. If molecular species are present in the electrolyte, then these can adsorb within the potential-determining layer or within the dense part of the double layer and alter the distribution of ions in the diffuse layer such that a new potential at the plane of shear is established. The number of ions in the diffuse layer which may be released when the electrolyte flows past the surface depends on the concentration of the electrolyte. Dilute solutions allow release of greater numbers of ions than do concentrated electrolyte solutions.

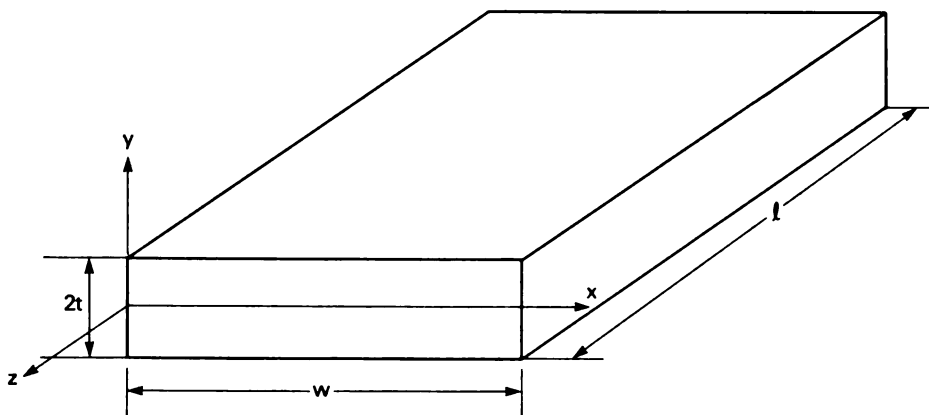
The potential at the plane of shear, the zeta potential, can be determined from a measurement of the streaming current or streaming potential which results when an electrolyte flows past the surface. The streaming potential is often orders of magnitude greater than the streaming current at physiologic salt concentrations; hence it is more readily accessible to measurement. Although techniques have been reported which allow the streaming current to be accurately measured, these techniques involve the sinusoidal flow of the electrolyte through

a capillary and thus are not generally suitable for research involving the adsorption of proteins. The solutions make multiple passes through the capillary rather than the single pass desired in such experiments.

As described by many authors, certain requirements must be met in order to use the streaming potential to measure the zeta potential of a surface. These requirements are

1. The flow must be laminar, and
2. The flow path must have a surface-to-surface gap which is large compared to the Debye length.

The derivation of the equation for the zeta potential starts with a description of the streaming current which comes about due to the ions displaced from the surface by a moving electrolyte. Consider a rectangular flow cell having the dimensions shown in figure A9.5.



Sketch of Rectangular Flow Cell (w , Width of Cell;
 $2t$, Thickness of Gap; and l , Length of Cell)

Figure A9.5--Rectangular flowcell.

The velocity profile between the two parallel flat plates is given by

$$v = \frac{P}{2\eta_0 l} (t^2 - y^2)$$

where

- v is the linear velocity of flow of the liquid in the gap
- y is the distance from the center of the gap
- P is the pressure drop across the entire length of flow
- η is the viscosity of the electrolyte
- l is the length of the flow cell
- t is half the thickness of the gap

This equation indicates that the velocity profile is parabolic in the y direction of the flow cell gap. The width of the gap, w , is taken to be large compared to the thickness, $2t$, so the flow is approximately uniform in the x direction. Ideally, there would be an extra equation

to describe the velocity profile in the x direction; however, for simplicity, this will not be considered in this development.

The velocity gradient very close to the flat flow cell walls is given by

$$\frac{dv}{dy} \Big|_{y=t} = -\frac{Pt}{\eta_0 l}$$

If the Debye length, s , is much less than the thickness of the gap, $2t$, then the velocity of the electrolyte at the plane of shear, measured from the solid wall is given by

$$v = \int_0^s \left(\frac{dv}{ds} \right) ds = \frac{Pts}{\eta_0 l}$$

where a change of variable has been made with $s = t - y$. The streaming current I_{str} is a product of the charge carried by the moving electrolyte times the velocity of this electrolyte at the plane of shear. Taking the charge in a thin laminar layer, $2w$, near the walls of the rectangular flow cell, then the streaming current is given by

$$I_{str} = 2w \int_0^t \rho v \, ds$$

In this equation, ρ is the charge density in the electrolyte. This charge density is related to the potential, Ψ , in the Poisson equation:

$$\nabla^2 \Psi = -\frac{4\pi\rho}{D}$$

Substituting the expression for v and Ψ (in one dimension) in the equation for the streaming current gives

$$I_{str} = -2w \int_0^t \frac{D}{4\pi} \frac{d^2\Psi}{ds^2} \frac{Pts}{\eta_0 l} \, ds$$

The assumption is made that D and η_0 are constants equal to their values in the bulk solution. This simplifies the streaming current equation to

$$I_{str} = - \frac{Pwt}{2\pi} \frac{D}{\eta_0 l} \int_0^t \frac{d^2\psi}{ds^2} s \, ds.$$

The integral can be integrated by parts using the relation

$$\int \xi \, dz = \xi z - \int z \, d\xi$$

with

$$\xi = s \quad \text{and} \quad dz = \frac{d^2\psi}{ds^2} \, ds$$

Having done this, the following expressions are derived:

$$\int_0^t \frac{d^2\psi}{ds^2} s \, ds = s \frac{d\psi}{ds} \Big|_0^t - \int_0^t \frac{d\psi}{ds} \, ds = s \frac{d\psi}{ds} \Big|_0^t - \psi \Big|_0^t$$

Considering the potential to be short ranged, it is usually safer to assume that the potential and its gradient are zero at the center of the flow cell. These boundary conditions are

$$\frac{d\psi}{ds} \Big|_{s=t} = 0 \quad \text{and} \quad \psi \Big|_{s=t} = 0$$

Under these conditions, the expression for the streaming current, rewritten below

$$I_{str} = - \frac{PwtD}{2\pi\eta_0 l} \left\{ \frac{sd\psi}{ds} \Big|_0^t - \psi \Big|_0^t \right\},$$

is reduced to the following expression where the potential $\psi|_{s=0}$ is taken to be the potential at the plane of shear or the well-known zeta potential, ζ .

$$I_{str} = - \frac{PwtD}{2\pi l \eta_0} \zeta$$

This expression for the streaming current indicates that for a negative zeta potential (this does not mean a negative surface potential, because potential determining ions on the surface could alter the potential away from the solid surface), the current in the direction of flow would be positive. That is, it would consist of positive counter ions displaced from the plan of shear by the moving electrolyte. This would

make the streaming potential, as discussed below, positive in the direction of fluid flow. The reverse would be true for a positive zeta potential.

This streaming current gives rise to a streaming potential E_{str} . I_{str} is balanced by a back current I_b conducting the charge released from the shear plane back through the electrolyte and the flow cell itself. When there is appreciable conduction through the bulk of the flow cell, because the flow cell is constructed of conductive materials, the equations which are developed must explicitly include this bulk conductivity. Under steady state conditions, the streaming current and this back current exactly balance. Therefore, the following expression is obtained:

$$\frac{P\omega D}{2\pi\eta_0 l} \zeta = E_{str} \frac{2\omega t}{l} \left[K_e + \frac{lK_s}{t} \right]$$

K_e = Conductivity of electrolyte

K_s = Conductivity of flow cell

Solving for ζ in this equation, we find

$$\zeta = \frac{4\pi\eta_0}{D} \frac{E_{str}}{P} \left[K_e + \frac{lK_s}{t} \right]$$

In the case where $K_s = 0$, the expression derived is

$$\zeta = 4\pi \frac{E_{str}}{P} \frac{\eta_0}{D} K_e ,$$

which is identical to that given in many texts for a simple circular glass capillary.

ELECTRODES

The electrodes recommended for most electrochemical characterizations of biomaterials are Ag/AgCl systems encased in an agar salt bridge. The salt bridge is placed in contact with flowing electrolytes, thereby protecting the electrodes from exposure to the solution and also reducing the possibility of their drying out between experiments.

During the preparation of such electrodes, silver wire is inserted through an insulating sleeve and then brushed with steel wool to remove the surface oxide. At the same time, a plate of silver, approximately 1" x 2", is also cleaned with steel wool. This silver plate along with a spare piece of silver wire is then placed in a 0.1 N HCl solution. The silver plate is then made positive and a potential applied between it and the silver wire such that 5 ma flows. After 5 minutes, the silver plate and the silver wire are removed and the silver plate which

is now black is once again cleaned with steel wool. The operation just described is to clean the plating solution prior to making the Ag/AgCl electrodes.

With the silver plate and all the silver wires to be made into Ag/AgCl electrodes rinsed free of any steel wool, the silver wires and plate are mounted in the beaker of cleaned electroplating solution. In the case where multiple electrodes are to be used, all silver wires to be plated should be connected together and plated simultaneously by making them positive with respect to their silver plate. A useful plating current is 3 ma, applied for 3 minutes. The Ag/AgCl electrodes so formed are bluish purple in color.

Prior to removing the electrodes from the plating bath, the potential between them should be checked with a high impedance volt meter. Freshly made electrodes typically have a voltage difference of only a few tenths of a millivolt between them. At this point, the electrodes are susceptible to damage if exposed to air. Therefore, all solutions for forming the salt bridge should also be prepared before removing the electrodes for insertion into the electrode holders.

A typical agar salt bridge is made by placing 0.75 gm of purified agar in a beaker containing 25 ml of boiling water. This beaker is within another beaker of boiling water, thereby forming a double boiler arrangement. When the agar is fully dissolved, a measured quantity of KCl is added to form a 0.1 M KCl solution. This mixture is stirred until it is clear. A heated syringe with a fine needle is then filled with this prepared solution and the solution is injected into the space between one of the Ag/AgCl electrodes which has been inserted into the electrode holder and the inside of the electrode holder. (The Ag/AgCl electrodes should be briefly rinsed in distilled water prior to insertion into the electrode holders.) All electrodes should be prepared in the same way, being careful to place the syringe filled with the prepared agar in hot water to prevent the agar from solidifying.

Electrodes prepared in this way when tested in a saturated KCl solution exhibit very small potential differences between them. (This is typically only tenths of a millivolt.)

Appendix 10

VIBRATING ELECTRODE SYSTEM FOR MEASUREMENT OF SURFACE PROPERTIES

Measurement of the surface charge and surface potential of bio-materials will aid in improving the reproducibility of the samples. It will help identify batch variations, and surface contamination, as well as the intrinsic surface potential of various biomaterials. There is no proven relationship between the potential of a surface in air, and the potential of this surface in blood; however, there may be a useful correlation between the surface potential in air and the subsequent observed blood compatibility. Collecting data for various biocompatible materials (and noncompatible materials) may add another factor for comparison with blood compatibility.

A device for measurement of surface charge and potential has been built based upon the Kelvin technique of vibrating an electrode perpendicular to the test surface. If the surface potential and the electrode potential are the same, then the electrode is vibrating in a field-free region and no electrical signal is generated. If there is a potential difference then there will be a signal whose phase depends upon the sign of the surface charge and surface potential.

A nulling voltage, V_B , is applied to the test material and adjusted to give an A.C. null on the vibrating electrode. For insulating materials, the back side of the sample is coated with a conducting paint, and then bonded to a conducting substance which is connected to the nulling voltage.

The relationship between the surface charge density, σ (in coulombs /m²), and the nulling voltage, V_B (volts), is given by

$$\sigma = \frac{K \epsilon_0 V_B}{d}$$

where K is the dielectric constant of the sample

ϵ_0 is the permittivity of free space
[= 8.854×10^{-12} Farad/m (MKS)]

d is the insulator sample thickness (in meters).

As an example, the average charge separation corresponding to a surface potential of -0.1 volts can be calculated for polyurethane.

The dielectric constant reported for various polyurethanes ranges from 5.4 to 7.6; a mean of 6.5 will be used for these calculations. Dielectric constants for polypropylene range from 2.2 to 2.6, while values for polyethylene range from 2.25 to 2.35.

Assume a polyurethane thickness of 4 mils (1.02×10^{-4} meter) and $V_B = 0.1$ volt

then $\sigma(\text{coulomb/m}^2) = (0.1)(6.5)(8.854 \times 10^{-2})/(1.02 \times 10^{-4})$

and $\sigma(\text{coulomb/m}^2) = 5.56 \times 10^{-8}$

or $\sigma(\text{coulomb/cm}^2) = 5.56 \times 10^{-12}$.

The charge of an electron is 1.602×10^{-19} coulomb; therefore the surface density of electrons is $n = 3.54 \times 10^7$ (cm^{-2}).

The average spacing between electrons of the surface charge is $n^{-1/2}$ and is 1.682×10^{-4} cm or 1.682 microns, for the case of 0.1 volt surface potential on 4 mil polyurethane and for the assumptions made.

The operational block diagram of the surface charge measurement system is shown in figure A10.1. The relative surface potential as seen by the vibrating measuring electrode is processed by a high impedance preamplifier which also serves as a filter turned to the frequency of vibration. This low level signal is further amplified and filtered to produce a sine wave which contains all information needed for synchronous detection.

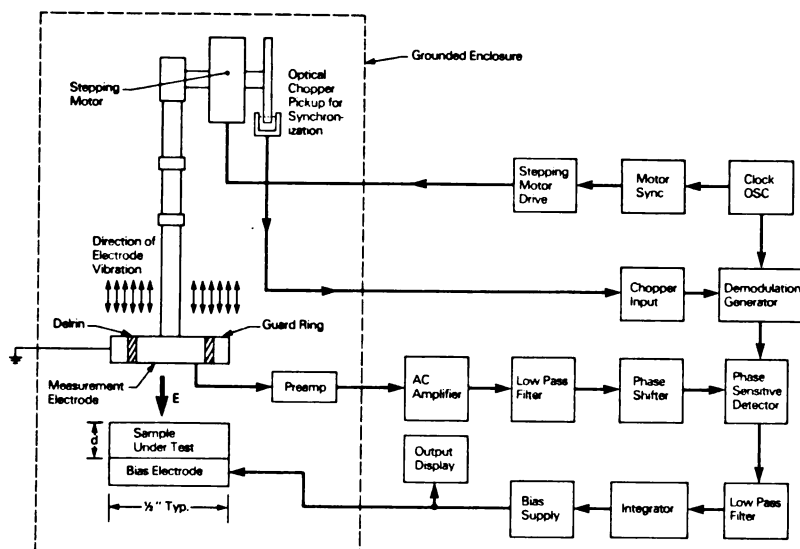


Figure A10.1--Instrumentation for measuring surface charge.

Meanwhile, an optical chopper coupled to the stepping motor generates a phase locked square wave for synchronous demodulation. After demodulation, the resultant output is further processed and is fed to a bipolar integrator. The integrator, which is driven either positive or negative in response to the vibrating electrode, controls the back bias potential applied to the specimen under test. When the vibrating electrode senses no net surface charge, its output drops to zero causing the integrator to hold at its present level. In this way,

the surface potential of the specimen under test is automatically brought to a null state.

This automatic nulling system should be of particular value in determining the state of surface charge on slowly varying surfaces. Resolutions of 10 mV have been achieved, corresponding to a surface charge sensitivity of 10^{13} coulombs/cm² for most polymeric surfaces of about 4 mil thickness.

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